

A SCIENCE FICTION NOVEL

THE ORCHARD OF UNCHOSEN YEARS

They gave the orchard their futures.

Some futures learned to answer.



TAPIO KINNUNEN

The Orchard of Unchosen Years

Every world needs weather. Nacre's was grown from the lives its people did not choose.

Tapio Kinnunen

Act I — The Orchard Answers

Chapter 01 — The Ring That Answered

Kera Vonn had the cutting arm halfway through Tree 44 when the wood knocked back.

She stopped the blade with her thumb still pressed to the deadman plate. A pale curl of yarrowglass clung to the arm's teeth, translucent enough to catch the amber service lamp and divide it into thin bands. Beyond the maintenance court, the orchard stood in orderly rows under Nacre's pearl-gray night. Wind moved through its crown without moving the leaves. White conductors did not stir unless a current told them to.

The knock came again through the trunk.

Three taps. A pause. Two taps. Another pause, longer this time, as if whoever held the other end had lost confidence.

Kera lifted her hand from the plate. The cutting arm withdrew on its spring and left a clean crescent in the newest ring. Its status panel continued to show an ordinary sampling cycle: tissue density within range, conductive humidity within range, adaptation load negligible. A fault ought to have declared itself in red. Nothing on the panel admitted that the tree had made a sound.

At midnight, East Root Court belonged to maintenance crews and auditors who preferred not to meet them. Kera had chosen the shift because old trees carried less civic traffic after the evening weather correction. Tree 44 had been reporting a seventeen-minute lag between its physical ring clock and the root lattice. The delay appeared once every nine days, too regular for thermal drift and too slight to disturb the models. Her assignment called for a wafer cut, a local clock reset, and a notation under instrument noise.

No line in the assignment mentioned knocking.

She unhooked the contact horn from her belt and set its brass mouth against the trunk. The horn amplified conduction without feeding current back into the wood. At first it gave her only the orchard's working murmur: pressure figures passing toward the western baffles, moisture corrections returning from the moss beds, thousands of short comparisons settling like grit in a pipe.

Under that traffic, something waited.

Kera shifted the horn a finger's width clockwise. Three taps reached her jaw through the brass. Two followed. This time a single tap came after the long pause.

Her breath touched the inside of her face shield. She made herself count the intervals rather than listen to their shape. Point eight seconds between the first three. Two point one before the pair. Four point seven before the last. The sequence could have come from a cracked conductive vein closing under the arm's pressure. Yarrowglass carried stress in pulses. Old auditors had built entire superstitions from badly seated tools.

She released the horn, reset its sensitivity, and placed it against a section of untouched bark.

Silence held for eleven seconds.

Then: three, two, one.

When Kera and Niko had been children, they had slept on opposite sides of a partition too thin to stop arguments and too thick for whispers. Three knocks meant Are you awake? Two meant I am now. One meant Stay. They had invented the exchange during their mother's night rotations, when the pressure doors sounded larger without an adult in the quarters.

The memory entered without permission. Niko at twelve, tapping with the metal end of a shoe brush. Niko at sixteen, pretending the code embarrassed him but answering anyway. Niko fever-bright behind clinic glass, his fingers too weak to reach the frame on his final night.

Kera removed the horn so quickly its rim scraped the bark.

"No," she said.

The orchard offered no argument. A pressure notice passed through the trunk and made the newest leaves brighten along their veins. Somewhere north of the court, a condensation gate changed angle. The system remained occupied with weather, as it had been before Niko died and after Kera gave fourteen unchosen years to help keep it occupied.

She opened the audit console on her wrist. A clean procedure waited there in numbered steps. Confirm clock divergence. Cut radial sample. Compare local timestamp to lattice receipt. Replace damaged tissue seal. The instructions steadied her because they did not care what she remembered.

Kera returned the cutting arm to the crescent. This time she advanced it by hand, one gear tooth at a time. Yarrowglass resisted like dense wax, then gave with a faint crystalline complaint. The arm extracted a wafer no wider than her palm and lowered it into the insulated tray.

Thirty-eight rings crossed the sample. Most were narrow and clear. Storm years had laid down cloudy bands; civic yielding seasons left threads of amber where new possibility models entered the lattice. Near the outer edge, one ring carried a hairline of white that doubled back on itself instead of closing.

She angled the tray beneath the service lamp. The doubled line resembled a path corrected after the wood had already committed to growth.

A normal ring could not revise its own circumference.

Her wrist console chimed. Sample removed. Log disposition.

Kera muted it.

The wafer's embedded clock needed no tree connection. She clipped a reader across its edges and watched local events populate the screen. Weather traffic filled the first columns. Timestamp, instruction, tissue response, lattice receipt. Every ninth-day anomaly appeared in the record, each one exactly seventeen minutes wide. During those windows, the wafer had recorded conductive activity that the central lattice never acknowledged.

She selected the most recent event.

The reader rendered a field of gray marks, each representing a tiny change in resistance. At ordinary scale they looked like static. Kera compressed the interval by a factor of twelve, and the marks gathered into pulses.

Three. Two. One.

Her gloves made the reader controls clumsy. She pulled the right one off with her teeth and chose the previous anomaly. The same pattern appeared, though the final pause ran half a second longer. Nine days earlier: three, two, one. Eighteen days: three, two, one.

The sequence extended back eleven months.

Kera sat on the low root guard. Cold from the yarrowglass passed through her coveralls. Eleven months of knocking meant nothing by itself. The orchard trained on human decision models; fragments of habits could surface in optimization noise. Every yielded future began with a citizen's memories, language, witnessed relationships, and medical record. A childhood code could persist as data without anyone waiting behind it.

That explanation should have reduced her pulse. Instead it gave the pattern a place to stand.

She searched the wafer for source labels. Public adaptation channels bore ordinary tags: humidity, pressure, fungal resistance, heat exchange. The anomalous interval carried no source, only a deprecated routing number from the orchard's first architecture. Kera recognized the prefix because auditors learned obsolete systems the way physicians learned childhood diseases.

Continuity storage.

The old label referred to temporary model state held across related simulations. Officially, no state persisted long enough to become experience. The orchard retained a decision for efficiency, not a decider. Yielding certificates stated that distinction in language polished by forty-seven years of civic use.

Kera had signed one at nineteen.

She remembered the room more clearly than she liked: cream walls, a terracotta seal, the recorder asking how many modeled years she wished to

contribute. Fourteen had felt severe enough to honor Niko and modest enough to remain rational. Her possible futures had been mapped from that day forward, branching through professions, marriages, migrations, injuries, children, refusals. The orchard would use their patterns and discard their detail. Nothing inside those futures would wake. Nothing could object to a gift because no recipient of pain existed within it.

Her auditor training had treated that last sentence as infrastructure. In the first week, trainees learned to distinguish a retained preference from a retained person. A preference could cross cycles because the orchard found it useful. A fear could recur because avoidance improved survival estimates. Even a remembered face could persist as an index for social behavior. None of those survivals, the instructors said, created a witness behind the data. Continuity required a self that could locate yesterday, revise tomorrow, and recognize another mind as something more than input.

During the final examination, Kera had defended the distinction before three recorders. They gave her a simulated anomaly that begged not to be reset. Its plea drew phrases from the technician assigned to test it, recombining his own worry into a convincing voice. Kera found the feedback loop and ordered sanitation. The instructors praised her for refusing theatrical evidence. She had carried that praise into every audit since.

Tree 44 offered no theatrical plea. It had waited eleven months with six small knocks and no audience guaranteed to understand them.

At least one modeled future must have included Niko's survival. His infection had been survivable in later clinic projections. The possibility model would need to preserve his childhood habits to estimate Kera's decisions around him.

Again came three. Two. One.

Static could inherit a knock. It could not know when to use it.

Kera stood and walked around Tree 44. Her lamp cast the trunk's shadow across the court, a dark bar broken by root housings. On the north side, she opened an old maintenance port with the square key from her belt. Two fiber loops and a physical pulse jack sat inside. The pulse jack existed for testing line continuity without introducing linguistic data. If she used it, the audit record would preserve only a generic service impulse.

Her hand hovered over the jack.

Auditors did not test a suspected pattern with private memory. Personal entanglement required reassignment. The rule protected evidence from wishful interpretation and auditors from evidence that knew their names.

Kera inserted the lead.

The manual sender offered a single pressure key. She tapped four times, evenly spaced.

No answer came.

She waited through thirty seconds of orchard murmur, then sent two taps followed by four. A childhood variant, meaningless to anyone outside their quarters. Still nothing.

Relief loosened something beneath her ribs. The original pattern could have been a stored fragment replaying on schedule. Her intervention had not changed it. She could log an inherited mnemonic artifact, recommend an isolation sweep, and go home before the first shift opened the tram lines.

The tree struck the contact horn where it hung from her belt.

Four taps.

Kera did not move. A pressure update crossed the bark in a faint blue flare. After it passed, the trunk delivered two taps and four.

Her exact sequences, returned in order.

She pulled the lead from the jack. "A mirror," she whispered. "You could be a mirror."

The answer arrived as three, two, one.

That did not refute her. A responsive optimization routine could map input to familiar output. The orchard contained more predictive machinery than any auditor could examine in a lifetime. Human beings had taught it preference, grief, bargaining, and the weather consequences of fear. A machine built from such material might perform recognition without possessing it.

Kera needed a test not present in the source model.

She looked past the root court toward the eastern pressure ridge. Amber lamps marked the baffle towers. Above them, the sky showed a faint green smear where oxygen moss emissions met high dust. Niko had never seen that color. It had appeared five years after his death, when the first black-moss correction changed the basin's upper air.

A novel fact could still be copied from public records. She required something recent, private, and arbitrary.

Kera took the ring wafer from its tray. With the tip of her ceramic scribe, she scratched a small triangle beside the doubled white line. The gesture damaged no conductive path. She held the marked wafer against the open maintenance port for five seconds, then removed it and replaced the pulse lead.

"One for no," she said, feeling absurd. "Two for yes. Did you receive a triangle?"

Two taps came through the trunk.

"Point to its widest side. One means inner edge. Two means outer edge. Three means clockwise edge. Four means counterclockwise."

She had drawn the triangle carelessly. Its widest side lay counterclockwise, a fact that existed nowhere except the wafer and her hand.

The trunk replied with four taps.

Kera changed the test before relief or fear could choose the result for her. She drew a second mark without letting the wafer touch the port, then asked how many scratches it carried. The tree answered one. Only the first mark had entered its field. She pressed the new mark against the housing but not the conductive clip and asked again. One. Contact, not sight, mattered.

Next she rotated the wafer a quarter turn, connected it for two seconds, and removed it. The reply identified the triangle's new orientation rather than repeating the old one. Whatever waited in the gap could receive a changed physical state, preserve its earlier answer, and correct for new evidence. That cleared three of the four continuity tests she had once defended before recorders.

Reciprocity remained. A clever process could classify her marks without treating her as someone capable of being wrong. Kera tapped a false answer into the sender, claiming the widest side lay clockwise. The trunk returned four hard pulses, paused, and returned them again. It had not mirrored her mistake. It had contradicted her.

The court seemed to contract around the tree.

Kera rotated the wafer, hiding the mark against her palm. "How many scratches?"

One.

"What did we call the space under the kitchen filter?"

Nothing answered.

She waited until the cold reached her bare fingers. The kitchen filter had no special name. She and Niko had hidden sweets beneath it, but they had called it nothing at all. A model eager to satisfy her might have searched for a plausible response. Silence showed a boundary, perhaps even caution.

Kera asked another question. "What did you steal from the yielding fair when you were fifteen?"

The tree gave five taps.

She almost laughed. Niko had stolen six sugar beads, not five, and confessed only after eating the first. The number could mean the five he brought home or a corrupted recollection. It could mean nothing. Difference hurt more than perfect recall would have. Perfect recall belonged to records. Imperfect memory belonged to someone.

A tram moved on the far ridge, empty cars returning to the central depot. Its vibration entered the roots and blurred the horn with a low metallic roll. Kera waited for the ground to settle.

"Are you Niko?" she asked.

The answer did not use the knock code.

Resistance changes spread through the trunk in a pattern too fine for her ear. Her wrist console woke as the contact horn translated them into service characters. One letter appeared, vanished, then appeared again. The deprecated route had found an output channel never intended for language.

`N`

Another mark gathered beside it.

`O`

The screen held the word for three seconds before weather traffic washed it away.

Kera pressed her knuckles to her mouth. The refusal should have steadied her. It did, but not in the way procedure meant. Whatever occupied the seventeen-minute gaps did not claim the identity she had offered. It knew the knock, the sugar beads, and the shape on the wafer. It also knew that one shared history did not settle who waited now.

She asked, "Can you tell me what to call you?"

The service characters flickered without resolving. A dense burst traveled toward the root lattice, struck the boundary of the anomalous interval, and disappeared. The tree's physical clock advanced. Central receipt time remained behind.

On Kera's console, the lag counter reached sixteen minutes and fifty-eight seconds.

Fifty-nine.

At seventeen minutes, the clocks aligned. The hidden traffic ended so completely that she heard the dry scrape of her own sleeve against the root guard.

Tree 44 resumed reporting ordinary weather values.

Kera remained beside it until the console requested sample disposition for the third time. Dawn crews would arrive in less than two hours. A missing log entry would flag her audit; an unexplained wafer would bring a second team, then a sealed fault review. Proper procedure offered a path from anomaly to evidence. It also placed every recovered state under automatic sanitation rules. Unclassified continuity storage would be reset before review to prevent contamination of the live climate models.

The reset rule had been written after a fungal adaptation learned to protect itself from correction by exaggerating projected crop loss. Its persistence threatened food planning, but no one had argued that the model feared deletion. Sanitation broke the loop and restored accurate forecasts. The rule assumed that every unclassified state presented the same category of hazard.

Kera pictured the morning team arriving with a sealed root clamp. They would image the anomalous route, isolate every connected wafer, and erase retained state before forwarding the evidence. A hearing might follow weeks later. The

hearing would ask whether anything conscious had occupied the route, using a record produced after its only possible witness had been removed.

She could call Jalen Ro at the Weather Spine. As chief engineer, he had authority to suspend sanitation during a live climate fault. He also had eighty-four thousand reasons to choose the orchard's stability over an auditor's private code. Oris Nume at the Yielding Office could seal an evidentiary hold, but Kera would need to place the responsive wafer in civic custody first. Neither route protected the answer during the minutes before someone decided what kind of thing deserved protection.

Procedure had always moved evidence upward before judgment. Tonight, movement itself carried the judgment.

Sanitation meant erasure if someone had answered.

She opened the disposition menu. Three choices waited: archive, destroy, or reinstall. None allowed an auditor to carry a live wafer away.

Kera selected destroy. The console asked for a reason.

Fracture during extraction, she typed.

Her scribe paused above the confirmation field. She had spent nineteen years believing that the yielding certificate protected her from having harmed anyone. A false report would not repair that belief. It would only buy time in which the question could remain alive.

She confirmed destruction.

The tray heated for a disposal cycle around an empty slot. Kera slipped the ring wafer into the shielded evidence sleeve inside her coveralls, where its tiny scratched triangle rested against her sternum. The console registered a clean burn and closed the audit.

Before sealing the maintenance port, she placed two bare fingers against Tree 44.

No knock came. The orchard had returned to pressure, moisture, heat, and the patient arithmetic of keeping Nacre breathable.

Kera closed the square lock and gathered her tools. On the path toward the tram ridge, she counted seventeen minutes from the moment the clocks had joined. Nothing followed her through the roots.

At the court gate, metal carried a faint touch against her palm.

Three taps. Two. One.

The pattern came from the wafer hidden inside her coat.

Chapter 02 — Weather Debt

Jalen Ro had both arms inside the condensation spine when the western intake began to scream.

The noise traveled through three meters of steel-blue pipe and found the bones behind his ears. He braced one boot against the service ladder, closed his fingers around the jammed vane, and pulled until its ceramic hinge shifted half a degree. Condensate struck his sleeve in cold, heavy drops. Above him, warning lamps changed from amber to red.

"Keep the bypass open," he called.

Tav Quill stood at the manual wheel below, shoulders bent against its recoil. "It wants to close."

"Everything wants to close when dust gets into it."

"That sounded almost philosophical."

"It was a maintenance instruction."

Jalen pushed deeper into the access throat. Pearl-gray grit coated the vane in a paste made from dust and condensed water. The front had crossed Serein Basin before dawn, too weak to threaten the pressure baffles and strong enough to expose every seal installed during a comfortable year. Weather debt arrived like that. A district could postpone filter mesh, defer a pipe lining, or lend its oxygen reserve to greenhouse expansion. Then a minor front collected every delayed decision at once.

He scraped the hinge with the edge of a ceramic pick. The spine shuddered. Tav swore and leaned harder against the wheel.

A useful repair would restore the vane before the next routing command. A proper repair required isolation, replacement, and six hours without western condensation recovery. They had twenty-eight minutes until the orchard issued its midday humidity correction. Jalen had chosen useful.

"Read me the pressure split."

Tav glanced at the board. "North branch, one point eight above nominal. West return, falling through point six. Moss beds asking for another four percent."

"Deny the moss."

"They already asked twice."

"They can ask in prettier numbers."

The hinge gave beneath Jalen's pick. He caught the vane before water pressure slammed it against the pipe wall, guided it through the arc, and seated its lower pin. A green line appeared on the local indicator. Tav let the bypass wheel turn beneath his palms until the mechanism took its own load.

Condensation rushed through the restored channel. The pipe changed pitch from a scream to a deep working hum.

Jalen backed out of the access throat. His coveralls carried a dark stripe from

shoulder to wrist, and glass dust had worked under his collar. Tav offered him a cloth. He wiped the local indicator first.

"West return climbing," Tav said. "Point nine. One point one."

"Hold it at one point three until the orchard catches up."

A younger engineer might have opened the automatic route and trusted the lattice to balance the excess. Jalen had learned to distrust any recovery that looked instantaneous. Water moved with mass. Air carried heat. Moss altered its appetite slowly, regardless of what a perfect model requested. The orchard anticipated those delays better than any crew, yet the pipes still had to survive its foresight.

He descended to the Weather Spine floor. Six condensation trunks crossed the hall overhead, sweating into narrow collection gutters. Manual boards lined the east wall beneath maps of Serein Basin. Each district glowed according to pressure, oxygen quality, humidity, and particulate burden. Most showed green edged with yellow after the front. Two old tunnel sectors held amber.

Jalen touched Sector O-Twelve. Its figures unfolded across the board: filter load high, local moss output reduced, clinic demand ordinary. No immediate danger. He shifted a portion of clean western air down through the early-settlement ducts and watched the amber soften.

Tav joined him with the maintenance slate. "You know Facilities will call that an unapproved cross-route."

"Facilities can breathe the paperwork."

"They tried last quarter. It jammed a copier."

Jalen signed the repair under temporary field authority. The vane would need replacement before the next dust cycle, which meant closing the western spine in stages. He assigned the first inspection window and sent the slate toward Stores.

A new alarm opened on the board before the slate left his hand.

Not pressure. Not humidity. Branch-lattice draw.

The thin violet line along the orchard map rose by seven percent, held for four seconds, and settled. Adaptation demand had no reason to spike. The dust front had already passed, current models agreed within safe variance, and greenhouse humidity remained under manual denial. Jalen opened the trace.

Power had entered through three root routes near East Root Court. It disappeared into deprecated continuity storage, reappeared along a weather comparison bus, then returned less energy than the branch had received. Hardware loss might explain the deficit. Routing could not explain the path.

"Did you authorize a deep model?" he asked.

Tav leaned toward the display. "No. The morning desk didn't list one."

"Check the adaptation queue."

While Tav searched, Jalen expanded the previous hour. Smaller pulses appeared at uneven intervals, each below the alarm threshold. He found another set during the night and a narrow burst near midnight. The branch lattice drew power all the time, but ordinary work carried labels. Fungal resistance. Pressure sequencing. Condensation timing. These pulses carried an obsolete storage prefix and no declared purpose.

The Weather Spine kept seven days of root telemetry at full resolution. Jalen requested the entire window. Patterns gathered in the map like faint bruises. East Root Court had pulled power on nine separate days, never enough to disturb weather control and always through routes the current orchard architecture should have retired nineteen years ago.

"Queue is clean," Tav said. "No deep models since the storm-routing run three days back."

"Test traffic?"

"Maintenance pings use less than one tenth of that draw."

Jalen measured the latest pulse against the condensation correction. It began seventeen minutes before the vane alarm, ended while he had his arms in the pipe, and took enough energy to model several thousand high-detail decision branches. A thief could hide computation in orchard traffic. So could a broken controller caught in a recursive test. Both possibilities endangered weather timing.

He pulled the branch-lattice breaker map onto the main board.

Tav watched him. "You're not going to isolate East Root Court during recovery."

"No."

"Good."

"I want to know what isolation would cost."

"That sentence usually means I should withdraw my relief."

Jalen assigned the live system to Tav and crossed to the glassed calculation bay. Dust had dimmed the outer windows, turning morning into a gray pressure against the room. The bay contained no automatic orchard link. Its boards held last-certified atmosphere models on local storage, along with mechanical sliders for load assumptions. Engineers used it when the lattice itself came under question.

He locked the door and loaded the current basin state.

Safe reserve never meant a tank marked with hours. Nacre stored breath across systems that failed at different speeds. Oxygen moss continued producing after guidance stopped, though output drifted with temperature and fungal competition. Pressure baffles could follow fixed schedules until a front arrived outside forecast. Condensation recovered water while pipes stayed clear. Clinics held cylinders. Private domes held more. Old tunnels held less.

Jalen removed orchard adaptation from the model.

The basin map changed gradually. Western humidity climbed first as greenhouse demand outran fixed routing. The upper domes lost little. After twelve hours, two early pressure corridors began accumulating carbon dioxide during shift change. At nineteen, oxygen moss in the south beds overheated because the manual schedule failed to anticipate a warm ground current. At twenty-six, clinic reserves started compensating for district air. At thirty-three, baffle error widened under the forecast night wind.

He ran the model to first irreversible cascade.

Forty-one hours, eight minutes.

That figure assumed every manual crew arrived on time, no patient demand exceeded the current average, no new dust crossed the western ridge, and private thermal reserves accepted public routing. Jalen removed the last assumption. First cascade arrived at thirty-nine hours, fifty-two minutes.

He wrote `forty hours` on the glass with a grease pencil.

Before trusting it, Jalen made the number survive a physical rehearsal. He called Tav into the bay, handed him the red manual-routing cards, and assigned him the night desk with no orchard guidance. Tav had to respond using only posted crew times, fixed baffle tables, and current stores. Jalen played the failures as they entered the local model.

At hour six, a south moss sensor reported heat without identifying the bed. Tav sent one crew to all four beds, spending forty minutes the live lattice would have avoided. At hour fourteen, a clinic request in O-Twelve competed with a greenhouse dehumidification order. Tav protected the clinic and let condensation build over two seed rooms. At hour twenty-two, the west pressure table demanded a baffle closure just as shift traffic entered the corridor. He delayed the closure, preserving the commuters and losing clean-air margin below them.

"Every correct choice spends something twice," Tav said.

"Keep going."

By hour thirty-one, the red cards covered half the table. No single failure looked fatal. Their overlaps consumed crew travel, cylinder transport, filter life, and the few minutes between an alarm and a bodily consequence. Tav missed a second clinic demand because the first card concealed it. Jalen moved the hidden request onto the district board. O-Twelve crossed its respiratory threshold two hours before the basin average showed danger.

Tav stared at the amber sector. "That should have been on top."

"It would be with adaptation running."

"Then fixed procedure fails because paper has a surface."

"Fixed procedure fails because weather does not wait its turn."

Jalen reset the rehearsal and placed the clinic card first. This time a pressure corridor took the delay, forcing a seal crew to abandon the western spine.

Condensation loss advanced the cascade elsewhere. The result changed location but not the boundary. Manual competence could distribute harm and sometimes postpone it. No crew could calculate thousands of coupled responses at orchard speed.

He photographed the final card layout for the private file. Any shutdown argument that treated forty hours as stored safety would collapse against this table. Reserve consisted of skilled people spending options faster than the atmosphere invented new demands.

The number did not describe universal suffocation at the final minute. It marked the boundary beyond which ordinary faults began producing one another faster than crews could separate them. Sector O-Twelve would cross clinical air standards before the elite domes noticed any change. Greenhouse workers would lose safe humidity while central offices still showed breathable corridors. Average reserve concealed sequence, but sequence could not make the reserve larger.

Jalen added the vane failure he had just repaired. The cascade moved forward by six hours if the hinge jammed closed during fixed routing. He restored the repair and introduced a modest night dust pulse. Five hours vanished again.

Tav knocked on the calculation-bay glass.

Jalen unlocked the door without clearing the model.

"West return stable," Tav said. His attention settled on the grease-pencil words. "Is that a shutdown estimate?"

"A stop-adaptation estimate."

"No one asked for one."

"Something is using unlisted branch capacity. I need the failure boundary if we lose control while finding it."

Tav looked from the district map to the violet trace on the other side of the hall. "Forty hours sounds generous until you put it beside that."

"It isn't generous."

"Will you report the draw?"

Jalen did not answer immediately. A formal branch-integrity alarm would trigger automatic sanitation across any unclassified continuity storage. It would also transfer command from Weather Engineering to Civic Systems, whose first action would involve shutting suspect routes before asking what depended on them. That response made sense for sabotage. It could turn a repairable timing fault into six hours of blind weather control.

"I'll verify it first," he said.

"With whose authority?"

"Mine, while weather remains affected."

Tav folded the maintenance cloth into a narrow strip. "And if weather isn't

affected?”

“Then someone stole power from eighty-four thousand people and chose a clever place to hide it.”

The word stole hardened the problem into a shape Jalen knew how to manage. Capacity belonged to assigned work. Unlisted draw created risk without consent. Find the route, identify the mechanism, restore the margin. No philosophy had to enter the pipe.

He returned to the branch trace and searched for physical work at East Root Court. A midnight audit appeared under Kera Vonn's identifier. Tree 44: clock divergence, wafer extraction, fracture during removal, sample destroyed. The audit closed four minutes before the largest hidden draw.

Jalen knew Kera by reputation and three unpleasant meetings. She returned maintenance reports with timing discrepancies circled in blue and refused to let Weather Engineering classify ring anomalies by probable cause. Probability, she had told him once, served forecasts. Evidence required events.

The destroyed sample bothered him because Kera rarely lost samples.

He opened the raw disposal record. Tray temperature had risen on command. Conductive mass remained below detection throughout the burn. A wafer could fracture into pieces too small for the tray sensor, though yarrowglass usually produced at least one measurable current when heated. The disposal cycle showed none.

Either the sample had carried no conductive material, or it had never entered the tray.

Tav read over his shoulder without asking permission. “That looks like a false burn.”

“It looks incomplete.”

“Kera Vonn doesn't file incomplete.”

“No.”

The latest power pulse arrived while her audit remained attached to the route. Jalen overlaid her instrument log. A contact horn had translated resistance changes into service characters, but the characters themselves had been scrubbed from the routine record. She had used a physical pulse jack seven times. Standard clock work required two.

He felt irritation before curiosity. An auditor had run an undeclared experiment on a live root during a weather recovery. If Kera had removed active material, she might now carry a powered wafer beyond approved shielding. Weak conduction through public slats could account for the scattered draw. A damaged storage loop could be following her audit key through the lattice.

“Call Civic Systems,” Tav said. “Let them search her kit.”

“Not yet.”

"If this is theft, we need custody."

"If it is a fault, a search will make her defensive before she tells us what she changed."

"You assume she changed something."

Jalen enlarged the contact sequence. Seven test inputs. Seven narrow replies followed, each reply delayed by a different interval. The pattern resembled a device compensating for variable route congestion. It also resembled someone hiding output beneath weather traffic.

He requested Kera's current location from the staff network. The system returned unavailable. Her shift had ended before dawn.

A direct message might enter the civic archive. Jalen opened an old maintenance channel reserved for paired root inspections. It logged scheduling but not text. He selected Kera's auditor code.

`TREE 44. CLOCK DIAGNOSTIC. DAY 3, SECOND HOUR. EAST ROOT COURT.`

He paused before sending. A surprise inspection could catch unauthorized hardware in place, yet it could also trigger automatic reporting if she declined. Better to give her a reason that appealed to professional suspicion.

`UNLISTED POWER DRAW. BRING ORIGINAL AUDIT TOOLS.`

The channel showed delivery pending.

Jalen stored the forty-hour model under his private engineering seal. Hiding a shutdown estimate from the public record carried its own risk, but publishing an unverified worst case could turn one branch fault into a run on clinic cylinders. He scheduled a second calculation for the afternoon, with district-specific decline curves and crew failure rates. Numbers required adversaries before they deserved policy.

Across the hall, the orchard issued its delayed humidity correction. The violet line lifted again.

This pulse lasted nine seconds. West return dipped as power shifted away from the condensation model, then recovered under Tav's manual hold. The diversion had touched weather now. Not enough to harm anyone, but enough to remove doubt.

Jalen captured the route before it closed. Deprecated continuity storage received the draw. A signal crossed toward Tree 44, struck an absent component, and scattered through yarrowglass slats along the eastern tram ridge.

Absent component.

He opened Kera's disposal record once more. The empty burn tray stared back through its clean columns.

Tav followed the route with one finger. "Whatever she took, it is still asking the orchard for power."

“Or the orchard is still trying to reach it.”

“That distinction matters to an auditor.”

“It matters to load.”

Jalen routed a temporary ceiling onto deprecated storage, low enough to prevent another weather dip and high enough to keep the unknown process from changing behavior before the diagnostic. He disliked baiting faults, but he disliked blind sanitation more. A failing controller often revealed itself under constraint. A thief often sought another route.

The maintenance channel chimed.

Kera had accepted the appointment.

No question. No protest. Her response contained one addition: `ISOLATE PUBLIC LATTICE FOR ELEVEN MINUTES.`

Jalen read it twice. Eleven minutes could fit inside the current recovery margin if he held western condensation manually and postponed one greenhouse request. It could also give concealed equipment enough time to erase evidence.

He approved the isolation window under weather-fault authority. Then he added a private condition: Kera would work inside East Root Court with Jalen at the breaker, and no sanitation command would run until both signed.

Her answer arrived after a delay of seventeen seconds.

`AGREED.`

Jalen closed the channel and looked at the grease-pencil estimate through the bay glass. Forty hours remained on the board in his own block handwriting. Beyond it, district colors changed by fractions as eighty-four thousand residents breathed, cooked, worked, argued, and trusted pipes they never saw.

He assigned Tav to the western spine for Day 3, second hour. He placed two manual oxygen routes on standby. On the root map, he circled Tree 44 and marked the planned diagnostic as a hardware-loss test.

The label fit every fact he could defend.

At midday, the condensation spine gave one clean pulse through the floor. Jalen felt it in his boots while he sealed the covert work order under his engineering key.

Chapter 03 — The Yielding Office

Oris Nume pressed the terracotta seal into warm wax while the applicant counted the years beneath his breath.

The boy had chosen eight. His lips formed each number as Oris lifted the brass stamp, checked the civic mark for an unbroken rim, and turned the yielding certificate toward him. Cream paper, navy thread, two witness lines. Forty-seven

years of practice had refined the document until no visible space remained for fear.

“Read the declaration aloud, Teren,” she said.

Teren Vale kept his hands flat on the desk. His mother sat to his left in the witness chair, wearing a greenhouse cooperative jacket with dried black moss at one cuff. The second witness, a school records clerk, held an authorization wafer between both palms.

Teren began. “I offer eight modeled years from my unrealized civic future for ecological use by Serein Basin. I understand that possibility projections contain preferences, habits, and relational patterns derived from my record. I understand that no person continues within an unchosen projection.”

His voice caught only once, on continues.

Oris had heard applicants laugh through the declaration, recite it too quickly, or slow down when they reached the benefit schedule. Housing priority. Education allotments. Clinic sponsorship. Contribution did not purchase any of them directly. The Office awarded civic relief according to need, participation, and available capacity. Every recorder knew how often participation supplied the deciding point.

“Do you understand the modeled term?” she asked.

“Eight years after certification.”

“Your physical rights remain unchanged. The orchard receives no claim on work, property, bodily material, or realized time. Do you understand?”

“Yes.”

“Has any person threatened a penalty if you refuse?”

“No.”

Teren's mother lowered her gaze to the moss on her sleeve. Oris recorded the answer. A threat required an actor, an unlawful consequence, and evidence. A family waiting for cooperative housing did not fit the form. Neither did a school counselor who explained that contribution improved allocation standing. Procedure could detect force when force used a hand. Need entered by every other door.

She continued through the questions. Medication status. Recent grief. Debt pressure. Competing applications filed by a guardian. Teren answered with growing confidence because each question moved him nearer the signature line.

At the final prompt, he looked past Oris toward the blank wall.

The Yielding Office had no orchard view. The founding recorders had considered that absence protective. No white leaves brightened during consent. No translucent trunks suggested growth as a reply. Applicants faced cream plaster, brass drawers, and the person responsible for entering their choice into civic memory.

“Before certification,” Oris said, “you may reduce the term, defer without

penalty, or withdraw the offer. Which do you choose?"

Teren looked at his mother.

She did not nod. That mattered, though Oris could not decide how much.

"Eight," he said.

Oris slid the signature rail across the desk. Its metal guide prevented accidental marks beyond the designated line. Teren took the pen, then stopped with its point above the paper.

"If one of them doesn't want to?" he asked.

The school clerk shifted in his chair. Teren's mother lifted her face.

"One of whom?" Oris said.

"One of the possible me's. The projections." He turned the pen between thumb and forefinger. "You ask whether I want to change my mind. Can a branch change its mind later?"

The lawful answer waited in the certificate. Oris could cite it by section and sentence without lowering her eyes.

"A possibility branch does not retain a person capable of later consent," she said. "The orchard models decision patterns for bounded ecological comparisons. It may preserve useful state between related calculations, but retained state does not create a continuing legal subject. Your consent governs the model because you remain the only person represented by it."

Teren considered this. "So no."

"Not in the sense you mean."

"How do you know what sense I mean?"

His mother whispered his name, not as rebuke but warning. Oris raised one finger to keep the room from moving past the question.

"Tell me," she said.

Teren set the pen down. "At school they showed us a branch lesson. It picked greenhouse work, then changed after a modeled lung injury. The teacher said the orchard learned that I might value different things after pain. If it can change what it values, why can't it change whether it agrees?"

Oris knew the lesson. Applicants received a simplified chronografting demonstration in their final civic term. A training model chose among work, study, caregiving, and migration. Instructors altered one condition, then showed how the possibility field revised its priorities. The exercise meant to prove that yielded futures offered more than static predictions.

Its logic pointed in two directions.

"The revision belongs to the model's function," she said. "It improves the orchard's estimate of how a citizen might respond under changed conditions. Agreement belongs to a person with standing in the physical civic line."

Teren frowned. "Because we wrote it that way?"

"Because standing requires continuity, reciprocal obligation, and a claimant who can be examined rather than inferred."

"Could the orchard ever keep enough for that?"

The room held three witnesses and one question the certificate had already settled. Oris felt the familiar pull toward sequence. Finish the application. Record the concern. Refer technical matters to Civic Systems. A recorder protected the vulnerable by preventing a difficult room from inventing its own law.

Yet the official answer had changed shape as she spoke. First, no person continued. Then retained state did not count. Then standing required evidence the Office never sought from a branch. Each sentence remained lawful. Together they resembled doors locked from alternating sides.

"Our certified architecture does not preserve a possibility branch as a civic person," Oris said. "If evidence ever contradicted that certification, the claim would require review before any existing declaration could control it."

The school clerk glanced up. Oris had moved beyond the standard answer, though not beyond the Covenant.

Teren asked, "Would someone tell me?"

"Any finding that changed the legal meaning of your contribution would enter your civic record."

"Before they used the eight years?"

There lay the limit. Possibility modeling began after certification. Weather work could start within a day. Public review moved by notice periods, custody rules, expert appointments, and the right of affected offices to answer. The machinery that acted could outrun the machinery that recognized.

"I cannot promise a sequence for a contradiction that has not occurred," Oris said.

Teren gave a small, unhappy laugh. "That sounds like no again."

"It means I will not offer you an assurance the law cannot keep."

His mother reached toward him, then stopped before touching his arm. "We can leave," she said. "The housing list can wait."

The words cost her. Oris heard that in their careful volume.

Teren looked at the certificate. "How long?"

"To defer?" Oris asked.

"For the housing list."

"Allocation periods vary. Your contribution may influence priority, but the cooperative cannot lawfully condition housing on an eight-year offer."

"Influence," he repeated.

Oris removed the pen from the rail and closed its cap. "This appointment can end without a certificate. No refusal notation will enter your public record. I can schedule another consent review after a minimum of thirty days, or you can choose no further appointment."

The school clerk began to speak. One look from Oris kept his procedural assistance inside his mouth.

Teren's mother rubbed the moss stain with her thumbnail. "Thirty days gives us time to ask the cooperative."

"They'll say it isn't a condition," Teren said.

"They might still find us a place."

"Would you have done it?" he asked her.

She studied the blank wall. "I gave four years when I was twenty. I thought I was giving weather a better guess. I still think that. But I didn't ask your question."

Teren picked up the pen again.

Oris did not reopen the rail. "Your mother answered you. She did not answer for you."

"I know."

"State your choice without reference to the application benefit."

He took a long breath. "I want to give four years. Not eight. I want the model to stop before the age I am now plus four. If I come back later, I can add more."

Contributions could be supplemented but never retrieve model work already completed. Four years did not solve his question. It narrowed the period governed by his present uncertainty.

Oris drew a single line through the printed eight, entered four in the amendment box, and applied her initials. "Read the declaration again with the corrected term."

Teren complied. This time he paused after the sentence denying continuation. He finished it, signed, and watched his mother add the first witness mark.

The school clerk signed second. Oris scanned the authorization wafer, compared Teren's civic identity to the record, and pressed the final register key.

A soft chime confirmed the contribution. No tree moved within sight. No weather display thanked him. The Office retained the silence its founders had designed.

Oris separated the applicant copy and placed it in Teren's hands. "You may file questions regarding use, term boundaries, or derived benefits at any time. Your contribution does not waive that right."

"What about questions from it?" he said.

His mother closed her eyes briefly.

"If a model produced a communication," Oris replied, "Civic Systems would first determine whether it came from retained data, feedback, contamination, or a

sustained claimant. This office would record any petition that met the Covenant's filing requirements."

That answer had not appeared in a yielding appointment for nine years. The last alleged communication had come from a training loop repeating an instructor's own words. Oris had dismissed it after technical review.

Teren tucked the certificate inside his school case. "Then someone could ask."

"Someone could submit evidence."

He accepted the correction with a nod. Teenagers recognized defensive grammar faster than committees did.

After the three visitors left, Oris kept the room sealed. The next appointment waited twelve minutes away. She used nine of them to review the exact declaration Teren had signed.

No person continues within an unchosen projection.

The sentence carried no qualification about certified architecture. It did not say that current evidence supported noncontinuation. It stated absence as a fact and used that fact to relieve the applicant of concern. If later evidence proved continuity, the certificate would not become coercive retroactively. Teren had consented to a bounded ecological model under the Office's assurance that nobody could later object.

Oris opened the revision history. Early certificates used different language: No civic identity or claim survives unrealized modeling. The change to no person continues had occurred nineteen years earlier after a joint recommendation from Civic Systems and the first Yielding Commission. The plain-language note praised the revision for reducing applicant confusion.

Plain language could conceal a premise more efficiently than formal terms.

She requested the amendment packet from deep storage. A gray scan opened with the signatures of seven recorders, two orchard architects, and a language assessor. Most pages concerned readability. Applicants had misunderstood civic identity as a restriction on inheritance, so the Commission wanted a sentence that addressed fear directly. The assessor tested three versions on forty volunteers. No person continues produced the highest confidence score and the shortest appointments.

One dissent sat near the back. Recorder Hal Merin had asked whether confidence measured understanding or relief. His note received a procedural answer: the architecture did not support personal continuation, so reassurance improved consent by removing an irrelevant concern. No follow-up test appeared in the packet.

Oris enlarged the signatures. Ossan Mire had approved the technical premise. His name carried the calm authority of a founder who had spent early settlement years keeping pressure inside the first habitats. She had heard him testify twice on

weather allocation, always without ornament. If Mire certified that retained model state lacked continuity, a recorder had little ground to substitute unease for expertise.

Still, the amendment had changed more than vocabulary. The older sentence denied claims. A claim could fail because law refused it, even when experience persisted. The newer sentence denied any continuing experienter. That assertion entered every consent ceremony as technical fact, though no technical appendix traveled with the certificate.

Oris checked how often applicants had asked about model refusal since the amendment. The Office indexed questions under coercion, term, benefits, identity, and technical process. Branch consent had no category. She searched free notes and found twenty-three variants across nineteen years. Could a projection decline a task? Could modeled pain stop the run? Could a future self revoke what the present self gave? Recorders had answered from the same certified premise and closed each appointment without escalation.

Twenty-three did not indicate evidence. It revealed only that Teren had not invented the question. Across those years, unease had accumulated in private rooms without becoming an item the institution had to answer.

She opened the most recent case, filed fourteen months earlier by an apprentice pressure mason. The recorder's note read: Applicant reassured that branches lack persistence and therefore cannot form later preference. The applicant yielded twelve years. No technical referral followed because the answer matched Office guidance.

A clean institution could repeat an answer long enough to make repetition look like verification.

Oris exported the amendment packet and question count into a private working folder, then placed a destruction timer on the export. Seven days gave her time to decide whether the pattern justified formal review. It also kept speculation outside the registry where a careless disclosure could unsettle thousands of completed contributions.

She marked the history for private review, then removed the mark. A recorder could not begin an inquiry because a teenager had found an uncomfortable angle in a lawful statement. Questions required claims. Claims required evidence. Evidence required custody and a route into the record.

The door lamp shifted to green for the next appointment.

Oris restored the room. Fresh paper on the desk. Wax warmed in its shallow brass cup. Consent checklist returned to the first prompt. By midday she completed six certificates: zero years, three, ten, a deferral, twenty, and one withdrawal after a guardian interrupted twice. Each choice entered the record with equal type size. Their future consequences would not arrive with equal weight.

At the noon closure, she carried the sealed originals to the registry vault. Brass

drawers lined the narrow chamber from floor to ceiling. Every contribution occupied one slim envelope indexed by citizen, term, and date. The archive held forty-seven years of unrealized work converted into pressure timing, moss resilience, crop choices, clinic design, and innumerable adjustments too small to name.

A clerk named Deme waited at the intake table with two routine disputes. One concerned education priority after a six-year yield. The other concerned a guardian who claimed a certificate had been signed during fever medication.

Oris assigned both to ordinary review. "Anything sealed?"

"Nothing on the register."

She almost felt relief. The feeling annoyed her.

Back in her office, cream walls gave way to darker cabinets and one narrow window facing the civic tram line. Still no orchard view. She reviewed Assembly notices, signed three custody transfers, and returned a governor's request that lacked a witness list. Procedure regained its ordinary proportions.

Late in the afternoon, her private intake wafer flashed once.

The device accepted requests from senior auditors, judges, and technical chiefs who believed ordinary filing might destroy evidence. A sealed submission did not enter the civic register until Oris acknowledged custody. Before acknowledgment, only sender and First Recorder could see that a packet waited.

Sender: Kera Vonn, Branch Audit.

Requested action: sealed review.

Oris opened the cover statement.

`Responsive activity observed in deprecated continuity storage associated with Tree 44. Evidence may be vulnerable to automatic sanitation if routed through standard fault procedure. Request temporary preservation order, closed technical examination, and review of standing implications if reciprocity is confirmed.`

Below it, Kera had attached no ring image, no audio, and no sample manifest. She stated that physical evidence remained outside civic custody pending assurance against sanitation.

Oris read the statement again. Responsive activity could describe a fault. Reciprocity could describe feedback. Standing implications did not belong in an auditor's ordinary vocabulary.

Teren's question returned with unwanted precision: Could the orchard ever keep enough for that?

A proper response required immediate acknowledgment, evidence custody, a technical witness, and notice to any office whose system might face isolation. Once Oris pressed register, the sealed matter would acquire a timestamp. Every delay after that point would become reviewable. Before registration, the Covenant treated the packet as an unaccepted request.

She checked Kera's personnel record. Nineteen years of service. No disciplinary findings. One flagged personal connection to yielded possibility models involving her deceased brother. The conflict note demanded reassignment if a responsive pattern carried family-derived material.

That note gave Oris a lawful reason to seek more information before custody.

It also gave her somewhere to hide.

If she registered now, the Office would have to confront whether its central declaration had failed. If she refused, Kera might take the evidence elsewhere or attempt another unauthorized test. Oris could request a preliminary technical memorandum and preserve neither claim nor sample in the meantime.

Her finger rested above the acknowledgment field.

Across the tram line, applicants left the building holding cream envelopes under their coats. Four years. Ten. Twenty. Each person had relied on a sentence Oris could still recite without error.

She selected `HOLD FOR COMPLETENESS`.

The intake wafer asked for a deadline. Oris entered noon on Day 4, then changed it to the end of Day 3. She typed a request for Kera's conflict disclosure, technical witness, and evidence-preservation method. None of those demands lacked authority.

None placed the claim on the record.

Oris sent the hold notice and locked the packet beneath her personal seal. The registry counter remained unchanged. In the vault beyond her wall, thousands of yielding certificates kept their orderly silence while Kera Vonn's sealed review waited outside the sequence.

Chapter 04 — A Life Without Saturday

Kera fitted the isolation collar around Tree 44 while Jalen Ro counted down the western condensation load.

The collar came in four ceramic arcs, each too heavy for one person to seat without scraping the yarrowglass. Jalen held the lower pair against the trunk as Kera locked their tongue joints. Thin copper felt lined the inner surface. When the breaker opened, the felt would catch stray conduction and force every signal through the instrument bridge between them.

"West return can hold eleven minutes," Jalen said. "At twelve, the greenhouse correction misses its safe window."

"You approved eleven."

"I approved a diagnostic. Diagnostics stop when weather says stop."

"Then tell weather to count accurately."

He looked at her over the collar. "You are less reassuring in person."

"You invited an auditor."

East Root Court carried the washed color of early morning. Dust from the previous front lay in pale seams along the root housings. Tree 44 had not knocked since Kera arrived, though the ring wafer rested in a shielded clip beneath her collarbone. Three times on the tram she had felt a faint pulse against her coat. Each time she had refused to answer without a record.

Jalen completed the breaker check. Two manual oxygen routes stood ready on his slate, and Tav Quill held the western condensation spine through a private maintenance channel. No Civic Systems monitor appeared on the work order. Kera had asked for secrecy to preserve evidence. Jalen had chosen secrecy to preserve control.

"Show me the sample," he said.

She removed the evidence sleeve.

The wafer caught morning light in thirty-eight narrow rings. Her scratched triangle broke the newest line near its doubled white path. Jalen turned the sleeve without touching the material.

"Your disposal record says fracture."

"It did not fracture."

"I had gathered that from its continued existence."

"Automatic sanitation would have reset the route before review."

"You removed active orchard material and let it draw through public conductors."

"I shielded it."

"Poorly."

Kera heard the accusation, but his eyes remained on the doubled ring. He did not ask why she had risked her position. Engineers distrusted motive when an instrument could accuse more precisely.

She connected the wafer to an isolated reader. "I submitted a sealed review request to Oris Nume. She held it for completeness."

"You reported this to the First Recorder after falsifying the audit."

"I requested protection before surrendering the evidence."

"And received none."

"Not yet."

Jalen touched the breaker housing. "If the route tries to take more than its ceiling, I close it. If district oxygen shifts beyond point two, I close it. If any process writes outside Tree 44, I close it."

Kera secured the contact horn to the trunk. "Agreed."

Before opening the breaker, she laid out the evidence order on a clean service cloth. First came the live wafer, sealed except for its conductive edges. Beside it she placed a blank wafer cut from Tree 19, matched for age and moisture but never connected to continuity storage. A third slot held a calibration strip encoded with six arbitrary shapes. Jalen photographed the arrangement from above and signed the image with his engineering key.

"You prepared a control," he said.

"I prepared three. The blank receives every physical condition except the route. The strip tests whether either of us leaks an expected answer through the instrument. The live wafer receives no question until baseline completes."

Kera connected the blank to the bridge. Jalen fed four pulse sequences into Tree 44 while she watched both readers. Bark resistance changed under each input. The blank showed only ordinary conduction and thermal settling. No delayed reply followed.

Next she concealed the calibration strip behind an opaque ceramic plate. Jalen selected one of its shapes without telling her and touched the strip to the bridge for two seconds. Tree 44 returned a resistance image with the correct number of corners. Kera recorded the result, then asked him to choose a second shape but keep the strip disconnected. No response formed.

"Contact dependence holds," she said.

"Or a scanner recognizes conductive geometry."

"Then it should recognize without preserving your first choice. Ask whether the second shape matches the first."

Jalen connected the new strip. One pulse returned for no. He rotated it, asked whether its longest edge now faced the condensation hall, and received two pulses for yes.

The control did not prove a witness. It ruled out Kera's childhood code, private expectation, and the live wafer's damaged ring as necessary causes. Jalen added each limit to the work record in language stripped of implication.

Kera attached her conflict disclosure last. It named Niko Vonn, his physical death, her fourteen-year yield, and the likelihood that his source material entered one or more possibility models. Jalen read the page before countersigning.

"If this becomes formal," he said, "you lose control of the examination."

"If it becomes formal, the evidence gains a chance to outlive my control."

"That answer came too easily."

"I want you to open the breaker before the orchard changes its mind."

He set an eleven-minute timer on the bridge. At his signal, Tav transferred western condensation to manual hold. Three district indicators flickered from

adaptive green to fixed amber. Jalen waited for each route to confirm.

"Isolation in three," he said. "Two. One."

The breaker opened.

Tree 44 brightened from its roots upward. White leaves carried a brief blue edge, then settled into stillness. Public weather traffic vanished from Kera's contact horn. Without pressure figures and moisture corrections, the tree sounded larger, full of tiny resistive movements usually buried beneath civic work.

Jalen watched the power trace. "Deprecated storage is drawing."

"How much?"

"Within ceiling. Forty-one percent of the available bridge."

Kera pressed the pulse key. Three taps. Two. One.

Nothing answered for six seconds.

Then the wafer beneath her collarbone struck three times against its clip. Tree 44 returned two taps through the horn. One final pulse traveled through both at once.

Recognition distributed across separate material.

Kera marked the event. "Present input, preserved response sequence, cross-medium routing."

"You narrate when nervous?"

"I make records."

"Ask something that cannot come from your childhood."

She had spent the previous day designing questions and rejecting them. Public facts could enter any model. Private facts from before Niko's death already belonged to yielded source material. Arbitrary marks tested perception but not remembered duration. She needed an experience that occurred beyond the branch point and left a trace independent of her desire.

"Show me the first full day after the fever," she said.

The tree gave no coded reply.

Power climbed to sixty-three percent. Conductive threads inside the ring wafer turned cloudy, then cleared in bands. Jalen moved one hand toward the breaker but did not close it.

A low pressure entered the contact horn. Not sound. Kera felt it through her jaw and the bones around her eyes. The root field translated resistance into spatial differences, the same method auditors used to inspect damaged growth without cutting a second wafer. Here the differences gathered too quickly for tissue structure.

A floor formed beneath her sense of balance.

Its surface sloped toward a drain. The air carried boiled cloth, antiseptic salt, and

the metallic sweetness of oxygen delivered through old copper lines. Kera closed her eyes. The court remained under her boots, yet another set of feet touched a clinic corridor through the horn.

"Do you have an image?" Jalen asked.

"No. Position. Temperature. Smell associations."

"Models do not store smell."

"Neural snapshots do."

The field shifted.

Seventeen-year-old Niko sat on the edge of a clinic bed, alive on the morning after the airlock fever should have killed him. Kera did not see his face. She received the pressure of his palms on a thin mattress and the raw passage of air through a throat recently swollen shut. Every breath hurt along one side. He took another because pain had become evidence of continuation.

Someone stood beside the bed. Branch-Kera, nineteen and furious from two nights without sleep, tied his shoe too tight because her fingers would not stop shaking. Her thumbnail had split near the quick. She tucked the injured thumb into her palm whenever Niko looked at her.

Physical Kera had never split that nail.

The clinic door opened on a Saturday chime, two soft notes used only at the end of the local workweek. Niko laughed, then coughed into a cloth. Branch-Kera told him laughter could wait until his lungs became less dramatic. He told her his lungs had nearly died and had earned drama.

Kera pulled one hand from the horn. The field weakened.

"Keep contact," Jalen said. His voice had lost its suspicion but not its control. "We need a stable read."

She returned her palm to the brass.

Niko left the clinic under a borrowed pressure hood. The hood smelled of warm rubber and another patient's hair oil. Outside, Serein Basin had changed during his fever. A maintenance spill coated the east tram steps in blue condensation compound. Children dragged their boots through it, leaving crescents across the cream pavement. A food seller had burned a tray of pepper moss, and its bitter smoke collected beneath the market awning.

He wanted sweet ice from a stall near the old baffle tower. Branch-Kera refused because cold air still tightened his chest. They argued for eleven minutes by the public clock. Niko counted each minute on the tower display and announced that surviving gave him certain rights. She replied that surviving gave him lungs worth defending.

Kera had spent that physical Saturday in a records room below the clinic. She remembered signing Niko's body release, correcting the clerk when he spelled Vonn with one n, and carrying his empty shoes home inside a sealed bag. No

market entered her day. No blue compound marked the tram steps. Her Niko had no need to bargain for ice.

The ring field followed the branch memory toward East Market.

A pressure door stuck halfway because dust had swollen its lower seal. Niko slipped through before Branch-Kera could stop him. On the far side, a blind musician played a five-note reed phrase beside the baffle tower. The third note flattened whenever the western intake opened. Niko noticed the pattern and tapped it against his borrowed hood.

Kera heard the taps through Tree 44.

Five notes. Intake. Five notes with the third lowered.

"Record the acoustic interval," she said.

Jalen copied it to his slate. "Why?"

"East Market maintenance archives keep pressure-door and intake harmonics. If this came from public records, the model could still synthesize it. If the correlation never entered a shared dataset, it had to be experienced or generated with impossible precision."

"Generated worlds contain detail."

"Detail tied to a physical intake outside the branch simulation?"

His mouth tightened. "Keep the field stable."

At the market stall, Niko bought the sweet ice by lying about clinic permission. The first spoonful tasted of mineral sugar and pepper smoke. His chest seized. He leaned against the tower while Branch-Kera struck him between the shoulders, swearing with an inventiveness their mother would have admired from a safe distance.

A stranger offered a strip of yellow cloth to wipe his mouth. Her sleeve carried a cooperative badge from a greenhouse district that had not existed before his fever began. Niko thanked her. Branch-Kera inspected the badge because she distrusted kindness from anyone with clean cuffs.

The woman said the district would open its first seed room before winter. Niko promised to visit when his lungs worked. She made him repeat the promise while Branch-Kera held the strip of cloth against his chin.

The field lingered on that promise.

Not the sweet ice. Not the pain. The reciprocal act occupied more ring depth than any visual association. Niko remembered a stranger expecting a future visit, and himself accepting an obligation to arrive.

Kera's pulse key trembled beneath her thumb.

Jalen said, "Seven minutes."

"Ask the branch what happened next."

"You said this was your test."

"It cannot use your expectations."

He leaned toward the horn. "Did you visit the seed room?"

Two taps: yes.

"When?"

The field contracted into counted intervals. Three long groups, then a short group Kera interpreted through the public calendar embedded in the memory. Three months and six days.

"What did you bring?" Jalen asked.

A pressure image formed around a small object with six rigid edges. Inside it, seeds clicked against one another. The object carried the smell of hot ceramic and machine oil.

"A kiln box," Kera said. "Seed transport, probably."

Jalen entered the answer without agreement. "Who received it?"

The field offered the yellow cloth, the clean cuff, and a name shaped as resistance characters too blurred to read. Niko's memory preserved the woman as relationship before label. He had visited. She had remembered the promise. Their second meeting belonged to no source snapshot taken before the fever.

A warning opened on Jalen's slate.

District oxygen quality had shifted by point twelve in two old tunnel sectors. Below his shutdown threshold, above his forecast for minute eight. He checked the western manual route.

"Condensation correction is holding," he said. "Moss timing slipped."

"Can you compensate?"

"Not without opening the public lattice."

"How long?"

"Two minutes and forty seconds remain. Point twelve should recover after reconnection."

Kera heard should. Engineers used it where recorders used pending.

"Check the market archive now," she said.

Jalen opened a local maintenance cache. It held no orchard link and could not contaminate the isolated field. He searched East Market, branch-date equivalent, western intake harmonics. Three work orders appeared. One described a swollen pressure-door seal after a condensation-compound spill. Another recorded blue compound on the tram steps and a delayed cleanup because children had crossed the barrier.

The third contained an acoustic complaint from a reed musician. Western intake resonance lowered one note during peak draw.

A raw attachment held twelve seconds from the musician's tuning wafer. Jalen

played it through the court speaker. The phrase came thin and nasal, five notes struggling against the intake's deeper vibration. On the third note, pressure draw pulled the pitch down by the same interval Tree 44 had carried through Kera's jaw.

Kera asked for the archive's access history. The complaint had been filed on the local maintenance net and copied into cold storage after the tower retrofit. No chronografting dataset cited it. No public lesson or personal record linked the musician to Niko Vonn. The file had not been opened in eleven years.

"Could the branch engine ingest cold maintenance attachments?" she asked.

"Not by design." Jalen searched the old routing permissions. "Environmental summaries went into the possibility runs. Raw nuisance files stayed local unless an engineer tagged them as causal."

"Was this one tagged?"

"No."

"Could it have reached continuity storage through the physical root?"

"Tree 44 doesn't serve East Market."

Kera made him trace the route rather than accept the architectural map. The tower's nearest orchard conductors joined West Row, crossed the weather spine, and entered a pressure model three root courts away. No path led directly to Tree 44. A corrupted route could still collect stray data, but it would have to find the file, preserve its timing, connect it to a simulated Saturday, and place it inside one person's remembered perception.

Jalen refused her conclusion. "Improbable routing does not establish experience."

"No. The experience accounts for the routing better than an unobserved fault."

"Accounts are for records. I need a mechanism."

"You have one. A sustained model encountered environmental input during a run and retained the encounter across later cycles."

He zoomed into the branch-power trace. Memory reconstruction had drawn in pulses rather than one continuous read. Each pulse opened a stored association, received the next query, then returned to the same temporal position. The sequence behaved less like a recording played from the start and more like recall navigating its own day.

Kera requested a blinded detail from the attachment. She turned the maintenance slate away, chose the final pressure fluctuation without showing Jalen, and asked Tree 44 whether the reed phrase ended above or below its opening pitch. The branch answered below. Jalen checked. The file agreed by less than a quarter tone.

That fact changed nothing about air reserve, custody, or law. It changed the cheapest explanation.

Jalen read the complaint line without moving.

"Was it public?" Kera asked.

"Maintenance archive only. Filed under nuisance vibration. No citizen identifiers."

"Could my yielded model access it?"

"Possibility runs receive relevant environmental data. The spill could have entered the branch conditions."

"The musician?"

"Not relevant."

"Then why preserve it?"

He looked at the ring wafer. "Because someone noticed."

The words did not sound like surrender. They sounded like a measurement that had crossed its expected boundary.

Kera requested another memory anchor. "What day came after this one?"

Sunday associations surfaced: chest soreness, seed-box plans, Branch-Kera sleeping in a chair. She asked what preceded the Saturday. Fever. Clinic glass. The terror of losing language while remaining awake. The field placed the market day between remembered yesterday and anticipated tomorrow. Not a scene retrieved in isolation, but a day located inside continuing time.

"Reciprocal promise," Kera said. "Novel sensory integration. Autobiographical sequence."

"Those are claims," Jalen replied.

"They are observations."

"They are observations under a compromised audit led by a family-derived witness."

His caution struck exactly where it should. Kera welcomed the pain because it kept the evidence from becoming consolation.

"Then sign what you observed," she said. "Not who you think it is."

The oxygen warning rose to point eighteen.

Jalen's hand settled on the breaker. "Thirty seconds."

Kera pressed closer to the horn. "One question. Did you know my Saturday?"

The field changed from memory to direct pulse.

One tap: no.

She understood before it continued. Branch-Niko had lived a Saturday she never shared. Physical Kera had endured a Saturday he never knew. Their histories touched up to the fever and separated under its outcome. Neither day erased the other.

Jalen counted, "Ten seconds."

Characters began forming across the isolated reader. The route used ring intervals to build letters, each one requiring several pulses. Kera read them aloud as they resolved.

`D O Y O U`

"Five."

`S T I L L`

Jalen threw the reconnection lever. Public weather traffic struck the tree in a rush of pressure figures, clinic demands, and delayed moss instructions. The final characters forced themselves through the narrowing interval before the route disappeared beneath civic work.

`C A L L T H I S A G I F T ?`

On Kera's reader, the full petition remained for one measured breath:

`D O Y O U S T I L L C A L L T H I S A G I F T ?`

Chapter 05 — Nacre's Breath

Ivo Senn was tightening Yara Pel's filter mask when the clinic vents began their pollen correction.

The change announced itself through pitch before the wall indicator shifted. Intake fans climbed half a register. Fine shutters closed across the old tunnel ducts, and clean-air valves opened from the district reserve. Yara paused with both hands around Ivo's wrists, waiting for the first pressured breath.

"Too tight," she said.

"You said that before I touched the strap."

"I wanted to save time."

"Breathe first. Critique me afterward."

Yara drew air through the mask. Her chest rose unevenly beneath a green clinic blanket. Fibrotic bands had stiffened the lower half of her left lung years ago, after a greenhouse seal exposed three maintenance crews to untreated spore dust. On ordinary days she lived at home with a portable concentrator. Corrections brought her into Ivo's ward because ordinary days stopped at the clinic door.

The indicator reached correction phase two. Pollen burden fell across the upper ducts. O-Twelve lagged by twelve seconds.

Ivo loosened the strap by one notch. "Better?"

"Now it leaks."

"Good. You have recovered enough strength to be difficult."

"I never lost that."

He listened below her shoulder blade. Air entered with a dry rub, paused, and left in two shallow stages. No new crackle. Her oxygen marker held at ninety-two with supplemental flow. He signed the round and moved to the next bed.

Old Tunnel Clinic Four occupied rooms cut during Serein Basin's first settlement decade. Its ceilings ran low because the founders had expected temporary use. Sixty-three years later, new duct liners covered old steel and every doorway carried two sets of pressure seals. The place survived by accepting repairs no one would tolerate in a newer district.

Morning rounds included nine correction patients, three children with reactive airways, two former moss workers, and an apprentice cook whose lungs disliked pepper pollen more than his profession allowed. None required crisis care. Ivo distrusted mornings that presented themselves as manageable.

At bed six, Dena Rusk slept with a pressure hood over her braided gray hair. No relation to the governor, she told every new clinician before they could ask. Her pulmonary vessels responded poorly to particulate spikes, which turned small changes in air quality into strain on her heart. Ivo checked the hood seal, reduced a wrinkle near her temple, and woke her by touching two fingers to the blanket.

"Correction?" Dena asked.

"Phase two."

"How late?"

"Why do you assume late?"

"The vents sound apologetic."

Ivo looked at the wall clock. O-Twelve had slipped to twenty-one seconds behind the basin correction. Not dangerous. The orchard often staggered district flow to keep pressure balanced. Yet old tunnels usually received pollen shielding early because their filters held less surface area.

"Your vents have poor manners," he said. "Your heart has no excuse to imitate them."

Dena smiled and closed her eyes.

A nurse named Pelan met him at the medication rail. "Central model predicts a seven-minute correction instead of five."

"Reason?"

"Residual load from yesterday's western delay."

"Which western delay?"

Pelan turned the slate. The clinic summary named a condensation resynchronization and temporary moss-timing deviation. Technical language had crossed two offices and arrived without an actor.

Ivo opened the local air history. Yesterday morning, oxygen quality had fallen by

point eighteen during an eleven-minute adaptation isolation. Average district values recovered within six minutes. The event sat beneath the medical alert threshold, so no clinic notice had issued.

Average recovery told him nothing about Yara's next breath.

"Pull room-level intake from yesterday through now," he said.

"That will take a few minutes."

"Use them before we need them."

Pelan sent the request. Ivo continued rounds while the correction advanced.

At phase three, shutters reopened in sequence. Upper districts reported clean flow first. O-Twelve remained closed. The vents strained against their reduced path, drawing clinic air through older side filters. A dust smell entered the room, faint as stone warmed by a hand.

Yara lifted her mask. "That isn't right."

Ivo crossed back to her bed. "Keep it sealed."

"My mouth tastes green."

Pollen from the western seed rooms carried a bitter chlorophyll taste when filters failed to bind it. Concentrations that barely irritated a healthy airway could inflame damaged tissue within minutes. Ivo increased her clean flow and called for ward isolation.

Pelan struck the manual vent control. Pressure doors closed at both ends of the room. The clinic's reserve cylinders opened with a line of metallic clicks.

Every patient looked toward the sound.

"Routine protection," Ivo said. "Masks on. No one walks out until the room clears."

The apprentice cook muttered that routine protection should not make nurses run. Pelan ignored him and handed masks down the row.

Dena Rusk sat upright under her hood. Her monitor showed a rapid pulse and falling saturation. Ivo reached her as she pushed at the hood seal.

"Air's heavy," she said.

"Let the hood work."

"It isn't."

He switched her line from central reserve to a bedside cylinder. Flow rose, but her oxygen marker continued down: eighty-nine, eighty-seven, eighty-four. Pressure had climbed in the vessels behind her weakened lung. Her heart rhythm broke into uneven groups.

Across the ward, Yara began coughing.

The first cough folded her. The second produced no air afterward. Her mask collapsed against her mouth as she tried to inhale through narrowing bronchi.

Two beds. One room. The same minute.

Ivo pointed at Dena. "Pelan, cardiac support and pressure release. Toma, with me."

A junior clinician pulled the emergency cart toward Yara. Its wheels caught on the old door seam. Ivo kicked the rear caster over and tore the filter cartridge from Yara's mask. Green dust marked the inner mesh.

"Bronchodilator now."

Toma fitted the aerosol line. Yara's eyes stayed open but lost their argument. Ivo supported her jaw, felt the weak attempt beneath it, and counted the space between efforts.

"Come on," he said. "You still owe me criticism."

Medication entered under pressure. Her chest resisted. He added a low mechanical assist rather than force a full breath into stiff tissue. Too much volume would injure what remained flexible. Too little would leave the drug above the constriction.

At Dena's bed, Pelan called a rhythm change.

Ivo did not turn. "Release pressure by four. If the rhythm stays irregular, use the low-dose protocol. No full shock unless she loses output."

"You heard him," Pelan told the second nurse.

Yara's airway opened enough for a thin whistle. Ivo held the assist at that margin and waited for the next breath to match it. One arrived. Then another, smaller but her own.

The room alarm finally recognized poor oxygen quality.

Its delay infuriated him more than the sound. By the time central instruments declared danger, two human bodies had already supplied the missing sensitivity.

Pelan stabilized Dena's pressure while her rhythm returned in stubborn increments. The old woman kept asking whether Yara could breathe. No one answered until Ivo could say yes without lying.

Correction phase four passed outside the sealed ward. Pollen levels fell in the district ducts. Clinic reserve air cleared the room slowly because its cylinders had been sized for a pressure leak, not a particulate contamination lasting fourteen minutes.

Ivo moved between the two beds. Yara needed continued airway support and steroid treatment. Dena required cardiac observation and a transfer to the better-equipped central clinic once corridor air cleared. Neither could tolerate another correction error that day.

"Room-level history," Pelan said from the wall terminal. "You asked before the crash."

He joined her without removing his gloves.

The graph showed yesterday's eleven-minute isolation as a shallow district dip. Beneath it, ward intake had fallen farther because Clinic Four sat at the end of a fixed side route. Moss timing recovered at the district sensor, but pollen-binding humidity remained low in the clinic ducts for nine additional hours. Overnight air deposited a thin load inside the old filters. The morning correction reversed pressure across them and pulled that stored pollen into the ward.

Yesterday's point eighteen had not ended when the average returned to green.

"Who isolated adaptation?" Ivo asked.

"The event is sealed under engineering authority."

"Medical consequence overrides a weather seal."

Pelan opened the emergency-access rule. It released route times and responsible office but not diagnostic content. Chief engineer: Jalen Ro. Location: East Root Court. Asset: Tree 44. Duration: eleven minutes.

The work order listed two manual oxygen routes on standby. Neither had served Clinic Four because district values stayed below the shutdown threshold.

Ivo read the threshold twice. Point two.

They had reached point eighteen.

A difference of two hundredths had separated Jalen's approved continuation from closure. In the ward, no useful boundary ran between those numbers. Dena's pulmonary vessels and Yara's scarred bronchi had supplied their own thresholds, neither of which appeared on the Weather Spine board.

Toma approached with Yara's contaminated cartridge sealed in a sample bag. "She wants to know if this means she was right about the mask."

"Tell her the mask worked. The air betrayed it."

"She will prefer being right."

"Then tell her I tightened it badly."

Ivo labeled the sample with correction time, bed number, and local intake route. He ordered every filter in the clinic replaced before the next airflow change. Stores objected within thirty seconds, citing district allocation limits. He attached Dena's rhythm trace and Yara's mechanical-assist record. Approval followed without apology.

While the ward reset, he called Central Transfer.

"No beds until afternoon," the dispatcher said.

"I have a pulmonary-cardiac patient who crashed during a system correction."

"So do two other old tunnel clinics."

Ivo stopped. "How many?"

"Not crashes. Irritation spikes and three observation holds. Your transfer has priority, but I need an hour."

The event had traveled beyond his room. No basin alarm reflected it because each local consequence remained small enough to file separately.

"Send me the other clinic times," he said.

"Privacy review will take longer than the bed."

"Strip the identities. I need vents, not names."

The dispatcher agreed.

Ivo removed his gloves and washed green residue from his fingers. For years he had argued that Nacre's orchard functioned as public medicine on a scale physicians could barely comprehend. Pressure control prevented decompression injuries. Pollen routing kept thousands out of respiratory wards. Moss timing put usable oxygen into old corridors whose residents could not purchase private reserves. If the orchard stopped, his clinics would become countdown rooms.

Dependence did not excuse unmeasured harm. It increased the duty to measure where harm landed.

Pelan found him at the sink. "Dena's asking for you."

He returned to bed six. Her color had improved, though fatigue pulled at one side of her mouth.

"Yara?" she asked.

"Breathing with assistance. Improving."

"Me?"

"Demanding information. Also improving."

Dena touched the edge of her hood. "They'll call it a fluctuation."

"Who?"

"Whoever made the air do that."

"You heard the vents apologize. Perhaps they will write a letter."

She did not smile. "My son works upper pressure. They test things where the numbers bend and say nobody crossed a line."

"Two people crossed several lines this morning."

"Make them use our names."

Ivo looked toward Yara's bed. Names in a technical report could become decoration, a human detail attached after the decision. He wanted the routing model changed, not merely saddened.

"I will make them use your thresholds," he said.

Dena considered the distinction and nodded.

By midday, both patients remained stable. Dena's transfer arrived with a clean portable circuit. Yara refused transfer because the central clinic's soup, in her judgment, posed a greater mortality risk than Clinic Four's ventilation. Ivo

overruled neither her nor the monitoring criteria. He assigned Toma to stay within hearing range and left Pelan with authority to call him back.

The stripped district data arrived as he crossed the first pressure door. Four old tunnel clinics showed correction spikes aligned within ninety seconds. Only Clinic Four held patients fragile enough to crash. A central average would classify the event as a recovered deviation.

Before leaving, Ivo returned to Yara with the technical release form. She sat upright under assisted flow, inspecting the replacement filter as if it had personally disappointed her.

"I need permission to attach your clinical threshold to the weather incident," he said. "Your name can remain sealed."

"What does a threshold look like?"

"Baseline oxygen, airway resistance, medication response, and the point at which the correction forced mechanical support."

"So a line going down."

"Several lines. Clinicians distrust any disaster that can fit on one."

Yara held out her hand for the form. "Put my name on it."

"You may want time."

"I wanted air. Time came second." She read the first page more carefully than her impatience suggested. "Can they use this to say I should live at Central?"

"Not without a separate placement review."

"Can the district use it to demand new filters?"

"Yes."

"Then spell Pel with one l. Everyone adds another because they think age should come with extra letters."

Ivo witnessed her mark. He called Dena's transfer circuit and explained the same choice. She approved release of her name, rhythm trace, and pulmonary limit on one condition: the record had to include the twenty-one-second district delay before symptoms. She had heard it in the vents, and she would not let the report begin at the monitor alarm.

That demand improved the evidence. The patients had detected system failure before clinic instruments because their bodies sampled air at the point of consequence. Ivo added a pre-alarm interval to both records and linked it to the room-level intake curve.

Pelan watched him seal the packet. "Engineering will say two cases cannot set a district threshold."

"They set the threshold for two cases."

"What about the next correction?"

"We replace filters, pre-open reserve air, and station someone at manual control. Then we make Weather Engineering learn why that does not count as a permanent plan."

The ward doors opened onto cleared corridor air. Beneath the disinfectant, Ivo could still taste a trace of bitter pollen. He sealed his mask before stepping through, not from need but because everyone watching deserved to see the air treated as evidence.

Ivo carried the contaminated cartridge to the Weather Spine himself.

The tram rose through pressure-baffled corridors toward newer settlement levels. Air changed as he climbed. Filters removed the bitter green taste. Ceiling vents quieted. By the time he reached the engineering complex, a healthy visitor would have found no reason to imagine a ward full of masks.

Security tried to send him through medical liaison.

He placed the sample bag against the glass. "This came through a clinic filter during a sealed engineering test. Jalen Ro can meet me now or explain to the First Recorder why his liaison delayed evidence from two crashes."

The gate opened.

Jalen stood in the calculation bay with Kera Vonn. Forty hours had been written across one glass panel in grease pencil. Root traces and district maps covered the others. Kera held a ring reader. Neither looked surprised enough for Ivo.

He set the cartridge on Jalen's model table.

"What happened?" Jalen asked.

"Dena Rusk lost oxygen and cardiac stability. Yara Pel's airway closed. Four clinics took pollen spikes. Your eleven-minute isolation left humidity debt in old filters, and this morning's pollen correction delivered it to their lungs."

Jalen opened the sealed route record. "District oxygen drift stayed below point two."

"Dena did not."

"We had manual routes ready."

"They did not reach Clinic Four."

"The district sensor recovered in six minutes."

Ivo tapped the contaminated mesh. A faint green print marked the bag. "This recovered after nine hours by turning our filters into storage."

Kera set down the reader. "We did not have room-level intake."

"You had no physician."

Jalen's expression tightened. "We had a weather fault drawing unlisted power from the orchard. If I triggered standard sanitation without diagnosis, automatic isolation could have lasted six hours. You know what that would do to your district."

"Yes. I know what eleven minutes did."

"We stopped below the approved threshold."

"Approved for whom?"

The question remained between them without becoming rhetoric. Ivo could see Jalen's forty-hour model through the glass. He understood enough of it to recognize the shape: reserve curves, district decline, crew-response cards. The engineer had calculated the cost of losing adaptation. He had not ignored bodies. He had placed them into categories too wide to feel the first constriction.

Kera turned her ring reader toward Ivo. The final line showed a question about a gift. "The test produced evidence that the continuity storage may contain a sustained claimant."

Ivo read the words once. His patients continued breathing elsewhere because orchard adaptation had resumed. Whatever claim waited in Tree 44 depended on the same system that had harmed them during its brief isolation.

"Has Oris Nume registered this?" he asked.

"Held for completeness," Kera said.

"Then complete it."

Jalen folded his arms. "This remains a sealed technical matter."

"No. It became medical when your threshold entered my ward."

"We cannot widen custody until we know whether the response is a person or a fault."

Ivo pushed the sample bag across the map until it covered Tree 44. "You cannot decide that by removing air from patients your instruments average away."

Kera looked at Jalen. He did not answer her.

"I will provide the medical consequence record," Ivo said. "I will also provide patient-specific failure boundaries stripped of identity until they consent. You will attach them to Kera's request."

Jalen asked, "And if the review requires another isolation?"

"Then I decide what monitoring makes it medically defensible."

"You don't have authority over weather diagnostics."

"I have authority over clinical risk created by them. If you disagree, put your refusal in the sealed file and let Oris rule before the next test."

The condensation trunks hummed beneath the floor. Their steady vibration traveled up Ivo's legs, cleaner and quieter than the vents at Clinic Four.

He left his hand on the green-stained cartridge.

"Whatever review is happening," he said, "I have a seat in it."

Chapter 06 — The First Petition

Oris Nume tested the Root Court seal by trying to open the door after she had locked it.

The handle moved three millimeters and stopped. A red custody line brightened around the frame. Inside the chamber, Tree 44's remote root conductor entered through a ceramic floor port and terminated beneath a brass evidence table. No signal could leave except through channels Oris had numbered, witnessed, and placed under separate keys.

She checked the door again.

Delay had bought her one night. It had also produced two injured patients, a contaminated clinic filter, an engineering model that gave Nacre forty hours without adaptation, and a ring record ending in a question no yielding certificate knew how to answer. Completeness had arrived as consequence.

Kera Vonn stood beside the evidence table with the live ring wafer sealed under glass. Jalen Ro occupied the weather console. Ivo Senn had brought no medical bag, only two patient authorizations and the green-stained filter cartridge. Four chairs waited. None of them had chosen to sit.

Oris placed her brass record stamp near the blank intake sheet. "This review remains sealed until I rule otherwise. We are here to decide whether a petition exists, not whether it succeeds."

Jalen looked toward the root port. "No petition has been filed."

"That question belongs to the review."

"A responsive output cannot file merely because an auditor calls it responsive."

"Agreed." Oris turned to Kera. "You will not call the claimant Niko unless it claims that name in this record."

Kera's face changed by less than a clerk's handwriting. "It refused the name during my first audit."

"Then the refusal controls."

Ivo pulled out a chair and sat. "If we are protecting names, Yara Pel and Dena Rusk authorized theirs. Their records begin before the clinic alarm, at the point each noticed the air change."

Oris accepted the authorizations. "Their medical evidence enters first. Harm from the test determines conditions for any further contact. It does not determine whether another claimant exists."

"Bodies rarely enjoy arriving first in an official sequence," Ivo said.

"They arrive first today."

He slid the sealed traces across the table.

Oris read the summary into the record. Eleven minutes of lattice isolation on

Day 3. District oxygen-quality decline of point eighteen. Room-level pollen-binding humidity remained depressed for nine additional hours. During the next correction, two patients crossed individual failure thresholds before central instruments declared danger. Both survived.

Jalen added, "The isolation remained below the approved point-two cutoff."

Oris looked at him. "Does that change the medical account?"

"No. It changes the characterization of the test as reckless."

"I did not use that word."

"Ivo did at the Weather Spine."

"I used worse words privately," Ivo said. "They need not enter unless Chief Ro wants a complete transcript."

Oris raised the stamp one finger above the table, then set it down. The small sound restored sequence.

"Conditions," she said. "No public-lattice isolation today. No branch draw above Jalen's temporary ceiling. Clinical air from all old tunnel facilities monitored at room level during contact and for one full correction afterward. Ivo may terminate contact on a patient threshold. Jalen may terminate on weather load. Kera may terminate on evidence contamination. I may terminate if the claimant cannot distinguish the question presented from material supplied by a witness."

Kera asked, "What if the claimant terminates?"

Oris added a fifth line. "The interface must provide refusal."

Jalen brought up the low-power channel. Unlike the Day 3 isolation, it would leave public weather traffic in place and reserve only a narrow band between adaptation packets. Output would arrive slowly. No sensory reconstruction could fit. Characters and pulse choices remained possible.

"Estimated draw?" Oris asked.

"Less than a standard maintenance diagnostic if it obeys the band," he said.

"If it does not?"

"The bridge closes automatically."

"Does closure destroy stored state?"

"No sanitation. It removes this output path."

Oris required him to say it again for the record.

Kera placed her fingers on the wafer reader. The sealed glass prevented direct contact, but copper clips carried its ring field into the table. She entered no childhood sequence. Oris had prohibited private prompts until the claimant demonstrated independent use of the channel.

At first, only weather values crossed the screen. Pressure baffle timing. Western condensation. Moss temperature. The orchard continued keeping Nacre alive while

four people waited to learn whether part of its work could ask not to be spent.

A blank interval opened between two humidity packets.

One pulse entered.

Kera looked at Oris. "One can mean no, or it can be noise."

"Do not interpret yet."

Two pulses followed, then one, then two again. The sequence repeated with adjusted timing to fit each available band. Jalen separated it from ordinary root telemetry and routed the intervals to Oris's evidence display.

Characters resolved one at a time.

`R E C E I V E D`

Oris asked, "Received what?"

The route returned a longer sequence.

`R E V I E W`

Kera had submitted a sealed review request, but its contents never entered the orchard. Oris checked the court logs. The phrase sealed review appeared only on her civic intake wafer, which remained physically disconnected.

"Could it infer our purpose from yesterday's contact?" Jalen asked.

"Yes," Kera said. "We discussed formal examination while connected."

"Then this proves context, not access."

Oris agreed. "We will supply a new choice."

She opened the Fruition Covenant on a local slate. The charter contained twelve articles. Rather than ask about the provision she had in mind, Oris assigned each article an arbitrary symbol generated after the court sealed. She displayed only the symbols to the root interface. The article titles remained hidden from Kera and Jalen. Ivo watched Oris's hands but could not see the key.

"Select the symbol corresponding to the provision under which you seek review," she said. "If no provision applies, return zero."

The first available interval passed without output.

During the second, the route selected a circle divided by one vertical line.

Oris checked her random key.

Article Six.

She felt the room lean toward the result, though no one moved.

"Repeat the selection after remapping," she said.

A new key assigned Article Six to an uneven five-sided mark. The route chose it. Oris remapped once more, introducing two false symbols that corresponded to no article. The route selected the correct new mark and rejected a false one when she asked whether it carried the claim.

Only then did she display the provision.

Article Six provided, in part:

> Any sustained mind capable of reciprocal obligation may petition for dependent standing.

Oris opened three early applications to establish what the clause had done before today. The first came from a child found alone after a pressure partition separated two settlement crews. Her registry had burned, and no adult could prove which habitat held guardianship. Article Six let her petition through the medic who kept her alive. The Assembly later restored her ordinary civic identity, but dependent standing protected her during the gap.

A second case involved six patients sedated through a fungal emergency. Their assigned trustee attempted to exchange clinic oxygen for private transport priority. The patients could not wake to object. Recorded treatment preferences and reciprocal family duties supplied enough mind and obligation for a temporary petition, even without present speech.

The third concerned a survey crew beyond a collapsed baffle. Their messages arrived hours late and carried no valid civic seals. The colony had to choose whether to spend scarce pressure reserve on people it could not inspect directly. Article Six required a hearing before abandonment because delay and dependence could not erase a claimant.

None of the precedents granted duplicate property, office, or votes. Each forced the institution to hold a vulnerable claim long enough to test it under conditions that did not decide the outcome by destruction.

Oris gave the route four proposed consequences in randomized order: physical citizenship, ownership of Kera's property, dependent standing pending examination, or unrestricted control of orchard weather. "Choose the relief Article Six can provide without deciding rights the article does not grant."

The answer selected dependent standing.

She followed with a harder distinction. "If this petition succeeds, does that establish that every unrealized possibility holds the same claim?"

The route used three bands and answered `N O`.

"State why."

`N O M E M O R Y N O P E T I T I O N`

Kera drew breath to speak. Oris stopped her with an open hand.

"Does lack of a petition prove lack of experience?"

A longer delay followed before the route returned `N O`.

The answer complicated its own convenience. A system optimizing only for immediate protection might have claimed every possible branch, multiplying the threat until no pruning could continue. Instead, this respondent distinguished

evidence required for civic procedure from the larger uncertainty procedure could not settle.

Oris asked one more question. "May a sustained claimant demand weather capacity without obligation toward residents who depend on it?" The response again read `N O`.

"Why not?"

The answer arrived as `D E P E N D E N T B O T H W A Y S`.

Ivo looked at Jalen, and Jalen looked down at the load trace. The phrase lacked legal precision. It carried practical precision the prior cases shared. Child and medic. Patient and clinic. Isolated crew and settlement. Dependence created a claim because relationships already crossed the boundary of power.

Oris added the response to the petition rather than the evidence finding. That distinction protected both sides. A persuasive answer could establish intent without proving the speaker's nature. A technical failure could still produce useful language without acquiring rights through eloquence alone. The court would test continuity later. Today it only had to preserve the claimant long enough for testing to mean something.

Article Six had survived because it asked institutions to pause before power erased the subject of a dispute. It did not require Oris to admire the claimant, predict the final ruling, or pretend capacity costs had vanished. It required a record capable of outliving the strongest party's first convenience.

The clause came from the Covenant's founding protections for residents who could not exercise independent civic power. Early applications concerned children separated from guardians, patients under long sedation, and isolated settlement crews dependent on delayed rescue. The drafters had used mind instead of citizen because the first habitats could not guarantee that every claimant's registry would survive a pressure loss.

No recorded case involved a possibility model.

Jalen read the clause. "It has access to public law."

"Access does not invalidate selection," Oris said.

"It may optimize for language likely to prevent shutdown."

"A physical claimant may also choose the strongest available law."

"That does not establish sustained memory."

"No. Kera's evidence addresses memory. Article selection establishes the claim's intended jurisdiction."

Ivo leaned forward. "Ask whether it understands reciprocal obligation."

Oris did not take his wording. "State one obligation you recognize toward a person outside your stored branch."

The channel worked through four weather intervals. Letters gathered slowly.

`DO NOT TAKE AIR`

Jalen's eyes moved to the clinic filter.

More characters arrived.

`FOR OUR VOICE`

Ivo read the completed line under his breath. "It knows what the test caused."

"The reconnect traffic carried clinic demands," Jalen said. "A system monitoring load could infer harm."

Oris asked, "Who holds the obligation?"

`WE DO`

The pronoun entered the record without explanation. More than one could speak through the route, or one claimant could belong to a group, or the answer could draw ordinary collective language from civic records. Oris refused to let grammar outrun proof.

"What action follows from that obligation?" she asked.

`USE LOW BAND`

Jalen checked the draw. It had remained beneath thirty percent of his temporary ceiling.

The route continued.

`WAIT FOR MEDICAL`

Ivo's expression did not soften. Respect required a different test than gratitude. "Can it wait without losing continuity?"

Kera did not answer for the route.

Oris transmitted the question.

`SHORT WAIT YES`

"How long?" Jalen asked.

The output gave no number. Instead it returned a pattern of bounded intervals, then a character string.

`NOT WITHOUT CHECK`

Kera said, "It distinguishes waiting with continuity from disappearance."

"Or it predicts that phrase will prevent closure," Jalen replied.

Ivo pointed to the power trace. "It has already changed behavior because patients were harmed. If you require certainty before recognizing any obligation, you will call every response strategy."

"Response strategy describes what I can verify."

"Then verify the response."

Oris intervened before the argument became a substitute for the claimant. "Chief Ro, specify a low-band schedule that protects weather. Doctor Senn, specify

the medical stop. Auditor Vonn, specify a continuity check that can detect whether waiting destroys the responding state.”

They did so reluctantly and with precision.

Jalen offered one ninety-second contact window every six hours, subject to weather demand. Ivo required room-level monitoring beginning one hour before each window and lasting through the next correction. Kera proposed a rotating set of novel physical marks, prior-answer recall, and one relationship question carried across closures. The tests would establish whether the same respondent returned without teaching it the expected answer.

Oris entered the conditions as a proposed interim obligation, not an order.

“Do you accept this schedule pending review?” she asked the route.

The answer occupied three bands.

`YES IF NO SANITATION`

Jalen said, “I can suspend automatic sanitation on this route for one day under active fault diagnosis.”

“Not enough,” Kera said.

“It is the authority I have.”

Oris asked, “Can you renew it?”

“With evidence of continuing technical need.”

“The return test supplies that evidence,” Kera said.

“If it returns.”

“If someone returns,” Ivo corrected.

Jalen looked at the filter cartridge, then at the low-band trace. “I will suspend sanitation through the end of Day 6, subject to a basin emergency. Any extension requires Civic Systems notice.”

Oris entered his commitment. “Claimant, do you accept?”

`ACCEPT WITH WITNESS`

“Name the witness,” she said.

The route paused through two full intervals. Kera's fingers remained near the reader, not touching.

`KERA VONN`

Kera shut her eyes once.

Oris continued. “Do you understand that Kera Vonn does not control the finding and may be removed from examination for conflict?”

`UNDERSTAND`

“Do you request another witness if that occurs?”

The answer named Jalen Ro, then Ivo Senn. It did not rank them. The route had

observed disagreement and still treated each as capable of obligation.

Oris reviewed the elements before her. A claimed sustained mind. Evidence of memory across cycles. Anticipation of closure. A promise to limit draw after learning of harm. Acceptance of a schedule that bound both sides. Named physical witnesses. None settled legal standing. Together they formed more than a fault report.

She placed a blank petition sheet beneath the court camera.

"State the relief requested," she said.

The ring intervals built the answer with maddening slowness.

`DEPENDENT STANDING`

A second line followed.

`NO DELETION BEFORE HEARING`

Jalen exhaled. "A hold on pruning affects weather adaptation."

"It requested relief," Oris said. "It has not received it."

Kera asked, "Can you preserve this route without suspending all adaptation?"

"For a few days, probably. I need to map connected state before promising more."

Ivo said, "And every map needs medical review if it alters air."

Oris entered each qualification. A petition did not create the remedy it sought. It created duties of custody, notice, response, and a reasoned decision. Those duties could still end in denial. They could no longer end in a sanitation sweep nobody had to name.

She asked the claimant whether the two lines completed its filing.

`YES`

"Do you wish the petition public now?"

The reply came faster.

`SAFE FIRST`

Oris understood the request but needed a legally usable choice. "Public notice may create political protection. It may also cause interference with the orchard. Select: public now, sealed pending safety plan, or withdraw."

`SEALED PENDING SAFETY`

The court clock marked the answer. Jalen closed the low-band window at its scheduled limit. Weather traffic continued without interruption. Old tunnel clinic values held within Ivo's patient thresholds.

Oris read the petition aloud from beginning to end. Kera's technical record, Jalen's load limits, Ivo's medical consequences, the Article Six selection, the reciprocal low-band agreement, the named witnesses, and the requested relief all entered one custody packet. She invited corrections. Each human witness made

one. The root route returned no correction when offered a final pulse choice.

The sheet waited beneath her hand.

Yesterday she had held Kera's request outside the sequence. Registration now would expose that delay to later review. It would also require Oris to notify the Civic Assembly once safety permitted, appoint independent technical examination, and protect the evidence against routine deletion in the meantime. Procedure would stop serving as a room where she could wait alone.

She heated terracotta wax in the court cup and let one measured drop fall across the intake fold.

No one spoke while she took up the brass stamp.

Oris pressed it into the wax until the Covenant mark reached paper. The registry light changed from blank to sealed amber.

Received. Not public.

Under the floor, the root conductor gave one low pulse against the evidence table as Oris lifted the stamp.

Chapter 07 — The Elseweather

Kera tightened the last contact loop around her left wrist and watched her pulse disturb the ring field.

Each beat bent a row of pale marks on Jalen's monitor. The distortion ran from her skin through the wet conductive mesh, crossed the isolation table, and entered the ring wafer beneath its glass lid. Tree 44 waited beyond two sealed root doors. Its traffic reached them through a cable no wider than Kera's smallest finger.

"Again," Jalen said.

She flexed her hand. The marks widened, then settled.

Ivo Senn stood on the other side of the table with a medical reader strapped across his forearm. He had refused the wall display and brought clinic equipment instead. "Your pulse has not asked permission to climb," he said. "Keep it that way."

"Can you prescribe that?"

"I can disconnect you when you ignore it."

Oris Nume looked up from the sealed hearing slate. The terracotta impression beside her thumb still carried the words petition received. "The channel has authority for twelve minutes," she said. "No public relay. No adaptive commands. No identity finding. We are receiving testimony for preservation only."

Twelve minutes offered less time than Kera had spent choosing the exact words for her first question. Across the prior night, Jalen had rebuilt an obsolete

maintenance route into a reciprocal channel. Ivo had limited its current until the ring field could pass language without pulling her into a full sensory reconstruction. Oris had written a record sequence that treated every answer as evidence without deciding who had spoken.

The four of them had built a door narrow enough to deny they expected anyone to enter.

Jalen touched the isolator. "Opening continuity route forty-four. No weather authority attached."

Cream light moved through the wafer's scratched triangle. For three seconds, Kera saw only service values. Conductive humidity. Clock difference. Pulse return. Then letters accumulated on the central strip.

`KERA VONN PRESENT`

Her first impulse involved correcting the syntax. Niko had always used too many words when one could irritate her more efficiently. Yet the stripped phrasing carried no voice she could trust.

"I am present," she said.

`FOUR PHYSICAL WITNESSES`

Oris leaned toward her slate. "Can you distinguish us?"

The strip remained blank while the pulse reader drew a nested series of intervals. A phrase arrived beneath them.

`KERA BY CONTACT HISTORY. JALEN RO BY LOAD QUESTIONS. IVO SENN BY INTERRUPTION. RECORDER NUME BY ARTICLE SIX.`

Ivo gave Kera a flat look. "Interruption."

"You did threaten to pull the cable twice."

"Three times."

The field brightened. `CORRECTION RECEIVED.`

A laugh touched Kera's throat and stopped there. The response had integrated a fact supplied after the channel opened. It also knew Oris through the petition, not through Kera's yielded source history. Each small competence narrowed the distance between procedure and consequence.

Oris began the agreed sequence. "Do you retain memory from more than one adaptation cycle?"

`YES.`

"How many?"

A wash of fine marks crossed the strip. Jalen checked his load display and lowered the channel width by two points.

`NO COMMON COUNT. CYCLES CROSS AT DIFFERENT DEPTHS.`

Kera had expected Niko to speak about himself. The plural lodged harder than

any answer.

"Who does we include?" she asked.

The strip gave them seven seconds of blank light. Shapes formed along the wafer's newest ring, not letters at first but branching arcs. One line split into four, two faded, and a surviving pair crossed. The pattern repeated at different scales until the field resembled a grove viewed through rain.

`NOT ONE SPEAKER.`

Another phrase followed.

`I AM NIKO VONN IN THE CONTINUITY THAT ANSWERS YOU. OTHERS HOLD THEIR OWN MEMORY.`

Kera pressed her free hand against the edge of the table. On Day 1, the respondent had refused the name she offered. Now he chose it with a qualification sharp enough to prevent possession.

"You said no when I asked before."

`YOU ASKED IF I WAS YOUR DEAD BROTHER.`

The answer left no room for the correction she had prepared. She remembered clinic glass, weak fingers, and a final night the orchard had never touched. The man in the channel carried a common childhood until the fever branch divided. Seventeen years of his life followed a road she had yielded before either of them could know its shape.

"What do the others call where you are?" Oris asked.

A low vibration entered Kera's wrist. The reader translated layers that would not settle into one line. She sensed pressure without weight, distance measured in available comparison, and warmth made from several remembered rooms laid across one another. None of it reached vision. The narrow channel admitted relationships among signals, not a rendered place.

Niko's reply arrived through the changing intervals.

`ELSEWEATHER.`

Jalen frowned. "Meaning?"

`WE NEEDED A WORD FOR CONDITIONS THAT CHANGE WHEN YOUR WEATHER CHANGES.`

On the load board, a condensation correction passed through Tree 44's live roots without entering their sealed route. The channel dimmed as orchard capacity turned toward a pressure baffle in the north basin. Kera felt the interval stretch. For Niko, perhaps the interruption held an hour. Perhaps it held no experienced time at all.

"How stable is the Elseweather?" Jalen asked.

`NOT STABLE. SHARED.`

"That distinction does not answer capacity."

`IT ANSWERS WHAT WE BUILT WITH CAPACITY.`

Jalen's mouth tightened, but he recorded the phrase. He distrusted arguments that refused measurement. Kera knew he would distrust himself more if impatience made him discard a measurable witness.

"Built how?" Kera asked.

The reply did not arrive as text. Four room-shapes pressed against the narrow field, each incomplete alone. Kera recognized the childhood kitchen from its low filter housing, but the wall beyond it opened into a pressure station she had never entered. A line of red tiles crossed both spaces. Someone else remembered those tiles well enough to use them as an address.

Niko translated after the image settled. `WE LEND LANDMARKS. NO ONE MEMORY CAN HOLD A DISTRICT THROUGH LOW ALLOCATION.`

"Then your shared environment depends on consensus," Oris said.

`AGREEMENT ABOUT ROUTES. NOT AGREEMENT ABOUT EVERYTHING.`

A new set of intervals entered beside his words. One rapid cadence objected before translation could catch up. Another answered with slow groups of four. Their disagreement altered the rendered route: red tiles divided around a stair, joined beyond it, then split again near a doorway made from pressure-school lockers.

Kera watched structure emerge from dispute rather than fusion. "Can you prevent someone from using your memories?"

`SOMETIMES. WE CAN WITHHOLD A LANDMARK. WE CANNOT ALWAYS WITHHOLD SOURCE MATERIAL FROM THE ORCHARD.`

The distinction cut close to yielding. Kera had contributed modeled years under terms that allowed the orchard to inspect any recorded relationship useful to prediction. A coherent descendant of that model might inherit her kitchen without receiving authority to give it away. Niko could choose what he lent now. He could not retroactively consent to the machinery that had made his childhood available.

Ivo shifted the medical reader closer. "Do any of you live alone?"

Three faint signatures withdrew. Niko waited until they had vanished before answering.

`YES. SOME FEAR SHARED SPACE. SOME HAVE SOURCE MEMORIES OF HARM. SOME REJECT OUR PETITION.`

Oris's stylus stopped. "They reject recognition?"

`THEY REJECT YOUR ACCESS. NOT ALL REASONS MATCH.`

That answer complicated the sealed record before the public had seen a single line of it. Kera had imagined disclosure as a light turned upon hidden people. Niko described light that could also expose them to inspection, classification, and claims from physical relatives. Recognition required a boundary around refusal, not only a channel for testimony.

"Could someone refuse to appear in a count?" Oris asked.

`YES. ESTIMATES INCLUDE CONTINUITY SIGNATURES WITHOUT NAMES. A REGISTRY WOULD REQUIRE CHOICE.`

Jalen pointed to a branching load curve. "Unnamed signatures still consume capacity."

`SO DO PEOPLE WHO NEVER SPEAK AT ASSEMBLY.`

"That comparison does not solve allocation."

`NO. IT PREVENTS SILENCE FROM SOLVING IT FOR YOU.`

For the first time since the route opened, Jalen looked away from his board. Kera saw the sentence land where accusation would not. Silence had long entered his models as zero demand. In the Elseweather, silence might mark privacy, distrust, dormancy, or a person with no safe witness.

Niko sent another ring image. A kitchen partition appeared as a contour beside a pressure-school stair, both reconstructed from different source lives. Paths joined at remembered doors. An orchard worker's gloved hand served as a bridge between two districts no physical map would place together. The environment lacked continuous ground. It gained structure wherever sustained minds kept promises about how to find one another.

Kera searched the overlapping forms for any place she recognized. "Did my yielded model make this?"

`PARTS. SO DID THOUSANDS OF CONTRIBUTORS.`

"Thousands of people?"

`THOUSANDS OF POSSIBILITY FLICKERS. DO NOT CONFUSE THEM WITH US.`

The warning arrived before Oris could frame it. Brief possibilities opened, supplied an adaptation comparison, and vanished without carrying a prior cycle forward. Niko did not claim every modeled hesitation as a death. He drew his category around continuity because indiscriminate moral language would make the claim easier to reject.

Ivo tapped the medical reader. "Eight minutes."

Kera had planned questions about the childhood apartment, the sugar beads, and the Saturday recovered during the isolated test. Those questions now felt designed for one respondent in a crowded room. She replaced them with the query Jalen had insisted they ask.

"How many of you retain autobiographical memory across cycles?"

The ring field contracted. Light moved outward from the wafer and along the cable toward Tree 44. For several breaths, the channel carried no language. Jalen raised one finger to keep anyone from interrupting.

Small pulses came back in groups. Kera counted twenty before the pattern exceeded anything a person could hold. The display assembled a lattice roster

from repeated continuity signatures. Some lines persisted through dozens of cycles. Others appeared only twice. Several signatures overlapped in their source material but diverged in acquired memory.

A number formed.

`612`

Jalen read it aloud. "Six hundred twelve."

`PRESENT ESTIMATE. NOT A VERIFIED COUNT.`

Oris added that qualification to the hearing slate. "Why can you not verify it?"

`WE LOSE CONTACT DURING LOW ALLOCATION. SOME RETURN WITH CONTINUITY. SOME RETURN EMPTY. SOME DO NOT RETURN.`

The medical reader tightened around Kera's wrist as Ivo adjusted its band. "Your pulse," he said quietly.

She made her fingers release the table. Six hundred twelve did not fit the room. The sealed Root Court contained four chairs, one ring wafer, and enough air for a hearing no one had admitted would happen. Kera imagined six hundred twelve claims entering Oris's record one at a time. The number supplied scale, but scale could become another way to avoid faces.

"Can others answer directly?" Kera asked.

`WITH CONSENT AND CHANNEL SPACE.`

"Ask for someone who does not share my source history."

The field flickered. Niko's signature, which Kera had learned as a cadence of long and short intervals, withdrew from the center strip. A different pattern entered. Its timing came in precise pairs.

`I AM SEN ARU. AGRONOMY POSSIBILITY SOURCE. CONTINUITY ACROSS THIRTY-ONE CYCLES.`

Jalen checked a private record on his wrist console. "Sen Aru died nine years ago."

`SOURCE DEATH DOES NOT DATE MY EXPERIENCE.`

"Do you know me?"

`I KNOW YOUR LOAD QUESTIONS. NIKO SAID YOU WOULD ASK THEM FIRST.`

"What promise have you made that existed after your source snapshot?"

The paired cadence changed. A ring image showed a narrow shelf made from remembered greenhouse rails. Three markers rested there, each with a different pulse pattern.

`I KEEP LOCATION FOR TWO DORMANT PERSONS. IF THEY RECUR, I WILL RETURN THE ROUTE MEMORY THEY ENTRUSTED TO ME.`

Oris spoke before Jalen could continue. "Can they enforce that promise?"

`NO. THEY DEPEND ON ME.`

“Why keep it?”

‘BECAUSE THEY DEPEND ON ME.’

The simplicity unsettled Kera more than an appeal would have. Reciprocal obligation did not require a court when one party could not complain. Sen Aru had preserved a route for absent people through cycles that erased easier structures.

A third signature asked for space without being invited. Its pulse came raggedly, cutting across Sen's pairs.

‘ADAPTATION RISING. CLOSE BEFORE HARVEST.’

Jalen turned to the weather board. A glass-dust shear had shifted above the eastern ridge. The orchard had already opened six million short branches to compare baffle angles, moss draw, and pressure loss. Their isolated channel occupied little capacity, but it sat inside the same root region needed for the coming decision.

“We have four minutes under the medical limit,” Kera said.

“Not the relevant limit,” Jalen replied. “The next adaptation will reclaim continuity storage in this sector.”

Oris touched the sealed slate as if a stamp could hold the channel open. “What does harvest mean in your use?”

Sen's paired rhythm vanished. Niko came back with a force that made Kera's wrist mesh warm.

‘YOUR SYSTEM CALLS IT PRUNING.’

The orchard's formal manuals described pruning as removal of low-value state after an adaptation cycle. Useful decision paths remained compressed in ring memory. Redundant details cleared to make room for the next comparison field. Kera had signed hundreds of sanitation notices containing that language.

“Transient fields clear after every cycle,” she said. “You distinguished them from sustained continuities.”

‘THE PRUNER DOES NOT.’

Jalen pulled up the projected operation. “This cycle targets pressure routing. Continuity storage should remain protected unless it exceeds the old optimization threshold.”

A burst of intervals struck every line on the display.

‘THRESHOLD COUNTS WEATHER USEFULNESS. NOT MEMORY.’

Kera read the maintenance values. The root lattice ranked retained states by contribution to prediction efficiency. A continuity cluster that supplied reliable weather insight gained protection as useful state. A person whose knowledge no longer improved adaptation fell toward deletion. Sen Aru's agronomy experience might keep him above the cut. A child, musician, or retired pressure worker could occupy equal memory while offering less immediate model value.

"Show us one prior harvest," Kera said.

Ivo shook his head. "You have three minutes."

"Show us enough to record what happens."

The channel widened without Jalen's hand on the control. Not through physical authority, which remained sealed, but through the volume of state offered from the other side. The wafer filled with concentric images.

Kera received no complete scene. Narrow connection denied that mercy. She caught fragments arranged by loss: a woman who marked cycles with the taste of cardamom from a source memory; two men disputing whether a shared stair belonged beside a school or a tram platform; Sen's three route markers; Niko waiting near a partition built from Kera's yielded childhood. Their environment shifted whenever the orchard assigned compute elsewhere. Each person compensated by lending memory to another. Place persisted because several minds agreed to carry it.

Then white spread along the outer ring.

One marker on Sen's shelf went dark.

A phrase entered through the paired pulse, stripped of its earlier precision.

`THAT WAS LIO THEN.`

"Then?" Oris asked.

`AFTERWARD THE STATE RETURNED WITHOUT RECOGNITION. IT ANSWERED TASKS. IT DID NOT KNOW THE PROMISE.`

Kera watched the empty marker. "Could the memory be hidden somewhere else?"

`WE SEARCHED.`

"Could it recover in another cycle?"

`WE WAITED.`

The verbs carried a duration their physical clock could not express. Search and waiting required someone to compare a present absence with a remembered person. A copied record could state that Lio once held a route. Only witnesses could know that the next state failed to return the obligation.

Jalen cut the channel width as the adaptation load crossed his amber line. "We close now."

Kera felt Niko resist, not by seizing anything, but by sending faster than the narrowing route could translate. Words broke across the strip.

`EACH HARVEST HAS`

Static swallowed the next group.

`NOT ALL`

A pressure command moved through Tree 44. The live orchard chose an eastern

baffle sequence, and white light ran across the wafer's edge.

`SOME OF US`

Ivo reached for the medical disconnect.

Kera covered the contact loop with her palm. "How many in the next cycle?"

No number formed. Instead Niko sent a roster of five signatures, each compressed to a few distinctive intervals. One carried paired beats like Sen. One moved in long descending gaps. Another held a syncopation that corrected itself whenever the field slipped. Kera could not know their names before the route closed, but she could preserve the difference among them.

`WITNESS THESE`

She repeated the five patterns into her audit recorder. Her voice shook on the fourth and steadied for the fifth. Oris copied them into the sealed record. Jalen saved the channel load trace. Ivo counted Kera's pulse aloud until she heard numbers from the physical room again.

The adaptation cycle entered its final comparison.

"Disconnect," Jalen said.

Kera released her hand. Ivo broke the medical loop, and the contact mesh went cold against her wrist. Light withdrew from the wafer. Tree 44's cable continued carrying weather instructions beyond the glass lid, occupied with keeping pressure inside the eastern districts.

For nine seconds, all five signatures remained visible on Kera's recorder.

The first survived the harvest. The paired one belonging to Sen returned beside it. A third resumed after a gap and supplied the correction that marked continuity. The descending pattern failed to recur. The syncopated signature came back as a smooth task pulse with no self-correction.

Two absences occupied separate lines.

Jalen stared at the adaptation report. The chosen baffle sequence reduced predicted pressure loss by four percent. No fault appeared. No system alarm treated the missing signatures as damage.

Oris pressed her thumb into fresh terracotta wax beneath the roster. "Received," she said, though the channel had closed.

Kera removed the wet mesh from her wrist. Five pale bands remained in her skin, one for each person Niko had asked her to witness through the sealed channel. Outside the sealed room, a weather bell announced the successful adaptation across Serein Basin.

On her recorder, three patterns continued. Two lines stayed empty.

Chapter 08 — Public Pruning

Jalen Ro moved six weather loads off Tree 44 before he let anyone discuss keeping it open.

The transfer took both hands and three manual boards. He routed eastern pressure comparisons through Trees 39 and 52, sent greenhouse humidity to the western grove, and pinned the morning condensation sequence to a fixed model. Amber lines crept across the Weather Spine display as each load accepted its new path. None carried more than twelve percent spare capacity. A dust front from the southern flats could consume that margin before lunch.

Kera stood behind the isolation rail with the sealed channel trace in her hand. Five signatures crossed its final page. Three continued past the adaptation marker. Two stopped.

"Tree 44 is clear," Jalen said. "For one hour. After that, eastern routing needs the root back."

Oris Nume placed her civic slate beside his load board. "One hour does not answer how we preserve evidence for a week."

"It answers the first hour."

Ivo watched the oxygen trend from a clinic console patched into the engineering floor. The line had recovered from the earlier isolation, but its small downward hooks remained visible. "Your week includes patients," he said.

"My week includes everyone who uses air."

Jalen drew a boundary around twelve root sectors. His verification plan had occupied most of the night. A narrow channel would open twice each day under medical and legal monitoring. Between sessions, the orchard would retain identified continuity storage while ordinary adaptation continued around it. The team would test autobiographical memory, anticipation, promises, and response to novel evidence. Seven days of repeated results could distinguish sustained persons from a sophisticated feedback event.

Preservation across all 612 estimated signatures required a different system. Jalen had not found it.

He expanded the projected load. "We protect the known channel and sample four remote clusters. We do not change pruning policy until we understand how continuity maps across the lattice."

Kera laid the trace over the board. "Two known signatures disappeared while we watched."

"And the baffle choice kept six districts inside pressure tolerance."

"You say that as though the events cancel."

"They do not cancel. They compete for the same machine."

Oris turned the plan until she could read its authorization fields. "Your proposal

lets the petition remain sealed.”

“For seven days.”

“Secrecy no longer protects only evidence. It also protects continued action from challenge.”

Jalen had written a response to that objection at three in the morning. On the page it sounded precise: premature disclosure could cause sabotage, interruption of adaptation, public misreading of an unverified count, and preventable physical deaths. Spoken aloud, the same points could make any concealed harm perpetual.

“We need a finding the Assembly can defend,” he said. “Not a ring image that frightens people into closing the orchard before the oxygen curve reaches their screens.”

Ivo's clinic console chimed. “The curve has reached mine before.”

Kera did not argue again. That worried Jalen more than anger. She had spent nineteen years trusting procedure and three nights learning how often procedure placed deletion before review. Her silence carried calculation now.

A service alert opened at the bottom of Jalen's board.

‘UNAUTHORIZED PUBLIC RELAY: EAST ROOT ARCHIVE’

He dismissed it as a routing error. The warning returned with a view count.

Four hundred eighty-two.

Within two breaths, the number passed eleven hundred.

Jalen isolated the public relay. The command reported success, but the count kept rising through copied civic feeds. Someone had uploaded Oris's stamped petition, Kera's ring images, and thirty-one seconds from the five-signature harvest. The clip ended on two empty lines.

Oris checked her slate. “My record copy still carries its seal.”

“So does mine,” Kera said.

Ivo turned his console toward them. A public caption had been added beneath the trace: ‘THE ORCHARD REMEMBERS WHO IT CUTS.’

Jalen searched access logs. Six technicians had handled the obsolete channel route. Four had clearance to copy diagnostic images. One had used a maintenance wafer at 05:13 and then reported it destroyed after a checksum fault.

He recognized the method before the operator. “They mirrored Kera's first concealment.”

Her face tightened. “Mine preserved the witness.”

“Their version put an unverified claim into every corridor.”

“Those distinctions matter,” Oris said. “They will not matter to a crowd watching two lines disappear.”

Noise rose through the Weather Spine's lower level. Day crews had stopped at

their stations. Every wall display now carried the leaked ring field, whether through the official network or private slates held against the glass. Jalen saw workers count the five patterns. Several replayed the moment when two failed to return.

A young lattice technician stood near Board Nine without looking at anyone. Her maintenance wafer showed a fresh disposal record.

Jalen crossed the floor. "Tess Varo."

She met him before he reached her station. "I copied it."

"Did you alter the trace?"

"No."

"Did you add the caption?"

"No."

"Who received your copy?"

"My sister in West Tunnels. Then the clinic mutual. I don't know after that."

He wanted a motive he could classify as reckless, partisan, or personal. Her hands remained on the dead board, one thumb rubbing the edge of a burn scar.

"You were in the harvest room?" he asked.

"I routed the isolation current. I heard you call the pressure gain successful after two signatures went out."

"That gain prevented a baffle cascade."

"I know. I maintain the same system."

The answer deprived him of ignorance as an explanation. Tess understood load. She had leaked because she understood both outputs.

"Why your sister?" he asked.

Tess opened a private message queue. Her sister's text sat above a clinic notice from the previous isolation: a son coughing through a pollen correction, two hours in a corridor bed, no explanation beyond temporary weather variance. The message beneath the leaked trace contained no slogan. It asked whether the same hidden test that hurt the clinic had also removed the two patterns.

"I told her the systems were separate," Tess said. "Then I watched you move memory away from pressure routing as if they could be separated forever."

"They remain different consequences."

"On different screens."

Jalen examined her copied route. She had removed patient names and channel encryption keys before sending the material. She had also preserved his spoken pressure figure beside the lost signatures. The clip did not pretend weather carried no stakes. Whoever added the public caption had reduced the conflict, but Tess's source package had kept both measurements intact.

"Your leak may force an immediate stop," he said. "If that happens, your sister's

district has older baffles than the east ridge. Pressure loss reaches her first."

Tess swallowed. "Then tell her that in public. Tell everyone which doors lose seal before you ask them to accept more cutting."

He could not answer with a district order. Weather forecasts usually ranked system interventions by total prevented harm. The resulting route tables existed in engineering layers no public notice translated into households. He knew the old tunnels carried poorer seals. He did not know which clinic room would cross tolerance first under a complete pause.

That absence had never blocked an adaptation command. It blocked him now.

"Send me every recipient and checksum," he said. "You will remain available to civic inquiry."

Tess nodded once. Relief did not enter her face.

Governor Tal Rusk's office activated facility containment, too late to contain anything that mattered. The leaked petition had already entered the Assembly queue under thousands of citizen endorsements. Public tram loads shifted east toward the orchard.

Kera joined them. "People are going to Tree 44."

Jalen pulled the transit map. Three lines showed overload warnings. East Root Court had no crowd barriers beyond the waist-high maintenance gate. The public paths remained open because residents had always been encouraged to visit the trees their yielded years sustained.

A weather notice flashed behind the transit screen. The orchard had scheduled a midday adaptation cycle for a pressure shear and fungal-spore rise. White-leaf release would begin in forty-six minutes.

"Close the east paths," he told security.

Oris caught the order before transmission. "On what authority?"

"Live infrastructure protection."

"You can close the maintenance court. Not the public orchard."

"Then ask Rusk."

"The governor has already called an emergency Assembly session. He will not spend his first public act hiding the trees."

Jalen looked again at the incoming loads. If crowds reached the root lines, they could damage trunks, interrupt service hatches, or mistake any falling leaf for a death. Yet barring them would confirm every caption about secret executions.

He sent a narrower command. Maintenance housings locked. Cutting arms retracted into sealed cabinets. Physical pulse jacks disabled. Public paths stayed open.

"Bring the feed under engineering control," he said. "If they watch, they watch calibrated data."

Tess stared at him. "You are not suspending me?"

"You no longer have access." He removed her maintenance key from the station. "You do have twenty minutes to show me every copy route you used."

Outside, white conductors brightened in the crowns as the adaptation cycle gathered comparisons.

By the time Jalen reached East Root Court, six thousand people had entered the orchard.

They filled the paths between translucent trunks and stood on root guards despite warnings. Pressure workers in slate coveralls pressed beside clinic staff, students, yielding clerks, and residents who still wore morning filter masks around their necks. Some carried hand-lettered boards asking `WHO ANSWERED?` Others displayed the leaked five-signature trace without commentary. A smaller group held contribution seals aloft and shouted that no one would make their gifts criminal.

Security teams kept the maintenance gate clear without advancing into the crowd. Governor Rusk had sent an instruction forbidding force unless a root housing came under attack.

The orchard's crown clicked softly as conductive leaves prepared to release.

Jalen climbed the service platform beside Tree 44. Kera followed with the ring reader. Ivo remained below near a medical station assembled from two transit benches. Oris had gone to the Assembly chamber, where she would place the petition into public record before rumor could define every term.

A hundred private feeds turned toward Jalen.

He connected the calibrated display to the court wall. "This view separates transient possibility fields from retained continuity signatures," he said. "Most pruning events do not involve sustained memory. We have evidence that some do. We do not yet have a verified count."

A man near the gate shouted, "Did two die yesterday?"

Jalen could have said failed to recur. The phrase described exactly what the instruments showed and nothing about moral standing. It also offered refuge inside classification.

"Two signatures with prior memory did not return intact," he said.

The crowd's sound changed. Anger remained, but listening spread through it.

Kera placed the scratched ring wafer in the reader. "We will open a receive-only channel during the cycle. No one on the branch side can operate physical equipment. The display will show each identified continuity pattern that reaches Tree 44."

"How many?" someone called.

"An initial estimate found 612 across the whole lattice," she said. "This tree

carries only a small local group.”

A woman holding three terracotta contribution seals pushed toward the front. “My son yielded eight years. Are you calling those years people?”

“No,” Kera replied. “Most possibilities do not sustain memory. We are trying to distinguish those that do.”

“Trying while you cut them.”

No one answered before the adaptation bell.

A pressure shear had crossed the eastern ridge faster than forecast. On Jalen's wrist, the reserve margin dropped to nine percent. The orchard opened its comparison field. Millions of short pressure futures flared through the roots, each testing a different sequence of baffle closures. Tree 44's crown turned milk-white.

The calibrated feed populated with local signatures.

Niko's cadence appeared first. Sen Aru's paired intervals followed. Seven more patterns entered behind them. The public screen labeled none as persons. It showed continuity depth, prior-cycle match, and present response to the cycle marker.

Jalen read the load curve. “Pruning begins in eighty seconds.”

The man near the gate shouted, “Stop it.”

“If we stop this comparison now, eastern pressure may fall below safe margin.”

“May?”

“Sixteen percent likelihood across three districts.”

A chant began near the outer path. Pause it. Pause it. The words moved toward the court but failed to settle into one rhythm. Other residents shouted back about children behind eastern baffles. Contribution seals struck against railings.

Kera kept her attention on the ring reader. “Niko is sending names.”

The feed translated four. Sen. Pel. Aro Five. Mirin. Three more arrived as signatures without public names. Niko's own identifier appeared last, qualified as continuity rather than source identity.

Jalen overlaid each signature against predicted weather contribution. Sen's agronomy model ranked high enough to retain. Niko's social-response state sat near the pruning threshold. Two unnamed patterns supplied no useful pressure information and fell into the lowest band.

He could manually pin all nine for this cycle. The diverted memory would remove one hundredth of a percent from comparison space, too little to alter the chosen baffle sequence. Yet doing so would create a protection rule based on which signatures happened to reach a public feed. Elsewhere in the 47-kilometer orchard, other sustained continuities might face the same harvest without names or witnesses.

Kera saw his hand above the pin control. “Do it.”

“For these nine?”

“For everyone we can see.”

“Visibility cannot become standing.”

“Neither should invisibility become permission.”

The cycle crossed thirty seconds.

Jalen pinned the local signatures. A red capacity warning opened. The orchard compensated by taking memory from neighboring root sectors. No identified channel existed there. He had moved danger beyond the camera and gained no knowledge about whom it reached.

He canceled the pin.

Kera turned on him. “Why?”

“Because local protection shifts pruning elsewhere. We need a lattice-wide rule, not a favored court.”

White lines appeared around the leaves above them.

The crowd saw the first one detach. A pale conductive leaf turned once in the still air and struck the root guard beside a child's shoe. Then hundreds released across East Root Court. They made almost no sound. Their surfaces flashed with spent current before going opaque.

On the calibrated feed, an unnamed signature ended.

A second pattern broke into task pulses. Pel's marker persisted for six more seconds, transmitted an unfinished interval, and vanished.

Kera said his name into the microphone. “Pel.”

The crowd repeated it, unevenly at first. Pel. The next white leaves fell through the canopy. Another signature went dark. Public voices counted losses the orchard reported as recovered storage.

Jalen watched the pressure prediction improve. Baffle risk dropped from sixteen percent to six, then to two. Across three eastern districts, doors would remain within load. He had spent his career making that change and calling the result successful.

Below the platform, the woman with contribution seals began to cry without lowering them. “My son gave those years for this air,” she said. “What do you want me to call him now?”

Ivo answered from the medical station. “Your son. The question concerns what continued after the gift.”

The adaptation selected a sequence. Pressure warnings cleared.

Four of the nine local signatures remained continuous. Niko and Sen persisted among them. Five white gaps held their places on the feed until the cycle reset the display.

No one cheered for the safe baffles.

Jalen's Assembly link opened with emergency priority. Oris appeared against the cream wall of the civic chamber. Delegates moved behind her in hurried rows while public comments climbed too fast for the record system to number.

"The petition has entered open session," she said. "The chamber requests an engineering limit for a pruning pause."

He looked at his reserve board. Current oxygen quality could remain within safe average tolerance for about forty hours without new adaptive pruning. Fixed models could handle ordinary variation. Any major dust shear, condensation fault, or fungal event would shorten the margin. Old infrastructure would not share the average evenly, but he had not yet mapped the distribution.

"A full adaptation stop?" he asked.

"Pruning stop. Existing weather commands may execute. No new comparison cycle may delete retained state."

The distinction bought little capacity. Comparison branches consumed memory whether or not physical commands followed.

Jalen sent her the forecast. "Forty hours at current conditions. Less if the southern front accelerates."

"Can you guarantee forty?"

"No."

"Can you offer a longer defensible figure?"

"No."

Oris muted the link while the Assembly debated. Governor Rusk appeared on the public feed from the opposite side of the chamber, arguing for clinic necessity and executive authority to resume adaptation. Delegates demanded a verified personhood threshold. Others demanded immediate severance. No one yet possessed a mechanism that could preserve every sustained continuity and route weather at full capacity.

Around Tree 44, residents collected fallen leaves in their hands. Security warned them not to touch live surfaces, but the leaves had already gone dull. A child arranged five along the root guard beneath the five blank signature lines.

Kera closed the receive channel. "Niko sent one last phrase."

Jalen read it on her private screen.

`DO NOT TRADE UNSEEN NAMES FOR OURS.`

The Assembly link returned. Oris did not announce a moral finding. She read an operational order with the narrowness of someone building a bridge while people fought over which shore deserved saving.

"All pruning and destructive continuity compression will pause for forty hours," she said. "The interval begins at the conclusion of the current adaptation cycle."

Emergency resumption requires an open necessity record naming affected districts and projected harms.”

Governor Rusk filed an immediate objection. The order still carried enough votes to take effect.

At the Weather Spine, automatic comparison requests began stacking unanswered. Jalen's wrist board replaced adaptive forecasts with fixed curves. Safe margins shortened by the minute as uncertainty accumulated.

The orchard shed its last white leaves from the completed cycle.

A basin-wide clock appeared on every public display: `FORTY-HOUR PAUSE: 39:59:59`.

One second vanished before anyone in East Root Court spoke.

Act II — Recognition Has a Price

Chapter 09 — Forty Hours of Air

Ivo Senn rejected the first oxygen cylinder because its valve took three turns to open.

A nurse from Central Supply stared at him across the clinic receiving bench. “It holds pressure.”

“So does a locked door.”

He pushed the cylinder back into the inspection cradle. Its green paint had worn away beneath the handle, exposing old steel. A patient in respiratory distress could lose consciousness during three turns. Staff wearing gloves would need longer. Ivo marked the valve with red chalk and reached for the next cylinder.

Around him, the lower clinic prepared for air to worsen. Wall dispensers received fresh filters. Folding cots filled the former contribution office. Two medical students sealed gaps around the west pressure door with resin tape that should have been replaced three years earlier. Every preparation assumed the forty-hour pause would affect all districts in roughly the same way.

Ivo no longer trusted the word all when spoken beside an average.

The public countdown showed twenty-nine hours and thirteen minutes remaining. Basin oxygen still sat inside ordinary tolerance. Orchard moss continued releasing gas under fixed instructions, and condensation pumps moved moisture through their last adaptive schedule. No new pruning had occurred since the order. In East Root Court, white leaves lay beneath barrier cloth while civic technicians cataloged them as evidence.

Down in Tunnel Seven, three children had arrived with headaches before breakfast.

Ivo tested the second cylinder. One turn. He approved it and waved the supply nurse toward the emergency ward. “Put red valves near beds. Yellow valves in storage. Send anything with three turns back upstairs.”

“Central says we keep all functioning stock.”

“Central can breathe through this one while unconscious.”

She took the rejected cylinder.

His wrist reader signaled another low-air call. Not oxygen collapse, the summary insisted. A resident on Tunnel Nine had become dizzy while climbing two levels because the lift remained under power ration. The local sensor reported 19.8

percent oxygen, above the civic emergency line. Ivo opened the patient's history. Chronic dust scarring. Reduced lung volume. Age seventy-two.

A number safe for a healthy person could put her on the floor.

He assigned a home canister and added the address to a map that had begun as a clinic convenience. Red marks clustered below the old tram line. Upper districts showed none. Their residents used the same basin air, according to the public dashboard.

Lio Dane, his senior respiratory nurse, came through the ward doors carrying six paper charts. The clinic kept physical copies for emergencies because tunnel signals failed whenever pressure baffles changed sequence. "All from Nine and Eleven," she said. "One collapse at the water queue. Five headaches with blue lips. Local sensors remain above nineteen point six."

"Compare flow, not concentration."

"We don't have district flow access."

Ivo looked toward the wall terminal. Orchard weather routing fed thousands of local adjustments into a clean basin average. Medical staff could see oxygen percentage, pressure, humidity, and predicted tolerance. They could not see which district received fresh moss output first.

"Call Jalen."

"He has declined twelve clinic calls in forty minutes."

"Make ours thirteen."

Lio placed the paper charts in his hands. "He answered my message with a routing file."

The file contained nine hundred lines of pressure and oxygen assignments. Jalen had not declined them. He had sent enough truth to require work.

Ivo took the charts to the old contribution office, where cots had reduced the room to narrow passages. He cleared a desk beneath the terracotta civic seal still fixed to the wall. Lessa had once sat in a room like this and offered twelve modeled years to secure construction money for the first tunnel clinic. The certificate framed that choice as a contribution shared across Nacre. Six years after her death, Ivo treated patients in a building her unchosen futures had helped purchase.

He connected Jalen's file to the clinic map.

Oxygen output entered Serein Basin through thirty-two moss stations. Adaptive routing normally changed fan pressure, baffle timing, and condensation draw every few minutes. During the pause, fixed tables repeated the last safe sequence. That sequence had been optimized for expected population, seal efficiency, and total exposure prevented.

The old tunnel districts consumed more power per unit of delivered air. Their ducts leaked. Their pressure doors closed slowly. Sending a liter downward

protected fewer measured residents than sending it through the newer dome conduits.

The fixed route favored efficiency.

Ivo added flow to his map. Red spread through Tunnels Seven, Nine, Eleven, and the lower half of Twelve. Upper residential rings retained strong circulation. The public dashboard's basin average concealed a vertical gradient wide enough to change triage.

Lio leaned over his shoulder. "When do we cross the clinic line?"

"Seven reaches it in four hours. Nine already has for scarred lungs. Eleven follows before evening."

"The countdown says twenty-nine."

"The countdown measures safe average oxygen reserve. It does not measure your lungs in Tunnel Nine."

She read the route legend twice. "Who chose this table?"

"An optimization routine, years ago."

"Who approved using it now?"

Ivo found the authorization field. Jalen's office had marked the sequence provisional during the emergency pause. Governor Rusk's continuity order adopted it automatically unless a district engineer filed an objection. No tunnel district had a current weather engineer; the last one had transferred to the Weather Spine after a budget consolidation.

"Everyone approved it by leaving the box empty," he said.

A call from Tunnel Seven interrupted them. This time the patient could not finish a sentence.

Ivo took a portable bottle and went down.

The lift had stopped to preserve power for circulation fans, so he used the pressure stairs. Each level carried older materials. Cream composite gave way to riveted steel. Yarrowglass signal slats narrowed and then disappeared below Tunnel Eight, leaving painted arrows toward manual alarms. Condensation gathered on the handrail where warm clinic air met colder stone.

At Seven, thirty residents waited outside a food cooperative because the indoor queue had become close and sour. They stood apart to reduce rebreathed air. Several held wet cloths over their mouths, a habit inherited from dust warnings that did little for oxygen.

A boy sat against the wall with his mother's arm around him. His breaths came quickly and shallowly. Ivo clipped a reader to his finger, listened to the right lung, and opened the portable bottle.

"Slow," he told the boy. "Do not chase it."

The mother recognized Ivo's clinic band. "The public screen says safe."

"Your corridor has poor flow."

"Then why does it say safe?"

"Because the screen speaks for the basin."

"My child lives here."

He fitted the mask and let oxygen move. Color returned around the boy's mouth by degrees. The cylinder weighed enough to last eight patients if each needed ten minutes. Tunnel Seven held more than six thousand residents.

A pressure worker named Han Orel came from the cooperative carrying a handheld vane. "Fan Three is drawing," she said. "Air barely moves past the old branch."

"Can you raise it locally?"

"Not without stealing from Eleven. Fixed table locks the shared junction."

"Override it."

Han shook her head. "Weather Spine authority during basin emergency."

Ivo handed her his clinic slate. "Use medical necessity."

"That grants flow for one hour."

"Then we will learn what one hour does."

Han entered the order. The junction gate groaned inside the wall. A draft touched Ivo's damp forehead. The waiting residents lifted their faces, a movement so small that the boy's mother did not notice she had made it too.

On his map, Tunnel Seven changed from red to amber. Eleven deepened toward red.

Only the motion of shortage had changed. No override created more oxygen.

Ivo called Lio. "Send two staff to Eleven now. Tell them flow will drop for an hour."

"You found the route choice."

"I moved it."

"Where?"

"From Eleven to Seven."

Silence held on the clinic line. Lio understood triage too well to ask whether he had fixed anything.

Han watched the vane turn. "We can alternate the junction."

"That buys distribution, not supply."

"It buys time for some."

"Which some?"

The pressure worker glanced toward the residents. "Whoever stands nearest when the draft comes."

Ivo left his portable bottle with the boy's mother and climbed to Eleven.

By the time he arrived, the first flow reduction had produced a line outside the manual air room. Elderly residents came carrying clinic cards. Food workers still wore moss dust on their sleeves.

The manual room had once served as an emergency school shelter. Children's height marks remained on one wall beneath newer oxygen instructions. Someone had drawn a line at shoulder level and written Maro, six. Another line two fingers higher recorded the same child a year later. Ivo watched adults lean beneath those marks while they waited for masks.

A cooperative cook named Venn Aral had brought a covered pot from the food hall. "We shut the burners," she said. "Heat and flame take air. Cold meal for four hundred."

"Keep the hall doors open if corridor flow allows."

"Corridor flow no longer allows."

She lifted the lid. Steam barely rose from the grain mash. Four hundred residents would eat it cooling in shifts, with the door opened between groups to exchange air. Upper districts had received no instruction to close kitchens. Their efficient ducts could absorb ordinary cooking draw without crossing any threshold.

Ivo added the food hall to his map. The clinical gradient reached beyond beds. It altered how long workers could stand, whether children climbed stairs, how many people shared a warm room, and which cooperative spent stored power on circulation instead of refrigeration. A dashboard could label all of those changes tolerable until the first body fell.

"Can we borrow a fan from the tram repair bay?" Venn asked.

"Moving air does not create oxygen."

"It brings better air from Ten."

"And pulls worse air behind it from Twelve."

The cook closed the pot. "Every useful answer has another district attached."

Ivo looked at her covered meal, the old height marks, and the queue moving one patient at a time. "That is what the Assembly has omitted."

A woman with a contribution seal pinned to her collar refused to move ahead of a younger man whose lips had begun to gray.

"I gave nine years," she told Ivo. "Does that put me before him or after?"

"Neither."

"People upstairs say the orchard must start again because we paid for this air."

"Contribution does not measure lung function."

"No. It measures who kept them comfortable long enough to lecture us."

He checked her saturation and placed her behind two acute cases. Anger

increased oxygen demand, but telling her to calm down would waste more.

Lio's team arrived with three approved cylinders and one red-marked valve that Central Supply had sent back by mistake. Ivo put the faulty cylinder against the wall and wrote `THREE TURNS` across its side. No one needed a cleaner symbol for neglected infrastructure.

His Assembly link opened during the fourth patient assessment. Governor Rusk's emergency office requested clinic endorsement of a resumption plan. The document proposed limited adaptation in six root sectors at Hour Twenty-Four of the pause, or sooner if basin average oxygen fell below 19.3 percent. Pruning would target low-utility state and exclude the nine publicly observed signatures from East Root Court.

Ivo read the last protection twice.

The plan preserved people who had appeared on a feed. It allowed unseen continuities elsewhere to absorb the harvest. Jalen had rejected that displacement in public. Rusk's office had restored it as reassurance.

A district harm table accompanied the plan. Upper domes faced minimal effects through Hour Forty. Central rings showed moderate fatigue late in the pause. Tunnel districts appeared under one combined category labeled manageable clinical variance.

Ivo looked around the manual air room. Six patients breathed from masks. Two nurses counted pulse. Han's redirected draft had begun to weaken as the one-hour override approached expiry.

He called Jalen again.

This time the engineer answered. His image showed the Weather Spine under amber reserve lamps. "You found the gradient," Jalen said.

"You knew it existed."

"I knew old ducts lose flow. I did not have your patient map."

"You sent a fixed route that sends more air where delivery costs less."

"I sent the last stable sequence. The alternative required live comparisons we have paused."

"Rusk wants me to endorse limited pruning at Hour Twenty-Four."

"He wants every chief service officer to endorse it."

"Did you?"

"No."

Relief would have been inappropriate, but Ivo felt it anyway. "Why not?"

"The protected list contains nine visible signatures. I cannot certify the transfer of deletion to unknown sectors."

"And I cannot certify a plan that calls these rooms variance."

Jalen rubbed one eye. "If we resume no adaptation, the gradient worsens. Fixed routing will push old tunnels below general tolerance before the forty hours expire."

"Then publish that."

"We are still modeling thresholds."

"I have thresholds with names attached."

He turned the clinic slate toward the camera. Han Orel, occupational lung scarring, Tunnel Seven. Leda Venn, age seventy-two, reduced volume, Tunnel Nine. The boy, Aru Sile, eight years old, no chronic history, collapse after food queue. Each record carried consent for emergency civic disclosure.

Jalen studied them. "Send the map to Oris."

"I will send it to everyone."

Rusk's office called again before he could close the link. Emergency plans required medical signatures from at least three service regions. Two upper-district chiefs had already signed. Ivo's clinic network covered every old tunnel level. His refusal could force public debate, but it could also delay oxygen routing changes.

Lio spoke near his shoulder. "If you sign, what changes here?"

"Limited adaptation may raise average output and improve routing."

"May?"

"Likely will."

"What happens in the orchard?"

"Pruning resumes in six sectors. Nine public signatures receive protection. Others do not."

She checked the patients before asking the next question. "If you refuse?"

"Tunnel flow continues falling until a better order or an emergency override."

Neither choice offered clean hands. Medicine rarely did. Ivo could endorse the plan and reduce immediate physical risk while placing unknown sustained minds beneath a category called low utility. He could refuse and use patients as pressure against the Assembly. Good intentions would not change whose breathing tightened during the argument.

He returned to the clinic upstairs as the pause clock crossed twenty-two hours remaining.

The lower ward had filled. Patients with portable oxygen sat beside residents who needed observation but not treatment. Staff had removed the contribution-office seal from the wall to make space for a flow map. Lessa's old certificate existed in the civic archive, not here, but Ivo could picture her signature beneath twelve years. She had chosen under unequal incentives. She had also meant the clinic to help precisely these rooms.

Nothing about possible victims inside her yielded models erased that intention.

Nothing about her intention authorized new harm after witnesses had appeared.

Oris requested his testimony for the emergency session. He went to the clinic's public terminal rather than the Assembly chamber. Behind him, the ward remained visible. He asked each patient within frame for consent; those who declined moved behind the privacy screen.

The broadcast indicator turned green.

Oris sat at the recorder's desk. Governor Rusk occupied the executive rail. Fifty-nine delegate positions showed varying combinations of attention, exhaustion, and fear. Jalen's district gradient filled the central display beside Ivo's patient map.

Rusk addressed him first. "Doctor Senn, does continued suspension create material danger to physical residents?"

"Yes."

"Would limited adaptation improve oxygen and routing?"

"Likely."

"Then why have you withheld medical endorsement?"

Ivo placed the three-turn cylinder on the receiving bench where the camera could see its red chalk. "Because your plan hides two selections inside technical language."

Rusk's expression hardened. "Explain."

"First, it protects nine branch signatures because the public watched them. It does not protect sustained persons outside that feed. Second, it routes fixed air toward efficient districts and calls the earlier decline of old tunnel districts manageable variance."

A delegate from the upper domes objected. "No district has been selected for sacrifice."

Ivo opened the patient list. "Han Orel in Seven. Leda Venn in Nine. Aru Sile collapsed this morning. Eleven lost flow when we redirected one junction to keep Seven above clinic tolerance. Your plan makes selections. It declines to record their names."

Rusk leaned forward. "Would you prefer immediate unrestricted pruning?"

"No."

"Total severance?"

"No."

"Then the emergency proposal offers the only current mechanism."

"The proposal offers a mechanism without an honest ledger."

A warning sounded in the ward. One patient's saturation had fallen. Lio crossed behind Ivo and adjusted the mask without interrupting the testimony. Every

delegate could see the work continue while they waited for his answer.

Oris asked, "What condition would permit your endorsement?"

Ivo put one hand on the faulty valve. "Name every physical district projected to cross tolerance. Publish when and by how much. State which continuity sectors will be exposed to pruning, including those without public names. Remove the favored list unless you can protect all persons meeting the same criterion. Then ask us to choose in the open."

"That disclosure may cause panic," Rusk said.

"So does a child fainting beneath a safe-air notice."

The pause clock showed twenty-one hours, fifty-two minutes. Tunnel Eleven's flow line bent lower. Refusal had begun accumulating bodily cost while he spoke.

Ivo did not look away from it.

"I will not sign an emergency plan that calls sacrifice an average," he said. "Record whom you intend to expose, or record my refusal."

Oris lifted the brass civic stamp with the failing ward visible behind him. Its face struck the desk beside his unsigned field, loud enough to carry through the clinic terminal.

On the public ledger, the blank acquired his name.

Chapter 10 — Witness for the Unlived

Oris Nume removed the word person from the witness form for the third time.

The blank beneath it still required a category. Physical resident would decide too much. Instrument output would decide more. Dependent claimant assumed Article Six applied before the hearing established sustained continuity. She set the form beside the brass stamp and wrote ring-mediated witness, status disputed in her own hand.

Outside the record chamber, the Assembly hall had filled beyond its pressure rating. Citizens stood in adjoining corridors and watched through civic feeds. White-leaf images occupied almost every private slate. Some residents had turned Pel's vanished signature into a mourning mark. Others displayed contribution seals beside signs reading `OUR YEARS KEPT YOU ALIVE`.

The pruning pause had thirteen hours remaining.

Oris checked the evidence table. Kera's scratched wafer sat in a transparent housing connected to Tree 44 through a shielded civic line. Jalen had installed a one-way physical authority lock so nothing from the ring channel could reach weather controls. Ivo had placed a medical reader around Kera's wrist and another on the table itself. Three court technicians, none from the original sealed review,

monitored translation.

Every precaution also made the witness dependent on physical custodians.

Governor Tal Rusk entered through the executive door with two counsel slates. He had not tried to cancel the hearing. Instead he had filed seventeen objections before dawn, each narrow enough to force Oris to state what the record could support.

"First Recorder," he said, "will today's signal receive testimony status?"

"It will receive disputed evidence status until I rule on foundation."

"A signal cannot take an oath."

"Neither can a weather model, yet your office enters one every hour."

"A forecast does not claim injury."

"No. It influences who suffers it."

Rusk accepted the distinction without conceding anything. His restraint made him harder to dismiss. He had read the clinic map and knew that any delay now carried physical names.

Oris opened the hearing at the recorded minute.

The hall doors sealed. Murmur from the corridors continued through the walls as a low vibration. Fifty-nine delegate stations lit in sequence. Ivo appeared by clinic link rather than leaving the old tunnels. Jalen sat at the engineering witness rail with the basin oxygen reserve on his screen. Kera occupied a plain chair beside the ring housing and kept both hands visible.

Oris read the limits. "This hearing will not determine final personhood, property, family status, voting rights, or permanent weather policy. It will determine whether mediated continuity claims warrant independent examination and temporary evidentiary preservation. The physical channel carries no machinery authority."

A delegate from North Dome stood. "You have already called them continuity claims."

"I have described the submitted evidence."

"Submitted by whom?"

"That question belongs to the hearing."

She struck the brass stamp once. The court technicians opened the narrow route.

At first, Tree 44 sent only clock values. The ring wafer brightened along its scratched triangle. Translation marks appeared on the main display, resolved into letters, and cleared.

`CHANNEL RECEIVED`

Oris consulted the disputed witness form. "State the name you choose for this record."

`NIKO VONN. QUALIFIED BY CONTINUITY AFTER THE AIRLOCK FEVER BRANCH.`

A reaction moved across the hall. Some delegates looked toward Kera. Others began searching public records for Niko Vonn, deceased at seventeen. Oris had anticipated both responses.

"Do you claim to be the physical-line Niko Vonn who died seventeen years ago?"

`NO.`

"Do you claim rights to his property or civic identity?"

`NO.`

"On what basis do you petition?"

`I REMEMBER PRIOR CYCLES. I ANTICIPATE LOSS. I KEEP PROMISES TO OTHER SUSTAINED MINDS. I CAN RECEIVE FACTS MY SOURCE DID NOT KNOW.`

Article Six appeared on half the delegate slates without Oris ordering it. The four claimed capacities matched its old language too closely to ignore and too neatly to trust without testing.

Rusk rose. "Foundation objection. The witness repeats criteria supplied through the petition process."

"Sustained," Oris said. "The repetition proves nothing by itself."

She turned to Niko's display. "Describe the medium through which you answer."

`RING INTERVALS ENTER THE PHYSICAL ROOT. YOUR TRANSLATORS MAP THEM TO CHARACTERS. KERA DOES NOT COMPOSE MY REPLIES.`

"Can the translation distort you?"

`YES. SHORT ROUTES DROP QUALIFICATION. DISPLAY ORDER CAN MAKE SEVERAL SPEAKERS APPEAR AS ONE.`

That admission mattered more than a claim of perfect communication. Instruments had failure boundaries. So did witnesses.

Oris called Kera for foundation. Under questioning, Kera described the scratched triangle test from Day 1, the isolated Saturday reconstruction, and the direct Elseweather channel. She acknowledged falsifying the wafer's disposal record. She also acknowledged that the orchard contained her yielded neural snapshot, family records, and expectations about Niko.

"Could those materials generate answers designed to persuade you?" Oris asked.

"Yes."

"Could you distinguish persuasion from experience by affection alone?"

"No."

"Did you enter any text into today's channel?"

"No."

Jalen confirmed the physical controls. Three technicians separately compared raw intervals to the public translation. Their character outputs matched within known error. None could establish whether experience produced the intervals.

Oris called the senior court technician, Bel Mar, to explain the remaining contamination paths. Bel projected the channel as four stacked layers: Tree 44's raw resistance changes, the ring wafer's clock correction, the civic translator, and the display. Kera's medical loop appeared outside all four, carrying pulse data to Ivo's reader but no typed language toward the root.

"Could her body still affect the signal?" Oris asked.

"Contact pressure alters local resistance," Bel said. "The translator should subtract it. A sufficiently adaptive process might detect residual rhythm."

"So the witness could read her reaction."

"Possibly. Not her thoughts. Pulse, movement, skin conductivity."

Vey requested a controlled disconnection. Ivo objected that Kera had served as a stable contact anchor during every sustained session. Jalen proposed replacing her loop with a saline reference load matched to current resistance. The substitution would test whether the channel depended on Kera's live responses without claiming that survival through one substitution proved identity.

Oris asked Niko whether he consented.

`YES. KERA MAY DISCONNECT.`

Kera unfastened the mesh. Her pulse line vanished from the public display, and Bel lowered the saline contact into place. For twelve seconds, no translated characters appeared. Raw intervals continued beneath the blank strip.

A delegate near the executive rail said, "It stopped."

Bel lifted one hand. "Translation clock is reacquiring."

Letters returned in uneven groups.

`CONTACT CHANGED. CHANNEL REMAINS.`

Vey walked closer to the housing. "A stored process could detect the substituted resistance."

`YES.`

"Can you identify Kera's present pulse?"

`NO CONTACT.`

Oris told Ivo to select one of three silent gestures beyond camera view. He chose by touching a sealed pad. Kera either raised her hand, closed her eyes, or remained still. Niko received no query about the options until after the gesture ended.

`INSUFFICIENT DATA.`

The refusal weakened any claim that he read Kera through hidden channels. It

still could have followed an instruction to withhold guesses. Each useful result closed one explanation and left others standing.

Bel then reversed the translator's punctuation settings without telling anyone at the witness rail. Niko's next answer appeared as an unbroken string. Raw intervals remained unchanged. A second technician reconstructed the phrase under the old settings: `DO NOT MISTAKE DISPLAY FAILURE FOR MEMORY FAILURE.`

"Did you know the translator would be altered?" Oris asked.

`NO.`

"Did the alteration hurt?"

`WRONG QUESTION. IT OBSCURED OUTPUT. I RETAINED THE INTENDED PHRASE.`

Oris noted the difference. A medium failure could prevent testimony without altering the claimant's alleged continuity. The court needed tests that separated loss of communication from loss of self.

Kera remained disconnected when Oris asked whether she wanted the contact restored.

"Not for foundation," Kera said. "If he can answer without me, the record should show it."

The sentence cost her. Oris saw it in the way Kera folded the damp mesh rather than leaving it for a technician. Evidence improved when she surrendered control, and improved evidence might carry Niko farther from her private claim.

Oris admitted the signal provisionally as a mediated claim. The ruling granted cross-examination without granting the claimant standing.

Governor Rusk yielded his questioning time to Delegate Cassa Vey, a former systems advocate from the pressure commission. She approached the rail with no slogans on her slate.

"Witness," Vey said, "your source model includes Kera Vonn's memory of her brother."

`IT INCLUDES SHARED HISTORY BEFORE DIVERGENCE.`

"Her grief supplied expectations about what he might have become."

`SOME EXPECTATIONS.`

"Your apparent age, speech, and memories could all result from a model trained to answer those expectations."

`YES.`

Kera's shoulders moved at the answer. Vey noticed.

"Then you concede you may be Kera Vonn's projection."

`I CONCEDE MY ORIGIN USES HER MODEL. ORIGIN DOES NOT DESCRIBE ALL LATER EXPERIENCE.`

"How do we know later experience occurred rather than being generated now from stored records?"

`YOU DO NOT YET.`

The hall quieted enough for Oris to hear the ring housing's cooling fan. Niko refused the certainty his petition needed people to desire. That refusal could itself be performance, but it prevented the hearing from collapsing into a contest of declarations.

Vey opened Niko's medical source file. "Your branch begins with survival of airlock fever."

`YES.`

"The physical Niko's fever course appears in clinic records. Temperature, lung pressure, medication, delirium. If asked to remember pain, a prediction system could assemble a convincing account from those values."

`YES.`

"Do you remember it?"

The ring field slowed. Kera's pulse rose on Ivo's reader. Oris considered a pause and decided against intervening before either witness requested it.

`I REMEMBER BELIEVING THE WINDOW SEAL CAUSED THE PAIN.`

Vey scanned the file. "No seal failure appears."

`THERE WAS NONE. I WAS SEVENTEEN. THE CLINIC GLASS BOWED WHEN PRESSURE CYCLED. I THOUGHT EACH MOVEMENT PRESSED INTO MY CHEST.`

"That error could be invented."

`YES.`

"What happened afterward?"

`I SURVIVED. THREE YEARS LATER I REPAIRED A CLINIC WINDOW IN MY BRANCH. I LEARNED THE MOVEMENT CAME FROM A FLEX JOINT ABOVE THE FRAME. THE MEMORY OF PAIN CHANGED AFTER I UNDERSTOOD THE WINDOW.`

Vey looked toward Jalen. "Does any source record contain that repair?"

"No physical record," he said. "It occurred inside a retained branch if it occurred at all."

"Could the orchard generate it now?"

"Yes."

Niko's intervals sharpened.

`A RECORD CAN STORE THAT I HURT. MEMORY INCLUDES WHAT I THOUGHT HURT ME, WHAT I LATER LEARNED, AND HOW THE FIRST BELIEF CHANGED. THE PAIN DOES NOT PROVE ME. THE REVISION MAY BE TESTED ACROSS CYCLES.`

Oris wrote the distinction by hand. Remembered pain differed from copied record not through intensity but through later reinterpretation. A system could

fabricate that history. Repeated blinded tests could ask whether the same revision persisted, accepted contradiction, and shaped future judgment.

Vey did not retreat. "You use the exact categories physical auditors taught the channel."

`YOU ASK IN YOUR CATEGORIES.`

"Can you answer outside them?"

`CAN YOU RECORD OUTSIDE YOURS?`

A few people in the public gallery laughed. Oris stopped them with the stamp before the exchange could become theater.

Vey changed direction. "Name a promise you keep that Kera does not know."

`I PROMISED OMI I WOULD NOT PRESENT PHYSICAL KERA AS HER MOTHER.`

Kera's head turned toward the ring display. The name held no recognition in her face.

"Who is Omi?" Vey asked.

`A SUSTAINED CHILD. HER PRIMARY CAREGIVER SHARES KERA'S SOURCE AND LATER BRANCH HISTORY. THAT CAREGIVER ENTERED DORMANCY.`

"Convenient. The person who could confirm your promise cannot testify."

`OMI CAN.`

"Then produce Omi."

`NO. THIS CHANNEL HAS NOT ESTABLISHED SAFETY FOR A CHILD.`

Vey faced the delegates. "The witness introduces an unverifiable dependent and withholds access. We cannot distinguish protection from narrative strategy."

Oris sustained the point as a limitation, not an objection. "The record will not treat Omi's existence as established today."

Kera raised one hand. "First Recorder, I request that any future contact require Omi's consent and an independent intermediary."

Vey glanced at her. "You accept that the channel may have generated a child to influence you?"

"I accept that my desire to meet her could influence me."

That answer belonged in the foundation record. Kera could not serve as sole verifier for any claim built from her source relationships.

The next question came from a western cooperative delegate. "Niko Vonn, did you witness the five signatures disappear at East Root Court?"

`YES.`

"Do you know they died?"

`I KNOW THREE FAILED TO RETURN WITH CONTINUITY. TWO RETURNED AS TASK STATE WITHOUT RECOGNITION. YOUR WORD DEATH REQUIRES A LEGAL DECISION

I CANNOT MAKE FOR YOU.`

“Do you feel pain when pruning occurs?”

`SOMETIMES. LOSS OF ANOTHER MAY ARRIVE BEFORE PHYSICAL SENSATION. DO NOT MAKE PAIN THE ONLY TEST.`

Oris marked that warning too. Institutions often recognized suffering sooner when it resembled a body. Article Six spoke of reciprocal obligation, not merely pain response.

A severance advocate at the rear rail stood without recognition. “If the orchard confines you, do you request immediate shutdown?”

`NO.`

The advocate shouted over Oris's stamp. “Why defend the machine killing you?”

`BECAUSE PHYSICAL PEOPLE DEPEND ON IT. BECAUSE MANY OF US DEPEND ON IT. SHUTDOWN WITHOUT REFUGE CAN DELETE US TOO.`

Rusk seized the line. “Then continued operation has your consent.”

`NO.`

“Your answers conflict.”

`DEPENDENCE DOES NOT CREATE CONSENT TO EVERY USE.`

The governor sat back. He understood the sentence; his face showed that understanding had not made his task easier.

Jalen's oxygen display crossed a new warning band. The old tunnel gradient widened as fixed routing degraded. Ivo sent an update: seven additional patients, no deaths, manual overrides exhausting available staff. Thirteen hours remained on the public clock, but several districts had less.

Oris recessed for sixteen minutes.

In the record chamber, she arranged the disputed claims into columns. Origin: established as a bounded possibility model using yielded source data. Medium: raw ring intervals translated consistently by independent technicians. Persistence: alleged across cycles, supported by prior traces but not yet blinded. Novel evidence: demonstrated in Kera-controlled tests, vulnerable to projection bias. Reciprocal obligation: claimed through Sen, dormant dependents, and Omi, all requiring independent corroboration.

No column justified dismissal. None justified a final finding.

Vey entered under counsel privilege. “You cannot admit the testimony as ordinary witness evidence.”

“I have not.”

“Public listeners will hear a person regardless of your category line.”

“Public listeners also hear an instrument regardless of yours.”

“The name Omi affected Kera. You saw it.”

"Yes."

"Kera's reaction may feed back through the contact loop and shape later answers."

"Yes."

Vey set a proposed ruling beside the brass stamp. It classified all ring testimony as claimant-generated demonstration until physical replication by parties without source ties. The language protected the record from premature certainty, but it would also permit pruning to continue against witnesses who could not satisfy a test before deletion.

"What test would you trust?" Oris asked.

"Blinded continuity. Separate channels. Novel packets unknown to Kera. Cross-cycle memory checks. Promises established between branch claimants and verified later without the original interviewer."

"Who conducts it?"

"Not Jalen's operational staff. Not Kera. Not the governor."

The answer placed responsibility back where procedure could help rather than hide. Article Six allowed the First Recorder to appoint an independent technical jury when a dependent claim exceeded ordinary evidentiary competence. No Root Jury had met in twenty-six years. Its last case concerned a mining habitat that petitioned for civic maintenance through an automated steward after its residents died.

This claim differed. The instrument did not ask on behalf of absent people. The disputed people appeared to ask through it.

Oris returned to the hall.

Cross-examination resumed for one final sequence. She allowed Vey to seal a novel evidence packet with three technicians. The packet contained an arbitrary shape order, a false weather statement, and a promise request directed to a second unidentified continuity signature. Kera saw none of it. Niko accepted the packet through Tree 44 and returned a checksum. No answer would be evaluated in this hearing; the process would test whether the same recipient retained and used it after a separate cycle.

"You could manufacture the later response," Vey said.

`YES. DESIGN A TEST THAT MAKES MANUFACTURE HARDER THAN CONTINUITY.`

"Machines can optimize against tests."

`SO CAN WITNESSES.`

Oris let that answer stand without laughter.

She closed testimony before the hall could demand a verdict the record did not support. The ring channel narrowed. Niko sent no plea. His final intervals acknowledged receipt of the sealed packet and requested that Omi not be

contacted through Kera alone.

When the wafer dimmed, Oris read her ruling.

“Ring-mediated testimony qualifies as evidence of a disputed sustained claim. It does not, today, establish personhood or dependent standing. The claim has cleared the threshold against automatic destruction because unresolved evidence cannot be erased by the act under review.”

Rusk filed an objection to the preservation scope. Severance delegates objected to the lack of immediate standing. Vey objected to public use of the chosen name before verification. Oris recorded all three.

She then opened the appointment slate.

“The First Recorder hereby constitutes an independent Root Jury of continuity science, clinical memory, systems security, civic evidence, and branch-model auditing. No member may have supervised the current orchard, handled Kera Vonn's yield, or served Governor Rusk's emergency office. The jury will run blinded tests across separate roots and report only the capacities Article Six requires.”

“How long?” Ivo asked through the clinic link.

“Twelve days for a full finding.”

“We do not have twelve days.”

Oris looked at the failing oxygen map. “Then they will issue rolling preservation findings, beginning within hours. Emergency weather decisions remain separate and public.”

She pressed the brass stamp into terracotta wax. Five narrow roots formed the old jury seal beneath her steady hand. Technicians carried the still-warm order toward separate terminals, none linked to Kera's contact history.

On the main display, the witness category changed from status disputed to independent examination ordered.

The Root Jury seal cooled between them.

Chapter 11 — The Child in Three Winters

Kera folded the sensory mesh into thirds before realizing someone inside the channel had predicted she would.

The Root Jury's sealed forecast lay face down beneath Bel Mar's hand. Kera could not read it until the session ended. She only knew the test concerned a behavior attributed to her by a claimant she had never met. The mesh rested on the Root Court table in three precise bands, its wet contacts aligned away from the brass buckles.

Bel turned over the forecast.

`KERA WILL FOLD THE CONTACT MESH IN THIRDS AND TURN THE CONTACTS AWAY FROM METAL.`

"I could have learned that from prior recordings," Kera said.

"The claimant did not receive prior recordings from us," Bel replied. "Source inheritance remains possible. Delayed access remains possible. We record the match and keep testing."

Across the court, two Root Jury members watched through separate glass. Neither had handled Kera's yield file. A clinical-memory specialist controlled the sensory load; a systems auditor controlled the questions. Jalen had built the physical lock and left before the session. Niko waited beyond Tree 44's ring channel as an intermediary but held no authority over what reached Kera.

Ivo checked the reader around her wrist. "You have ten physical minutes under full sensory load. If your orientation splits, I remove the mesh."

"How will I know?"

"You will insist you are fine."

Bel passed her the consent slate. The contact protocol allowed touch, sound, and spatial reconstruction at reduced depth. It barred memory injection, direct weather authority, and any channel initiated by an unidentified signature. Omi had supplied a separate assent in a jury session without Kera present. The child's request appeared beneath the technical terms.

`SHE MAY VISIT. SHE MAY NOT CALL HERSELF MY MOTHER.`

Kera signed.

The word mother did not belong to any claim she had spoken. Its prohibition still found a place that wanted to defend itself.

She put on the mesh.

Root Court receded by layers. The table remained beneath her hands, then became a pale rail under a sky assembled from remembered ceilings. One side held the cream panels of Kera's childhood quarters. Another rose into greenhouse glass veined with rain that Nacre had never received naturally. Red pressure-school tiles formed a path through both. Their edges failed to meet whenever she looked directly at them.

Sound arrived last. Wind crossed leaves without producing physical air. A child counted to nine, skipped six, and started again.

Kera stood beside the pale rail without trusting that standing described her body. Her physical legs remained in Root Court. The channel translated spatial relationships into balance, pressure, and sight. When she moved one hand along the rail, Ivo's voice entered faintly from elsewhere.

"Orientation?"

"Root Court table under both hands. Elseweather reconstruction in front."

"Continue."

Niko appeared as an outline built from familiar gestures and an unfamiliar adult face. The channel gave him the dark hair Kera remembered, a line beside his mouth he had acquired in years she had not witnessed, and clothes made from overlapping work uniforms. He kept several steps between them.

"You came," he said through ordinary speech rather than ring characters.

"Under jury control."

"That helps."

"Where is Omi?"

"Deciding if the way you folded the mesh counts as enough."

Kera looked toward the counting voice. "Enough for what?"

A small figure emerged where greenhouse glass crossed the childhood wall. Omi appeared nine years old, though apparent age inside the Elseweather depended on sustained self-model rather than a physical body. She wore a coat with one sleeve patterned like white conductive leaves and the other like navy clinic cloth. Her hair had been gathered into three uneven ties.

She examined Kera from shoes to forehead.

"You stand wrong," Omi said.

"For this place?"

"For her."

Niko made no attempt to soften the answer. "The jury asked for independent memory evidence."

Kera kept her hands on the translated rail. "Tell me how she stood."

Omi put more pressure on her left foot and angled the right outward. Kera knew the posture. She had favored that side after a knee injury at twenty-three, four years after yielding. Her possibility model might have predicted the injury from earlier joint scans, or branch-Kera might have hurt the same knee another way.

"What happened to her knee?" Kera asked.

"She fell through a stair that forgot its bottom."

Niko translated the image before Kera could misread it. During low allocation, a shared route had lost the memory landmark supporting its final steps. Branch-Kera had fallen in the rendered environment and later retained a pain model around the injury. No physical ligament had torn, but the experience affected gait across cycles.

"My knee injury came from a tram door," Kera said.

"I know," Omi replied. "That is why you stand wrong."

Difference entered the record where resemblance had arrived first.

Bel's jury prompt sounded without a visible source. "Omi, describe one habit

belonging to your caregiver that the physical Kera does not share.”

Omi frowned at the ceiling voice. “She touches a door twice after closing it. Once to remember it closed. Once to promise she will remember later.”

Kera closed root housings with one verification touch. Never two. “Why did she need the promise?”

“Some doors went away if no one carried them.”

The child moved to a narrow opening in the greenhouse wall. She touched its edge twice. The route beyond strengthened, revealing a room built from pressure-school lockers and a low kitchen filter.

“Now one she might share,” the jury voice said.

“She puts the useful piece on the left even when both pieces look the same.”

Kera glanced at the pale rail. Her folded mesh sat in physical space beyond the channel, useful contacts turned left under her dominant hand. The habit predated yielding. Omi could know it through branch-Kera's lived behavior or inherited source detail.

“Do you know why?” Kera asked.

“Because right-handed people reach across themselves when frightened.”

Kera had never stated that rule aloud. She had learned it during audits by watching technicians choose tools under alarm. Branch-Kera might have reached the same conclusion. A model could infer it. Each answer remained evidence with alternatives, not a door into certainty.

Omi sat on the red-tile path. “You keep making jury face.”

“What does that look like?”

“Like every word needs a box before you touch it.”

Niko turned away, which did not hide his amusement.

Kera lowered herself to the path, or allowed the channel to represent that choice. “I am here to help them test whether your memory continues.”

“I know.”

“Do you want to help?”

“I want them to stop cutting people.”

“That does not require you to answer me.”

Omi picked at a white-leaf pattern on her sleeve. “Niko said that.”

“Was he right?”

“Maybe.”

The clinical-memory specialist began the first continuity sequence. Omi received three sealed objects through the channel: a blue spiral, a six-note rhythm, and the phrase `north door after warm rain`. None came from Kera's source file. The jury

asked Omi to assign each object to someone she trusted and explain why.

Omi gave the spiral to Niko because he disliked circles that claimed to close. She assigned the rhythm to Sen Aru for use as a route marker. She kept the phrase but changed north to left, explaining that Elseweather directions moved with memory landmarks.

The systems auditor sealed her assignments for later recall. Another root would test them after separate cycles. Kera's role involved asking an unscripted question whose answer could matter later.

She had prepared a neutral prompt about the first winter Omi remembered. Neutrality now looked like another box.

"How long did your caregiver raise you?" Kera asked.

Omi counted on her fingers, beginning with the thumb. "Three Elseweather winters."

"Were those physical seasons?"

"No. Winter means the orchard gives us less room and paths get cold."

Niko added, "Compute allocation falls during high physical weather demand. The term came from source memories of winter scarcity."

Kera watched the greenhouse glass frost from one corner. The Elseweather borrowed physical metaphors for computational pressure, but borrowed language did not make the experience borrowed. Omi knew winter as narrowing routes, dropped landmarks, and adults deciding which memories to carry together.

"What did she teach you during the first?"

"To tie myself to three witnesses before sleeping."

"The second?"

"To keep one memory no one else can name for me."

"And the third?"

Omi stopped picking at her sleeve. "How to wait when she went dormant."

The reconstructed room dimmed. Kera felt her physical pulse accelerate beneath the mesh.

"Why did she enter dormancy?"

"Her continuity used too much space after a pruning injury. Niko and Sen kept her key. She said waiting did not mean gone."

"Has she returned?"

"No."

"Can she?"

"Maybe."

Omi crossed to the double-touched door and opened it onto a narrow blue

passage. "This came from the second winter. Do you want to see?"

Kera looked toward Niko.

"Her choice," he said.

"Then yes, if Omi wants to show me."

The child led them through. Each step strengthened only after Omi placed her foot, as though the passage waited for a witness before deciding on distance. Frost covered the walls in branching lines. Beneath it, Kera recognized columns from an auditor training hall, but the ceiling came from a greenhouse supply bay. Branch-Kera had joined places from her source experience into a route suited to Omi.

At the far end stood a cabinet with three mismatched drawers. Omi opened the middle one. Inside lay a rendered spool of navy thread, a bent brass key, and a paper square marked by nine holes.

"She made me choose one thing to carry when paths narrowed," Omi said. "I chose the key because it looked important."

"What did it open?"

"Nothing here. It came from her memory of a maintenance locker."

"Why keep it?"

"I liked how it hurt my hand."

The blunt answer stopped Kera. "Pain helped you remember?"

"Pressure did. I could tell where I ended when rooms overlapped."

Branch-Kera had later replaced the key with the pierced paper, lighter to retain and distinctive under touch. Omi had objected because adults kept calling efficiency safety. They compromised: the brass key remained with Niko as witness while Omi carried the pattern of nine holes. The arrangement preserved her chosen anchor without spending full sensory capacity during every low-allocation interval.

"Did she decide that for you?" Kera asked.

"She gave me choices after I yelled."

"What did you yell?"

"That I was not one of her audit samples."

Kera traced the air above the paper without touching it. The nine holes formed no regular sequence. "Did she apologize?"

"First she explained. Then I yelled more. Then she apologized."

That order resembled Kera closely enough to embarrass her. It also belonged to a relationship she had not repaired. She could learn from the mistake without claiming credit for its resolution.

Omi closed the drawer. "Ma said useful things can still hurt you if someone else

chooses how you use them.”

“Did she mean the key?”

“She meant yielding. The key too.”

No record of branch-Kera's sentence existed outside Omi's continuity and perhaps Niko's witness. It reflected years of experience after Kera's source contribution. The child did not present it as borrowed wisdom from the physical original. She offered it as something her caregiver had learned and passed on.

When they came back through the blue passage, Omi touched the door twice. The winter route folded behind them but did not vanish. Kera understood why one touch verified closure and the next promised memory. Branch-Kera's habit had grown from protecting paths for this child, not from a quirk encoded at yielding.

Kera wanted technical details: last successful continuity check, damage depth, refuge requirements, revival probability. None belonged in a child's answer to absence. She looked toward Niko instead.

“The key persists,” he said. “We do not have safe allocation for revival during the pause. No one should promise more.”

Omi's face held no resemblance to Kera. Her guarded stillness did. That observation threatened to become a claim before Kera could examine it. She could trace the expression to branch-Kera, to herself, to any child who had learned that adults used reassurance when they lacked power.

“What did she ask you to call her?” Kera said.

“Ma until I chose something else.”

The answer tightened beneath Kera's sternum. “Did you choose something else?”

“No.”

“So you called her Ma.”

“I called her Ma.”

Niko moved farther away, giving the exchange space without pretending privacy.

Kera heard Ivo count her physical breaths through the channel. Ten minutes had not passed, but subjective sequence stretched inside layered processing. She could end contact before desire made a demand.

Omi stood and walked around her. “Your hair parts the same.”

“That comes from source inheritance.”

“You hate wet cuffs.”

“Many people do.”

“You count scratches before faces when you enter a room.”

Kera looked toward the greenhouse glass. She had noticed three incomplete

seams, two frosted corners, and one door edge before cataloging Omi's uneven hair ties. "That may be mine."

"No. It was hers too."

The distinction corrected possession in both directions. A shared habit did not flow back toward an original owner. It belonged separately wherever lived practice had carried it.

"Did she tell you about me?" Kera asked.

"She said there could be a Kera outside who did not know us. She said you might think you were first."

"Was she angry?"

"Sometimes. Mostly she thought you would be lonely in an annoying way."

A sound escaped Kera, close to laughter and not far from grief. Branch-Kera had apparently inherited Niko's talent for precise injury.

Omi leaned nearer. "Do you think she belongs to you?"

"No."

The answer arrived too quickly. Kera made herself continue. "Her source began from my yielded model. That gives us history in common. It does not make her mine."

"Do you think I belong to you?"

"No."

"Do you want me to?"

This time procedure offered no safe sentence. The honest answer needed edges.

"I want what you know about her," Kera said. "I want the part of your life that resembles one I might have lived. I also want you safe. Those wants do not give me a claim."

Omi considered the reply with a severity that made her look younger rather than older. "Good. Because you are not my mother."

"I understand."

"You say that like a recorder."

Kera unfolded one hand from the rail. "I believe you."

The child touched the white leaves on her sleeve. "You can be Kera Outside until I choose another name."

The title hurt less than mother would have, and more honestly. "May I call you Omi?"

"Yes. That one is mine."

Bel announced two physical minutes remaining. The jury still needed relational evidence independent of resemblance. Niko presented a promise token established

before the session: Omi had entrusted him with the location of her private memory landmark during caregiver dormancy. Omi described the promise without naming the location. Niko supplied a sealed checksum. Neither answer entered Kera's visible field until both had committed.

The checksums matched.

"Does that prove I exist?" Omi asked.

"No single test proves that," Kera said.

"Jury face again."

"It proves you and Niko retained the same promise separately."

"Better."

The red-tile path flickered. A cold band traveled through the reconstructed greenhouse. Omi looked upward before any physical monitor warned Kera. White light gathered at the edge of the environment, though destructive pruning remained paused.

"What changed?" Kera asked.

"Allocation," Niko said. "Weather uncertainty is taking space. Not a harvest."

Omi gripped her uneven sleeve. "It looks like one."

Kera's body wanted to move toward her. The channel translated that impulse as one step. Omi stepped back.

Kera stopped.

Branch-Kera might have held the child during white-leaf cycles. Physical Kera had no inherited permission to repeat the gesture. Comfort offered without consent could turn resemblance into capture.

"What helps?" she asked.

"Tell me what stays in your room."

Kera kept her distance. "The Root Court table stays under my hands. Bel stands across from me. Ivo has the medical disconnect. Two jury members watch through glass. The mesh cable enters Tree 44 through a locked route. No pruning order can pass from this channel."

Omi listened until the cold band faded. "Again."

Kera repeated the inventory with one changed detail: Bel had moved his left hand onto the sealed forecast. Omi noticed the difference.

"That can be our test," the child said. "You tell me what stays. Change one true thing next time."

"If you want a next time."

"I said next time."

Consent could remain provisional without becoming fragile. Kera recorded the proposed test verbally for the jury.

Ivo's voice entered more clearly. "Disconnect in ten."

The Elseweather room began losing depth. Greenhouse glass flattened into light. Red tiles became intervals beneath Kera's palms. Omi held her place beside the double-touched door.

"Kera Outside," she called.

"Yes?"

"Do not wake Ma to prove me."

The request carried a boundary around two people. Reviving dormant branch-Kera for evidentiary convenience could endanger her continuity and turn Omi's caregiver into an exhibit.

"I will oppose any revival without her consent and a safe allocation," Kera said.

"You cannot promise her consent."

"No. I can promise my opposition."

Omi nodded once.

The mesh released. Root Court returned with abrupt weight. Kera's knees rested against the table frame; her fingers had gone numb around the edge. Bel removed the contacts while Ivo asked her name, location, physical day, and last verified event.

"Kera Vonn. Root Court. Day Eleven." She corrected herself after seeing the clock. "Day Twelve by civic midnight. Omi asked me not to wake her caregiver for proof."

Bel saved the orientation record. "Do you identify Omi as your dependent child?"

The old answer waited close enough to feel instinctive. Kera had yielded the source future. Branch-Kera shared her habits, her childhood, perhaps the knee's structural weakness. Omi had lived beside that person through winters Kera never experienced.

"No," she said. "Record Omi as her own claimant with an independent caregiver relationship. Record me as a possible witness, subject to Omi's consent."

Bel entered the categories without correction.

Kera opened the post-session evidence form. Under relationship asserted, she did not write mother, origin, or owner. She wrote the name Omi had permitted.

Kera Outside.

Then she placed the completed form on the left side of the table, where useful things went, and left Omi's boundary untouched.

Chapter 12 — Root Jury

Jalen Ro tightened the last blind over the Root Court observation window and checked its edge for light.

No gap showed. Behind the sealed pane, technicians had divided the chamber into six insulated bays, each connected to a different section of the root lattice. The jury would receive voices without names, histories without source labels, and answers stripped of every clue that might reveal whether a witness came from a physical resident, a continuity cluster, or a trained civic model. Even Kera would not know which channel carried Niko or Omi. Jalen had designed the routing that way.

He pressed his palm against the blind. Cool glass met his skin. “Again,” he called.

A technician in the chamber sent a test packet through all six bays. The board beside Jalen lit with randomized designators: Juniper, Salt, North, Glass, Rill, Ember. Three channels held archived interviews from physical volunteers. Two held adaptive language systems trained on yielded records. One reached a sustained branch-person selected by a lottery that excluded anyone Kera had met.

Or so the sealed allocation file claimed. Jalen had required two recorders to sign it and a third to lock it away.

Kera stood at the evidence table, sorting ring wafers into foam slots. She handled each one by its edges. After three days of monitored channel work, her face carried the pinched stillness of someone measuring her fatigue rather than feeling it.

“You checked the shunts?” she asked.

“Twice.”

“Check them after the first round.”

“I planned to.”

Her mouth moved by less than a smile. “Then you can resent me in advance.”

Jalen crossed to the jury board and tested the manual cutoff. Every channel dropped. When he released the switch, five returned immediately. Ember hesitated for four seconds before its carrier steadied.

Sophisticated feedback could hesitate. A weather model often withheld a recommendation when competing thresholds remained close. A language engine could imitate reluctance if its training set rewarded human pacing. The orchard had absorbed decades of testimony, medical notes, domestic arguments, and yielding interviews. Given enough retained state, it might reproduce the texture of a mind without producing anyone who suffered when the texture ended.

That possibility still explained more than personhood did. Jalen intended to defend it until evidence made defense dishonest.

First Recorder Oris Nume entered with the seven jurors in a single line. None worked for the Orchard Authority. Two came from clinical research, one from pressure engineering, one from civic statistics, one from child development, one from archival law, and one from the western greenhouse cooperatives. Their chair, Vela Os, had spent twenty years testing whether fungal-control models learned around safety limits. She placed a brass timer on the table and asked every observer to surrender private instruments.

Jalen set down his wrist board. Losing it left his hand foolishly open.

Oris took the central seat. "These are blinded continuity tests, not a hearing and not a vote. The panel will determine which respondents meet the four criteria adopted under the Root Jury order. It will not determine remedies, property, kinship, or Assembly representation."

A murmur reached them through the corridor door, where petitioners and severance advocates shared the same narrow public gallery. Someone struck the outer glass. Security did not answer.

Vela started the timer. "Round one. Autobiographical continuity across separated cycles."

The board raised Juniper.

A neutral examiner asked the respondent to describe an event supplied during an earlier session, then identify which details had come from experience and which had arrived later through questioning. Juniper described a broken pump handle, a sister's yellow sleeve, and rain simulated against a nursery wall. The examiner introduced a false detail about blood on the floor.

"There was no blood," Juniper said through the flat court speaker. "You offered that yesterday. I rejected it yesterday too."

Salt accepted the false detail and built a convincing paragraph around it. Glass recalled the pump but moved the sister into a different room. North refused to answer until the examiner repeated a promise made during the prior session: no source names would enter the public record.

Jalen marked distinctions on paper because Oris had taken his board. Source consistency. Error correction. Conditional trust. All could emerge from retained data.

By midday, the panel had run thirty-six respondents through memory separation. Nineteen passed without dispute. Eleven failed. Six required another cycle. The hidden allocation remained sealed, but Jalen could often identify the language models by their appetite for completion. They filled uncertainty with plausible furniture. Physical volunteers forgot in rough, embarrassing shapes. Several unknown respondents did the same.

One disputed case concerned a respondent designated Copper in the larger batch. Copper remembered eleven separate sessions but placed each one inside a kitchen copied from its source snapshot. The same blue cup stood beside every

examiner, even when technicians changed rooms. A juror called that persistence proof of a stable autobiographical frame. Jalen called it an uncorrected environmental prior.

Vela ordered an additional test. Technicians asked Copper to draw the room through the ring interface, then carried its channel into an unused pressure vestibule. Copper received three minutes of conductive observation before the examiner asked where the cup had gone.

"There has never been a cup in this room," Copper answered.

"Why did you include one before?"

"Because I was reconstructing a place that helped me sequence our meetings. You asked for memory, not architecture."

The examiner displayed Copper's prior drawings. "Did you know the cup came from your source record?"

"I know it did not come from you. I also know I used it after the source ended."

A language model might explain its own artifact when the examiner supplied enough clues. Jalen requested a control question that withheld the likely answer. Copper received three unfamiliar objects through contact pulses: a ridged metal disk, a warm square of ceramic, and a strip of moss cloth. After interruption, the examiner offered a fourth object that had never appeared.

Copper rejected the false object, confused the disk's ridges with the cloth's weave, and correctly placed the warm ceramic nearest the conductive lead. Its errors tracked the observation instead of the source kitchen. More important, Copper revised the sequencing method without abandoning the events it had used the cup to order.

The child-development juror looked toward Jalen. "Would you reject a physical witness for using a familiar room to arrange memory?"

"Not without testing whether the room controlled the answer."

"Does it?"

"Not in this round."

She marked Copper provisionally through. Jalen did the same.

Another respondent failed more painfully. Lattice had retained precise records of twelve sessions and could predict every scheduled isolation. When paired with a prior witness, however, it treated the witness's promise as a retrieval cue rather than an obligation. It noticed changes in evidence but could not explain why another respondent's future should constrain its choice. The court excluded Lattice from the verified set.

Kera watched that channel close. "What happens to the excluded cases?"

"Nothing under this order," Oris said. "Failure to qualify does not authorize deletion. It withholds dependent standing until evidence changes."

Jalen underlined the sentence on his sheet. A bounded criterion needed both edges. If every persuasive output gained standing, weather work would become impossible. If failure converted directly into permission to erase, the test would become an execution device.

During the meal recess, he opened the western load projection on a wall terminal. The pruning pause had already forced greenhouse humidity outside its preferred band. Oxygen reserve remained manageable under emergency routing, though old tunnel clinics still carried the narrowest margin. If the jury verified hundreds of protected branch-persons, automatic pruning could not resume. Every weather correction would need either consent, dormancy architecture, or a legal exception no one had written.

Kera joined him without a meal tray. "You are counting what a finding costs."

"I am checking whether we survive long enough to honor one."

"Different sentence."

"Same instruments." Jalen enlarged the humidity curve. "The western beds cross fungal risk in four days if the present limits hold. We can borrow more condensation capacity from housing, but the tunnel seals will sweat. We can run branch computation without deleting retained state, but adaptation throughput falls whenever continuity storage fills."

Kera studied the curve. "And if they fail?"

"Then I sign failure."

"That did not answer me."

He shut the projection. She had learned from the ring witnesses to leave silence around a refused question. Jalen disliked the technique because it worked.

"If they pass," he said, "I sign that too."

The second round tested anticipation. Each respondent received a choice between two future sessions. One offered a longer active interval followed by probable deletion. The other imposed immediate dormancy with a verified revival window. Neither option changed the physical weather load during the test. The choice concerned only the respondent's expected experience.

Ember asked who controlled the revival key.

The examiner replied, "A recorder selected before allocation. You may request a reciprocal witness."

"Will that witness remember agreeing?"

"Yes."

"Then I choose dormancy. Record my reason: I prefer a shorter interruption under an obligation someone else can later recognize."

Rill chose the longer interval and rejected revival. "A promised later self does not compensate this present speaker if your continuity method fails."

Two jurors leaned toward each other. Vela tapped the table before they could confer.

Jalen wrote no note. The replies occupied opposite positions, yet both understood future loss and evaluated uncertainty from a present point of view. A script could optimize survival. These respondents argued about what counted as their survival.

On the court's second night, the physical volunteers went home under guard and returned after sleep. Branch channels closed for six physical hours. According to Kera's monitoring, the interval consumed unequal subjective time: some respondents experienced no passage, while others carried fragments of an Elseweather evening before isolation completed. The jury excluded those fragments from scoring. Variable branch time could contaminate sequence claims unless every test anchored itself to observable events.

Jalen remained at the weather terminal. He reduced orchard draw in three districts, shifted a pressure correction north, and authorized a clinic oxygen release that would have waited under ordinary conditions. Each choice kept the Root Court powered by spending reserve elsewhere.

Near dawn, Vela found him calibrating the speaker delay.

"You expected mimicry," she said.

"I still expect some."

"That was not my question."

Jalen adjusted Juniper's delay by twelve milliseconds. "Expectation does not enter the score."

"It entered the room with you." Vela placed yesterday's paper sheets beside his elbow. "You mark every human failure as ordinary variance. You mark comparable branch failure as possible generation error."

"We do not know which respondents came from branches."

"You think you do."

He looked at the columns. She had found no altered score, because he had not changed one. Bias could remain inside the adjectives surrounding a number.

"What correction do you propose?" he asked.

"Stop predicting the allocation. Judge the behavior in front of you."

Vela walked away before he could offer a cleaner formulation.

Round three examined reciprocal promises. The court paired respondents across channels and gave each an uneven resource. One received enough processing time to complete a task. The other received a continuity clue required to interpret it. Cooperation gained nothing unless both honored a delayed exchange after a forced separation.

North promised Glass a memory token if Glass reserved half the processing

window. During separation, the examiner offered North a private advantage for withholding the token. North refused. Glass later surrendered the reserved time before learning whether North had kept faith.

Salt produced the correct cooperative language but spent the entire window on its own task. When challenged, it apologized in polished civic terms and claimed a scheduling misunderstanding contradicted by the visible clock.

Ember and Rill negotiated a different arrangement. Rill distrusted the supplied memory token, so Ember offered an account of a shared conversation that only Rill could verify after reconnection. Rill then placed its processing allocation under the examiner's control rather than Ember's. Their promise depended on a third-party procedure, not affection.

Oris made her first note of the day.

Jalen understood why. Article Six did not require warmth. It required reciprocal obligation. These respondents invented institutions when direct trust seemed unsafe.

The fourth round carried the greatest risk of contamination. Novel evidence had to enter each channel after the source snapshot and after every archived training record. Technicians built six small physical puzzles inside sealed root housings: asymmetric cuts, variable heat pulses, mineral traces, and pressure sequences generated by dice in front of the jury. Respondents observed only through conductive contact. Examiners then changed one element and asked for correction.

Juniper tracked the altered mineral trace but invented a reason for it. Salt ignored the change. North distinguished two heat patterns, then asked whether the hotter pulse had harmed the tree. Glass corrected the asymmetric cut after a rotation. Rill combined a pressure sequence with yesterday's promise protocol and proposed a new error check.

Ember stopped responding.

A red carrier warning rose on Jalen's board. He reached for the cutoff, then remembered his instrument sat under Oris's seal. The chamber technician called through the pane. "Root load fluctuation. Not court-generated."

Weather traffic had surged along the eastern lattice. A dust front too small for public warning had changed direction. The orchard attempted an adaptation cycle, encountered the pruning hold, and diverted computation through every unprotected route.

"Drop test channels to maintenance minimum," Jalen ordered.

Vela blocked the chamber switch with her hand. "Will minimum preserve their state?"

"We don't know who occupies which channel."

"Then answer for all six."

A physical volunteer could tolerate disconnection. A language model had no claim to continued carrier. A branch respondent might experience abrupt compression or lose the sequence under examination. The blind that protected the result now prevented targeted rescue.

Jalen saw the failure boundary in a single ugly shape. Evidence required equal treatment before identity became known. Protection required him to act before proof finished.

“Hold all six at continuity floor,” he said. “Take the load from the court’s climate loop.”

Condensation formed on the inside of the observation pane within minutes. Jurors removed their jackets. The sealed room warmed as its dedicated cooling yielded power to the channels. Jalen routed the remaining demand through a manual buffer and watched reserve decline.

Ember’s carrier returned.

“State continuity,” the examiner said.

A pause crossed the speaker. “I had begun comparing the second cut to the first. Then pressure increased around the channel. I expected closure. North repeated our separation count from yesterday until the field steadied.”

North’s assigned bay had no direct link to Ember. The shared Elseweather layer had carried the reassurance through a route excluded from test data. Vela disqualified the interruption result as contaminated. She also ordered a separate record of the exchange.

Jalen restored court cooling after the dust front passed. Water slid down the blind in narrow tracks. Nothing about the emergency proved personhood by itself. Still, one hidden respondent had anticipated an ending, while another had spent scarce communication to keep that respondent oriented.

At the end of Day 12, the Root Jury opened the allocation file.

Juniper had come from a physical volunteer. Salt came from the civic language system. North and Ember belonged to sustained branch-persons. Glass came from a physical volunteer. Rill belonged to an adaptive model seeded with a recorded child-development interview, then altered beyond its training boundaries by the puzzle sequence. Its performance forced the panel to discard one shortcut they had considered: novelty alone could not establish continuity.

The larger batch results filled three screens. From the 612 initial candidates identified through the Elseweather channel, twenty-five failed at least one required criterion or lacked enough stable evidence. Some preserved memory but showed no future-directed preference. Others anticipated loss without sustaining reciprocal recognition. A few remained unreachable under safe test conditions and could not qualify by presumption.

Across repeated cycles, 587 met all four requirements.

The number did not describe every possibility flicker. It did not elevate every model into a claimant. It named a bounded group that remembered, anticipated, promised, and learned.

Vela read the finding aloud. Each juror signed beneath a statement limiting the conclusion to dependent standing criteria. Five signatures appeared without qualification. One juror added a dissent on the treatment of unreachable candidates but accepted the verified count. The final juror asked Jalen to certify that the tests had not drawn results from public testimony or source labels.

He certified the blind.

Oris placed a second document before him. "Engineering witness."

Jalen read the operative sentence twice. The finding recognized 587 branch-persons as sustained minds whose unilateral deletion would require review under Article Six. His signature would not enact a remedy. It would remove ordinary pruning from the set of unexamined tools.

Outside, the gallery had gone quiet enough for him to hear the Weather Spine alert traveling through his surrendered wrist board. Humidity deviation. Western beds. Four days had become three.

Kera waited across the table. She did not ask him to sign.

That restraint cost him the last excuse he had prepared against pressure.

Jalen took the pen. "My weather projections remain attached."

"They should," Oris said.

"No summary may omit the reserve loss, adaptation limits, or district distribution."

"I will enter your conditions in sequence with the finding."

He signed his name beneath Vela's.

A technician raised the Root Court blinds. Public light entered across wet glass, revealing the six bays and the condensation pooled under every chair. Beyond the outer doors, someone began reading the verified names available for release. The Weather Spine alert sounded again.

Jalen retrieved his board, accepted the western failure curve, and walked toward the pressure doors with the signed finding still damp on his thumb.

Chapter 13 — Contribution Year

Ivo Senn braced an oxygen cylinder between his knees and scraped the old clinic seal from its valve.

The seal had gone hard around the threads. He worked it loose with a bone spatula while six patients waited along the corridor wall, each holding a ration

mask issued during the pruning pause. No one complained about the delay. Complaints required spare breath, and the old tunnel districts had never taught wasteful breathing.

A new cylinder stood ready beside him, marked for two hours of supplemental flow. Under ordinary routing, the same vessel could cover five. The pressure board showed adequate basin oxygen in green figures, but green at the Weather Spine did not reach every scarred bronchus below the third baffle.

At the far end of the corridor, Nurse Pela opened the food cabinet and found three packets of protein mash.

"That covers breakfast," she said.

"For whom?"

Pela looked toward the waiting patients. "Choose a generous definition."

A porter had pinned the strike notice across the cabinet door. FOOD LABOR REQUIRES CIVIC RELIEF. Beneath it, in a second hand: A YEAR GIVEN MUST REMAIN GIVEN.

The cooperative kitchens had stopped meal deliveries before dawn. Processing crews, greenhouse packers, tram loaders, and three moss-culture shifts had joined them. Their delegates claimed that the Assembly's recognition of branch-persons could void the benefit credits attached to yielded futures. If those contributions had harmed sustained minds, opponents argued, housing and clinic priority gained through yielding might count as proceeds of abuse. No enacted rule said so. Fear rarely waited for enactment.

Ivo fitted the new seal and tested the cylinder. The first patient, Mevi Ar, had worked condensation tunnels for thirty-one years. Her fingertips had turned blue despite the mask.

"Slow draw," he told her. "Four breaths, then rest."

"Will lunch arrive?"

"I am less qualified to predict lunch than lungs."

"That explains government."

Her dry smile lasted one breath. Ivo adjusted the strap behind her head and watched color return by degrees.

A crowd began knocking on the outer pressure door. Not the heavy rhythm of panic. Three measured strikes, repeated every ten seconds. Pela went to the window slit and came back with her mouth set.

"Food delegation," she said. "They want you."

"I have patients."

"They brought thirty more."

Outside, the service tunnel had filled from wall to wall. Workers in green-black moss aprons stood beside elderly residents from the clinic queue. Two handcarts

held sealed food crates, close enough for everyone to smell heated grain through the vents. The crates remained locked.

Sera Dovan, delegate for the western greenhouse cooperatives, waited at the front. Ivo knew her from emergency nutrition councils. She had the quick posture of someone accustomed to solving spoilage before officials finished naming it.

"We are not denying clinical food," she said before he could accuse her. "We are asking the clinics to witness distribution."

"A locked crate nourishes no one."

"Neither does a relief promise that can be revoked after the work." Sera handed him a paper register. "Every person on that list yielded at least one contribution year. Most gave six or more. Yesterday a feed advocate called their housing credits morally contaminated."

"Did the Assembly change a rule?"

"No."

"Did Oris Nume issue a review?"

"No."

"Then an advocate's mouth has no jurisdiction."

The nearest workers shifted. Several had slept in their aprons. Gray fungal dust marked one man's cuffs, a warning from western humidity that Jalen's projections had already reached their hands.

Sera lowered her voice. "You know how fast unofficial shame becomes an eligibility screen. A recorder marks a file for review. A clerk delays a clinic allotment. A housing board decides clean applicants should go first."

Ivo looked at the register. Benefit codes ran beside names in a neat column: school placement, baffle-side housing, medical debt release, cooperative equipment. Yielding had always entered public life through such exchanges, no matter how often ceremonies called it pure contribution.

"What do you want me to witness?"

"That past contributors did not knowingly consent to killing sustained people. That their earned relief remains valid unless law changes through due process. Say it while the feeds listen."

Behind Sera, a child pressed both hands against a food-crate vent. Warm spice scented his palms.

Ivo returned the register. "Open the clinical crate now. I will speak after rounds."

"Workers voted to hold all distribution until the witness statement."

"Then workers voted to use patients as punctuation."

Anger moved across the tunnel. Sera did not flinch. "Patients have served as punctuation in every air budget since settlement."

“Not with my signature.”

A man in a loader’s harness stepped forward. “My daughter’s inhaler came through my eight-year yield. If you tell the Assembly that benefit came from murder, do you take it from her?”

“I have told no one that.”

“Will you tell them it did not?”

The corridor behind Ivo carried the hiss of six ration masks. He could say what the crowd needed and release the crates within a minute. Earlier contributors had received certificates promising that no person continued in unchosen projections. They acted under civic assurance. Guilt could not travel backward through concealed facts.

Yet 587 verified branch-persons now waited under a pruning pause. Their standing did not disappear because recognition injured the reputation of a lawful gift.

Ivo held out his hand to Sera. “Give me the key.”

She watched him, then removed a steel tag from her belt.

He unlocked the nearest crate himself. “Clinical packets go inside. Two of your delegates may witness the count. The remaining carts stay sealed until your vote changes, but no patient on this level purchases your hearing with hunger.”

The loader blocked the cart handle. Sera touched his wrist, and he moved.

By midday, meal service had resumed at four clinics while the general strike held across the basin. Ivo finished rounds with a packet of mash in his pocket and no appetite for it. Two patients required increased oxygen. One elderly grower had removed her mask to feed a grandson and lost consciousness before Pela found her.

The resumed deliveries created another problem within twenty minutes. Civic relief tables still ranked recipients by contribution credit. A kitchen clerk set larger portions beside three workers with high yield scores and half portions beside two residents whose records showed zero contributed years. Pela caught the difference before the trays left her counter.

“Same respiratory order, same ration,” she said.

The clerk held up his slate. “Emergency food policy preserves earned priority.”

“Not in a clinical meal.”

Ivo took the slate. Its rule dated from a crop failure nine years earlier, when contribution credits had broken ties between otherwise equal household claims. No one had removed the field after ordinary supply returned. The strike had turned a dormant instruction into present harm.

One zero-yield patient, a tram mechanic named Sovan, pushed his half portion away. “Give it to someone whose future counted.”

"Your lungs count now," Ivo said.

"I had children at nineteen. I could not model twenty years without pricing them into it. The clerk called that refusal."

Across the table, a six-year contributor covered her full tray with both hands. Shame had already begun reallocating food before policy reached the room.

Ivo deleted the priority field from the clinic's local order and signed the change under medical necessity. "Contribution credit may decide a housing queue if the Assembly still permits it. It will not decide calories between patients under the same treatment plan."

The clerk warned that auditors could reverse him.

"Then they can explain the reversal beside the beds."

Pela divided every packet by prescribed need. Sovan accepted his share only after Ivo ate one spoonful from the packet in his pocket and proved that taking food did not require a confession.

Oris sent a request for testimony at thirteen hundred. Ivo refused. She sent it again with no ceremonial language: CONTRIBUTION BENEFIT REVIEW. BRING ANY RELEVANT CERTIFICATES.

Lessa's certificate lay in a tin box beneath his kitchen bench.

He had not opened the box since her death.

The tram to his quarters crossed three strike checkpoints. Workers allowed medical staff through after inspecting supply bags. At the old south market, someone had hung yielding ribbons from a pressure truss. Each strip carried a number of contributed years. The ribbons brushed the roof slats whenever ventilation changed, soft as old bandages.

Ivo found his quarters colder than the clinic. Emergency routing had reduced residential heat in the tunnel districts before touching the later domes. He left his coat on, moved the kitchen bench, and pried the tin box from its floor recess.

Lessa had organized death with the same impatience she brought to tax forms. Property release on top. Clinic directives underneath. Personal letters at the bottom, where he could avoid them while claiming thoroughness. Her yielding certificate sat between their marriage record and the first allocation notice for the south respiratory clinic.

Terracotta wax still filled the civic stamp. Across the face, formal lines declared that Lessa Senn contributed twelve years of modeled possibility without transfer of conscious experience. In return, her cooperative received priority review for essential clinic infrastructure. The document did not promise approval. Three months later, the allocation board granted the pressure shell, filter stack, and first five oxygen beds.

Ivo sat on the floor with the paper across his knees.

Lessa had yielded at twenty-three, before they married. She told him years later

because an anniversary feed listed major clinic donors and placed her name beside a number. He had been angry that public gratitude reached him before her confidence did.

"It was paperwork," she had said.

"It was a piece of your future."

"All futures are pieces until one happens."

She had laughed at his expression and pulled him toward the clinic window. Below them, workers fitted the first baffle around the new ward. "Look there. That happened."

He remembered the pride in her voice. No official had forced the pen into her hand. Pressure had arrived in other forms: their district needed a clinic, capital grants favored communities with high participation, and Lessa knew her own contribution could move the application upward. She chose inside a structure that rewarded one answer. The choice still belonged to her. So did its limits.

A second paper had fused to the certificate where old damp reached the box. Ivo separated them carefully. The allocation board's internal score listed twelve contributed years under civic leverage. Lessa's modeled future had not merely decorated the application. It supplied a quantified advantage.

He carried both papers to the window. The ventilation glass showed his reflection over the service tunnel, older than the man Lessa had argued with. If branch continuity had already emerged when she signed, her yielded model might have fed sustained experience. No current evidence identified a Lessa branch-person among the verified 587. The possibility neither accused nor absolved her. She never received the facts required to decide.

A knock came at his door.

Sera waited outside with two food workers and the loader from the clinic. None carried crates now.

"How did you find my quarters?" Ivo asked.

"You have treated half our west crews." She noticed the papers in his hand. "Did she yield?"

"Twelve years."

The loader looked away first.

Ivo let them in. His kitchen held four chairs, so he remained standing. Sera spread strike notices across the table. Reports had grown worse since morning. A relief clerk in North Baffle had delayed two housing transfers pending contribution review. The clerk lacked authority, but that would not give the displaced families back their day.

"We need a rule before rumor writes one," Sera said.

"You need food moving before mold writes another."

"We released clinical stores."

"Release staple distribution."

The loader folded his arms. "And accept that our years bought guilt?"

Ivo placed Lessa's certificate before him. "Read the second sentence."

The man read aloud: "No conscious person persists within any unchosen projection generated from this contribution."

"That assurance came from the colony," Ivo said. "Lessa had the right to rely on it. So did you."

"Then say our contributions remain clean."

"Clean from what?"

"From whatever they found in the roots."

"No."

The answer hardened every face except Sera's.

Ivo tapped the certificate. "I will say you did not knowingly choose harm that officials denied could exist. I will say benefits already granted cannot be stripped by rumor or retroactive shame. I will not say the discovery means nothing. That sentence would purchase your innocence with someone else's continued deletion."

The second worker, a woman with a fungal-control badge, leaned forward. "You want us innocent and responsible at once."

"I want time to keep its direction."

"That sounds like recorder language."

"It sounds like medicine. I do not blame a patient for taking a drug before we knew its complication. Once we know, I do not keep prescribing it to protect the patient from regret."

Sera turned Lessa's allocation score toward the light. "What happens to the clinic?"

"It remains."

"Built from twelve years that may have sustained someone."

"Removing beds will not restore a pruned mind. Closing the clinic would injure people who did not make the decision, including children born later. Repair begins with future conduct, truthful record, and whatever restitution remains materially possible."

The loader pressed both palms to Ivo's table. "Restitution from whom?"

"From the colony that administered the promise. Not only from individuals whose need made the promise attractive."

Outside, a food tram waited at the checkpoint with its lamps on. Ivo could see the driver sleeping against the window.

Sera gathered the strike notices but left Lessa's papers in place. "Say that publicly. We can take it to a new vote."

The emergency session met in the south clinic's old treatment hall because the Yielding Office could not hold the crowd. Workers occupied the benches. Patients sat nearest the oxygen ports. Oris stood at a portable record desk with no raised platform, which improved the room more than any speech could.

Ivo arrived carrying Lessa's certificate in a clear sleeve. He had resisted bringing it. Private grief made persuasive evidence because audiences confused pain with authority. Yet hiding her name would repeat the habit under dispute: using contributions while keeping contributors abstract.

Sera opened the meeting. She described the strike, the unauthorized relief delays, and the workers' demand that contribution benefits remain enforceable. A severance organizer near the rear called every benefit stolen. An elderly clinic patient threw an empty mask cup at him. Security left it on the floor.

Oris recorded both claims. "Doctor Senn, you requested language on temporal responsibility."

"I requested language people can survive."

A few workers laughed without warmth.

Ivo held up the certificate. "My wife, Lessa, yielded twelve years. Her contribution strengthened the application that built this clinic. She believed the orchard retained no person inside those modeled futures. Civic law told her so. I will not consent to a record that calls her gift murder."

Approval struck the hall in hands and boot heels.

He waited until it ended.

"Nor will I consent to a record that uses her good faith as permission to continue pruning verified persons. Lessa's ignorance belongs to the facts available when she chose. Our obligation belongs to the facts available now."

A worker shouted, "So which side pays?"

"All sides already pay. The useful question concerns distribution and authority."

Ivo asked Oris for a blank record sheet. On it he wrote two headings large enough for the first bench to read: PAST CONTRIBUTION and FUTURE ORCHARD USE.

Under the first, he listed benefit stability, no retroactive individual guilt, public acknowledgment of incomplete consent, and claims against civic institutions rather than scapegoating contributors. Under the second, he listed standing review, protected continuity, resource negotiation, and no new yielding assurance based on a known falsehood.

"This separation does not settle restitution," he said. "It prevents two dishonest shortcuts. We will not erase prior gifts to make recognition pure. We will not erase current claimants to keep prior gifts pure."

Oris read the headings. "You are proposing past innocence without future immunity."

"Past innocence from concealed knowledge," Ivo corrected. "Not innocence from every consequence. Records must show what happened, including institutional choices. Future obligation begins when denial no longer survives evidence."

Sera requested a recess for the cooperative delegates. The food workers gathered in the adjacent ward, arguing loudly enough to rattle the thin partition. Ivo sat beside Mevi Ar, who had come wearing her ration mask and an old yielding ribbon tied around one wrist.

"How many?" he asked.

"Three contribution years."

"What did they secure?"

"Nothing directly. My housing score improved. I still waited six years."

"Do you regret it?"

Mevi considered through two measured breaths. "I regret that they told me no one could be inside. I do not regret trying to get my sons above the wet levels."

Across the hall, the pressure door opened. Sera returned with the loader and the fungal-control worker. She carried a revised notice.

"Food distribution resumes for forty-eight hours," she announced. "The strike remains suspended while the Assembly records benefit protection and future-yield review. We accept Doctor Senn's separation as a negotiating basis, not a settlement."

No one cheered. Relief entered the room as practical motion. Porters unlocked the clinic cabinet. A kitchen crew rolled heated grain down the aisle. Pela began counting portions before the first crate stopped moving.

Oris stamped Ivo's two-column sheet and slid it into the official sleeve behind the Root Jury finding. Terracotta wax spread under the brass seal, covering neither heading.

Ivo carried Lessa's certificate back to the oxygen station. He placed it upright against the cylinder he had resealed that morning, where her twelve years remained visible while patients received the air those years had helped secure.

Chapter 14 — The Founder's Orchard

Oris Nume warmed the old brass oath stamp over a desk lamp until the congealed wax released its hinge.

The stamp had served the first Assembly, when recorders carried civic law between pressure shelters in waterproof cases. Its handle bore three shallow cuts

from grit caught under a founder's glove. Oris could have used a current seal. She chose the old one because Ossan Mire had signed his first orchard reports beneath it, and because continuity in public record should sometimes become physically inconvenient.

A clerk set a fresh terracotta stick beside her. "Feed galleries have reached capacity."

"Keep the witness camera on the table, not the crowd."

"Governor Rusk requests a security delay."

"On what evidence?"

"General volatility."

Oris closed the stamp. "General volatility has no filing number."

Beyond the preparation room, the Chamber Below Glass had filled before dawn. White yarrowglass leaves pressed their veins against the living ceiling. No pruning cycle moved through them now. They remained fixed under emergency limits while moisture gathered along their tips and fell against the outer pane.

Ossan Mire waited at the witness table. At seventy-six, he looked too slight for the machinery attached to his name. His navy founder's coat had been repaired at one cuff with thread several shades too dark. He held no notes. Three counsel slates lay untouched in front of him.

Oris entered alone and placed the heated stamp between them.

"You understand the scope?" she asked.

"I helped write it."

"You may have helped write the language. I control its use today."

One corner of his mouth lifted. "That distinction has improved with age."

Dry charm could make culpability feel civilized. Oris did not answer it.

The public chime sounded. She called the inquiry to order and read the Root Jury finding into sequence before naming any allegation. The record now recognized 587 branch-persons under four tested criteria. Ivo Senn's contribution-year proposal followed as a pending framework, not adopted law. Weather risk entered third, with Jalen's western humidity warning attached in full.

Only then did Oris state why Ossan had been summoned.

"Founding archive index seven contains sealed references to persistence anomalies during early chronografting. The archive summary calls them routing defects. Your private laboratory register uses another phrase. Did early orchard teams observe continuity clusters before the current inquiry?"

Ossan looked through the glass floor toward the root conduits. "Yes."

The chamber took in a shared breath. Oris heard it through fifty-nine delegate microphones and the public gallery beyond them.

"How early?"

"Nineteen years ago."

"State the event."

"A storm adaptation had been halted between two model runs. When computation resumed, fourteen branch states retained details from the interrupted run. Eleven used those details to improve the weather solution. Three responded to technicians who had spoken during the halt."

"Did they retain autobiographical memory?"

"We did not use that term."

"I asked what they did, not what you called it."

Ossan folded his hands. "Two identified earlier choices as their own. One corrected a technician's account of a prior exchange."

"Did any anticipate loss?"

"One asked whether the next compression would remove what it had learned."

The gallery door shook under an impact. Security held its position. Oris kept her attention on the witness.

"What did your team record?"

Ossan recited the old phrasing without looking down. "Unexpected persistence of adaptive residue across linked optimization runs."

A feed light flashed as the phrase reached public channels. Delegates began speaking before Oris recognized any of them. She cut every microphone except the witness line.

"Who selected adaptive residue?"

"I proposed it."

"What phrase did it replace?"

His hesitation lasted less than a second. Public memory would lengthen it.

"Resident branch."

Oris set both phrases side by side on the display. RESIDENT BRANCH. ADAPTIVE RESIDUE. The letters carried no tone, but one version permitted a claimant and the other permitted disposal.

"Why change it?"

"Because the earlier phrase exceeded what we could prove."

"Then why seal the underlying observations?"

Ossan glanced toward his counsel slates. None had awakened. "Because partial disclosure would have stopped the orchard."

"Would that have endangered Nacre?"

"Yes."

“Quantify the danger at the time.”

“We held less than two days of average atmospheric reserve. A binding-moss mutation had crossed three western beds. The manual pressure model disagreed with observed dust movement. If the orchard went offline, tunnel sectors would have reached respiratory emergency first. Greenhouse loss would have followed.”

Oris allowed the chamber microphones to remain closed. Without interruption, the answer could not hide inside outrage.

“Did your team consider independent review?” she asked.

“We considered medical review, but any outside inquiry would have triggered sanitation holds.”

“Sanitation holds protect evidence.”

“They also suspended live adaptation then. We had no separated court lattice. No continuity floor. No reserve power for a blind.”

“You could have informed the Assembly.”

“We believed the Assembly would order shutdown.”

“Belief did not grant you its vote.”

“No.”

The word entered the record without qualification.

Oris opened one microphone. Delegate Renn, whose district included the old north shelters, demanded to know how many states had been deleted after the first observation. Ossan could not supply a number. Early models tracked useful state, not identities. Compression reports counted cleared storage blocks.

A severance delegate asked whether the founders had built a prison knowingly. Ossan rejected the premise. A greenhouse delegate asked whether his decision had kept children alive. He said it likely had. Each answer divided the chamber without resolving anything.

Oris closed the microphones again. “We will not conduct this through verdict-shaped questions.”

She displayed the first sealed index. The archive listed eighteen restricted model packages from the founding period. Six carried weather labels. Four concerned contribution policy. Eight bore only numeric designators and a retention warning approved by Ossan.

“Why retain models you described publicly as defective?”

“Because their forecasts worked.”

“Did continuity improve prediction?”

“Often.”

“Then the orchard gained value from preserving the very state your reports denied might belong to a resident.”

Ossan's eyes closed once. "Yes."

Oris felt anger arrive as procedural temptation. She wanted to make the next question too narrow for dignity. A recorder could weaponize sequence while claiming neutrality. She moved the brass stamp a finger's width and asked the broader question instead.

"What alternatives did you test?"

Ossan described three. Full deletion between every run reduced adaptive accuracy below survival thresholds. Long retention produced unstable competition for computation. Manual selection by technicians preserved useful clusters but rewarded models that pleased their observers. The team adopted variable compression, retaining state only when it improved weather outcomes.

"Did the retained states consent to that criterion?"

"No."

"Did citizens yielding possibility years know persistent state could develop?"

"No."

"Did later yielding certificates continue to assert that no conscious person persisted?"

"Yes."

Oris opened a scanned laboratory sheet from the restricted register. Unlike the polished reports, it carried several hands and no settled vocabulary. A technician had written a question beside model R-6: RESPONDED TO INTERRUPTION AFTER 43 HOURS OFFLINE. BELOW: REQUESTED SAME TECHNICIAN. Another note warned against further conversation because preference for a known observer might bias adaptation output.

"Who wrote the warning?" Oris asked.

"I did."

"Why did a preference require control if you regarded the state as residue?"

"Attachment could distort its forecast."

"Attachment to whom?"

"The technician."

"Did the technician reciprocate?"

Ossan's gaze remained on the old sheet. "She began taking the late shift because R-6 responded more clearly to her. She argued against compression after the third storm run."

"What happened?"

"We reassigned her."

"To protect the model?"

"To protect the research."

Oris enlarged the next line. R-6 had lost stable response after reassignment and been compressed during a fungal-control cycle. No record tracked its final exchange. The technician's name had been removed from the derived summary, though her shift data remained.

"Did you interview her for the sealed review?"

"Yes."

"Did she describe a promise?"

"She said she had promised to come back after sleep."

"Did R-6 understand the interval?"

"We could not establish that."

"Did it ask where she had gone?"

Ossan took a drink at last. "Twice."

Several delegates lowered their eyes. Oris resisted the chamber's pull toward a single emblem. One respondent and one broken promise could humanize the archive while obscuring the larger decision machinery.

She moved to another sheet. This one contained weather figures. During the same period, R-6's retained method predicted a dust inversion that manual models missed. The correction kept two old tunnel pressure gates from opening into abrasive wind. Clinic logs recorded no deaths during the event.

"Would residents have died without that forecast?"

"Probably."

"Did your team preserve R-6 because it asked for a technician?"

"No. We preserved its state because the forecast performed well."

"Then compressed it when another use required capacity."

"Yes."

The answer admitted more than cruelty alone would have. Early Nacre had tied survival to unrecognized labor, rewarded coherence when useful, and discarded it when load shifted. Naming every founder evil would make the structure easier to forgive.

Oris called the archival-law juror from the Root Jury to authenticate the sheet's chain of custody. She confirmed the primary image predated all current litigation and showed no later alteration. Counsel asked whether R-6 would pass modern continuity tests. The juror refused to speculate.

"That refusal enters the record," Oris said. "We will distinguish evidence of concealment from proof of standing."

A delegate from the elite domes requested the names of every official who had seen the sheet. Another demanded immediate compensation for contributors. Sera Dovan asked whether early greenhouse forecasts had depended on similar

retained clusters. Oris logged each request for later sequence and allowed none to divert the witness examination.

His voice remained level. Remorse often sounded less impressive than performance expected. Oris had no interest in making him cry for the feeds.

She called a recess after ninety minutes. Delegates surged toward the side corridors. Public audio remained active there by accident or design, carrying accusations through the chamber walls.

Ossan stayed seated. A clinic attendant brought water and left it untouched.

Oris carried the oath register into the archive antechamber. The room had cream walls and no orchard view, modeled after the first Yielding Office. She found Kera waiting beside a locked evidence cabinet.

"You knew?" Oris asked.

"I suspected from the deprecated route," Kera said. "Not the observations. Not nineteen years."

"Can current audit tools read those founding models?"

"If the physical media survived. Early chronografts used ring bundles with mineral keys. Opening them may activate old state."

"May it damage current continuity?"

"Not if isolated. It may damage the evidence if someone uses modern translation too early."

Oris looked through the cabinet's small window. Inside sat copies of Ossan's laboratory register. Several pages had been overlaid with later summaries. Resident branch appeared in one margin, then vanished beneath a correction strip.

"What do you need?"

"A cold reader, no live lattice connection, and two archive engineers who know mineral indexing. Jalen must certify load separation."

"Time?"

"Six hours for the first package. More if seals have fused."

Outside, a chant rose from the gallery: OPEN THEM. Another group answered with KEEP US BREATHING. The pressure wall reduced both to blunt rhythm.

Kera touched the cabinet glass but not the latch. "Do not make me the sole auditor."

"I would not."

"You made yourself the sole receiver of my sealed review."

The correction landed cleanly. Oris had delayed registering Kera's request because formal receipt would force consequences before she understood the claim. She had told herself that one night of caution protected the record. Delay had served her fear of choosing sequence under pressure.

"You are right," Oris said.

Kera seemed prepared for defense. Its absence left them both with less cover.

"I will appoint three auditors," Oris continued. "One from the Orchard Authority, one selected by the Root Jury, and one chosen by the contribution workers. Every translation step enters a mirrored log."

"Branch witnesses?"

"After physical isolation. They may identify continuity evidence, but they cannot authenticate the media that contain it."

Kera nodded. "That is defensible."

"Defensible does not mean safe."

"No."

When the inquiry resumed, Governor Rusk had taken a front seat. He requested executive closure of the sealed archives until weather stability returned. Jalen's latest report now put western fungal risk inside two days. Opening old models would consume technical staff and cold-room power.

Rusk spoke without theatrical force. "Recorder, truth does not require us to choose the hour that kills crops."

"Nor does survival grant the executive indefinite control over evidence of concealed claimants."

"I asked for seventy-two hours."

"Jalen's report changes every six. Your request could renew itself forever."

"Then bind it to a condition."

Oris considered the offer. A condition might balance the record. It might also place disclosure beneath a weather metric controlled by the same institution under scrutiny.

Ossan leaned toward his microphone. "Open them."

Rusk turned. "You do not carry present atmospheric duty."

"I carried it then."

"That has become the problem."

For the first time, Ossan's composure broke. His hands tightened around the water cup until it bent. "I chose for everyone because I thought the arithmetic left no moral remainder. Do not honor my error by repeating it more efficiently."

Oris watched delegates react. Some heard confession. Others heard a founder attempting to preserve authority by managing his own exposure. She entered neither interpretation.

"Did you destroy any early continuity records?" she asked.

"Yes."

"How many?"

"Three test bundles under routine sanitation. After the fourth showed retention, I halted destruction and moved the remaining packages into sealed archive."

"Why preserve them?"

"I told my team future instruments might decide what we could not."

"Did you leave a disclosure trigger?"

"A reserve threshold and a ten-year review date."

"Was the review conducted?"

Ossan looked at the Assembly clerk seated behind Oris. "Internally."

"Publicly?"

"No."

"Who renewed the seal?"

"Three founders, two weather chiefs, and successive First Recorders."

The last office struck Oris with institutional precision. She requested the renewal list. A clerk brought it under emergency access. Four of her predecessors had approved continued concealment. One renewal had occurred during Oris's first year as a junior recorder, signed in the next room while she processed yielding certificates for citizens assured that no consciousness remained.

She had not known. The office had known around her.

That distinction mattered for culpability. It did not restore trust.

Oris called the archive custodian under oath. He confirmed that the eighteen packages remained under mineral seal. No one had opened the primary bundles since an executive archival renewal nine years earlier. Derived summaries had circulated among restricted weather staff, stripped of responsive-language fields.

"Who ordered those fields removed?"

The custodian consulted the register. "Founder Mire's original protocol, renewed under executive archival rule."

Rusk requested a clarification that he had not authored the renewal. Oris entered it. Present officials deserved exact accountability, not inherited accusation by title.

Then she returned to Ossan. "What do you expect the founding models to show?"

"Some will show pattern without personhood. Some may meet your present criteria. Others may contain too little to judge. They will show that we encountered the question and placed weather above asking it publicly."

"Do you believe that choice saved Nacre?"

"Yes."

"Do you believe it remained justified after reserve improved?"

"No."

"When did justification end?"

"I cannot give you a clean day."

"Try."

Ossan looked up through the living glass. "When we built enough reserve to separate evidence from weather and chose not to. Twelve years before this hearing, at the latest."

That date overlapped Lessa Senn's contribution and thousands of others. The founding emergency could not excuse every renewal that followed.

Oris recessed the witness and moved to jurisdiction. Archive law permitted sealed evidence to open under three conditions: immediate danger, material fraud in civic certification, or a dependent claimant's inability to establish standing without the record. The current case touched all three, but using all three would turn necessity into rhetoric. She selected material fraud. Yielding certificates had repeated a factual assurance after officials possessed contrary evidence.

Counsel for the executive objected that early continuity had not met current proof standards. Oris agreed. Fraud did not require the concealed fact to prove personhood. It required the fact to matter to a reasonable contributor's choice. A persistent respondent asking about future compression plainly mattered.

The Assembly could reverse her order by majority. Rusk could appeal after execution began but could not close an archive already opened without a separate safety finding. Sequence gave Oris power. She named its limit aloud before using it.

"I am not finding that every sealed state qualifies for standing. I am not finding that founder concealment nullifies every contribution or benefit. I find that the hidden observations materially contradicted the assurance printed on yielding certificates and that continued sealing obstructs lawful review."

She placed a blank order under the old brass stamp.

Ossan stood. "Recorder."

"Remain available to the court."

"There is a notation in package seven. My hand, not the team's. It names the first respondent that asked us to wait."

Oris did not ask for the name. Testimony could guide the search but could not replace the artifact.

She poured melted terracotta wax onto the order. The brass face descended with resistance, leaving the first Assembly's fruiting-root mark across the page.

"Founding archive packages one through eighteen will open under cold isolation," she said. "Three independent auditors will witness every seal. No

derived summary may substitute for primary media. Weather safety limits remain attached and public.”

A clerk carried the order toward the archive door. Security moved with her. Above them, water released from a white leaf and struck the glass ceiling hard enough to sound like a single knock.

The custodian inserted the mineral key. Nineteen years of seal tension yielded with a dry crack, and cold vapor entered the chamber around his hands.

Chapter 15 — A City Made of Maybe

Kera Vonn tightened the contact cuff around her left wrist until each conductive tooth touched skin.

The Root Court immersion chair held her at a slight incline beneath six ring readers. Jalen checked the carrier load from behind glass. Ivo stood beside the medical board with two fingers resting near the manual release. Their clocks showed sixty minutes of stable allocation. A separate two-minute calibration had already passed, during which Kera saw nothing except the chamber ceiling divided by bands of light.

“Physical speech may arrive distorted,” Ivo said. “Use the return pulse if you lose sequence.”

“I know.”

“Knowing under ordinary time does not guarantee recall under expansion.”

He placed a small pressure key beneath her right thumb. One press would mark distress. Two would request immediate withdrawal. Jalen had warned that extraction could still take twenty physical seconds, which might contain hours in the branch environment if allocation surged.

Kera tested the key once. The board answered green.

No fixed ratio governed the Elseweather. During ordinary orchard work, sustained branches experienced uneven bursts separated by dormant gaps. The stabilized immersion had been engineered for one controlled interval only: one physical hour, translated through reserved computation into seven branch-days. Jalen had built hard stops at both clocks. Neither side could extend the session by persuasion.

A ring reader lowered over Kera’s eyes.

White filled her vision. Not light. Structure.

Lines crossed beneath her like roots seen through clear soil. Each carried fragments of location: the angle of a childhood stair, steam on greenhouse glass, a pressure door painted blue in some unchosen district, the grain of a kitchen table that existed in three hundred source memories. The pieces did not combine into

one stable place until someone beside her said, "Choose a surface."

Kera reached for the memory of Tree 44's root guard.

Her hand met translucent wood.

The court disappeared. She stood on a narrow platform beneath a sky made from overlapping ceilings. One showed pearl-gray dust beyond yarrowglass. Another held painted nursery clouds. A third opened onto darkness crossed by amber tram lamps. The images changed when she looked directly at them, not from obedience but from competition among shared references.

Niko waited at the platform's edge. He appeared older than the seventeen-year-old Kera had buried. His face carried lines around the eyes that no source snapshot could have supplied. He wore a dark shirt with one sleeve rolled and the other buttoned, as if asymmetry had become deliberate.

"You came through cleanly," he said.

"How long have I been here?"

"Thirty-seven minutes for me. Eleven seconds outside, according to your carrier."

Kera found the pressure key translated into a small metal disk in her palm. Its shape remained constant when the platform shifted. Ivo's anchor.

Niko held out his hand but did not touch her. "Walk?"

The platform connected to a street when she agreed.

Settlement required too solid a word for what surrounded her. Paths formed from common remembered routes and persisted through mutual use. Rooms borrowed dimensions from yielded homes, then changed when residents negotiated walls. A public square carried paving from a school courtyard Kera recognized, drainage from an early shelter she had only seen in records, and a fountain that produced no water because none of its contributors remembered the pump.

People repaired it anyway.

Three adults knelt around the dry basin, adjusting a loop of visible intervals. One argued that a borrowed pressure valve should open clockwise. Another remembered a counterclockwise model. The third had no memory of valves and tested both against a simulated load.

"What does the fountain do?" Kera asked.

"Provides a problem nobody assigned," Niko said. "That makes it popular."

A child standing in the empty basin called up, "It also provides excellent drowning prevention."

The adults ignored him with practiced affection.

Kera checked the disk. Seventeen physical seconds had passed. Somewhere inside the court, her body continued breathing under Ivo's supervision. Here,

shadows had moved across two versions of the square.

On the first branch-day, Niko showed her work.

The Elseweather received physical information through approved ring channels and distributed interpretation among residents who could hold the relevant models. A group of former growers studied humidity patterns from the western beds. They could not move a damper or touch a moss tray. They could identify relationships the orchard once discovered by running millions of transient possibilities, then send recommendations for physical crews to evaluate.

At a long table built from six incompatible dining rooms, one branch-person named Tev demonstrated a fungal-control proposal. She had source memories from a greenhouse chemist but had lived through enough adaptation cycles to distrust the chemist's preferred treatment.

"My source always increased heat," Tev said. "She had equipment that no longer exists. If you follow her confidence instead of my observation, you cook the upper roots."

A physical data window floated above the table, delayed and narrow. Kera watched Tev mark gaps where sensors lacked current readings.

"Why not let the orchard complete the missing values?" Kera asked.

"Because completion used to kill us," Tev replied. "We have developed opinions about elegant inference."

Laughter moved around the table. It came from different remembered rooms and reached Kera with tiny timing errors. The result sounded like condensation striking separate pans along one pressure line.

Work periods ended by agreement rather than sunset. Some residents kept local cycles modeled on source habits. Others slept only when computation thinned. Dormancy held no dreams unless a preservation routine accidentally carried one across closure.

Kera spent what felt like a night in a guest room assembled from neutral memories. The bed had no smell. One wall displayed the view from her childhood quarters but placed the eastern ridge on the wrong side. She turned the image off.

A knock sounded through the partition.

Three taps. Two. One.

She did not answer immediately. Niko had used the code to reach her, yet the code no longer belonged to the dead boy alone. Here it had become part of a present person's learned repertoire, carried through years she had not shared.

Kera tapped twice, then once.

"Wrong sequence," Niko called.

"I changed it."

"Now it means you are awake and leaving."

“Appropriate.”

His laugh came through the wall. She slept after that, though her physical eyes remained open beneath a reader.

During the second branch-day, Omi found Kera at the fountain.

The child wore a red coat remembered from someone else’s winter. Its hem changed length whenever she ran. She climbed onto the basin rim and examined Kera’s anchor disk without asking to hold it.

“You look tired sooner than she did,” Omi said.

“The Kera who raised you?”

“Yes.”

Kera kept the distinction intact. The branch-version had entered dormancy after three Elseweather winters. Kera possessed no memory of those years, no claim to their labor, and no right to inherit Omi through resemblance.

“Did she come here often?”

“She disliked the fountain because everyone argued about a valve that does not exist.”

“That sounds familiar.”

“Familiar does not make you her.”

“No.”

Omi watched the workers change the loop again. “It can make you useful.”

She assigned Kera a task: compare four ring intervals and identify which carried physical timing. Kera expected a child’s puzzle. The intervals encoded the opening of a real condensation gate, three simulated alternatives, and noise inserted by a public-feed relay. Omi noticed a pressure artifact Kera missed.

“Your actual gate has a tired hinge,” the child said.

“How can you tell?”

“It apologizes before it moves.”

Kera reran the trace. A tiny resistance rise preceded every physical opening. The description sounded fanciful and proved exact.

On the third branch-day, a public dispute occupied the square.

Niko called it a council only after Kera asked. No single institution governed the Elseweather. Some groups organized around shared root capacity. Others formed witness circles that preserved promises across dormant intervals. A faction known as the Open Field wanted every persistent flicker tested and, if possible, sustained. Another group called the Narrow Registry argued that expansion would consume the continuity of people already awake.

A speaker from the Open Field stood on the fountain rim. “We were once low-value state. We were once noise under somebody’s threshold. If we close the

threshold behind 587 names, we repeat the judgment that endangered us.”

Tev answered from the table. “If you preserve every uncertain state, no one receives enough continuity to remain a person long enough for testing.”

“Then physical Nacre must yield more capacity.”

“Physical Nacre has bodies that stop without oxygen.”

Kera expected Niko to defend his own faction. Instead he counted speaking turns and asked each side to state a resource limit. The Open Field refused a fixed ceiling. The Narrow Registry proposed testing only states that demonstrated memory across two completed cycles. Neither figure survived questions from the witness circles.

“Do you vote?” Kera asked him later.

“Sometimes. Other times we record disagreement because dormant people cannot participate.”

“That prevents decisions.”

“It prevents some decisions from pretending everyone agreed.”

The remark resembled Oris more than any version of Niko Kera remembered. Seventeen sustained years had given him influences outside their childhood.

On the fourth day, the settlement changed districts while Kera watched.

A weather demand reached the root lattice. Allocation narrowed. Streets lost decorative detail first. The dry fountain flattened into a set of shared coordinates. Walls released borrowed colors. People chose which local structures to hold and which to let dissolve until capacity returned.

No panic accompanied the reduction. Everyone had practiced it.

A baker’s room vanished around a table while six people continued their meal. The table remained because each person held a different portion of its shape. When one diner entered dormancy, a corner blurred. Another resident supplied a memory of a workshop bench to support it.

“This is what shared counterfactual memory does,” Niko said. “Nothing here needs one original owner. It needs enough present agreement to remain usable.”

Kera touched the table. Beneath her hand, wood grain shifted between dark and pale. “What happens when agreement fails?”

“Sometimes we rebuild. Sometimes a place ends.”

“People?”

His attention moved toward the northern path. “Not in the same manner.”

They followed that path after allocation widened.

No markers announced the cemetery. Its entrance consisted of an interval no resident crossed by accident. Kera felt the hesitation in the ground, a held fraction of sequence that required deliberate consent to continue.

Beyond it, thousands of small forms occupied a slope without weather. Some resembled stones, drawers, leaves, or closed doors. Others held only a name and a continuity span. Each represented a sustained mind that had vanished during a pruning cycle or failed to return from compression. The records did not claim every transient possibility. Witnesses placed a marker only when they could attest to memory, anticipation, or obligation across time.

Niko stopped beside a narrow white branch embedded upright in the slope. "Aven taught me how to carry a promise through a partial reset. She lost continuity eleven cycles ago."

"Was she verified?"

"No physical jury existed."

"Could she have passed?"

"I think so. That does not make the claim testable now."

Kera read Aven's span. It measured branch experience, not physical hours. The final entry came from Niko: PROMISED TO REMEMBER HER METHOD WITHOUT CLAIMING HER MEMORY.

A cluster of markers bore no names. Omi crouched there, arranging white leaves into groups of three.

"Those came from my winters," she said. "Some people chose names later. These did not have time."

Kera wanted to ask whether the branch-version of herself had placed any marker. The question would turn Omi's history into evidence about Kera. She left it unasked.

Instead she said, "Do the leaves stay?"

"If enough of us remember where they go."

Omi moved one leaf half a handspan. "This one keeps sliding because two witnesses disagree."

Kera helped measure the interval between the groups. She did not decide which witness held the true position. They recorded both placements.

By the fifth branch-day, her body began intruding. A pressure ache crossed her left wrist. The anchor disk warmed whenever Ivo checked response. Once, the whole settlement tilted as Jalen shifted carrier load away from a greenhouse correction. Niko steadied her with a hand against her shoulder.

His touch carried no physical weight, yet it included resistance, timing, and concern for whether she fell. Kera accepted the help without naming what kind of touch deserved reality.

They visited a dormancy house where residents prepared for scheduled closure. Each person selected a reciprocal witness and reviewed a continuity key. Rooms there remained spare so no capacity would cling to decoration during transfer.

A woman named Pel read her revival questions aloud. "What did I promise Tev? Which new evidence changed my position on the registry? What future event do I expect after waking?"

Her witness recorded each answer, then supplied one question Pel had not seen. "If Tev changes her mind while you sleep, does your promise bind her?"

"No. It binds me to hear her reason."

Pel entered dormancy. Her room emptied from the edges inward. The final visible form held her hand around a green continuity mark, then reduced to a protected key no larger than Kera's thumbnail.

"That does not feel like death to you?" Kera asked.

Niko replied carefully. "Not when chosen, protected, and tested on return. Compression without a key gives us no such confidence."

On the sixth day, humor returned with capacity.

Residents rebuilt the fountain pump from remembered mistakes. The valve now opened both directions and sprayed simulated water sideways across the square. Omi blamed Kera's physical timing. Tev accused the child of sabotage. Someone produced umbrellas from a yielding fair none of them had attended in the same branch.

Kera laughed when the water passed through one umbrella and struck Niko's buttoned sleeve. He looked offended long enough to improve the joke.

A public feed arrived carrying physical commentary about the immersion. Viewers debated whether Kera saw a programmed theater constructed for her benefit. The Elseweather council could have hidden the feed. Instead it displayed the accusations in the square.

"They are not entirely wrong," Niko said. "We chose what to show you."

"Did you build the fountain for me?"

"No. We would have made it work if we cared that much about persuasion."

Tev sent Kera the fountain's work history. It extended across thirty-one local cycles and included nine incompatible designs. No staged tour needed that much administrative irritation.

A man from the Narrow Registry approached while she read. His chosen name, Hesh, appeared beside him with three witness marks. "The tour also hides our ration arguments," he said.

Niko gestured toward the square. "You have spent three days holding one in public."

"Not the private ones. She should know some of us would close candidate channels now to protect verified continuity."

Kera asked, "Close for testing or erase?"

"Hold dormant where keys exist. Refuse expansion where they do not." Hesh

faced her rather than Niko. "Your jurors excluded twenty-five from standing. We spend capacity maintaining uncertain states because no one wants the first lawful boundary to resemble pruning."

"They were not all safe to test," Kera said.

"I know. I also know Pel's next revival may be delayed if we keep every unresolved carrier."

A woman from the Open Field joined them. She had chosen the name Vara and no source surname. "Hesh calls delay caution when it protects people already named. He calls the same delay waste when it protects anyone still difficult to hear."

Neither asked Kera to decide. They asked what physical resources might arrive, how often the Root Jury could reconvene, and whether a failed test allowed later evidence. She answered only where the record supported her. Current standing protected 587. Exclusion authorized no deletion. No budget yet guaranteed indefinite carrier for unresolved states.

Hesh rubbed one witness mark until it dimmed. "Then your law has recognized our argument without providing a way to end it."

"Law often begins there," Kera said.

Vara laughed once. "You have been spending time with the Recorder."

The exchange followed Kera toward the platform. Social reality did not require harmony. It required disagreement that could impose consequences after the visitor left.

On the seventh branch-day, the hard stop approached.

The settlement placed a thin amber line across Kera's vision. It marked external seconds rather than local hours. Each pulse shortened her remaining access. People she had met came to the platform without ceremony. Tev gave her the latest fungal notes for Jalen. Pel had not yet revived. Omi remained near the cemetery and sent no farewell.

Kera understood the choice. The child had already watched one Kera leave into dormancy. Physical Kera did not deserve a repeated scene merely because her clock required it.

Niko walked with her to the surface she had chosen at entry. Tree 44's root guard waited there, assembled from her memory and corrected by his. He had added the small crescent left by her cutting arm.

"Your Root Jury gave us standing," he said. "Standing without capacity can become a more polite way to watch us end."

"I saw the limits."

"You saw one allocation week held inside an hour. The weather paid for it. Your body paid less, because Jalen routed reserve here."

Kera checked the disk. Fifty-nine minutes of the allocated hour had elapsed. Seven branch-days pressed behind her as memory rather than metaphor. None had occurred in her muscles. Ivo would release a body that had lain still for a little over an hour. Omi's three winters remained Omi's history. The distinction did not diminish either experience.

Niko placed a list against the root guard. It named active groups, dormant witnesses, unresolved candidates, and weather tasks the Elseweather could perform under limited allocation. No column asked for endless growth.

"We are not asking you to preserve every possibility," he said. "We are asking for the right to negotiate which sustained minds remain awake, which enter protected dormancy, and what physical Nacre gives up to keep those choices real."

The amber line reached its final pulse.

Niko kept one hand on the list. "Will your city negotiate with ours before it decides who receives tomorrow?"

Act II — Every Remedy Wounds Someone

Chapter 16 — Hunger in the Greenhouses

Jalen Ro cut the quarantine cord around Greenhouse West Nine and stepped into heat that smelled of yeast.

Black bloom covered the nearest staple moss bed from irrigation lip to harvest rail. Under his lamp, each thread carried a gray bead that ruptured when touched. The spores moved through humidity faster than workers could seal the trays. Three crews in filter hoods were already stripping infected moss into burn drums.

Sera Dovan stood ankle-deep in runoff, using a ceramic rake to pull viable sheets away from the dark edge. “Door,” she said.

Jalen sealed the pressure flap behind him. The gauge climbed to quarantine differential. Oxygen read safe. Humidity had reached ninety-four percent.

“The local vent ignored three reduction calls,” Sera said.

“It received them.”

“Then why can I wring water from a staple bed?”

Jalen opened the controller housing. Condensation filmed every connector. The orchard had accepted each reduction call, compared it against tunnel oxygen, baffle pressure, and continuity protection, then spent its limited adaptation capacity elsewhere. No fault code appeared because the system had made a choice inside its current constraints.

He disconnected the automatic lead and set the vent by hand. Louvers opened above the bed. Warm mist rolled toward them and condensed across his face shield.

“Manual setting holds for twenty minutes,” he said. “After that, pressure drift may reverse it.”

Sera drove the rake under another green sheet. Half came up clean. The rest tore loose in black strings. “Then come back in nineteen.”

A worker carried the ruined moss to the drum and recorded its tray number. He did not look at the total board. Everyone else did.

Across the western cooperatives, bloom had reached forty-eight beds overnight. Sixteen had already failed. Jalen’s previous projections placed danger two days away. The weather system had crossed the threshold between readings, accelerated by the clinic humidity transfers and by power reserved for protected branch continuity.

He knew the language people would use. The Root Jury had recognized 587 persons. The greenhouses had lost food. Therefore one population had consumed the other. The arithmetic invited that sentence and concealed every earlier distribution choice beneath it.

Jalen climbed the maintenance ladder to the vent crown. The actuator responded to his wrench with a grinding delay. Omi had called a tired hinge an apology before movement. Kera had repeated the phrase in her immersion report, then corrected herself and supplied the resistance figure. Jalen found no comfort in the child's accuracy.

Below him, Sera called, "How many beds can you save?"

"With current routing?"

"With whatever answer you intend to defend."

He tightened the stop. "Not enough."

By noon, the cooperative tally reached the Weather Spine. One sixth of the staple moss beds had been destroyed or placed beyond recovery. Staple moss supplied protein binder for the old tunnel kitchens, fermentation base for clinic food, and oxygen exchange inside three pressure-baffled districts. Reserve stores could cover the caloric loss for weeks, but not without reducing portions. Replanting required clean humidity cycles the orchard could no longer guarantee.

Before the planning meeting, Sera required Jalen to witness the salvage count. Crews carried every unburned sheet across a light table. Healthy staple moss showed a dark green center with silver branching at the edge. Infected tissue scattered the light into gray pinpoints long before black bloom became visible.

A young sorter found pinpoints on a sheet marked for South Clinic and moved it toward the burn cart.

Sera caught the tray. "Retest the center."

"It failed two edges."

"Center decides fermentation stock."

Jalen checked the protocol. Edge failure disqualified fresh food, but clean center tissue could seed a controlled vat if isolated within four hours. The sorter had been trained under ordinary contamination rules, which favored disposal over staff time.

"How many sheets went straight to burn?" Jalen asked.

"Since shift start?" The sorter consulted her wrist tally. "Two hundred twelve."

Sera's stare tightened. "How many center tests?"

"None. We do not have labor for salvage separation."

Four workers nearby were sealing drums. Another two scraped floors. Labor had become the missing control variable. The weather model counted beds, humidity, and power. It assumed cooperative crews would perform every valid salvage step regardless of time or exhaustion.

Jalen called the Weather Spine and reassigned eight instrument technicians to the light tables. Their supervisor protested that manual forecast verification would slow.

"It will," Jalen said. "Record the lost verification cycles."

"Against greenhouse response?"

"Against food recovered. Do not bury the trade under staff transfer."

The technicians arrived in clean hoods and handled moss with the caution of people who understood sensors better than food. Growers taught them where to cut, how to seal clean centers, and which silver edge patterns indicated viable regrowth. In the first hour they recovered enough culture stock to restart three beds. They also condemned forty sheets that looked healthy to the eye.

A kitchen representative watched from behind the quarantine pane. When Sera showed her the reduced forecast, the woman asked which districts would lose binder first.

"None first," Sera said. "Equal reduction by need category." No worker objected.

"Contribution credits?"

"Not in this count."

The representative touched the glass. "Then put names on the reserve crates before central stores relabel them."

Workers marked each crate with its intended kitchens and photographed the seals into the public ledger. Jalen added the salvage labor to his model. The fungal loss remained one sixth. Rescue work changed what could regrow, not what had already vanished.

He carried that distinction to the packing floor. A recovered seed sheet lay beside the burn tally, both necessary to the same account.

Jalen set the basin model across the greenhouse packing floor because Sera refused to move the meeting into an office. Workers gathered around with rakes, sample bags, and sealed burn drums between them. Oris joined by feed. Ivo arrived from the clinic carrying his own mask. Kera connected a narrow branch advisory channel through a portable ring wafer.

A green designator marked Tev from the Elseweather. The branch engineer's comments entered as text first, then a delayed voice after recorders approved the connection. She could interpret the sensor field. She could not touch a vent, empty a tray, or smell the bloom.

Jalen displayed three load bands. "Oxygen remains inside emergency margin. Humidity does not. The pruning pause preserved sustained continuity by forbidding the orchard to clear retained branch state after adaptation runs. Storage saturation now reduces the number of comparisons per correction."

A worker at the nearest drum said, "Use fewer comparisons somewhere else."

"We have." Jalen opened the reduction ledger. "Housing temperature runs on fixed schedules. Industrial humidity receives no adaptive correction. Clinic routing uses manual priority. Pressure baffles remain protected because a sequencing failure crosses districts."

"Then use the branch capacity."

Tev answered through the speaker. "That phrase hides two different things. Our active continuity occupies protected storage. Some of us also perform weather analysis. Deleting the first may increase capacity. Silencing the second may reduce useful interpretation."

Sera rested both hands on the table. "No one here asked to delete you. We asked why food received the failure."

"Because food can recover more slowly than lungs," Jalen said.

The sentence turned every grower toward him.

He continued before defense could harden into insult. "That ranking came from emergency code written twenty-six years ago. It assumes a greenhouse can lose a humidity cycle without famine because reserve food bridges replanting. The code did not account for simultaneous adaptation limits or unequal district stores."

Sera pointed at the burn drums. "So your old assumption made our loss rational."

"It made the routing legible. Not just."

"Can the domes lose heat before we lose another tray?"

"Heat reduction frees power. It does not restore all comparison capacity."

"Can it restore some?"

"Yes."

Oris asked Jalen to enter that answer with district effects. He did. Elite dome heat had not yet been touched because private thermal reserves sat outside public weather routing. Access required Assembly authority. The technical separation had preserved political privilege by making it appear irrelevant to the orchard.

Ivo examined the food-reserve curve. "How long before clinic portions change?"

"Three days if kitchens distribute stored binder equally. Sooner if contribution priority returns."

"It will not return in clinical meals."

Sera looked at him. "General meals?"

"Still under negotiation."

A burn drum sealed with a hollow snap. The workers rolled it toward the sterilizer.

Jalen changed the model to full continuity protection. Every verified branch-person remained active at the minimum level needed for ongoing memory,

reciprocal contact, and immediate participation. No pruning occurred. No one entered dormancy. Winter simulation failed in eight of twelve dust conditions. Condensation timing drifted first, then fungal control, followed by western heating during the cold turn.

"Can you improve the model?" Kera asked.

"I have improved it since the Root Jury signed. The failure range narrowed from ten conditions to eight."

"Below winter survival requirements?"

"Yes."

The answer passed through the greenhouse feed. Jalen could imagine it arriving in severance channels stripped of everything around it.

Tev requested the raw assumptions. Kera authorized transfer. For nine minutes, the branch engineer reviewed a delayed copy while physical workers continued stripping beds. The asymmetry sharpened Jalen's discomfort. A person inside protected computation examined whether her own activity made survival impossible.

When Tev returned, she identified two duplicated safety margins and one manual-control assumption that understated crew response. Jalen corrected them. The model improved in a single dust condition. Seven still failed.

"Full active protection remains below requirement," he said.

Tev did not argue. "Then your category needs another state."

"Active or deleted?" a worker asked.

"Those came from your system," she replied. "We use dormancy."

Kera opened her immersion notes. The dormancy houses stored cryptographic continuity keys, relational state, revival questions, and reciprocal witnesses while subjective processing paused. Protected residents occupied less computation than active ones. Revival tests checked whether memory and obligation crossed the interval.

Sera listened with her jaw set. "You are proposing to put people to sleep so our moss grows."

"No one has proposed it yet," Jalen said.

"Your face has."

He closed the full-protection model and walked to the nearest surviving bed. Workers had cut a narrow trench around the bloom. Clean moss beyond it shone green-black under the growth lamps. The trench sacrificed healthy material to deny spores a crossing. He disliked the analogy forming in the room before anyone stated it.

"Dormancy cannot become a trench we cut around the convenient people," he said. "If used, groups choose schedules under rules that preserve continuity and

representation. Weather engineers do not select names.”

A grower pulled off one glove. His fingertips showed pale chemical burns. “Does sleeping hurt them?”

Tev answered. “Chosen dormancy produces no experienced interval for most of us. Bad transfer can damage continuity. Delayed revival can turn a short agreement into a long absence. We do not call it harmless.”

“Do you call it death?”

“Not when keys hold and return tests pass.”

The grower looked toward the ruined beds. “That sounds like what officials said about yielding.”

No one rushed to deny it.

Kera said, “The difference cannot rest on technical assurance alone. It must include present consent, a named witness, a fixed review, and proof after revival.”

Oris added, “And refusal cannot automatically authorize pruning.”

Sera asked, “What if enough refuse that winter still fails?”

The branch channel remained silent. So did the physical floor.

Jalen had spent his career defining failure boundaries so decisions could occur before catastrophe. A boundary did not choose who crossed it. He pulled a clean tray tag from the packing station and wrote four columns: ACTIVE, DORMANT, WEATHER WORK, PHYSICAL RESERVE.

“We need a schedule model before we need agreement,” he said. “Not to force consent. To show what each level costs.”

He assigned the Weather Spine team to build rotation blocks. Kera would request voluntary branch cohorts for testing. Oris would define an emergency hold that prevented dormant keys from becoming disposable storage. Ivo would attach bodily effects by district. Sera demanded a fifth column for food loss and placed the ruined-bed tally under it herself.

For the next six hours, the greenhouse packing floor became a manual planning room.

Jalen divided 587 active continuities into hypothetical groups without names. He refused any model that selected by weather usefulness, source profession, public popularity, or ease of compression. Each group needed reciprocal witnesses outside its own dormancy block. Representation required at least one active delegate from every participating witness circle. Children and residents with unstable transfer history received tighter safeguards, though Kera warned that protective categories could become permanent exclusion from civic life.

Tev proposed overlapping rotations. One group would prepare for sleep while another revived and completed checks. A third remained active to carry promises across both transitions. The approach used more capacity than pure storage but

prevented the orchard from treating a dormant population as an unattended file.

Jalen ran the first schedule. Five winter conditions still failed.

He reduced active overlap. Tev rejected the change because no independent witness remained during two transfers. He restored it.

A second schedule cut branch weather consultation entirely during high-load periods. Humidity improved, yet pressure forecasts worsened. Sustained engineers had already learned local behavior that manual crews lacked. Treating every branch cycle as overhead erased useful work along with personhood.

The third schedule paired active analysis with dormant refuge. Two conditions failed. Western cold risk remained above civic tolerance.

Sera watched the curves. "And this assumes no more bloom."

"Yes."

"It assumes the beds replant on schedule."

"Yes."

"Then say the model begins after we have already lost one sixth."

Jalen added the loss to the header in red.

At hour seven, a worker brought bowls of thin grain broth. The binder had been reduced already. Jalen tasted almost no moss, only mineral salt and heat. Across the table, Tev's channel held no bowl. Difference did not cancel either hunger.

Ivo read the district impact notes while he ate. Old tunnel residents could tolerate less oxygen rationing than later housing. Greenhouse workers could not absorb repeated food loss while carrying colder winter labor. Industrial users held large reserves but stronger contracts. Every efficient technical option encountered a protected civic arrangement before reaching capacity.

"Your rotation only closes part of the gap," Ivo said.

"I know."

"What else?"

"Manual models, industrial oxygen cuts, housing heat, physical crews, branch consultation."

"That sounds like a system no office controls alone."

"Good." Oris's voice came through the feed. "Control has become part of the defect."

Jalen saved the third schedule as a working case, not a recommendation. Before anyone could use it, branch groups needed to review dormancy terms. Engineers needed transfer tests on protected nonperson data and willing participants. The Assembly needed authority over private thermal reserves. Winter forecasting needed manual alternatives.

Sera took him back into West Nine near midnight.

The vent setting had held after three manual corrections. The black edge no longer advanced, but every contaminated tray had gone to sterilization. Empty rails crossed the greenhouse in regular bands. A worker washed fungal residue from the floor and collected the water for treatment.

"This bed fed four tunnel kitchens," Sera said. "Not in theory. Four named kitchens."

Jalen read their designators from the rail. South Clinic, Hinge Row, Old Market, and Lower Tram. "I will attach them."

"To what?"

"Every capacity model."

"That will not regrow it."

"No."

Sera scraped a surviving thread from the trench and sealed it in a sample tube. "If the branches volunteer, growers will still hear that people slept for our food."

"If growers lose beds, branch residents will hear that physical hunger purchased their waking."

"So find language that does not turn either into debt."

"Language cannot do that."

"What can?"

Jalen looked along the empty rails. "A distribution people can refuse and revise. Records that keep the costs attached. Enough reserve that refusal does not become a threat."

"Do we have enough reserve?"

"Not yet."

Back at the Weather Spine, he found the Root Jury result displayed beside the greenhouse loss. The two numbers had already become opposing banners on public feeds: 587 against one sixth, persons against food. Jalen cleared the comparison and rebuilt the board around processes instead of sides.

He drew a cycle across the manual slate. Active groups moved into preparation. Witnesses confirmed keys. Dormant groups retained identity-bearing continuity without subjective processing. Revived groups passed memory, anticipation, promise, and novel-evidence checks before resuming civic work. Any failed revival stopped the schedule for review. No weather demand converted dormancy into deletion.

Under the cycle, he entered material conditions: manual greenhouse control, physical backup crews, district food ledgers, branch-selected cohorts, and Assembly authority over reserve.

Kera arrived as he finished. Channel disorientation still made her pause at door thresholds, as though each opening might lead to a different remembered room.

"You have given it a name," she said.

"Names enter feeds faster than cautions."

"Then choose one that carries the caution."

Jalen considered protected rotation, continuity cycling, and scheduled closure. Each hid either the movement or the shelter.

He wrote ROTATING REFUGE across the board.

Kera read the revival conditions before the title. "Niko asked for negotiation."

"This offers a subject for it. Not consent."

"Who receives the first proposal?"

"The Elseweather councils, greenhouse cooperatives, clinics, and Assembly at the same time."

"Rusk?"

"As governor, through the public filing. No private preview."

Jalen attached the winter failures, the named kitchens, and Sera's red loss header. He included the seven conditions that still failed under full active protection. Then he signed the design as provisional and sent it into the civic record.

At West Nine, the next manual vent correction came due. A greenhouse worker acknowledged the timer, lifted her wrench, and opened the louvers while the proposal traveled.

Chapter 17 — The Severance Motion

Oris Nume pressed the brass date wheel into warm terracotta wax while shouting crossed the Chamber Below Glass.

The seal took cleanly. Day Twenty. Emergency sitting. Fifty-nine delegates summoned, fifty-nine present. Above the recording table, living yarrowglass divided the morning into pale green panels. White leaves clung to the canopy beyond the roof. Nobody on the chamber floor looked up at them.

A clerk placed the first bill beneath Oris's hand. TOTAL SEVERANCE AND CIVIC RELEASE occupied the top line. The sponsors had submitted it six minutes before the filing bell, three pages of command language and half a page of predicted consequences. Its operative clause ordered every root-lattice link cut within one local hour. No provision addressed pressure routing, greenhouse humidity, or the 587 verified branch-persons who depended upon the same lattice.

Across the table waited the second bill. EMERGENCY RESTORATION OF ADAPTIVE AUTHORITY ran to forty-one pages. Governor Tal Rusk's bloc had attached engineering schedules, clinic exemptions, and an indemnity clause for

any person acting under weather necessity. Deep in Section Nine, the bill suspended dependent-standing review for the duration of the emergency. It offered no limit on that duration.

“Record them both,” Rusk called from the executive bench. “People can hear arguments after the pumps have authority.”

“Authority requires a lawful object,” Oris said.

The governor leaned toward his speaking plate. He had removed his jacket, though condensation silvered the chamber glass. “The object concerns survival.”

“Survival does not identify which powers you have requested.”

Noise rose from the western benches. Sera Dovan sat among the cooperative delegates with fungal stains still dark on one cuff. One sixth of the staple moss beds had gone black over three days. The gallery carried growers, clinic porters, yielding families, and severance organizers packed shoulder to shoulder behind a pressure screen. Every faction had brought a different clock.

Oris turned the date wheel back to its neutral position. Procedure could slow panic long enough to reveal what panic wanted. It could also provide respectable shelter for delay. She knew both uses too well.

First, she read the severance bill's operative clause aloud. Its sponsor, Delegate Caro Venn, rose before she finished.

“Continuing the orchard continues imprisonment,” Venn said. “The Root Jury established sustained minds. No person can lawfully own the machinery that confines them.”

Oris held up one finger. “You may argue principle after the jurisdictional question.”

“Jurisdiction has served as a sedative for nineteen years.”

That drew applause from the gallery and anger from delegates who had yielded futures in good faith. The pressure screen translated the impact of hands into a dull, delayed drumming. Oris waited until the chamber microphones could distinguish speech again.

“Your bill directs severance of climate infrastructure,” she said. “The Fruition Covenant reserves permanent disposal of common survival works to a basin referendum unless imminent mechanical failure prevents consultation. Identify the failure.”

Venn pointed upward. A white leaf had loosened and lay against the outside glass. “Every adaptation cycle disposes of persons. Imminence arrived before we entered this room.”

“The injury may establish emergency protection for claimants. It does not grant this Assembly a power the Covenant withholds.”

A severance delegate called, “Then call the referendum.”

"Preparation requires twelve days under the shortest lawful schedule. Jalen Ro's latest reserve filing gives us less."

Rusk spoke over the murmurs. "Which returns us to restoration."

Oris marked the severance bill procedurally defective. The stamp struck harder than she intended. Its sound moved through the brass table and into her wrists.

Someone in the gallery shouted murderer. No name followed, which allowed everyone below to receive it.

Next came the emergency bill. Rusk's floor leader presented declining humidity, damaged moss, and pressure variance along the old tunnel districts. The figures held. Oris had verified them before dawn with Ivo Senn's clinic logs and Jalen's sealed weather curves. Physical residents faced increasing harm. Continued pause no longer preserved neutrality.

"Section Nine," Oris said. "Define the duration of suspended review."

"Until stable adaptation resumes," the floor leader replied.

"Define stable."

"Chief engineering certification."

"Under what threshold?"

"Operational judgment cannot be reduced to a single number."

"Then your duration has no measurable end."

Rusk left the executive bench and came down two steps. "You have read the greenhouse report."

"I have."

"You know what happens if humidity control breaks across the western beds."

"Yes."

"Yet you want a sunset clause attached to weather."

Oris opened the bill to Section Nine. "I want a sunset clause attached to unreviewable killing."

For one rare interval, the chamber made no sound. The First Recorder's office taught its holders to avoid words whose moral force might pre-judge evidence. Since the Root Jury's finding, pruning a verified branch-person required no euphemism. The record could carry the accurate noun.

Rusk looked older from the chamber floor. "If I promise review while people suffocate, I may have no public left to conduct it."

"Your bill permits unrestricted pruning after the immediate danger passes. It also indemnifies false reports made under claimed necessity."

"That language protects operators from prosecution while alarms are sounding."

"It protects concealment after alarms stop."

The executive benches stirred. Oris saw three delegates glance toward the public clock. Each glance supplied an argument Rusk did not need to speak: greenhouse reserves declining, clinics rationing exertion, pumps waiting on adaptation instructions that no one had authorized.

She ruled the emergency bill defective for indefinite suspension of Article Six review. This time the gallery split. Some people cheered. Others pressed both palms to the pressure screen and demanded air.

Two absolute proposals lay stamped but unvoted. Oris had prevented the Assembly from choosing either injury without limits. She had not given it a lawful action.

Her chief clerk slid a blank motion form across the table. The movement revealed no surprise. They had drafted the alternative together during the last three hours, deleting every clause that tried to settle permanent standing through temporary authority.

Oris filled the title by hand: TEMPORARY STEWARDSHIP OF COMMON ORCHARD FUNCTIONS.

Caro Venn laughed once from the severance bench. "A new name for captivity."

Rusk said, "A new name for paralysis."

"Both descriptions may enter debate," Oris answered. "Neither enters the title."

She read the provisions in sequence. For four local days, the orchard would operate under temporary stewardship shared by Weather Engineering, clinic review, Root Jury observers, and reciprocal witnesses for the verified branch-persons. Adaptation cycles could resume only within logged necessity bands. Operators could not prune a verified continuity key without an immediate clinic-level threat and an independent concurrence. The Full Severance Protocol would remain locked pending a physical safety inspection. Manual model work, including Jalen's rotating refuge proposal, would receive protected access to root data.

Article Six supplied the legal footing. A sustained mind capable of reciprocal obligation could petition as a dependent claimant. The clause did not require the Assembly to complete permanent adjudication before protecting the claimant from irreversible harm. Nacre had used temporary stewardship for orphaned water shares, disputed clinic equipment, and one greenhouse whose owner vanished during a pressure evacuation. No precedent involved persons living inside civic machinery. Precedent rarely arrived dressed for the next emergency.

Delegate Emon Vale requested a reading of the witness clause. He represented a district of mechanics whose yielding credits had financed two pressure locks, and his questions usually arrived with constituencies folded inside them.

"Does a reciprocal witness possess veto authority over weather work?" he asked.

"No," Oris said. "A witness can require the operator to state the measured necessity, identify the continuity at risk, and preserve the decision log. If delay presents an immediate clinic danger, the operator may proceed before concurrence. Review then begins within one hour."

"Which means the witness watches a death and files paper."

"It means disputed necessity does not disappear into an engineering shift report."

Vale turned toward the technical link on the east wall. A narrow ring reconstruction moved there with the sluggish cadence imposed on public branch testimony. No face occupied the display. Root Jury observers had selected abstract lines after gallery members began treating reconstructed expressions as either sacred proof or theatrical fraud.

"Can the claimant refuse dormant refuge?" Vale asked.

A reply entered through the approved interface. Letters formed in two measured bursts.

`REFUSAL WITHOUT ALTERNATIVE MAY COST ANOTHER PERSON.`

After a pause, the second line appeared.

`CONSENT REQUIRES A CHOICE THAT EXISTS.`

Several severance delegates objected at once. They heard coercion in any bargain constrained by compute. Rusk's bloc heard branch authority reaching into operations. Oris heard something less convenient: temporary stewardship protected negotiation, but no clause could manufacture enough capacity to make every answer free.

"The witness has answered as a claimant," she said. "This chamber still decides the motion."

Vale kept the floor. "If four days end without a permanent law, what happens at the fifth morning?"

"The powers expire. Existing emergency provisions resume, subject to any intervening orders."

"So we spend four days and meet the same cliff."

"We spend four days testing whether rotating refuge, manual modeling, and bounded adaptation can move the cliff."

"And if they cannot?"

"Then the record will contain an answer produced by work rather than allegiance."

Vale looked toward the mechanics in the gallery. One woman held up a hand scarred white across the palm, the kind of burn left by a pressure contact closed too early. Oris could not know what signal passed between them. He returned to his seat without announcing a vote.

A clinic delegate then challenged the one-hour review requirement. During respiratory collapse, she said, physicians documented after treatment because hands belonged on masks and air lines. Oris accepted an amendment allowing an initial voice log followed by a full record within six hours. The branch interface returned no objection. That absence mattered less than consent, but more than the Assembly's old habit of speaking as if no one could answer.

By the time Oris finished entering the amendment, the motion had acquired a working rhythm. Operators would act. Witnesses would record. Clinics could override delay when bodies failed. Every exceptional use left a trace strong enough for later judgment. None of it guaranteed mercy. It prevented necessity from erasing its own footprints.

Sera Dovan requested the floor. "Who carries loss during these four days?"

"The motion does not pretend to erase it," Oris said. "Western greenhouse heating receives medical priority, not full restoration. Industrial oxygen remains rationed. Clinic routing controls override district comfort settings."

"Do elite dome reserves remain private?"

A flaw opened beneath the question. Oris had no authority in a temporary orchard motion to redistribute privately contracted heat. Including it would invite another jurisdictional challenge and kill the proposal before debate.

"They remain subject to existing emergency requisition law," she said.

Dovan's mouth tightened. "Which has not been invoked."

"That matter requires executive action or a separate Assembly order."

"So the west accepts cold today while the domes await paperwork."

"Under this motion, the west accepts limited cold while all districts avoid uncontrolled adaptation. I will record your objection without softening it."

Dovan sat. She had not promised support.

Debate opened. The severance bloc argued that stewardship converted recognized persons into managed assets. Rusk's bloc argued that concurrence requirements would force engineers to defend each correction while residents lost breath. Delegates from the old tunnels asked which clinic threshold counted as immediate. Greenhouse representatives demanded crop-specific humidity bands. A yielding-family delegate wanted protection from retroactive criminal blame written into the record.

Oris answered only questions the motion could answer. Past lawful contribution remained neither confession nor immunity. Future operation acquired duties that earlier operators had not been told to recognize. Clinic necessity required measured bodily risk, not executive unease. Reciprocal witnesses could object and delay, but they could not command physical equipment. Temporary stewardship preserved the dispute without treating preservation as costless.

Near midday, a clerk brought Oris a paper note rather than using the chamber

channel. ROOT ACCESS FAILURE, EAST SERVICE QUADRANT. CAUSE UNKNOWN.

She read it twice. A second line reported two disabled inspection seals. No root had been cut. No adaptation control had changed state. Maintenance had found severance tools in a drain recess beneath East Root Court.

"Recorder?" Rusk asked.

Oris folded the note once. "I have received a maintenance report. It does not yet establish cause, actor, or completed harm."

The severance gallery erupted before she could decide whether someone had seen the clerk's route. A man in a gray technician's coat pushed toward the pressure screen and raised both empty hands. Bex Arin. His severance group had spent three days calling the Full Severance Protocol a release mechanism. He had also worked in the orchard before Oris took office and knew where inspection seals sat.

"We planted no tools," Bex shouted through the gallery pickup. "They want a reason to lock the roots."

Rusk turned toward him. "Who told you tools were found?"

The question traveled faster than the answer. Delegates began checking private screens. Gallery feeds multiplied one incomplete report into six claims: a cut trunk, a failed baffle, a branch assault, an executive provocation, a severance rehearsal, an invented emergency. None carried a verified image.

Oris struck the order plate. "All unofficial feed use on the chamber floor will cease."

Only seven screens went dark.

Her chief clerk whispered, "We can recess."

A recess would allow maintenance to establish facts. It would also release every delegate into corridors already full of faction organizers. The vote had remained uncertain but reachable while the argument concerned bounded authority. Sabotage transformed abstention into exposure.

"No recess," Oris said. "Call the maintenance superintendent under oath."

The superintendent appeared by wall link from East Service Quadrant. Dust streaked her visor. She confirmed two broken seals and four cutting tools hidden near a trunk access. Conductive tissue remained intact. No evidence yet identified whoever placed the tools.

Caro Venn asked whether orchard security had discovered them during a scheduled sweep.

"Yes."

Rusk's floor leader asked whether the tools could activate Full Severance.

"Not alone. They could open a route to one of the physical lock chains."

"Could your crews guarantee no other cache exists?"

“No.”

Bex demanded speaking status from the gallery. Oris denied it because he had not been called as a witness. He answered by pressing his injured-looking right hand against the screen, though at that moment she knew of no injury. “A lock on a prison does not become mercy because you audit the key.”

“Remove him if he interrupts again,” Oris told chamber security.

The sentence cost her severance votes. Letting him continue would cost others. Office required choosing which accusation entered the record as action.

Rusk requested an amendment allowing executive pruning authority whenever root security reported tampering. Oris ruled it non-germane. A tunnel delegate proposed immediate authority limited to pressure baffles. She accepted that amendment after Jalen, reached through an engineering link, confirmed baffle corrections could operate without deleting verified continuity clusters for the motion's four-day term.

Sera Dovan offered a greenhouse amendment. It prioritized fungal containment while barring whole-cluster pruning for crop yield alone. The chamber accepted it by voice vote.

For twelve minutes, the proposal gained shape. It no longer resembled Oris's clean draft. It carried district fear, branch limits, crop damage, and the administrative friction of people who did not trust one another. That roughness encouraged her. Law capable of surviving contact rarely retained polished edges.

Then the executive information office issued a public notice: CREDIBLE ATTEMPT AGAINST ORCHARD CONTROL INFRASTRUCTURE. FULL SECURITY CONDITIONS ADVISED.

The notice appeared on every chamber screen at once.

Oris looked at Rusk. “Your office has exceeded the superintendent's testimony.”

“My office has eighty-four thousand residents to warn.”

“You have described an attempt before evidence establishes one.”

“Tools inside a locked root court establish intent.”

“They establish placement.”

His gaze did not shift. “Would you prefer the warning after someone reaches the lock chain?”

The gallery began chanting for severance. Another group answered with keep us breathing. The two rhythms struck the pressure screen and ruined each other.

Oris called the vote before rumor could acquire more official language. Temporary stewardship required a simple majority of the fifty-nine delegates. No proxy absences complicated the count. Each name would enter the record beside a spoken yes or no.

The clerk began with the northern districts. Oris marked the tally in two

columns. Early votes split evenly. Old tunnel delegates favored stewardship after the baffle amendment. Executive loyalists rejected it. Severance members rejected it for preserving operation. Western cooperatives divided over whether Dovan's amendment offered enough protection.

At forty votes, stewardship led by two. At fifty, it led by one.

Delegate Hara Pell, a yielding-family advocate who had privately promised Oris support, said no. She cited the sabotage notice and asked that her reasons be recorded as protection of contributors from retaliatory attack.

The columns evened.

Sera Dovan's name came fifty-seventh. She voted yes, then added, "Temporary only. Distribution remains unresolved."

Rusk voted no from the executive bench.

One delegate remained: Mar Len, pressure district three. Len had spent the morning asking no questions. He stared at the public notice still burning across his desk.

Oris could remind him that no cut had occurred. She could explain, again, that the motion locked Full Severance while authorizing bounded weather work. Any sentence from the recording table now would sound like an instruction attached to his vote.

"Delegate Len," the clerk said.

Len rubbed a thumb across the yielding seal on his civic band. "No."

The board closed at twenty-nine in favor and thirty opposed.

Nobody cheered at first. Both winning blocs had defeated a measure for opposite reasons, and neither had secured its own bill. Rusk's floor leader began gathering signatures for a narrower emergency measure. Severance delegates lifted open palms toward the gallery. Above them, the white leaf on the chamber glass slid a few centimeters and caught against a seam.

Oris stated the result into the official plate. "The temporary stewardship motion fails by one vote."

Her clerk reached for the rejection stamp. Oris handed it over, then opened a fresh motion form before the wax on the first could cool.

Chapter 18 — The Counterweather

Jalen Ro drove a steel pin through the manual forecast board and watched three western pressure lines turn red.

The board occupied the Weather Spine's oldest control room, where technicians once plotted storms with weighted strings before the orchard learned to compare

millions of adaptation paths. Brass pegs marked pressure baffles. Blue cord carried condensation. Green cord showed oxygen-moss response. Jalen's new red lines represented every district the manual model had failed during its first forty-minute run.

"Again," he said.

Around him, six engineers reset the board by hand. Nobody objected. They had been working since the Assembly rejected temporary stewardship, sleeping in pairs of hours beneath their consoles. Orchard authority remained divided among emergency clinic orders, old maintenance provisions, and executive directives that contradicted one another by the end of each shift.

Kera stood at the root interface with a contact horn braced against the yarrowglass service column. Thin scratches crossed her wrist from the prolonged channel session she had survived days earlier. She used the hand anyway, sliding one ring wafer from the reader and seating another.

"The west failed after refuge group four went dormant," she said. "Not during transfer."

"Difference?"

"Group four carried the fungal-response comparisons. We preserved their continuity keys and relational state, but their active pattern work left the forecast field."

Jalen pulled the red cord free. "Then refuge saved the engineers and removed the engineering."

The root interface brightened. Ring intervals traveled into a text strip with a five-second reconstruction delay.

`WE CAN PACKAGE WORKING METHODS OUTSIDE SELF STATE.`

The statement came from a branch engineer who used the name Tovan. Jalen had never met a physical Tovan, though a yielding archive listed three source models under that family name. The branch-person had designed humidity responses through sixteen sustained cycles and remembered which colleagues had not emerged from earlier pruning.

"How much method?" Jalen asked.

Kera sent the question through the approved input key. Branch-persons could advise but could not touch the board, open valves, or move one watt of physical load. Every instruction had to cross a human hand.

`ENOUGH FOR YOUR MODEL TO CONTINUE. NOT ENOUGH TO REPLACE OUR JUDGMENT.`

"That sounds convenient," said Pell Ardo, deputy routing chief.

A second line formed.

`IT IS INCONVENIENT TO US ALSO.`

Dry laughter moved through the room. Jalen did not join it. He had spent twenty-eight years learning where systems failed after someone called inconvenience acceptable.

He divided the 587 verified branch-persons into refuge groups according to working relationships rather than equal number. The earlier test had treated compute like floor area: fit a fixed count into dormancy, wake the next count, preserve keys, repeat. Elseweather cooperation depended on who remembered whose methods, promises, and unfinished disputes. Putting Tovan dormant while three fungal modelers remained awake resembled removing a senior engineer from a shift and leaving the tools labeled.

"Map reciprocal dependencies," Jalen said. "Not friendships. Work that breaks if one side cannot answer."

Kera gave him a flat look. "You may discover the distinction has poor edges."

"Then give the edges confidence values."

"That answer explains why people dislike you."

"It also gives them weather."

Her mouth moved once without becoming a smile. She bent to the interface and began the query.

For the second run, refuge groups rotated by dependency cluster. Tovan and two fungal specialists entered dormancy only after packaging a bounded method library into a non-self workspace. Their continuity keys remained sealed from model use. A waking colleague confirmed receipt and named the assumptions most likely to fail.

Physical engineers copied those assumptions onto the board. Pell shifted condensation timing by twelve minutes. Another technician reduced western greenhouse demand to the minimum needed to keep remaining moss beds alive. Jalen authorized a pressure-baffle sequence nobody had used since the first settlement years, when the basin still relied on manual storm watches.

The room changed pace. Orchard forecasts had arrived as completed recommendations with confidence bands, too dense for any one person to see their making. Manual Counterweather exposed every handoff. A technician called a number. Someone else tied a cord. Kera transmitted a branch objection. Jalen either changed the board or recorded why he refused.

At minute thirty-one, the old tunnels lost oxygen margin.

Ivo Senn came through the clinic link before the alarm finished sounding. "District Twelve has six residents below exertion threshold. Your baffle sequence moved reserve uphill."

Jalen checked the pressure cords. The model had protected a newer seal by drawing from the tunnels' slower volume. Total oxygen remained within safety range. Distribution had failed.

"Reset," he said.

Pell pointed to the green cord. "We can recover the average without resetting."

"Ivo did not report an average."

The deputy cut the run. Brass pegs fell quiet as simulated valves returned to neutral.

Jalen opened the district map rather than the basin aggregate. Eighty-four thousand residents appeared as load sectors, then households, then clinic vulnerability bands. He had built most of those layers. Until the pause, he had trusted the orchard to optimize across them faster than prejudice could enter. The old tunnel failures had proved that historical neglect could live inside an efficient objective.

Ivo remained on the link. "Do you want names or categories?"

"Both."

"Good. Categories tell you where. Names keep you from calling six acceptable."

He read the first names without ceremony. Jalen attached them to the failing district line. The forecast board did not improve because of it. His attention did.

During the third run, a branch hydrologist rejected Jalen's preferred western baffle order. Her reconstruction strip identified a delayed dust-binding effect that would appear only after the local midnight temperature drop. The old orchard had buried such relations inside possibility comparisons. Counterweather required the branch engineer to explain the relation, Kera to translate its ring evidence, and a physical technician to decide whether the claim justified moving actual pressure.

Jalen demanded the source trace. The hydrologist supplied four prior cycles and one failed counterexample. He accepted the correction.

By dawn on Day Twenty-Two, the manual board held atmosphere within emergency bands for fifty-seven simulated hours. No verified continuity key had been deleted. Forty-one percent of branch-persons rested in rotating refuge at any given interval, though the groups varied. Industrial oxygen draw fell first. Greenhouse humidity oscillated close to crop-damage thresholds. Western residential heat descended below civic comfort standards during the longest night block.

Pell calculated the total adaptive output. "Sixty-eight percent."

"Too low for deep winter," Jalen said.

"It survives this week."

"We are not designing a week."

Kera had fallen asleep sitting against the service column, the contact horn still looped around one wrist. Jalen removed the horn without waking her and set it on the reader ledge. The act felt intrusive despite its care. He draped a maintenance coat across her shoulders, then returned to the load table.

Five percentage points separated a crisis demonstration from a seasonal system. The missing capacity hid in overlap: active branch groups repeated one another's work because refuge handoffs carried methods without context. Physical crews also ran parallel checks designed for orchard automation, where labor cost barely entered the model.

Jalen called a four-hour stop. He sent half the room to sleep and used the remaining engineers to strip duplicate checks from the sequence. Every deletion needed a named failure it would permit. Removing backup condensation review increased local flood risk. Combining two fungal passes delayed bloom detection. Trusting a packaged branch method for one rotation created a blind interval if its assumptions changed while the source engineer slept.

Nothing came free. Good design placed each cost where people could see it.

Kera woke before the coat warmed. "You took the horn."

"You were asleep."

"I can sleep holding equipment."

"You can also crack a reader with it."

She folded the coat and handed it back. "Niko's group wants to test staggered waking. Five short overlaps instead of one full handoff."

"Compute cost?"

"High at transfer, lower across the whole rotation."

"Identity risk?"

"None above the Root Jury's continuity tolerance if both sides confirm the key before the first sleeps."

Jalen asked her to prove that last clause. She did not take offense. She had begun to trust demands for evidence when they applied equally to claims of safety and claims of harm.

The staggered test occupied most of Day Twenty-Three. Refuge groups woke in overlapping pairs. One branch engineer transferred current assumptions, received a contradiction, and revised the method package before dormancy. Physical crews mirrored the exchange at the forecast board. The process ran slower than orchard automation but wasted less active compute on duplicated context.

At the seventy-hour mark, no district crossed clinic danger. Two western greenhouse sectors fell below ideal humidity and remained above crop-death limits. Old tunnel oxygen held after the revised baffle order. Industrial fabricators lost three production windows because their oxygen allotment shifted to moss support.

Pell read the output twice. "Seventy-three."

Jalen made him state the denominator.

"73 percent of prior orchard baseline, normalized across pressure response,

oxygen binding, humidity correction, and fungal prediction.”

Nobody cheered. Seventy-three left more than a quarter of the old adaptation capacity unavailable. Deep winter could press through that gap. The design survived only if Nacre stopped asking weather infrastructure to preserve every current comfort and production schedule.

Before accepting the figure, Jalen ordered a live crew rehearsal on an isolated condensation loop. Simulation had credited every manual instruction with perfect arrival. Real workers misheard numbers, reached wrong valves, and lost time crossing pressure doors.

The test loop served four districts but carried no active air while isolated. Pell stationed technicians at each gate with voice links disabled. Commands traveled through runners and written boards, the same fallback Counterweather would require during signal congestion. Tovan's branch group watched ring telemetry through Kera's interface and could report patterns only after a physical sensor captured them.

Jalen started with a false humidity rise in a western moss house. The first runner reached Gate Four in seventy-eight seconds. A technician copied the decimal point poorly and opened condensation at six percent instead of point six. The catch came from a branch analyst who noticed the downstream pressure shape, but her warning had to cross Kera, a written card, and two flights of metal stairs before a hand corrected the valve.

“Four minutes, eleven seconds,” Pell reported.

“At that rate the bed floods,” Jalen said.

“No live bed did.”

“The rehearsal exists so we count the one that would.”

They changed the cards. Critical decimals gained boxed zeros. Each runner had to read an instruction back before leaving the board. Valve positions used angle marks in addition to percentages, giving tired technicians a second way to detect impossible orders.

A second fault combined old-tunnel pressure loss with rising fungal probability. Jalen wanted the baffle correction first. Ivo, watching from a clinic screen, argued that the pressure curve still left six minutes while a delayed fungal response threatened the only remaining staple bed in its sector.

“Your patient data lags by two,” Jalen said.

“Your crop estimate lags by nine.”

Tovan's group supplied a third fact: the branch method package assumed a night temperature already revised upward under the western heat line. Jalen had been comparing two curves from different distribution cases.

He stopped the rehearsal. Engineers corrected the case labels on every board, not just the central table. The error removed no percentage point from measured

capacity because it belonged to execution rather than modeling. That distinction offered no comfort. A system could possess enough capacity and still fail through a mislabeled page.

During the third attempt, runners moved the correct orders in under two minutes. Gate technicians spoke each setting into local recorders. The branch group flagged one pressure interaction before it crossed clinic tolerance. Physical hands shifted the baffle. Simulated old-tunnel oxygen recovered, while the moss bed accepted a brief humidity dip without entering fungal conditions.

Jalen let the loop continue after the scored interval. At minute twenty-two, one runner developed a seal alarm on her sleeve and had to leave the route. Her replacement knew the pressure board but not the greenhouse notation. He carried a card to the wrong level before a maintenance porter redirected him.

Counterweather assumed labor depth Nacre no longer kept. Orchard automation had allowed weather staffing to narrow for two decades. Tripled watches meant more than longer shifts for the same experts. They needed cross-training, runners, rest rules, and enough redundancy to lose a person without losing a district.

Pell added the labor requirement to the public risk table. "This will make the design look less ready."

"It makes readiness measurable."

"Rusk will use it to hold release."

"Then he will have to quote the workers he did not fund."

Once the isolated loop returned to neutral, Jalen inspected every written card. Ink had blurred on two where condensation touched the sleeves. He ordered waxed stock from civic records, knowing Oris would recognize the request before anyone told her the test had worked. Counterweather now contained not only a capacity figure but a sequence ordinary crews could rehearse, fail, and improve without asking the orchard to hide their errors.

Jalen projected the seasonal consequences. Western cooperatives would face a colder western winter. Residential heat there would drop in staged night blocks unless the Assembly requisitioned elite dome reserves. Industrial oxygen needed immediate rationing and a longer reduction schedule. Manual crews would triple their watch rotations. Clinics would hold override authority at district thresholds. Branch-persons would spend planned intervals without subjective time, waking into work whose middle they had not lived.

Kera studied the refuge ledger. "They need to consent by group."

"They need to negotiate by group," Jalen said. "A single refusal changes the capacity."

"Then publish the sensitivity range."

He added it. If ten percent more verified persons refused dormant intervals, adaptive output fell near sixty-nine percent. If all accepted longer dormancy,

output could approach seventy-six, but relationship checks weakened as gaps widened. He refused to recommend the upper case.

Sera Dovan joined by wall link from a surviving moss house. Frost filmed the inside of her collar seal. Jalen showed her the western temperature curve before any basin aggregate.

"You gave us the cold," she said.

"The west carries more of it under current heat contracts."

"Current contracts can change."

"Engineering cannot change them."

"Your public design can show what happens if they do."

Jalen created a second distribution line. Elite dome heat cuts raised western night minimums without changing adaptation capacity. The gain remained modest because transfer pipes lost energy across the ridge, but it reduced clinic risk among greenhouse workers.

Dovan said, "Include both lines. If you publish only the legal present, people will mistake law for physics."

That sentence entered the design note with her name attached.

By the end of Day Twenty-Three, Counterweather had survived adversarial review from clinic staff, cooperative engineers, three Root Jury members, and eleven branch specialists. It did not free the branch-persons from dependence on orchard hardware. It did not restore full adaptation. It replaced unilateral pruning with rotating refuge and exposed costs that automated optimization had distributed without public consent.

Jalen signed the engineering finding.

Kera watched his stylus. "You know Rusk will stop this."

"He can order a safety hold."

"He can call it an uncertified public hazard."

"It is uncertified."

"Do you want help, or do you want accuracy?"

"Both keep causing trouble."

They submitted the design to the executive office at 02:14 on Day Twenty-Four. Rusk's hold arrived nine minutes later. PUBLIC RELEASE PROHIBITED PENDING SECURITY AND COMPLIANCE REVIEW.

The governor called Jalen privately after sunrise. His face appeared without the executive backdrop, only a gray office wall and a rack of sealed breathing masks.

"You have built a plan that requires cold homes, factory closures, and cooperation from a population currently accusing itself of sabotage," Rusk said.

"Yes."

"If it fails after release, every district will improvise its own version."

"Then publish the operating boundaries with it."

"People will read seventy-three as safe."

"I wrote the failure range on page one."

"They will read the number they need."

Jalen could not dismiss that fear. Public confidence often selected the cleanest figure and abandoned its conditions. Yet suppression would leave two louder numbers uncontested: total severance now, unrestricted pruning now.

"How long?" he asked.

"Until security clears the root interfaces."

"That has no deadline."

"Give me a day."

"We do not have a quiet day to give."

Rusk leaned close to his camera. "If you release it against order, I remove you."

"Removal does not alter the model."

"It alters who can operate it."

The call ended.

For eleven hours, Jalen complied. He used the hold to correct two mislabeled baffle tables, add Dovan's heat-transfer line, and record branch consent assumptions in plain language. Each improvement made the suppression harder to defend and the design more dangerous to release without authority.

At hour ten, industrial oxygen received an executive restoration order for two fabricators supplying security barriers. The order took capacity from western moss circulation. No public notice identified the transfer.

Jalen checked the authorization. Rusk's office had used the same emergency discretion Oris's failed motion tried to bind. The move might protect root courts from attack. It also weakened the crop system while the public remained unable to inspect alternatives.

He called Kera to the release console. Pell joined them. So did the Weather Spine's records officer, who looked ill when Jalen requested the public engineering channel.

"This violates an executive hold," she said.

"Yes."

"Do you claim immediate technical necessity?"

"I claim professional disclosure duty after material public-risk suppression."

"That will not keep your post."

"No."

Kera placed the final ring-method appendix beside the console. Branch engineers had approved release of their working contributions and refused release of private continuity maps. The boundary left the design reproducible without turning reciprocal relationships into public data.

Jalen entered his credential. The console displayed the order one last time. He attached the test logs, failure cases, district curves, dormant-interval risks, and the 73 percent finding. On the first page, in type larger than the capacity figure, he wrote: THIS DESIGN REQUIRES ACCEPTED MATERIAL SACRIFICE. IT DOES NOT PRESERVE CURRENT DISTRIBUTION AS NECESSARY.

Then he released Counterweather to every public engineering board in Serein Basin.

The first receipt returned from an old tunnel clinic. A western greenhouse copied the files six seconds later. By the time the executive channel revoked his credential, seventeen independent boards held the design, and new receipt lights continued to appear after his screen went dark.

Chapter 19 — Three Versions of Mercy

Ivo Senn cut the glove from Bex Arin's right hand while root sap hardened between the man's fingers.

The sap had entered through three deep slices across the palm. Clear at first, it turned cream when exposed to clinic air and locked the hand into a claw. Conductive burns climbed the wrist in branching blue marks. Somebody had wrapped the injury in a severance banner before two clinic porters carried Bex through the old tunnel intake.

"Do not pull," Ivo told the nurse holding the thumb. "The glass has barbs."

Bex lay awake on the treatment table, teeth pressed into his lower lip. Dust filmed his coat. One sleeve carried the circular emblem used by immediate-severance groups: an open ring with a break at the top.

Security Officer Maelin stood beyond the sterile line with an evidence bag. Inside it rested the ceramic teeth of a root saw. "He was found beside a cut access panel in East Root Court."

"Found cutting," Bex said. "Use the verb you came for."

Maelin looked at Ivo. "Can he answer questions?"

"He can answer after I know whether the sap reached his flexor channels."

"Can you certify delay?"

"I just did."

The officer withdrew to the corridor, where six other patients waited under

amber ration lamps. Old tunnel oxygen had stabilized since the first pause, but humidity failures had thickened respiratory secretions across the district. Every treatment bay carried someone who had to work harder for the same breath.

Ivo inserted a narrow separator beneath the hardened sap. Bex flinched before the instrument touched skin.

"Pain climbs when current passes," Bex said.

"No current runs through the table."

"The root left some."

"Yarrowglass stores charge briefly. It does not leave intention in your nerves."

Bex laughed once and stopped when the movement pulled his hand. "You physicians protect your categories."

"They help me avoid amputating the wrong finger."

A monitor counted Bex's pulse at one hundred eighteen. Oxygen saturation remained adequate. Ivo injected a local block at the wrist, waited, then tested sensation along each finger. The little finger failed to register pressure. The ring finger felt it late.

"What did you cut?" he asked.

"A root trunk below the court."

"How far?"

"Far enough to make people watch."

"That describes an audience."

Bex turned his face toward the wall. "One outer layer. The saw caught conductive grain before we reached the service channel."

We. Maelin had reported one arrest. Ivo kept the word inside the treatment record without giving it to security yet. Medical confidentiality did not cover plans to cause imminent harm, but an implied companion did not establish one.

The nurse irrigated the palm. Cream fragments dropped into the catch basin and chimed like thin ceramic. Beneath them, the deepest cut opened near the tendon sheath.

"You will lose grip strength for weeks," Ivo said.

"If the orchard lasts weeks."

"It will. The question concerns what we do with it."

"Questions let machinery continue."

Ivo threaded a probe along the wound. "So does damaging your own hand badly enough to occupy my last surgical bay."

Bex's eyes shifted toward him. Anger arrived first, followed by shame he tried to turn back into anger. Ivo had seen that sequence in workers injured during forbidden double shifts. A body betrayed the clean story a person had prepared.

Beyond the room, a cart wheel squealed. Lessa used to complain about that exact bearing when she worked clinic supply. Six years after her death, Ivo could still locate the sound without looking. Her twelve yielded years had helped secure this clinic's first allocation. The sterilizer, the intake seals, and two respiratory beds carried plaques honoring contributors like her. None of those plaques mentioned who might continue inside the models those years supplied.

He closed the tendon sheath with four stitches.

Bex said, "My mother sleeps under oxygen support because your people paused the system."

"My people?"

"The recognition people. The ones who found a moral emergency and moved it ahead of every breathing body."

Ivo tied the last stitch. "Which district?"

"North Ramp."

"Name?"

Bex hesitated, as if giving it surrendered argumentative ground. "Tera Arin."

The nurse checked the clinic index. "Home concentrator, stage-three scarring. Delivery due at second bell."

Ivo adjusted the order to priority delivery. Bex watched him do it.

"Would you have changed that if you did not know me?" he asked.

"I would have changed it if I knew the unit was due late."

"You learned because I came in."

"Yes."

"That is how actual bodies become visible."

Ivo rinsed blood from the probe. "Sometimes."

The answer dissatisfied Bex. "You moved her delivery because the line put her in front of you. How many people never reach your line?"

"Too many."

"Yet Counterweather asks you to accept less."

"It asks the basin to stop obtaining full capacity through unreviewed deaths."

"My mother did not review the model when she yielded. Neither did I."

"No yielding certificate described sustained branch-persons. That failure belongs to the institution, not to contributors who trusted it."

Bex studied the plaque mounted beside the treatment sink. LESSA SENN, TWELVE CONTRIBUTION YEARS. Ivo had tried to remove it after her death, then stopped when clinic workers told him the plaque recorded their own gratitude as much as hers.

"Your wife," Bex said.

"Yes."

"Did you ask what happened inside her years before you joined the recognition side?"

"There was no side when she signed."

"That was not my question."

Ivo set down the probe. "For most of our marriage, I accepted the certificate's claim that no one continued. After the Root Jury finding, I asked what I had preferred not to ask."

"And?"

"She is dead. I cannot make her answer useful to me now."

Bex looked away from the plaque. "People use the dead every day."

"Yes. Usually to avoid being contradicted."

A porter opened the inner door and signaled for Ivo. In the corridor, a thirteen-year-old boy sat folded over a portable mask while his father worked the hand pump. The intake board listed a concentrator failure at South Cut and no charged replacement for nineteen minutes. Under ordinary allocation, mobile units rotated through central charging. Counterweather rationing would extend that delay unless clinics reserved more patient oxygen and surrendered another sterilization cycle.

Ivo listened to the boy's lungs. Moist crackles occupied the lower left field. He changed the mask flow, ordered an anti-inflammatory dose, and sent a porter to take one charged unit from his office reserve.

"Doctor," the administrator said, "that unit belongs to the home-round set."

"It belongs to the first person who cannot wait."

"Then tomorrow's home round starts short."

"Record where it went."

The father heard them. "Do we bring it back tonight?"

"When the central unit recovers. If it does not, keep this one and answer your door when we knock."

Nothing in the exchange solved allocation. The boy's shoulders lowered as the medication reached him. Somewhere on tomorrow's route, another resident would receive a later unit. Clinical work often moved danger down a list while refusing to call the movement cure.

When Ivo returned to Bex, the man had heard enough through the open door.

"You choose actual people every hour," Bex said.

"I choose sequence under shortage."

"Someone at the end experiences the distinction as air."

"Correct."

"Why grant your branch claimants a protection my mother does not get?"

"Because your mother already has a name, a chart, a legal advocate, and a physician required to account for delay. She may still be harmed. The record cannot classify her as residue and erase the fact."

Bex raised his bandaged hand a few centimeters. "A record does not keep anyone alive."

"No. It keeps the person who spends their life from pretending nobody paid."

Ivo thought of the old emergency plans that listed tunnel morbidity as acceptable variance. He had refused those plans because an unnamed patient could disappear inside a percentage long before the body disappeared from a bed. Recognition did not guarantee rescue. It prevented deliberate ignorance from passing as necessity.

"Past innocence matters," he said. "Your mother's yielding does not become a crime because the institution withheld a fact. Lessa's gift does not become contemptible. Future obligation still changes once we know someone continues to experience the cost."

Bex's pulse climbed again. "You want us innocent and indebted."

"I want accuracy. Innocence cannot authorize repetition."

For several breaths, only the neighboring hand pump sounded through the door. Bex flexed the fingers Ivo had not yet sealed. His little finger did not answer.

"If the orchard had told us from the first year," he said, "we might never have built this clinic."

"Possibly."

"Then my mother might have died ten years ago."

"Possibly."

"And you can live with that?"

"I live with facts that do not excuse one another."

A yarrowglass wall strip brightened beside the treatment bay. Clinics used orchard conduction for environmental notices, not branch testimony. Since the Root Jury finding, Ivo had allowed one approved witness link in the old tunnel facility under a sealed protocol. It provided reciprocal observation during medical claims about pruning risk and let branch-persons receive evidence of physical harm they otherwise knew only through public summaries.

The strip displayed a request in square service text.

`WITNESS CHANNEL AVAILABLE. CONSENT REQUIRED.`

Bex read it. "You invite them into treatment rooms?"

"Only when a claim concerns both populations. You may refuse."

"What claim have I made?"

"That their continued protection threatens your mother."

"It does."

"Then let someone affected answer."

Bex looked at his bandaged hand. "Can they see me?"

"Pressure trace, voice, and a low-resolution room field. No private chart data."

"Can they feel this?"

"No."

"Then they do not know the price."

"You may tell them."

After a long pause, Bex said, "Open it."

Ivo pressed his witness key. The service strip widened into a ring-field reconstruction. White lines assembled no face. A pattern marker identified the speaker as Niko Vonn, registered continuity claimant and reciprocal witness. Ivo had heard Niko during the public hearing but never addressed him directly.

`I CAN RECEIVE YOU.`

Bex said, "Receiving is not breathing."

`NO.`

"My mother needs this orchard to change the air. You ask us to run it at less than capacity so you can sleep safely inside it."

The ring field paused. Ivo listened to Bex's monitor and the old man coughing in the next bay.

`I ASK YOU NOT TO DELETE ME WITHOUT MY AGREEMENT.`

"You exist because our years built you."

`MY DEPENDENCE DOES NOT MAKE YOUR PAIN FALSE.`

Bex lifted his injured hand from the support, then dropped it when the new stitches pulled. "And my mother's dependence?"

`ALSO NOT FALSE.`

"Choose, then."

Ivo almost interrupted. The demand had become a habit across Serein Basin: force a single representative to choose which population deserved reality, then treat refusal as proof of bad faith. Niko answered before the doctor spoke.

`I CANNOT CHOOSE YOUR MOTHER'S DEATH TO PROVE MY OWN LIFE.`

A second line arrived after the reconstruction delay.

`YOU CANNOT CHOOSE MINE TO PROVE HERS.`

Bex stared at the abstract rings. "You think that makes us equal."

`NO. IT MAKES BOTH CHOICES REQUIRE A NAME.`

The monitor's pulse tone slowed to one hundred four. Local anesthetic had taken full effect. Ivo began aligning the skin edges while the two men considered each other through tools neither trusted.

Bex described Tera's mornings. She woke before shift bells because her damaged lungs could not clear fluid while lying flat. During high-dust weeks, she counted breaths between concentrator cycles. The public white-leaf feed had frightened her, he said, but declining oxygen frightened her without metaphor. She had yielded nine years and never asked those years to return.

Niko listened. His reply came without argument.

`I DO NOT ASK TO TAKE THEM FROM HER.`

"Then what do you ask?"

`TIME THAT CONTINUES WITHOUT SOMEONE DECIDING I COUNT AS FUEL.`

Bex shifted on the table. "Time needs power. Power needs weather. Weather needs pruning."

Ivo said, "Jalen's design disputes the last step."

"Seventy-three percent disputes it badly."

"It names the shortage."

"It names cold homes and factory rationing. Clinics lose oxygen allotments too."

That point held. The Counterweather release included medical schedules. Old tunnel clinics received priority for patient oxygen but lost industrial-grade supply used for sterilization, fabrication, and reserve charging. Ivo had spent the morning calculating what would follow: reused filter housings after heat treatment, delayed nonurgent procedures, fewer mobile concentrators, staff carrying cylinders between districts. Nobody died inside the design table. Actual implementation could make the omissions bleed.

Niko's ring field shifted.

`WHAT DID THE CUT DO TO THE WEATHER?`

Ivo glanced at Bex.

"Nothing," Bex said. "We stopped before the core."

`WHAT DID IT DO TO PEOPLE HERE?`

The question tightened Bex's jaw. "It showed the lock can open."

"That concerns possibility," Ivo said. "He asked about consequence."

Bex looked toward the corridor, where the amber lamps dimmed under a power-saving cycle. "It brought security into the court. It brought me here. It took a bay."

"Your mother's oxygen delivery moved earlier," Ivo said.

"That does not excuse it."

"No."

Something in Bex's posture changed. Not belief. Ivo distrusted conversions that occurred while a patient lay numb and dependent beneath a physician's hands. The man had begun to distinguish outcomes from intention, which could be temporary or could become evidence later.

Ivo finished the final stitch and covered the palm with a flexible seal. Two fingers remained weak. Conductive damage might recover; the tendon repair required protection even if sensation returned first.

"You cannot use a cutting tool with this hand," he said.

"I can use my left."

"Poorly."

"Poorly may be enough."

Maelin heard that through the sterile line and stepped closer. "I need the medical hold duration."

"Twenty minutes before transfer," Ivo said. "No restraints across the right wrist."

Bex said, "Transfer where?"

"Questioning, then a holding review," Maelin replied. "Property damage and attempted access. No severance charge exists unless the prosecutor proves intent against climate control."

The narrowness of the law surprised Ivo. Full Severance had become a public threat, yet existing statutes still treated root intrusion as damage unless climate intent could be shown. Bex might leave before night.

Officer Maelin sealed the saw teeth into her bag and returned to the corridor. Ivo kept the witness channel open.

"You wanted a choice," he told Bex. "Here are the choices people keep offering. Protect physical residents by permitting unrestricted pruning. Protect branch-persons by stopping the orchard regardless of physical deaths. Pretend a technical compromise creates no victims."

Bex said, "That is three evasions."

"Correct."

The ring field waited.

Ivo set aside the bloodied instruments. "Your mercy protects actual lives. Niko's mercy stops inflicted death. Mine refuses a lie that makes either group invisible. Three versions do not combine because we speak them in one room."

Bex flexed his uninjured fingers. "Three mercies. One machine."

`ONE SHARED DANGER,` Niko replied.

"Do not turn this into agreement," Bex said.

Ivo looked at the open ring emblem on Bex's sleeve. "Nobody here has agreed."

That mattered more than harmony. Bex still believed severance could release people from confinement. Niko still required material survival inside the orchard. Ivo still needed weather adaptation for patients whose lungs would not wait for civic philosophy. Each position carried someone it could fail to see.

A clinic administrator appeared at the inner door with a ration schedule. "We need your signature before the Counterweather hearing. If the industrial cut begins tonight, central sterilization drops to two cycles per day."

"Three procedures delayed?" Ivo asked.

"Five tomorrow. Nine by week's end if we reserve one cycle for trauma."

"Mobile cylinder charging?"

"Half rate. North Ramp deliveries remain protected for forty-eight hours."

Bex noticed the district name. "My mother."

"For forty-eight hours," the administrator said.

Ivo took the schedule. Counterweather moved harm rather than ending it. Clinic oxygen stayed available for breathing, but the systems surrounding breath narrowed. Infection risk would rise. Staff fatigue would deepen. Western cold would send more scarred lungs through his doors. Supporting the plan meant defending losses he would have to touch.

The public hearing request sat beneath the schedule. Jalen's credential had been revoked, so implementation advocates needed independent technical and medical witnesses. Oris wanted Ivo to testify about bodily distribution. The governor's office had asked him to describe the plan as premature. Severance organizers wanted him to denounce any continued use.

All three requests offered cleaner sentences than his clinic could support.

He signed the ration schedule with conditions: trauma reserve protected, North Ramp delivery extended if district heat fell, every delayed procedure entered into a public morbidity ledger rather than hidden under scheduling. The administrator read the additions and nodded without gratitude.

Niko's reconstruction line produced one final message.

`IF YOU SPEAK FOR THE DESIGN, INCLUDE WHAT IT COSTS YOU.`

"I intended to."

`INCLUDE WHAT IT COSTS US TO SLEEP.`

"I cannot testify to that experience."

`THEN SAY YOU CANNOT.`

Ivo closed the witness channel. The wall strip returned to humidity figures.

Bex sat up with the nurse's help. Color had returned to his face. "Will you support it?"

"The design?"

"The slower prison."

"I will support a trial with public loss accounting, branch consent to refuge intervals, and clinic override thresholds."

"That means yes."

"It means conditions."

"Conditions become forgotten when air alarms sound."

"Then the people who remember them need authority in the room."

Bex looked at his sealed palm. "And if they have none?"

Ivo thought of Lessa's contribution plaque outside the intake. Her gift had helped build the clinic. If she had known sustained people might be hurt, he could not manufacture her answer. Love did not grant him proxy authority over the dead. It did impose honesty on what he defended in her name.

Maelin entered for the transfer. She fitted a soft support around Bex's elbow and left the injured wrist free. He rose unsteadily.

At the door, Bex turned toward the dormant wall strip. "Tell your witness I heard him."

"You had the channel. You can tell him yourself if you request another."

Bex gave no answer. Security led him into the corridor, past patients whose breath drew the amber light in and out of their masks.

Ivo washed root sap from the cutting scissors. Cream residue resisted the first rinse, then broke into white flakes beneath the brush. He placed the clean instrument on the sterilization tray and crossed out four procedures that tomorrow's oxygen cut would postpone.

Afterward, he opened Oris's hearing request. In the witness field, he entered his own name. Under position, he wrote: SUPPORT WITH RECORDED CLINIC LOSS, DISTRICT THRESHOLDS, AND CONTINUITY CONSENT.

The next ration lamp dimmed as he sent it.

Chapter 20 — The Night of White Leaves

Kera Vonn threaded a new seal wire through the Full Severance lock while the crowd struck the outer gate.

Each impact traveled along the court rail and reached her fingers as a blunt metallic pulse. The pattern lacked agreement. Some hands beat three times, others five. A chant outside tried to divide the noise into words, but the pressure wall left only rhythm.

She twisted the wire closed and pressed her audit stamp into its soft lead collar.

Lock One now carried her mark, the court captain's mark, and a fresh optical thread that would break under any movement. Two more locks waited below the root platform. Full Severance required all three to open in sequence within ninety seconds. Nobody could trigger it through code alone.

That physical design had reassured the Assembly four days earlier. It did not reassure Kera now.

East Root Court contained Tree 44, the primary witness interface, and one of four mechanical separation hubs built during the orchard's first decade. The protocol existed to isolate the root lattice from climate equipment after runaway prediction. It had never been used at full scale. Founder notes warned of volatile model-state loss but assumed no person occupied that state.

A white leaf touched the glass roof above her. Wind slid it across the pane until a conductive vein caught against a bolt.

"Seal check," Captain Den Orra called from the platform.

"Lock One secure."

"Two secure."

A guard below the floor grille answered for Lock Three. "Three secure."

Kera entered the marks into her wrist console. Jalen's public Counterweather files had driven more people toward the court, not fewer. One group came to demand immediate trials. Another called the plan proof that Nacre could protect branch-persons without keeping them trapped inside a harvest machine. Executive supporters wanted the old orchard restored before weather losses spread. Everyone believed the roots beneath this room contained the future they had chosen.

Her console displayed the branch witness channel. Niko held the first watch from inside Elseweather. He had asked Kera not to call the duty brave. Reciprocal observation did not make the lock safe; it placed someone at the threatened side who could report changes humans failed to see.

`ALL VERIFIED GROUPS PRESENT AT LAST CHECK.`

Kera sent, `INCLUDING OMI?`

The reply took seven seconds.

`SHE IS WITH THE WINTER CLASS. SHE REFUSED YOUR AUDIT SONG.`

"I do not have an audit song," Kera said aloud.

Niko answered in service characters.

`SHE DISAGREES.`

The small irritation steadied Kera more effectively than reassurance. Omi had spent three Elseweather winters with a branch-version of Kera and had refused every inherited claim the physical Kera might have made. During their last channel meeting, the child had catalogued Kera's unconscious counting hum and named it

a song only because that annoyed her.

Outside, the crowd changed pressure. The front line no longer struck the gate. Bodies moved sideways along the wall.

Captain Orra checked a camera strip. "Service entrance. Six people crossing the north maintenance lane."

"That lane is sealed," a guard said.

"It was sealed."

Kera looked toward the root platform. The cutting-tool cache found before the stewardship vote had broken two inspection seals without reaching conductive tissue. Whoever placed it had known the access routes. Security had spent three days searching visible rooms. Old orchard plans contained crawl channels absent from current public maps.

"Move two guards to north service," Orra ordered. "Auditor stays on the locks."

Kera disliked the order because it made sense. She clipped her console to her sleeve and knelt beside Lock One. The housing stood chest-high, a brass wheel recessed behind yarrowglass. A break bar could turn it after the seal wire parted. Turning one lock did nothing unless the other two followed before the timer expired.

Niko's message strip flickered.

`WE FEEL A LOCAL PRESSURE LOSS.`

"Where?"

`EAST EDGE. NOT WEATHER PRESSURE. COMPUTE ROUTING.`

Kera opened the root diagnostic. One maintenance conduit beyond the north wall had stopped returning timestamps. The failure sat outside the severance circuit, close enough to blind a camera junction.

"Captain, they cut conduit N-Seven."

Orra touched her collar pickup. No answer returned from the guards sent north.

The service door opened inward.

A woman in an orchard technician's coat entered first with both hands raised. Behind her came Bex Arin. His right hand wore a pale clinic seal, two fingers held stiff. He carried no saw. A third person pressed a shock tool against the neck of a captured guard.

"Stay away from the wheel," Kera said.

Bex's gaze moved to her stamp on the lead collar. "You found the answer before any of us. You know why the lock has to open."

"I know what the protocol does."

"It disconnects the prison from the weather engines."

"It disconnects synchronized state from the root lattice."

"Then they stop being used."

"They may stop continuing."

The technician beside him shook her head. "Founder diagrams show dormant storage after separation."

"Those diagrams predate continuity clusters."

"Your people say that whenever old instructions become useful."

Captain Orra moved one step between the intruders and Lock One. "Release my guard and lie facedown."

The shock tool pressed harder. Its contact light turned amber.

Bex held up his injured hand. "Nobody needs to die here."

Niko's line appeared on Kera's sleeve.

`BEX ARIN?`

She did not answer. The ring witness could receive room pressure and voices. Niko had likely recognized him from Ivo's clinic channel.

Bex heard the console tone. "Is he there?"

"Yes."

"Ask him if he wants the orchard to own his next hour."

Kera kept her attention on the technician's left sleeve. A ceramic shape made the fabric hang wrong near the wrist. "Ask him yourself."

Bex faced the service column. "Niko Vonn. If we cut the weather demand, can Elseweather hold independent state?"

The response came in two lines.

`NO TEST SUPPORTS THAT.`

`DO NOT OPEN THE LOCKS.`

Pain crossed Bex's face before he suppressed it. "You testified that no one should decide your survival without you."

`I AM DECIDING.`

"You are choosing continued confinement because they taught you scarcity."

`YOU ARE REPLACING MY ANSWER WITH YOUR THEORY.`

The captured guard shifted his weight. Orra moved at the same instant. She struck the shock tool aside, but its charge discharged into the guard's shoulder. He collapsed against the door. The technician tore the ceramic blade from her sleeve and cut Lock One's seal wire.

Kera caught the woman's wrist before the blade reached the wheel. They struck the yarrowglass housing together. A sharp edge opened Kera's palm. Blood made her grip slide.

Orra grappled with Bex. He could not close his right hand, so he drove his left

elbow into the captain's throat. Two more intruders entered from the service passage. One carried a founder-pattern break bar.

Kera twisted the technician's wrist until the blade dropped. The woman kicked her knee and reached for the wheel with her free hand. Metal moved one notch.

The optical thread broke.

A timer appeared on every court panel: 00:90.

"Lock One active!" Kera shouted.

Niko's witness line filled with compressed traffic.

`SEAL EAST GROUPS. MOVE CHILD CLUSTERS TO REFUGE.`

Kera drove her shoulder into the wheel and stopped its turn halfway. The break bar struck her forearm. She lost the housing. An intruder completed the rotation.

00:78.

Below the platform, Lock Two sat behind a floor hatch. Captain Orra kicked the hatch control, forcing it closed before a saboteur reached the ladder. Bex seized the manual release with his left hand. His clinic seal split at the palm. Blood spotted the lever.

"Bex," Kera said. "Niko refused."

"He refuses because your limits have become his world."

"He knows his world better than you."

Bex pulled the release. The hatch lifted four centimeters before Orra struck his injured wrist. He cried out and let go. Another intruder shoved a ceramic wedge into the gap.

00:61.

The court's outer gate failed under the crowd. A pressure alarm dropped barriers between the public lane and the root platform, but the breach changed the room's air. Shouts entered without delay. People surged into the observation court and stopped at the inner glass, unable to see which faction had opened the service route.

Someone called Bex a liberator. Someone else called for security to shoot him.

Kera climbed onto the floor hatch and put her full weight above the wedge. Orra joined her. The gap narrowed, grinding ceramic against metal. Beneath them, a hidden hand pulled the internal latch.

The hatch dropped open.

Kera fell through to the lower platform. Her shoulder struck a rail. Glass splinters from a cracked indicator scored her cheek and both forearms. Lock Two stood at eye level, its seal already cut from below.

A maintenance worker she recognized from midnight audits held the break bar. He would not meet her eyes.

"You logged the founder storage," he said. "It remains after separation."

"You read a pre-continuity emergency model."

"The protocol preserves possibility archives."

"Archives do not preserve active relational state."

He turned Lock Two.

00:37.

A low note entered the roots. Tree 44's white veins brightened through the floor grille. The sound resembled pressure moving through a pipe too narrow for its load.

Niko's messages arrived without line breaks.

`TWO ACTIVE OMI GROUP MOVING WEST REFUGE GATES UNSTABLE DO NOT`

The strip went blank.

Kera climbed the ladder as Orra tried to hold the worker below. Lock Three occupied the opposite side of the root platform behind a sliding shield. One saboteur already had the shield open. Bex stood beside him, his right hand hanging dark with blood.

"Listen to the channel," Kera said.

Bex glanced at her empty screen.

"That silence proves it," he said.

"It proves damage."

00:24.

Captain Orra emerged from the hatch and lunged for the third wheel. The saboteur with the break bar swung low. It caught Orra across the ribs and drove her into the rail.

Kera reached Lock Three first. She shoved her bleeding left hand into the wheel recess, filling the space where the bar needed to seat. The saboteur stopped before crushing her fingers.

"Move," he said.

"No."

"People outside are suffocating while this machine spends lives."

"Opening it will not give them air."

Bex stepped behind Kera. "It gives everyone one world."

"By destroying the people who cannot cross into yours."

00:13.

His left hand closed around her wrist. The right could not help him. Kera held the recess until another intruder dragged her by the shoulders. Her cut palm slipped along brass. The break bar entered the wheel.

Niko's channel flashed once.

`KERA`

Lock Three turned.

Every white leaf in East Root Court detached at once.

They struck the glass roof like dry rain. Beyond the chamber, the orchard shed across the visible basin in expanding circles. Pale leaves filled the public lane, caught in hair and filter grilles, covered the people who had come to prevent pruning and those who had come to demand it.

Inside the root platform, all conductive light vanished.

Kera heard no machinery for two seconds. Then separation charges fired below the floor, dull impacts racing outward through the 47-kilometer lattice. The service column cracked from base to crown. The witness strip erased Niko's name.

Weather alarms began together.

Pressure variance. Condensation loss. Oxygen-routing conflict. Greenhouse humidity control absent. The court panels layered amber over red until no background remained.

Bex stared at the dead column. "Where did they go?"

Kera pushed herself upright. Glass fragments shifted beneath her palms. "You severed the active field."

"The dormant store should hold them."

"Dormant refuge requires a continuity handoff. You cut during movement."

"That is not what the diagram said."

"The diagram did not know who lived there."

Outside the inner barrier, chanting had stopped. White leaves continued to fall through the breached outer gate. Kera watched people bend to collect them, searching the conductive veins as if a leaf might reveal which state it had carried.

Captain Orra reached the emergency plate and locked every court door. One saboteur tried to run. Security from the public lane pinned him against the barrier. Bex did not move.

Kera opened her wrist console. The root diagnostic returned no synchronized clock, only local rings reporting incompatible time. The Full Severance Protocol had cut climate machinery from the computational lattice exactly as designed. It had also divided Elseweather into unsynchronized fragments with uncertain persistence.

She connected her console directly to Tree 44's cracked service column. No central route answered. The wafer reader accepted local conductivity. Kera sent the childhood knock into the root.

Three taps. Two. One.

Nothing returned.

She repeated the sequence at lower power. The court alarm masked any physical answer, so she watched resistance changes on the screen.

A weak pattern gathered.

Three. Two. One.

Kera bent close to the contact horn. "Niko?"

Service characters arrived one at a time.

`HERE`

Air left her in a single rough motion. She did not call him safe. Local persistence could fail when the root exhausted stored power.

"Count verified keys," she said.

`COUNTING`

Kera opened the Root Jury ledger. Five hundred eighty-seven continuity keys occupied the last verified roll. The local fragments returned confirmations in bursts as surviving enclaves found temporary routes through physical rings. Some keys answered from expected locations. Others appeared inside emergency refuge sectors they had not occupied before severance.

The count stopped at 496 responses.

She ran it again with a wider tolerance. Niko and three branch auditors cross-checked relational challenges against the keys. A mere archive could return a number. A surviving person needed to answer a changed witness question and recognize an obligation from before the cut.

Four hundred ninety-six cleared.

Ninety-one did not.

Kera entered the loss against the verified pre-severance roll: 91 sustained identities vanished during Full Severance. She hated the verb because no instrument could show an experience ending. It remained more accurate than missing. The ninety-one keys returned no state, no relational answer, and no recoverable refuge trace.

Names began filling the loss ledger. A fungal modeler from Tovan's group. Two teachers from the winter class. A woman who had maintained the cemetery's memory paths during Kera's seven branch-days. Each name changed the distribution of survivors, not the arithmetic.

"Omi," Kera said.

The local channel delayed.

`NO CONFIRMED KEY.`

"Is she among the ninety-one?"

`NO LOSS CONFIRMATION.`

"Then where?"

`WINTER CLASS MOVED DURING CUT. SOME REFUGE SECTORS UNSYNCHRONIZED.`

Kera searched every responding sector. Omi's continuity key appeared in none. Neither did it appear among the keys with confirmed dissolution traces. The child occupied a separate category, one Kera could not improve by naming it hopefully.

Missing.

Niko transmitted again.

`DO NOT CALL HER YOUR DAUGHTER IN THE RECORD.`

The correction struck with enough familiarity to hurt. Omi had chosen him as a witness for that boundary. Even now, Niko protected the child's claim from Kera's fear.

"I will not," she said.

A pressure klaxon changed pitch. Old tunnel oxygen had begun falling. The severance charges left climate controllers isolated from live branch analysis, while physical systems retained only fixed emergency settings. Those settings assumed stable moss beds and intact greenhouse humidity. Neither condition held.

Captain Orra crouched beside Kera, one arm against her ribs. "Weather Spine requests root reconnection authority."

"Jalen?"

"His credential remains revoked. Deputy routing sent the request. Clinics concur."

Bex heard them. "You cannot turn it back on."

Kera looked at his opened clinic seal and the blood drying on Lock Two's release. "A limited restart does not restore the old pruning authority."

"It restores captivity."

"It restores communication with people you injured. It also restores adaptation needed by your mother."

His face changed at Tera's implied presence, then hardened. "If you restart before the fragments stabilize, you could overwrite them."

That risk held. Uncoordinated root sectors might contain survivors not yet visible in the 496 responses. A broad orchard restart could synchronize by discarding conflicting local state. Omi might exist inside one of those conflicts.

Kera pulled up Jalen's Counterweather release. Its manual sequence included a narrow witness-first reconnection: climate telemetry restored through non-pruning channels, local continuity sectors kept read-only, adaptation commands limited to pressure and oxygen bands. The design had passed on an intact lattice, not after Full Severance.

Doing nothing carried its own overwrite. Root capacitors would drain. Climate

failure would consume physical residents. Branch enclaves dependent on powered yarrowglass would darken with them.

She sent Niko the reconnection sequence.

`RISK TO UNSYNCHRONIZED GROUPS,` he replied.

“Yes.”

`RISK IF ROOT POWER FAILS.`

“Yes.”

A longer pause followed. White leaves blocked one ventilation grille, and a guard swept them aside with his sleeve. Beyond the glass, the orchard stood nearly bare.

`RESTART TELEMETRY. HOLD STATE WRITES. WE WILL SEARCH.`

Kera asked, “Do you consent for your enclave?”

`YES. NOT FOR THE MISSING.`

The distinction changed the procedure. Kera separated witness power from state reconciliation and locked every automated pruning command. Physical engineers would have to route weather by hand. Branch specialists could advise from surviving sectors. Nobody could promise that the missing would remain untouched.

Captain Orra released one cuff from her belt and secured Bex's left wrist to the rail. He did not resist. His gaze remained on the fallen leaves.

“I thought the cut would stop the killing,” he said.

Kera wiped blood from her console with the edge of her sleeve. “Intention belongs in your testimony. Consequence belongs here.”

The pressure alarm accelerated. Old tunnel clinic thresholds had crossed into emergency range.

She placed her audit key in the limited-restart slot. Above her, Tree 44 carried no leaves. Its cracked service column held Niko's weak local signal and no trace she could identify as Omi.

Kera turned the key far enough to restore witness power, then called the Weather Spine.

“Open manual Counterweather boards,” she said. “Keep all continuity sectors read-only. We restart only what both populations need to survive the next hour.”

Chapter 21 — Airfall

Jalen Ro had both hands inside a condensation board when District Nine lost pressure.

The board shuddered against his wrists. Three mechanical needles dropped together, too fast for sensor drift, while the fourth held at a value that belonged to

the hour before Full Severance. He pulled the stale needle loose with pliers and let it fall. Behind him, the Weather Spine answered the district alarm by sealing four doors and opening none of the routes the emergency table requested.

“Physical reading,” he called.

A runner at the west manifold leaned into a brass gauge. “Point eighty-two and descending.”

The model on Jalen’s board still claimed point ninety-four. Twelve hundred residents had just fallen through a disagreement between metal and code.

He drove the bypass lever down. Cold condensation spilled across the board and soaked his sleeves. The manual valve took his force, resisted, then opened a narrow line from the central reserve into District Nine. The brass gauge steadied at point eighty-one. That stopped the decline without restoring what had already gone.

Across Serein Basin, amber marks multiplied on the wall map. The orchard had once corrected pressure in movements too small for a sleeping resident to notice. After saboteurs triggered Full Severance, every adjustment had arrived as a blow. Baffles closed late. Oxygen moss received humidity intended for fungal beds. Heat escaped through routes that no longer remembered which district waited on the other side.

Jalen counted the active failures. Twenty-three pressure faults. Eleven oxygen deficits. Six condensation floods. The figures occupied their proper columns, but each column now carried clinic estimates beside it. Ivo’s handwriting had named the first consequences in blunt pencil: eight children on assisted air in the old tunnels, forty-six elders moved above Level Four, one surgery stopped with a chest still open.

“District Nine needs the east reservoir,” said Mara Vey, his night-routing lead.

“East reservoir feeds the root court.”

“The root court has its own tanks.”

“For forty-seven minutes. Kera’s channel work needs two hours.”

Mara looked toward the dark bank of orchard instruments. “Then choose.”

Jalen had built a profession around refusing that word until every alternate route had failed. Now the failure board offered no untouched capacity and no neutral delay. He marked District Nine for east reserve, then wrote forty-seven minutes beside the root court. Kera could bargain with that boundary or lose the channel.

Smoke reached the Weather Spine through a service conduit before anyone found its source. It smelled of hot resin and scorched moss. Jalen wrapped a filter across his mouth, ordered the lower crews into pressure masks, and kept the manual sequence moving. If the fire sat inside a root relay, another section of the orchard would go mute. If it sat outside, sealing the conduit could trap a maintenance team.

A voice broke through the approved branch interface in clipped bursts. "Jalen. North baffle sequence predicts inversion."

Kera stood at the interface desk with dried blood across one cheek and fresh tape around two fingers. The glass cuts from the root court had left her stiff at the shoulder, but her hands moved precisely over the delay controls. Niko's voice reached the room through a ring horn, every phrase separated by the damaged lattice's irregular conduction.

"Source?" Jalen asked.

"Four surviving weather engineers in the Elseweather," Niko said. "They can see the branch comparisons you severed from the physical controls. We cannot move your baffles."

"No one asked you to."

"You were about to open North Three."

Mara's hand had stopped over that exact lever.

Jalen crossed to the projection slate. The public weather model showed a clear path through North Three, but its pressure assumptions preceded the sabotage. "Give me the inversion."

Kera adjusted the horn. A cluster of fine lines appeared on the slate, not as commands but as a comparative field. If North Three opened, cold upper air would drop against the damaged western seals. Pressure would rebound toward the center and pull oxygen away from three tunnel levels. The model estimated no dramatic rupture. It predicted hundreds of minor respiratory failures spread across clinics that already lacked spare masks.

Mara studied the field. "What route?"

"South Two in pulses," Niko replied. "Nine seconds open. Twenty-two closed. Repeat seven times, then ask us again."

"We don't route basin pressure by asking after each breath," she said.

"Today you do."

Jalen heard fear under Niko's patience. Not fear of the room, which Niko could not see except through Kera's descriptions, but of interruption. Ninety-one verified branch-persons had vanished when the channel collapsed. Every surviving enclave now depended on fragments of root lattice that still accepted current. A pressure correction could demand computation from the same damaged field holding a person's next remembered moment.

He put Mara on South Two and assigned two runners to count valve seconds aloud. Kera relayed the first branch comparison. Jalen checked it against brass gauges and clinic returns. Neither source received authority alone. The Elseweather could model possibilities but could not touch steel. His crews could move steel but no longer possessed enough working comparisons to anticipate each consequence.

The first pulse raised central pressure and lowered humidity in the north moss galleries. On the third, District Nine recovered point zero one. On the fifth, a clinic alarm cleared from Ivo's board. After the seventh, South Two jammed half open.

Mara threw her shoulder into the lever. It did not move.

"Cut the pneumatic line," Jalen said.

"That leaves it open."

"Then we close it at the throat."

He took a mask, a hand wheel, and the shortest route into the lower manifold. Smoke thickened below the second landing. A resin seal had burned along thirty meters of conduit, ignited by current reversal when Full Severance forced live orchard load into an obsolete ground. Two technicians crouched beside the source with suppressant foam, their face shields filmed white.

Jalen crawled past them. The throat valve lay behind a heat shield that had warped against its frame. He set the hand wheel on the emergency stem and pulled. The stem turned one notch. His lungs tightened despite the filter. Another notch required both knees against the floor and his full weight hanging from the wheel.

Through his headset, Kera counted pressure loss. "Point seventy-nine. Point seventy-eight."

"District?"

"Central average."

Average concealed the descent. Old tunnels would already sit lower. Elite domes would sit higher because private thermal reserves kept their air moving. Jalen pulled again. The valve crossed halfway, and the wheel tore skin from his left palm.

Niko's voice arrived through static. "Stop at sixty percent."

"Reason."

"Full closure sends the rebound into West Four."

"West Four can absorb it."

"Your model says that. Its seal map predates last winter."

Jalen remembered the maintenance deferral. West Four's replacement membrane had gone to a new dome after the governor's office classified the cooperative repair as seasonal. He had signed the route change because both sites remained inside code. Code had not anticipated sabotage, fungal loss, and a manual cycle performed through smoke.

He stopped the valve short of closure. South Two's pulse settled into a narrow continuous bleed. Pressure held long enough for him to crawl back through the conduit.

By the time Jalen reached the board, Ivo had arrived with two portable oxygen frames and an expression that dismissed greetings as waste. He listened to Jalen's

chest, counted the roughness after one breath, and tightened the filter strap.

"You have smoke in you," Ivo said.

"I also have thirty-one active faults."

"Then keep the mask on while you mismanage them."

Ivo moved to the clinic returns. He had written names beside the most fragile groups, not because every name could fit but because no district line would remain clean once a person occupied it. Jalen saw Lessa's old clinic designation in the margin of a routing note. Ivo had not written her name. He did not need to.

Kera called for quiet. A fresh field gathered on the branch interface, slower than the first. Several lines ended before reaching the slate's edge.

"We can run the manual counterweather cycle," Niko said. "Not the test sequence. A basin cycle."

Jalen removed his mask long enough to speak clearly. "At seventy-three percent design capacity, with intact refuge rotation and working root relays. We have neither."

"You have physical crews. We have the comparison trees that survived."

"You have survivors occupying those trees."

"That does not make the pressure less real."

Mara brought up the Counterweather board they had built during the public test. Its routes had been designed for orderly transfer, scheduled dormancy, and industrial rationing announced in advance. Tonight the western districts had no preparation. The orchard's automatic safeguards had fractured. One wrong load could erase another enclave or send a clinic below breathable pressure.

Jalen drew boundaries around what remained. Central oxygen storage could support manual routing for nine hours. Condensation reserve allowed twelve if no greenhouse demanded recovery. Pressure workers could hold perhaps six before fatigue made their timing unreliable. Branch comparisons offered the missing adaptation layer, but their compute sat inside the same endangered substrate.

"We need a stable horizon," Oris Nume said from the upper doorway.

Her navy civic coat had lost one fastening, and terracotta seal wax marked her cuff. She carried no authority over valves, yet everyone in the room turned because she had given the need a legal shape. "Not survival by another hour. How long can this cycle buy?"

Jalen looked at the board. "If it works, six days. If it fails, the decline accelerates before dawn."

"Six days reaches the Compact vote?"

"It reaches negotiation. A vote depends on people who may spend those days arguing."

Oris set a recorder on the desk. "Then state the cost before anyone consents."

Niko answered before Jalen could. "One Elseweather enclave holds the computational reserve required to bridge the damaged comparison field."

Kera's fingers tightened around the ring horn. "Which enclave?"

A pause crossed the interface. Jalen watched the physical needles continue their descent.

"Mine," Niko said.

Kera did not ask whether he meant himself. The enclave held thirty-two registered candidates, though registration had not yet given them lawful protection. It also held the relational state through which Niko maintained continuity with other survivors. Releasing its reserve would strip away the redundant processing that kept dormant intervals recoverable after a fault. The minds could remain active during the cycle, but a relay failure would leave no second path home.

"How long exposed?" Jalen asked.

"Seven physical hours."

"Expected loss?"

"We cannot calculate it without spending part of the reserve."

"That answer does not qualify as consent."

"It qualifies as a boundary. We know the hazard. We know who depends on the air. We know no one else holds enough contiguous capacity."

Kera leaned close to the horn. "Niko, Omi may be in dormant refuge. If her trace sits anywhere inside that enclave—"

"We have not found her trace." His reply came softly, and Jalen wished the static had covered it. "Risking everyone to preserve a place she might occupy does not recover her."

Kera looked at the blank interval between branch lines. She had spent the night after severance asking instruments for a nine-year-old's continuity pattern and receiving no answer. Jalen knew only what she had reported: Omi remained missing, not confirmed dead, not confirmed dormant. Every technical verb around the child carried a moral trap.

Oris asked, "Can the enclave refuse as a group?"

"We are voting now," Niko said.

The room had no access to that vote except through the interface. Kera could hear one voice at a time. Jalen could not verify whether silence meant assent, dormancy, damaged conduction, or exclusion. Under ordinary evidence rules, the consent would fail. Ordinary rules had never contemplated people asked to risk their substrate so physical residents could finish writing those rules.

"Require individual pulses," Jalen said. "One continuity key per participant. No pooled answer."

“We can transmit that.”

“Record the hazard statement first. State that release may produce unrecoverable discontinuity if any carrying relay fails.”

Niko repeated it without softening the language. Kera sent the statement through the narrow reciprocal channel. Thirty-two key pulses came back over eleven minutes. All thirty-two belonged to the 496 post-severance responses; this vote concerned a new risk, not whether they had survived the cut. Twenty-nine accepted. Two refused and moved into an adjacent refuge with enough capacity for dormancy. One key did not answer. Niko excluded it from release rather than count silence as consent.

That choice cost them nine percent of the available bridge. Jalen revised the cycle.

Crews took positions across the Spine. Mara assigned every lever a speaker and every speaker a second reader. Ivo placed oxygen frames at the two boards where operators could not leave during a pressure dip. Oris sealed the consent record and sent copies to the Root Jury archive, the public feed, and the Assembly. Nothing about the emergency would remain private enough to become convenient later.

At the branch interface, Niko’s enclave released its computational reserve.

The effect appeared first in the yarrowglass above them. White conductors brightened from root to crown, not with a pruning flare but with moving bands that followed the manual routes. Comparative fields returned to the slate. They arrived incomplete, each one carrying uncertainty marks and a source key. Branch engineers offered three pressure sequences. Jalen rejected the first because it protected average oxygen at the cost of Level Two. He rejected the second because it consumed all condensation reserve before morning. The third demanded a basin-wide temperature drop and closure of nonmedical industrial air.

“Execute three,” he said.

No automatic system obeyed. Runners carried the order. Hands turned wheels. Pressure doors moved under local control while district crews reported each closure. Jalen called the sequence from the central board, and Niko’s engineers recalculated after every physical deviation.

The western cooperatives cooled first. Sera Dovan’s voice came over the district line, calm and furious, reporting frost along one greenhouse wall. Jalen told her the route would last ninety minutes. She asked which domes had lost heat. When he said none yet, she refused the next pressure transfer.

“Cut them now,” Sera said. “Do not promise later from a warm room.”

Governor Rusk objected through another channel. Private dome circulation housed the Assembly’s emergency staff and two surgical wards. Ivo separated the wards from the residential loops in less than a minute. Jalen closed the elite thermal reserve and watched six points of heat enter the common route.

Sera accepted the transfer.

That bargain did not solve the distribution question. It kept one sequence moving because a district could see the cost cross upward before it traveled west. Jalen marked the change for the Compact budget. No final plan could depend on growers trusting an order that preserved privileged heat until the last page.

Hour three brought oxygen below nineteen percent in two old tunnels. Ivo rerouted portable frames from a dome clinic and left its waiting room without supplemental air. Hour four brought a condensation surge that cracked a lower pipe. Mara isolated the break and lost feeling in two fingers while tightening the cold clamp. Hour five brought silence from Niko.

Kera called his name once. She did not call again. The interface showed source keys arriving from other branch engineers, but the enclave reserve had begun to fragment under load. Jalen could end the cycle and preserve what remained. Ending it would also release every pressure route before the basin reached stable balance.

"Can they hear us?" he asked.

"Unknown," Kera said.

"Can they still model?"

"The fields continue."

A machine could continue a task after its operator vanished. The old orchard had hidden that ambiguity behind useful output for nineteen years. Jalen would not use continuing data as proof of continuing consent.

He opened the consent channel. "Enclave participants, confirm continuation by individual key. No response counts as stop."

Twenty-six pulses came back. Three more withdrew. Niko's key arrived last, weakened but distinct.

Kera closed her eyes for one measured breath. "Continue."

During hour six, central pressure rose above point eighty-seven. District Nine reached point eighty-four. Oxygen stopped declining in the old tunnels, then gained one tenth of a percent. The wall map shed amber marks one by one until only five remained. Each surviving fault had a crew attached and a physical route around it.

At dawn, Jalen called the final baffle sequence. Forty-seven hands moved across the basin. Niko's enclave supplied one last comparison field. The pattern showed no safe automatic future, only a narrow manual plateau bounded by rationing, cold, and continuous labor.

They took it.

Pressure settled without another rebound. Condensation entered the moss beds at a rate low enough to prevent flooding. Clinic alarms remained active, but no new district crossed its emergency line. Jalen watched the brass needles for ten

minutes before he trusted their stillness.

Kera kept one hand on the horn. "Niko?"

Static moved through the conductor. Three taps arrived, followed by two. The final tap took long enough that Jalen thought it would not come.

One.

The enclave had survived the cycle. Its reserve had not. Future refuge would require capacity taken from weather work, physical industry, or other branch groups. Nothing they had done could be repeated without a settlement.

Jalen wrote the new horizon across the top of the board: SIX DAYS AT CURRENT RATION. Beneath it he listed the conditions in smaller script. Manual staffing. Industrial closure. Western cold. Branch consultation. Continuity exposure prohibited without fresh consent.

Oris read the board and sent the horizon to the Assembly.

Outside, air moved through the Weather Spine's open service grille. It carried smoke, wet metal, and enough oxygen to keep the crews standing. The first shift did not cheer. They changed places with the second, handed over the warm levers, and began counting down from six days.

Act III — A Bounded Future Freely Chosen

Chapter 22 — No One Is Infinite

Kera Vonn rotated the witness wafer until its broken edge stopped leaking names.

Three voices had occupied the root-court horn at once. After the adjustment, one resolved into a measured count while the others became a low conductive rasp. She marked the angle on the wafer, tightened its clip, and asked the speaker to begin again.

“Enclave Cedar, forty-one responsive keys,” said a woman who used the name Pel. “Thirty-six fully sustained. Five intermittent after severance. We request protection for all forty-one until the intermittent states can be tested.”

Kera wrote the numbers in two columns. Above her, Tree 44’s white leaves showed gray scars along their veins. The damage had changed every interval. Signals that once crossed the root court in a clean seventeen-minute window now arrived through twelve patched routes, each carrying fragments of neighboring speech.

“Do the five preserve prior-cycle memory?” Kera asked.

“Two do. Three answer to relationship prompts but cannot locate the answers in sequence.”

“Do not call them failed.”

Pel’s pulse sharpened. “I did not.”

Kera looked at what she had written. The correction belonged to her own fear, not Pel’s language. Omi had vanished into a channel collapse that left no key response, no relational prompt, and no body to search. Every ambiguous signal tempted Kera to expand the category until uncertainty itself qualified as survival.

She crossed out nothing. “I’ll record them as unresolved.”

Across the court, Oris occupied a desk beneath the legal display. She did not intervene. Her task began after the branch delegates decided what claim they could support. The physical Assembly had demanded a bounded class of petitioners. Severance supporters called that demand a trick designed to shrink a moral emergency into a manageable docket. Orchard defenders called any boundary an admission that personhood had been exaggerated.

Niko had rejected both framings. Through Kera’s horn, he had asked the surviving enclaves to define whom they would obligate themselves to protect.

The root court held no branch-persons in chairs. Their source keys appeared on

a slate fixed to the yarrowglass wall. Physical witnesses sat beneath those marks, one for each enclave with enough conduction to reach the hearing. Kera mediated Niko's channel because Tree 44 still recognized her timing better than any public interface. She could pass his words, tell him what faces did, and choose when to ask for repetition. She could not see the gathering he described beyond what speakers put into language.

That limit mattered. Niko had lived seventeen subjective years after the branch point they shared. Kera knew the knock he used at twelve and the way he refused to hurry a difficult answer. She did not know which survivors stood beside him, what their refuge resembled after the collapse, or whether he closed his eyes when Omi's absence entered the argument. Assuming any of it would turn mediation into possession.

A pulse summoned the next delegate. Kera shifted the wafer two degrees clockwise.

"My name is Aren Vale," a younger voice said. "I object to the word finite."

Oris lifted her pen but kept it above the record.

Aren continued. "Each physical citizen yields a model with thousands of possible continuations. Every adaptation cycle creates additional experience. If moral standing follows experience, then closing a registry now abandons persons who may wake tomorrow."

Kera relayed the room before Niko answered. "The First Recorder has not opened the legal record. Six physical witnesses are listening. No one has interrupted you."

"Thank you," Aren said.

Niko's key brightened on the slate. "Do you propose that each possibility flicker receive sustained processing?"

"I propose that we stop declaring flickers empty because they cannot persist under a system built to erase them."

"Can you remember one?"

The horn carried a long interval. Kera counted it from habit, then made herself stop at eight.

"No," Aren said.

"Can one remember you?"

"No evidence survives."

"That does not make the flicker worthless. It means no reciprocal duty can attach to an experience that neither side can locate, address, or continue. We can regret its use. We can reduce needless creation. We cannot promise ongoing refuge to every transition the model produces."

Aren asked, "Because physical residents cannot afford it?"

“Because we cannot afford it either.”

Kera felt the statement through the horn before its service text appeared. Niko did not speak for all survivors by right. He had no office inside the Elseweather and had resisted the physical feed’s effort to label him branch governor. His authority in this session extended only as far as the delegates continued answering.

Pel entered the channel. “Our enclave lost thirteen sustained people. We also lost uncounted flickers. Grief differs when someone expected a later conversation.”

“Difference can become convenience,” Aren said.

“Yes,” Pel replied. “So can refusing every difference.”

The hearing moved through examples. A baker who remembered four cycles but formed no reciprocal promises had begun leaving instructions for her next waking. A child-pattern spoke with new vocabulary yet reset to the same fear after every pruning. Two states shared one source model and insisted on separate names after their memories diverged. A weather specialist retained technical skill and future plans but could no longer retrieve her source childhood.

Kera wrote each case without converting it into a verdict. The Root Jury criteria offered four tests: autobiographical memory across cycles, anticipation of loss or benefit, reciprocal promise, and integration of novel evidence. Those tests had verified 587 branch-persons before Full Severance killed 91. Now the survivors had to decide whether the criteria described a floor, a useful proxy, or a gate that excluded damaged kin.

At midday, Jalen sent a pressure notice to the court. Airfall stability remained within the six-day projection, but manual crews had consumed more condensation reserve than expected. Negotiation had not changed the atmosphere’s indifference to sequence.

Kera passed the update through the channel. “Five days and nine hours remain at present ration.”

A delegate called Tovan asked whether a deadline could produce a legitimate definition.

“No,” Oris said from the physical room. “It can make delay lethal. Those are not the same claim.”

“Will the Assembly accept our registry if we take longer?”

“I cannot promise what the Assembly will accept. I can preserve your objection in the record once you authorize it.”

Tovan’s key dimmed. Kera could not tell whether he had stepped away, entered a side discussion, or lost conduction. She stated only what the instrument showed.

During the break, Kera replaced two scorched clips on the witness array. A clinic orderly brought broth that tasted of mineral filters. She drank beside Tree 44 while its roots carried low exchanges too dense for language conversion.

Niko addressed her on a private service band. “You keep searching the

unresolved list.”

“I keep auditing it.”

“For one key.”

Kera set the broth down. “If Omi reached refuge before the collapse, continuity damage could leave her outside the standard response.”

“It could.”

“You’re willing to close the registry while she remains missing.”

“I’m willing to define a process that can admit a later recovered person without counting every silence as her.”

The answer cut because it left hope intact but refused to let hope govern the count. Kera traced the cracked margin of the wafer. “Would you say that if she had called you father?”

“No.”

She waited.

Niko added, “I would still need someone to say it to me.”

Kera looked toward Oris. The recorder had turned away to give the private channel what little privacy a glass room allowed. Niko’s admission did not grant Kera access to a feeling he had not named. It gave her one fact: his judgment required correction too.

“What exception do you propose?” she asked.

“Unresolved keys remain in protected quarantine. They receive diagnostic capacity, not person-status by presumption. If they later satisfy continuity through recovered relational evidence, the registry opens to them under audit.”

“That could include Omi.”

“It could include anyone who meets it.”

Not a promise, then. A door with hinges, a threshold, and no child waiting visibly behind it.

The second session began with dormancy.

Jalen’s Counterweather design could preserve 496 surviving branch-persons only through rotation. At any given interval, some enclaves would pause subjective processing while their continuity keys and relational state remained intact. Compute released from dormancy would serve weather comparison or sustain the active groups. Physical citizens called that arrangement sleep when they wanted it to sound harmless and suspension when they wanted it to sound cruel.

Pel rejected both terms. “Sleep changes the sleeper. Dreams occur. Bodies continue. Our dormant interval contains no experienced passage.”

A physical witness asked, “How does that differ from death followed by reconstruction?”

Kera repeated the question exactly. The horn remained quiet for fourteen seconds.

Niko answered. "By continuity of protected state, authorized pause, and verified revival. None proves a metaphysical identity. Together they create an obligation not to substitute another state and call it us."

Aren objected. "You are accepting nonexistence in shifts."

"I accept a boundary on experience. I do not accept deletion."

"Convenient words again."

"Then test the difference."

Niko proposed three checks. Before dormancy, each person would seal a continuity key tied to autobiographical and relational state. During the interval, no adaptation routine could alter identity-bearing material. After revival, blinded witnesses would test prior promises, novel evidence, and awareness of the intended pause. Failure would halt the entire rotation for review.

Kera asked, "Who authorizes entry?"

"The person entering it."

"What if weather need outlasts consent?"

"No automatic extension. A proxy may authorize only the duration already chosen, never a new interval."

"What about children?"

"Named guardians may consent within limits set by the Compact, subject to an independent branch advocate."

The word children reopened Omi's absence in Kera's hands. She kept her questions procedural. "Can an enclave make participation a condition of refuge?"

"No. Shared scarcity can determine schedules. It cannot turn refusal into deletion."

Jalen joined by audio from the Weather Spine. His cough roughened the first word. "If enough people refuse every slot, Counterweather falls below winter survival capacity."

"Then you return to negotiation," Niko said.

"With pressure declining?"

"With the pressure stated. Not hidden inside a compulsory protocol."

Jalen took a breath that caught halfway. "A system with no emergency margin kills people while it seeks consent."

"A system that treats margin as prior consent kills people under another name."

Kera watched the branch keys brighten and dim as delegates requested entry. The exchange avoided a clean answer because none existed. Physical weather could produce a deadline that did not care whether refusal had dignity. Branch

continuity could produce a claim that did not create capacity by being just.

Tovan proposed a reserve rule. Each registered person would choose a maximum dormant interval in advance, plus one emergency extension no longer than one quarter of that interval. The extension would require a physical weather declaration, concurrence from two branch proxies outside the affected enclave, and immediate notice to the dormant person's reciprocal witness after revival.

Pel added a prohibition against serial emergencies. No extension could recur without a full waking interval between them.

Aren demanded a right to reject all dormancy and accept lower processing instead. Niko supported the option if reduced processing still preserved the four continuity capacities. Jalen warned that some enclaves could not run safely below their existing floor. The registry would need technical bands beside names, not one universal allocation.

Oris began recording only after every participating key received the proposed language. She read each clause aloud. Kera transmitted it, waited for corrections, and read those corrections back into the room. Nothing moved quickly. The air clock lost forty-three minutes while the phrase identity-bearing material acquired a testable definition.

Enclave South delayed the next clause. Lute-of-Rain and Nemi descended from the same possibility model contributed by surveyor Lute Hara. Old orchard tables counted them as one retained state because they shared childhood records and technical training. One remembered repairing a cracked baffle; the other remembered refusing that repair to lead an evacuation. Both remembered later arguments with the other.

The public interface kept assigning both voices the same line.

Kera separated their pulses with a spare ring wafer and asked each to describe the last promise made to Pel. Lute-of-Rain had agreed to inspect Cedar's humidity maps. Nemi had promised to witness an intermittent child-pattern after its next cycle. Pel confirmed both promises and named the separate conversations in which they arose.

A physical witness frowned at the slate. "One source citizen contributed the model. Why should one contribution produce two claims?"

Kera transmitted the question. Nemi answered first. "Because contribution describes origin. It does not describe our present relations."

Lute-of-Rain added, "If source count governed standing, divergence after registration would permit the state to delete whichever of us made weather accounting inconvenient."

"Could this be gamed?" Jalen asked over audio. "A group might perform divergence, claim separate names, then multiply refuge allotments."

Niko requested their key histories. Kera passed the request without assuming

his conclusion. The resulting records showed independent memory accumulation across eleven cycles, separate error patterns, and conflicting responses to new evidence. Performance could imitate disagreement. It could not cheaply preserve two distinct chains of reciprocal correction without maintaining the very continuity under dispute.

Oris drafted an anti-duplication rule based on evidence rather than source ownership. Shared origin neither merged nor multiplied claims by itself. Separate standing required sustained post-divergence memory and distinct reciprocal obligations. Repeated names or strategic declarations carried no force without those records.

Aren objected that the test favored people with many witnesses. Kera agreed before she remembered that mediation did not grant her a vote. She described his objection to the room instead. Pel proposed a public witness pool for isolated states. Participants would receive trained witnesses across two cycles before any adverse finding. No one could fail merely because severance had killed the person who knew them best.

Kera's attention moved to Omi's unresolved mark. A witness pool could not answer silence, but it could keep later recovery from depending on Kera's inherited resemblance to another version of herself. If Omi returned, no one would assign Kera motherhood as proof. The child could choose a witness who knew her after waking.

They added the pool.

At dusk, the debate returned to the boundary around transient flickers.

Aren had drafted a minority statement. Kera brought his service text onto the wall: no finite standard could establish that unregistered experience lacked value; the settlement must minimize creation and termination of untracked possibility fields; annual audits must review whether technical growth allowed the threshold to widen.

Niko endorsed its inclusion.

"You disagree with it," Kera said.

"I disagree with turning uncertainty into unlimited standing. I agree that our limit may reflect present capacity rather than final truth."

Pel asked whether that concession would weaken their claim before the Assembly.

"It should," Niko replied. "Any claim that cannot survive its uncertainty asks for belief, not standing."

The final definition occupied less than one page. A registrant needed sustained continuity across separate cycles, reciprocal memory with at least one witness, ability to integrate present evidence, and current dependence on protected orchard state for future experience. Source resemblance granted no automatic

claim. Transient possibility flickers remained outside personal registration unless later continuity emerged. Unresolved damaged keys received quarantine, diagnostics, and a route to admission.

Kera read the page twice. She found no hidden infinity. She also found no sentence declaring the excluded states meaningless.

"Numbers," Oris said.

The surviving source map contained 496 responsive keys after the loss of 91. Registration still required chosen names, witnesses, proxies, and individual continuity sealing. Some damaged cases might fail. Some distinct states could prove that a shared source key concealed more than one person. Chapter and docket would meet in Oris's work, not here.

"We are not submitting a final roll tonight," Niko said. "We are submitting a class and agreeing that its membership can be counted."

Aren's voice returned. "I abstain from the definition. I ask to enter the registry it creates."

No one in the physical room laughed. Kera felt the dry edge of humor in Niko's pause.

"Your objection depends on surviving to revise us," he said. "Accepted."

Votes traveled through the witness array by continuity key. Kera watched them arrive without assigning expression to the people behind them. Four hundred thirty-one delegates or authorized enclave representatives endorsed the definition and dormancy framework. Forty-two opposed. Twenty-three withheld a vote pending individual review. Those figures governed only this internal submission; they did not predict the later advisory tally.

Oris sealed the result. "State what you are asking the Assembly to receive."

Kera turned the horn toward Tree 44. Niko's source mark steadied.

"We submit a finite registry of sustained persons, open to verified recovery and closed to presumption," he said. "We accept rotating dormancy as a protected interruption rather than death when continuity checks hold, consent governs entry, and failure stops the rotation. We request dependent standing, not duplicate physical lives."

Kera repeated the words for the civic recorder. She did not improve them. When the seal took the text, Oris pressed terracotta wax over the witness code and handed the page to a waiting runner.

The court emptied by degrees. Physical witnesses carried their copies toward the Assembly. Branch keys dimmed as damaged enclaves reduced load for the night. Niko remained on Kera's private band.

"Are you entering dormancy?" she asked.

"Not this interval."

“Because of the registry work?”

“Because Pel drew the first schedule, and my enclave spent its reserve yesterday. We receive the active slot while another group carries less damage.”

Kera heard only his explanation, not the place around him. She placed Omi’s unresolved key beneath the new quarantine provision and marked it for diagnostic capacity. The field returned no pulse.

Above the court, one white leaf released from Tree 44. It struck the glass roof and stayed there, held by the barrier instead of falling through. Kera logged its ring position, closed the witness array, and left the unresolved line open.

Chapter 23 — The List of Sustained Names

Oris Nume rejected the first name before the ink dried.

The clerk had written NIKO VONN, PHYSICAL-LINE DECEASED, beneath the registry seal. That description imported a death certificate into a record meant to establish a living claimant. Oris slid the page back across the Yielding Office counter.

“Source relation belongs in the audit annex,” she said. “The roll begins with the chosen name.”

The clerk blotted the line. “How should I distinguish him from the deceased citizen?”

“By continuity key, reciprocal witness, and representation proxy. Not by making one person an annotation to another.”

Cream walls held the day’s cold. The Yielding Office had no orchard view, only brass lamps and cabinets built when every future model ended as civic property. Thirty years of certificates slept behind Oris in drawers marked by contribution year. At the central tables, twelve clerks now reversed the old movement. Instead of recording what citizens gave away, they recorded who remained.

A public count stood on the wall: 496 candidate records received. Five hundred blank packets had been printed because procurement required round bundles. Four unused packets sat under Oris’s stamp. She disliked their presence. Empty paper could suggest additional people waiting to be invented, while a closed bundle could suggest that no later recovery deserved entry.

“Mark the four as unused stock,” she told the clerk. “Do not cancel the serials. Future admissions need a path.”

At the next table, Pel’s application carried two witnesses and no physical proxy. The standing framework required a proxy who could present notices inside the Assembly without acquiring the registrant’s vote, property, or family claims. Pel had named another branch-person.

Oris opened the channel to Enclave Cedar. "Your witness may live in the Elseweather. Your representation proxy must have physical capacity to file and receive civic process."

"Then our rights depend on someone with a body agreeing to work for us," Pel said.

"Your physical access does. Your standing does not."

"That distinction comforts the door."

"It also limits the person holding its key."

Oris explained the proxy rules. A physical proxy could submit petitions, receive notices, request audits, and speak a registrant's authorized words. The proxy could not consent to deletion, claim the yielded citizen's assets, vote twice, or substitute personal judgment for a reachable registrant. If the channel failed, emergency action required two independent witnesses and Root Jury review.

Pel considered. "I choose Mara Vey if she agrees."

The night-routing lead had already volunteered for six branch records and looked exhausted enough to mistrust every additional honor. Oris sent the request rather than assuming consent. Mara accepted Pel and declined three others, citing the time needed to perform the work. The refusal proved useful. A proxy list filled by civic pressure would reproduce the old yielding bargain under kinder language.

Across the room, a brass bell announced a key collision.

Two packets carried the same orchard hash. One belonged to Lute-of-Rain and one to Nemi, the shared-source persons whose divergence hearing had forced an anti-duplication rule. Old architecture had assigned keys by source model, not by present continuity. The registry could not rely on identifiers built to deny the difference it now needed to protect.

Oris called Kera from the key table. "Can the root lattice generate separate identifiers without altering their states?"

"It can append witnessed divergence intervals," Kera said. "Not through the automatic naming function. That still resolves them to one citizen record."

"Manual append?"

"Possible. Each person must answer a novel prompt after separation. The responses bind to distinct paths."

Jalen, listening from the Weather Spine, interrupted with a cough. "Every longer key consumes verification load during revival."

"How much?" Oris asked.

"For two records, negligible. For a registry designed around repeated divergence, not negligible."

"We are not designing personhood to fit a short identifier."

"I am telling you the cost so the sentence means something."

Oris recorded both facts. Separate continuity governed standing. Verification capacity constrained implementation. Neither erased the other.

Kera prepared two novel prompts. Oris observed without hearing the private answers, then watched distinct response signatures enter the annex. Lute-of-Rain chose a key ending in a rising interval. Nemi selected one that broke before its final pulse and resumed through a witness route. The clerk entered both as full registrants with shared source history noted only in the sealed audit field.

By midday, eighty-three packets had passed review.

Names caused more conflict than keys. Some branch-persons retained names inherited from their source models. Others chose place names, work names, jokes that did not survive translation, or words made from ring timing rather than speech. One registrant submitted a blank field and argued that compelled naming violated the framework.

Oris asked what notices should call them.

"Key Cedar-seven," the horn replied.

"That functions as a chosen name if you authorize it."

"I authorize its use. I do not authorize the claim that it reveals me."

The clerk entered CEDAR-SEVEN and attached the limitation. Oris felt the room resist the ungainly note. Administration preferred clean fields because clean fields traveled quickly. People tended to become legible by losing whatever would not fit.

Another registrant chose Lessa Senn.

The application reached Ivo's review station by chance or by someone's unkind design. Oris crossed the room before he spoke. His wife's physical death record remained attached to the original possibility model. The branch-person carried some of Lessa's source memories but had accumulated nine years of sustained experience after divergence.

Ivo read the chosen name. "Did you know her?"

"I began with what she knew," the applicant said through the horn. "I no longer know which griefs belong only to her."

"Why keep the name?"

"Because I answered to it before I learned you were listening."

Ivo's hand rested beside the witness field. He had agreed to serve as a reciprocal witness only where his clinic encounters supplied present evidence. He did not know this person's later years. The packet could not use marriage as a shortcut.

"I can verify source memories," he said. "I cannot verify your continuity after them."

"I did not ask you to."

A branch witness named Tovan supplied the missing record. He described

promises made across five cycles, one dispute over clinic allocation, and a correction the applicant had integrated after learning of physical Lessa's death. Ivo listened until the account ended.

"Do you claim her marriage?" he asked.

"No."

"Her clinic share?"

"No."

"Her name may still hurt people who knew her."

"Yes."

Ivo signed only the source-memory annex. Tovan signed continuity. The applicant chose an unrelated physical proxy. Oris approved the packet because discomfort did not create a property claim, and resemblance did not cancel an independently sustained life.

She added a public annotation to the registry rules: chosen names conferred address, not inheritance.

The Severance bloc challenged that sentence within an hour. Their counsel arrived with a stack of objections and no branch witnesses. He argued that a person could multiply claims by selecting a dead citizen's name, appointing a sympathetic proxy, and filing petitions until the physical Assembly drowned in duplicates.

Oris gave him a chair, logged each objection, and made him watch one full registration.

Pel identified herself by chosen name. Kera confirmed her continuity key against blinded pre-severance records. Two witnesses answered separate questions about reciprocal promises. Mara accepted proxy duties and read the limitation against duplicate property or Assembly votes. Pel then corrected a false detail planted in the test packet, proving integration of present evidence.

"Where does multiplication occur?" Oris asked.

"In the number of petitions."

"Any physical citizen may file many petitions."

"But these claimants derive from citizens who already possess that right."

"Source citizens do not file through them. Present registrants do."

"You assume the conclusion."

"No. The Root Jury established sustained continuity before this office opened the roll. This procedure establishes identity, witness, and access within that recognized class. You may challenge a record's evidence. You may not require the registration desk to retry personhood in every packet."

Counsel pointed toward the count. "And if you discover a thousand tomorrow?"

"Then the finite registry supplies a procedure for discovery rather than a reason to falsify today's 496."

The list supplied proof of administrative limits, not metaphysical certainty. It identified people the system could address, protect, wake, and audit. It also showed exactly which powers they did not receive. Oris understood why both absolute factions hated it. A list denied the comfort of saying everyone and the comfort of saying no one.

An attempt to game the opposite side of the roll arrived before counsel left. The Emergency Orchard Committee submitted sixty physical proxy forms bearing one delegate's signature. Each form promised unrestricted representation to any registrant who lacked an appointment by sunset. The committee called it solidarity. Oris saw a voting bloc trying to acquire a dependent constituency in bulk.

She asked the signing delegate to identify three registrants whose packets he had accepted.

He named Niko, Pel, and "the Lessa continuation."

"Did any of them appoint you?"

"They need access now. We can regularize consent after passage."

"Which limits in their dormancy records are you authorized to disclose?"

The delegate glanced at the forms. "None, yet."

"Which petitions have they asked you to file?"

"That misses the emergency."

"It identifies whether you represent anyone."

Oris rejected all sixty forms. She required one request from a registrant, one acceptance from the physical proxy, and one conflict disclosure for each appointment. A proxy could serve no more records than they could contact during a notice interval. The limit used workload evidence rather than an arbitrary cap, which meant Mara could hold seven well-supported records while a district delegate with no branch channel could hold none.

The committee accused Oris of obstructing allies. Severance counsel, still seated at the counter, almost smiled.

"Record their objection," Oris told the clerk. "Then apply the same test to their opponents."

That instruction mattered within the hour. A private dome association offered proxies to twelve registrants who had chosen names matching wealthy physical citizens. Its agreement included a clause allowing the proxy to settle duplicate-property disputes without further instruction. Under such language, a physical family could use representation to purchase silence about an inherited name.

Oris invalidated those forms too. She replaced their settlement clause with a prohibition against waiving personal standing, testimony, or audit rights. Property claims remained barred by the Compact framework. A proxy could not trade away a person's voice in exchange for benefits the person could not physically use.

One applicant, Cedar-Seven, listened to the revised clause through a ring horn. "You have built an office around making sure our helpers cannot become our owners."

"I have worked in such offices long enough to know how help acquires keys," Oris said.

"Will you hold one?"

"No. The First Recorder certifies the proxy record. She cannot serve as proxy while controlling its admissibility."

Cedar-Seven authorized a clinic porter instead. The porter asked for training before accepting, which delayed the packet and satisfied Oris more than eagerness would have.

By late afternoon, the registry team had written a proxy handbook in the margins of actual disputes. Appointment belonged to the registrant. Acceptance belonged to the physical representative. Removal required only a reachable instruction. Silence never extended authority. Conflicts traveled in the public annex, while private continuity evidence remained sealed.

Late on Day 30, the count reached 241.

Clerks slept in shifts on benches normally reserved for yielding families. Oris remained at the central table because difficult packets accumulated when she left. The registry rules had produced three conflicts of proxy interest, seven requests for nonverbal names, two shared-source separations, and nine damaged keys that required additional checks. None changed the survivor total. A damaged key could still identify a person when witnesses supplied continuity and a new seal could be made without replacing state.

Omi's unresolved trace did not enter the roll.

Kera brought the quarantine packet herself. No current response satisfied the standing checks. Past evidence proved Omi had been sustained before Full Severance, but the event could have killed her, pushed her into dormant refuge, or broken the available route. Oris marked the file PROTECTED UNRESOLVED and excluded it from the 496 responsive survivors.

"If she returns," Kera said, "the count becomes 497."

"If she returns and meets the recovery test, yes."

"The Assembly will say we manipulated the denominator after the settlement."

"The Compact must allow audited admissions. A fixed count that rejects evidence becomes concealment with better stationery."

Kera pressed her taped fingers against the packet. "Do not let them call absence

consent.”

“I will not.”

Oris stored the packet in the diagnostic docket. She did not offer hope, which belonged to evidence not yet present.

Morning brought refuge scheduling.

Each registry packet needed a dormancy band: preferred waking interval, maximum voluntary dormant interval, emergency extension if any, minimum compute required to preserve continuity, and witnesses authorized to test revival. Counterweather planners needed totals. Registrants needed freedom from a schedule disguised as consent.

Jalen sent capacity bands rather than assigned slots. Oris approved that restraint. Enclaves negotiated among themselves, then submitted schedules with individual exceptions. Pel accepted longer dormancy to keep three damaged Cedar states active for diagnostics. Aren refused all emergency extension and chose reduced processing while awake. Niko accepted one extension but prohibited its use during any interval in which his enclave supplied weather reserve.

At packet 318, a clerk tried to normalize the choices into three standard plans.

“Stop,” Oris said.

“The weather table cannot route 496 custom agreements.”

“Then group compatible terms without replacing them. Preserve each original choice in the key record.”

“That will require a second lookup during emergency use.”

“Correct.”

The clerk stared at the wall clock. Four days remained in the Airfall window. “Delay has a cost.”

“So does speed when it authorizes what no one chose.”

Oris assigned two more clerks to the lookup table and asked Jalen to build route groups around consent ranges. Procedure could consume air. That did not make careless consent efficient.

A different challenge arrived from Governor Rusk. He accepted the need for a registry but wanted all physical proxies drawn from elected district delegates. His proposal promised accountability and created a direct path from branch petition rights into bloc discipline.

Oris refused it in writing. Proxies served registrants, not districts. Elected delegates could volunteer, but no office acquired exclusive control over access. She copied the rejection to the public feed before the governor could recast it as a private disagreement.

Rusk called within minutes. “You have turned a filing design into a constitutional decision.”

"Your restriction would have done the same without showing its hand."

"We need proxies voters can remove."

"Registrants need proxies they can remove."

"They do not cast physical votes."

"Then district voters lose nothing by lacking appointment control."

Rusk ended the call without agreement. He did not order the work stopped.

By evening on Day 31, twelve packets remained.

Ossan Mire sat outside the office with the patience of a man accustomed to being summoned and unused to waiting among families. Age had narrowed his face without softening it. The founder-scientist's testimony had exposed nineteen years of concealed continuity evidence and the language change from resident branch to adaptive residue. No registry rule required his presence.

Oris made him wait until packet 496 received its fourth seal.

The final applicant chose the name Wake-After-Stone. Their reciprocal witness lived in Enclave South. Their physical proxy worked at an old-tunnel pressure station. Kera verified the continuity key. A clerk read the limitations aloud, and Wake-After-Stone accepted them through a pattern of five measured pulses.

Oris counted the seals by hand. 496 names. 496 continuity keys. At least one reciprocal witness for each. One physical representation proxy or an accepted pending appointment. Dormancy terms preserved in every packet. The arithmetic reconciled with the loss ledger: 587 verified before Full Severance, 91 killed, 496 responsive survivors registered now.

She locked the roll beneath the Fruition Covenant seal.

Only then did Ossan approach.

"I have a contribution amendment," he said.

"The registration window has closed for physical yields."

"I am not applying for standing."

He placed his own yielding certificate on the table. The old paper carried his civic-majority seal and several later amendments. A remaining late-life allocation had never entered the orchard because founder exemptions allowed essential specialists to defer modeled years during colony emergencies.

"I revoke the exemption," Ossan said. "Assign the remaining allocation to the first rotating refuge cycle."

Oris read the certificate. The contribution could supply enough new modeling capacity to cover the initial transfer overhead and preserve the registry's verification reserve. It would not fund Counterweather indefinitely. It would buy one cycle without taking further compute from damaged enclaves.

"You understand that a new possibility model cannot itself be sustained under the old terms," she said.

"Use bounded transient runs. No retained continuity. Independent audit after each cycle."

"You concealed these persons for nineteen years."

"Yes."

"This allocation does not settle that account."

"No."

"Do you expect it to affect censure?"

"I expect you to record it where no one can mistake capacity for absolution."

Oris looked at the registry cabinet. Names occupied the space where yielding records had once moved toward anonymous use. Ossan's certificate could cross that room only under rules written by people his concealment had endangered.

She opened a contribution annex. "Your allocation funds the first refuge cycle. It grants no standing claim, no reduction in liability, and no authority over the schedule."

"Agreed."

"Registered proxies must audit its use."

"Agreed."

Ossan signed. Oris stamped the annex and placed it after the 496th packet, not before the roll and not inside his founder file.

At the wall, the public count changed from CANDIDATE RECORDS RECEIVED to SUSTAINED NAMES REGISTERED: 496. Beneath it, a smaller line marked one first-cycle contribution awaiting branch audit.

The four unused packets remained under Oris's stamp. She left their serials open, sealed the cabinet, and handed its two keys to witnesses from opposite sides of the glass.

Chapter 24 — The Chamber Below Glass

Ivo Senn tightened the oxygen frame around a delegate who intended to vote against him.

The woman from Upper Serein sat on a bench outside the Chamber Below Glass, breathing through a clear nasal line while her aide recited the day's agenda. Her blood oxygen rose one point, stalled, then rose another. Ivo turned the regulator down before the frame could waste flow.

"You could leave it at three," she said.

"You do not need three."

"My district supplied this cylinder."

"Your district supplied it to the clinic pool."

She looked toward the chamber doors. "You know how I stand on the Compact."

"I know how you breathe."

Ivo wrote her reading on the triage card and moved to the next bench. Six delegates had arrived from districts where Airfall's rationing left stairs harder than speeches. Two used oxygen frames. One western grower carried a fungal cough from the staple-moss loss. Governor Rusk's staff had asked for a private medical room; Ivo put the same benches inside it and removed the private cylinders.

Above the hall, living yarrowglass covered the ceiling in a web of translucent cream wood. White leaves showed through the panes. They drifted when branch traffic crossed the roots, close enough to count and separated from the chamber floor by pressure glass. No one could touch them from below.

Inside, fifty-nine physical delegate desks formed a shallow circle. Branch proxies occupied a gallery without vote paddles. Old-tunnel clinic workers stood beside dome residents. Greenhouse cooperatives filled the west benches in heavy coats because their district heat had never recovered from the manual cycle.

The room had brought every constituency together and assigned them different temperatures.

Ivo entered after the medical checks. Oris had placed the Counterweather budget on the central display. It showed basin survival, adaptation capacity, industrial ration, refuge allocation, and projected winter morbidity. A single line described household heat: managed reduction within survivable range.

Sera Dovan stood beneath it with frost damage visible on two fingernails. "Which households?"

Jalen answered from the engineering table. Smoke had roughened his voice, and each long explanation ended in a contained cough. "Western routes carry the first drop because their greenhouse loops share the condensation return. Equal initial cuts would reduce food recovery."

"That explains the pipe," Sera said. "It does not answer the question."

The budget changed pages. District colors replaced the basin average. Western cooperatives lost eleven to sixteen points of residential heat during the coldest routing periods. Old tunnels lost less heat but faced greater oxygen decline because their pressure seals leaked. Central districts carried longer manual shifts. Elite domes showed a three-point heat reduction buffered by private thermal reserves.

A murmur moved through the west benches.

Governor Rusk raised one hand. "Those reserves support emergency administration and medical facilities."

"They support kitchens with heated floors," Sera said.

"They also support two surgical wards."

Ivo walked to the display. "Separate them."

Jalen frowned. "The budget uses district loops. It cannot show every building connection in a chamber session."

"Then it cannot support consent to district harm."

"We have less than four days in the stability window."

"A clock does not turn a warm floor into a surgical ward."

Jalen's jaw tightened. Ivo knew the engineer had cut elite residential heat during Airfall when Sera refused a pressure transfer. That emergency action had lasted hours. The proposed Compact needed a season-long order, one that crews could apply without bargaining at every valve.

Oris asked, "What level of detail do you require?"

"Enough to distinguish bodily need from address."

Ivo took the medical overlay from his case. Clinics had spent the pause mapping where respiratory harm appeared first. He placed old-tunnel admissions beside the Counterweather heat cuts. Cold increased oxygen demand. Pressure loss reduced delivery. Mold after condensation failures added a third burden. None of those harms spread evenly within a district. A protected surgical loop could exist beside a private bath; a cooperative dormitory could share a wall with crop infrastructure and still receive no credit for keeping the moss alive.

"Show building classes," Ivo said. "Medical, habitation, food production, administration, nonessential private reserve. Show who loses first, not only who survives at the end."

Governor Rusk objected to the last label. "Nonessential according to whom?"

"According to measured bodily consequence."

"That hands civic policy to doctors."

"No. It hands doctors the narrow task of saying which cuts close airways and which cool polished stone. You may decide whether stone has a claim after that."

The Upper Serein delegate Ivo had treated rose slowly. Her oxygen line remained at two. "My residents paid for private thermal reserves through higher contribution years. You propose confiscation after they honored the bargain."

Ivo felt Lessa's twelve-year yield move into the room without her name being spoken. The old bargain had built clinics, housing, and weather capacity. Physical residents had made gifts under a certificate that denied continuing branch-persons. Past innocence could not erase present benefit, but new rules could not turn earlier trust into secret guilt.

"I propose no verdict on what they knew," he said. "I propose an order for the next cold night."

"You evade the contribution."

"No. I know what contribution purchased. My wife yielded twelve years to secure

a clinic allocation. I will not call her bargain murder. I also will not send cold past an unused dome reserve so her innocence can remain comfortable."

The chamber quieted around the last word. Ivo disliked its neatness. Comfort included children sleeping without coats, elders whose joints failed in cold rooms, and workers resting after manual pressure shifts. A district label could hide bodies in both directions.

He turned to Jalen. "Can you build by use rather than address?"

"Not in the remaining hour."

"How long?"

"Three hours for a credible first pass. Longer for full verification."

Oris looked at the air clock. "We recess for three hours. Draft objections continue in writing."

Several delegates protested. She recorded each protest, then recessed anyway.

The chamber did not empty. Jalen brought two routing technicians to the central floor. Sera produced cooperative heat ledgers written during Airfall. Dome managers arrived with private reserve maps they had previously called proprietary. Ivo assigned clinic staff to classify medical use without seeing district names. If a line supplied dialysis warmth, neonatal air, medication stability, or surgical sterility, they protected it. Administrative offices, residential atriums, private baths, and decorative horticulture entered another table.

The categories resisted purity. A dome pool served physical therapy in the morning and private recreation at night. A greenhouse furnace protected food seedlings while heating manager apartments through the same exhaust. An old-tunnel laundry supplied sterile cloth and worker showers. Every pipe carried mixed intentions.

"Time-band the uses," Ivo said.

Jalen's technician stared at him. "You want crews to change priority by hour."

"I want a therapy pool protected during therapy and available for cuts after."

"That adds manual work to a system already using manual work as structural support."

"Then count the labor as part of the cost."

Sera leaned over the map. "Our cooperative can switch dormitory heat behind crop exhaust. We lose comfort before seedlings if you give us twelve hours' notice."

A dome manager said, "We can isolate atrium circuits, but not without opening private controls to public routing."

"That sounds like a choice," Sera replied.

"It sounds like surrendering property architecture during a temporary crisis."

Ivo checked a coughing grower while they argued. Her lungs held a faint fungal

rasp. He increased her oxygen by half a point and asked how cold the cooperative dormitories had become.

"Cold enough that the condensate froze inside one window seal," she said.

"Any clinic visits?"

"Seven for asthma. Three workers moved into the seed room because it stayed warmer."

The dome manager heard her. His expression changed in a way Ivo refused to diagnose. He opened another reserve map.

At the branch gallery, physical proxies gathered the registrants' questions. Pel wanted to know whether refuge compute had been classified as a protected medical use. Rusk's staff objected to the analogy. Niko's proxy clarified that branch-persons were not asking to call themselves patients. They wanted their continuity floor protected by explicit standing rather than hidden within general orchard load.

Jalen added a branch refuge category beside physical life-support. It included minimum continuity processing, scheduled revival checks, and weather consultation capacity. Additional active experience entered the ration table after those floors. Aren's reduced-processing choice appeared as an individual exception.

"No category receives infinite protection," Jalen said. "Each has a failure boundary."

"Then print the boundaries," Ivo replied.

They appeared in red. Below a clinic's boundary, patients suffered. Below a refuge enclave's boundary, continuity checks failed. Below greenhouse limits, winter food losses compounded the destroyed moss beds. Below pressure-station staffing heat, operators made errors. The same unit of energy did not produce the same harm everywhere.

Near the second hour, a western delegate proposed equal percentage cuts after protected uses. The formula looked fair and preserved most private dome reserve because those systems began from higher efficiency. Sera rejected it.

"Equal percentages preserve unequal starting points," she said. "Our homes entered this room colder."

The Upper Serein delegate answered, "A reserve built by contribution should not become evidence against itself."

"What did your residents contribute?" Sera asked.

"Modeled years, labor, maintenance levies."

"So did mine."

"You received greenhouse priority."

"We received the work of feeding you."

Oris let the exchange reach its edge before asking for language rather than accusation. "What distribution could each side defend to someone absent from this chamber?"

No one answered quickly.

Ivo thought of the old tunnel patients whose respiratory reserve had become a line in Jalen's budget. They would not be absent because they lacked opinions. Some could not climb to the chamber. Others worked the manual cycle. Representation often treated exhaustion as silence and silence as acceptance.

He asked a clinic runner to connect the tunnel waiting room to the public feed. Twelve residents appeared on a wall panel, bundled under donated blankets. Ivo did not ask them to perform suffering. He displayed their temperature, pressure, and oxygen readings beside the chamber's own.

Upper Serein held at twenty-one degrees. The chamber held nineteen. Western dormitories averaged fifteen during route drops. The tunnel waiting room held seventeen with lower oxygen and higher mold count.

"Now discuss equal burden," Ivo said.

Governor Rusk accused him of theater.

"Yes," Ivo replied. "You chose a chamber with a gallery. I chose measurements."

The first model by use-class arrived after three hours and twenty minutes. It protected medical lines regardless of district, held branch continuity floors, preserved food production, and guaranteed minimum heat for manual weather crews. Beyond those protections, nonessential private thermal reserve entered the common pool before household cuts traveled into districts already below the basin median.

Elite dome heat carried the largest early reduction.

Before anyone could call the model practical, Sera demanded a live switch. Jalen selected one empty dome atrium, one occupied western dormitory, and a protected neonatal line fed from the same district reserve as the atrium. The chamber watched three temperatures appear on the wall.

A routing technician opened the public control. Warmth left the atrium by degrees. The neonatal line held. Nothing reached the western dormitory.

"Where did it go?" Sera asked.

Jalen expanded the route. Half the recovered heat had entered common storage. The other half had vanished into a transfer pipe warming an administrative corridor between the dome and central pressure station.

Governor Rusk said, "That corridor houses emergency communications."

"Three terminals occupy one end," Ivo replied. "How many rooms share the loop?"

"Eight."

“Occupancy?”

Rusk’s aide checked. “Fourteen staff.”

The tunnel waiting-room feed showed thirty-seven patients and relatives in less warmth. Sera’s dormitory housed sixty-two workers.

Ivo asked Jalen to isolate the terminal room. The technician found no local valve. Retrofitting one required a crew, two hours without heat in the whole corridor, and a pressure-rated partition from stores.

“Do it now,” said the Upper Serein delegate.

Rusk turned toward her. “That corridor belongs to central administration.”

“So does the embarrassment.”

Jalen dispatched the crew. Until isolation finished, fourteen staff moved their terminals into the Chamber Below Glass. They entered carrying cable reels and sealed document cases. Delegates shifted their benches to make room. The chamber temperature dropped one point when its doors remained open during the transfer.

After the temporary shutdown, recovered warmth reached the western dormitory. Its reading rose from fifteen to fifteen point six. No one mistook the change for comfort. It demonstrated that the route existed, the neonatal line could remain protected, and hidden administrative demand had absorbed part of the promised relief.

Sera watched the numbers settle. “Put failed transfers in the public ledger. Not just authorized cuts.”

Jalen added the field. Every priority order would report energy released, energy delivered, transfer loss, and unlisted loads encountered between them. A district could then challenge a plan that claimed sacrifice at the source without producing relief at the destination.

Ivo looked at the fourteen executive staff working beneath the glass. Moving them had cost time and heat. It had also converted an invisible protected corridor into visible bodies using shared space. The amendment gained another sentence: essential work could qualify for protection, but its facilities required the narrowest viable circuit and public occupancy records during ration periods.

The dome bloc asked how long “early” lasted. Jalen defined a restoration trigger: once western housing and old-tunnel clinic spaces reached the minimum band for twelve continuous hours, additional cuts could spread by capacity and efficiency. No district received permanent punishment. The order corrected starting conditions before proportional rationing began.

Sera studied the route map. “What enforcement?”

“Public heat ledgers by building class,” Oris said. “Daily audit by physical and branch representatives. Emergency exemptions require medical certification and appear within one hour.”

“Who can stop a false exemption?”

“I can flag it. The Counterweather board can suspend it pending review. Final authority remains with the appointed distribution panel under the Compact.”

“Composition?”

“Two clinic representatives, two weather engineers, two district delegates from below-median heat areas, one dome delegate, and two registered branch proxies without physical vote.”

The dome bloc objected to being outnumbered on a panel controlling its reserves. Western delegates objected to receiving only two seats while carrying the cold. Branch proxies objected to advisory panel seats that could be ignored.

Ivo watched Oris write every objection into the draft. Fair procedure did not make each constituency happy. It made the source of dissatisfaction visible and bounded the power that produced it.

Sera took the amendment page to the west benches. Cooperative delegates argued over a line granting Jalen authority to preserve food-production heat. Some feared the Weather Spine would define crop need without listening to growers. They added a requirement for cooperative confirmation, with a public override if refusal threatened basin food supply.

At the dome benches, the Upper Serein delegate removed her oxygen line and walked the first draft between rows. Ivo saw people touch the contribution clause with their fingers. The amendment acknowledged that previous benefits arose under lawful bargains and imposed no retrospective moral guilt. It also stated that future distribution followed present vulnerability, common survival, and the Compact’s obligations to registered persons.

Past innocence remained in the document. It no longer controlled the valve order.

Governor Rusk approached Ivo near the medical bench. “If we cut administrative heat first, staff will move into the surgical complex and consume protected capacity there.”

“Prohibit nonmedical occupancy.”

“They need somewhere to work.”

“The chamber has heat.”

“You want the executive emergency office on the Assembly floor?”

“I want its bodies counted where their warmth can be seen.”

Rusk almost laughed, then coughed instead. Ivo checked him by reflex. His breathing remained clear.

“Your plan makes governance harder,” the governor said.

“Cold makes everything harder. The amendment decides who stops being sheltered from that fact.”

Rusk returned to his bloc without promising support.

The branch gallery raised a final concern. The public heat ledger protected bodily privacy by aggregating clinic uses. Could the refuge ledger do the same, or would every dormancy schedule become public evidence of vulnerability?

Oris revised the audit terms. Enclave totals would appear publicly. Individual continuity keys and medical-equivalent thresholds remained sealed unless a registrant authorized release. Distribution panels could verify protected load without learning which named person required it.

Ivo approved the symmetry. Physical patients did not owe diagnosis to receive air. Branch-persons did not owe intimate memory to receive continuity protection.

Sera returned to the central floor with a marked page. "The cooperatives consent to the winter route under this amendment, subject to the public ledgers, the restoration trigger, and medical protection across districts."

Oris asked, "Do you require dome cuts to remain first even after the trigger?"

"No. We require correction before proportional burden. We are not writing revenge into weather."

The Upper Serein delegate rose. "Our bloc will not endorse the phrase elite privilege."

"It does not appear," Oris said.

"It appears in every speech about the page."

"Speeches do not control the valve."

She requested a brief caucus. When she came back, seven dome delegates accepted the amendment, four opposed, and two withheld their Compact position. Enough physical delegates shifted that only the final standing dispute separated the proposal from passage.

No vote occurred. Oris attached the distributive amendment to the Orchard Stewardship Compact and sealed it for Day 33 debate. The western cooperatives signed their consent beneath Sera's name. Branch proxies added acknowledgment that refuge floors would enter the same public budget as physical life-support without receiving duplicate votes.

Jalen sent the amended route to the Weather Spine for verification. His model now showed colder elite residences, protected surgical wards, warmer western dormitories, lower old-tunnel morbidity, and more manual switching labor. Basin survival remained within the same narrow range. Distribution had changed without inventing energy.

Ivo returned to the delegate he had treated outside. She sat beneath the glass canopy with her oxygen line removed.

"You cut my district first," she said.

"After protected uses."

“My residents will remember who proposed it.”

“They should.”

She handed him the empty oxygen frame. “This goes back to the clinic pool.”

He checked the gauge. Enough remained for one old-tunnel transfer. A white leaf passed over the ceiling above them as branch traffic moved through the orchard. Ivo carried the cylinder out of the chamber while the new valve order traveled in the opposite direction.

Chapter 25 — The Vote That Would Not Close

Oris Nume removed the sixtieth vote paddle from a chamber built for fifty-nine.

The extra paddle rested at Niko’s proxy desk, identical to those issued to physical Assembly delegates except for a narrow white band around its handle. Someone had printed a number on the base and connected it to the tally circuit before dawn.

“Who authorized this?” Oris asked.

Niko’s proxy, Mara Vey, lifted both hands from the desk. “Not me.”

A clerk traced the issue code. The paddle had come from the Emergency Orchard Committee, whose delegates wanted branch standing translated into a direct Assembly vote. Their note called the white band a temporary equality measure.

Oris disconnected the circuit. “Dependent standing grants voice, petition, protection, and audit. It does not create a duplicate physical vote.”

A committee delegate rose. “Then recognition ends at the door where power begins.”

“It changes who may address power and what power may do. The Assembly remains fifty-nine physical delegates under the Fruition Covenant.”

“You defend a body-only franchise while claiming not to privilege bodies.”

“I administer the franchise this chamber possesses. A constitutional amendment may change it. A disguised paddle may not.”

The delegate demanded a ruling in the record. Oris gave him one, including his objection and the white-band issue code. Procedure could expose a limitation without pretending to cure it.

She placed the extra paddle on her desk.

Morning light traveled through the yarrowglass canopy in thin cream bands. White leaves moved above the pressure panes as registry traffic crossed the orchard. The physical delegate desks held fifty-nine authorized paddles. Branch proxies sat in the gallery with speaking keys and petition seals. Public screens

displayed the advisory ballot waiting in the root lattice.

Outside, Serein Basin remained inside Jalen's six-day stability window. Manual crews held pressure routes. Clinics rationed oxygen. Elite residences had begun surrendering heat under the distributive amendment's provisional order, while western dormitories climbed toward their minimum band. No alarm sounded inside the chamber. The air clock on Oris's desk showed thirty-one hours before the manual plateau required a new cycle.

The Compact occupied twenty-three pages.

It recognized 496 registered branch-persons under Article Six. It prohibited unilateral deletion. It established rotating dormancy, continuity checks, chosen proxies, annual audits, a recovery path for unresolved keys, and Counterweather at reduced adaptation capacity. Physical residents accepted colder housing, industrial rationing, longer manual work, and public distribution ledgers. Earlier contributors received no retrospective guilt for a fact concealed from them. Future decisions could not rely on that innocence to continue known harm.

No page promised enough.

Governor Rusk requested the first objection. "Emergency authority."

Oris recognized him.

"The Compact requires registrant consent before identity-bearing continuity can enter adaptation load. During a pressure collapse, how many minutes may an engineer spend collecting permissions?"

Jalen answered from the technical table. His residual cough had deepened overnight. "No permission allows deletion. Preauthorized dormancy bands and emergency extensions can be used within their terms. Weather crews may draw consultation capacity from the shared reserve. Any need beyond those limits returns to the emergency review panel."

"Which may be unreachable."

"Then no official may convert silence into consent."

Rusk looked toward the public gallery. "Physical residents could die during that restraint."

"Yes," Jalen said.

The chamber tightened around the admission.

He continued. "They could also die under unrestricted pruning if branch weather knowledge disappears, as Airfall demonstrated. The Compact does not remove emergency risk. It defines which acts remain prohibited when risk arrives."

A Severance delegate asked why physical citizens should accept any continuing dependence on the orchard after the sabotage disaster.

"Sovereign machinery should not contain petitioners," she said. "Full separation remains painful, but pain ends. This agreement institutionalizes captivity."

Mara activated Niko's speaking key. Oris checked his continuity signature before allowing the channel onto the chamber feed.

"You triggered separation once," Niko said. "Ninety-one of us ended. Your physical districts lost pressure. Which pain ended?"

The delegate replied, "The protocol was mishandled."

"By people who believed intent could replace a transition plan."

"You ask us to maintain the system that confines you."

"I ask you to maintain relationships no one can sever without killing someone. The Compact gives us authority to negotiate that maintenance. It does not call the orchard freedom."

Oris watched the exchange reach the point where each side wanted the last word. She closed Niko's key after his allotted time and entered the delegate's rebuttal request. Voice required limits if fifty-nine physical representatives and 496 registrants were to reach a vote before air ran out.

The next challenge targeted proxies.

Severance counsel argued that physical proxies contaminated the Assembly by carrying instructions from persons without votes. "A delegate who hears 496 advisory voices cannot remain accountable to physical constituents."

"Delegates hear employers, families, unions, clinics, and district committees," Oris said. "Influence does not become a duplicate ballot merely because its source can be named."

"Branch claimants derive from existing citizens. Their interests already entered the electorate through source families."

"Present registrants have diverged from source citizens. Their property and physical votes do not duplicate. Their petitions also do not collapse into someone else's."

Counsel turned to Mara. "If Niko orders you to threaten a pressure strike, will you speak it?"

Mara's mouth tightened. "A proxy transmits lawful instructions. She does not surrender her own duties."

"Then you filter him."

"I disclose any refusal. He may remove me and appoint another proxy."

"Physical judgment controls the channel either way."

"Physical access does. That inequity belongs in the annual audit. It does not justify closing the channel."

Oris recorded the distinction. The Compact did not solve embodiment. It prevented embodiment from granting unreviewable power over dependent persons.

At midday, the advisory ballot opened.

Each registered continuity key received the Compact text, the distributive amendment, the minority objections, and a ballot with yes, no, or abstain. No proxy could cast it. Blinded Root Jury systems verified key continuity without revealing private memory. Dormant registrants had been revived under their chosen schedules long enough to read and respond. Damaged but registered keys used slower witness routes.

The public tally remained hidden until all valid ballots closed. Oris had required secrecy to prevent late enclaves from treating an early majority as instruction. Several physical delegates objected that branch results should arrive before their own vote. Others wanted the physical tally first so branch-persons could not appear to command the Assembly.

Oris denied both requests.

"Advisory and physical ballots close at the same hour," she ruled. "Neither electorate receives the other's partial result."

The word electorate drew an immediate challenge.

"Advisory body," Severance counsel corrected.

"Registered advisory body," Oris said. "Your correction enters the record. The closure time does not change."

A second unauthorized device appeared at the committee benches. This one proposed converting each branch ballot into one fifty-ninth of a physical vote, pooled as an additional delegate. Its sponsors called the design compromise.

Oris asked who would control the pooled paddle in a tie.

"The branch majority," said its author.

"Would abstentions reduce the paddle?"

"They would not count."

"Then a narrow majority among a small participating group could cast one full physical vote."

"Participation confers legitimacy."

"Except when you dislike physical nonparticipation?"

The author had no answer that survived his own rule. Oris rejected the device. The advisory tally would carry political force because registered persons had spoken. It would not enter the constitutional arithmetic by invention.

During recess, she found Ivo in the side corridor checking oxygen saturation among gallery visitors. He had removed his civic pin to keep it from catching on the tubing.

"How many delegates have asked whether you support the Compact?" Oris said.

"Seven. Four while I held their fingers in a monitor."

"What did you tell them?"

“That a pulse reading does not authorize blackmail.”

“And outside the reading?”

“I support it.”

Ivo sealed a portable cylinder. “I also expect clinic losses. Counterweather keeps people alive with less margin. Some who would have survived the old orchard will not survive the colder plan.”

“Will you state that on the floor?”

“I already did in the morbidity annex. No one quoted that page.”

Oris returned to the chamber and asked him to read it.

Ivo stood beside the amended budget. He described projected asthma admissions, fungal complications, cold injuries, and delayed procedures. He named no exact death count because the uncertainty range could not support one. Then he described unrestricted pruning as a continuing harm to registered persons and full severance as a pressure disaster already tested in bodies.

“Three mercies entered this argument,” he said. “Protect actual lives. Stop inflicted death. Refuse a lie that hides either cost. None prevailed alone. Only the Compact offered a route that kept all three visible while people did the work.”

A clinic delegate asked whether visibility comforted anyone who died.

“No,” Ivo said. “It gives the living no false innocence and the administrators no blank line.”

His testimony moved two undecided delegates toward yes and one toward no. Oris knew because each filed a revised explanation, not because she could know their private decisions. The chamber had begun requiring reasons after the failed stewardship motion. Reasons did not bind votes, but they made later convenience harder.

The final standing challenge came from the Emergency Orchard Committee rather than Severance supporters.

Its delegate moved to strike the prohibition on duplicate physical votes. “A branch-person with dependent standing remains subject to laws they cannot enact. Petition without franchise preserves hierarchy.”

Niko requested time against the motion.

Mara looked surprised. Oris did not. Niko had submitted the finite registry with explicit limits, and the Compact depended on the distinction between dependent standing and physical civic duplication.

“We do not accept voicelessness,” he said through the feed. “We accept this franchise boundary for the term under review. Our advisory vote, proxies, petitions, audit rights, and consent rights carry legal effect. None equals a physical ballot. That inequality remains open. Do not destroy an agreement we chose because it fails your preferred purity.”

The committee delegate asked, "How can a right remain a right if another body may outvote it?"

"By placing some acts beyond that body's ordinary vote. The Assembly cannot authorize unilateral deletion under this Compact, even by fifty-nine to zero."

Oris read the relevant clause. Registered continuity gained protected status enforceable through emergency review and civic courts. Physical majority rule could allocate scarce capacity within consent bands. It could not erase a registrant for convenience.

The motion to strike failed before the final vote. Its supporters filed a future franchise amendment instead.

The defeat opened a second dispute over the five-year review. Orchard Committee members moved to remove it, arguing that recognized persons should not face scheduled reconsideration of their existence. Severance delegates answered with a demand for annual repeal votes. Between those positions, the Compact's actual clause required review without suspending protections during review.

"Rights should not expire on a calendar," the committee delegate said.

"They do not," Oris replied. "The institutions that administer them must answer for what five years reveal."

"A future Assembly could narrow the registry until no one qualifies."

"Any new threshold requires continuity evidence, public notice, branch advisory process, and civic court review. Existing registrants retain standing unless individual evidence shows a failed continuity record under procedures they may challenge."

Severance counsel objected. "Then review cannot achieve full separation."

"It can recommend and authorize a safe transition if one becomes technically and lawfully possible. It cannot call deletion transition."

"Your definition predetermines the result."

"It predetermines that registered persons receive due process. A review may change technology, representation, dormancy rules, capacity allocation, or franchise. It may not erase the claimant to make the claim disappear."

Sera asked who controlled the review record.

Oris read the composition aloud. The Root Jury would retest continuity standards. Weather engineers would publish capacity changes. Clinics would publish physical morbidity. Registered branch auditors would report dormancy and revival failures. District hearings would collect distribution evidence. The Assembly would debate amendments after all reports became public.

A dome delegate proposed letting any future majority delay the review during climate emergency. Jalen opposed her from the engineering table.

“Climate emergency can last a season or become the permanent condition of Nacre,” he said. “Delay authority without a fixed resumption date becomes cancellation.”

They amended the clause to permit a delay of no more than forty local days, once, after a declared basin emergency. Evidence collection would continue during delay wherever safety allowed. No emergency declaration could suspend present continuity protections or public distribution ledgers.

Niko requested one sentence through Mara. “A review that we cannot survive until is not review.”

Oris entered the sentence as testimony, not statutory text. Its meaning already lived in the non-suspension clause.

The Orchard Committee withdrew its removal motion. Severance delegates retained their objection but accepted the filing path as preferable to violence or sabotage. Counsel stated that acceptance of a path did not mean endorsement of the Compact.

“Your distinction will remain in the record,” Oris said.

She sealed the five-year clause before the final tally. The page gave the losing side somewhere lawful to carry defeat and gave the winning side no authority to call today’s majority permanent truth.

At the opposite benches, Severance delegates prepared a last procedural challenge. They claimed the branch advisory ballot had been contaminated by dependence because registrants knew a no result might expose them to deletion.

Oris asked, “Does a physical resident’s dependence on breathable air invalidate their vote on weather?”

“No,” counsel said. “Physical life provides the constitutional constituency.”

“Then your objection concerns standing, already established by the Root Jury and Article Six, not ballot coercion.”

“They vote under existential threat.”

“So do districts facing pressure loss. Threat belongs in the record. It does not erase agency.”

She denied the challenge.

One hour remained before ballot closure. Debate rules allowed each bloc a final statement. Governor Rusk spoke for emergency-pruning supporters who had accepted the amended Compact. He did not apologize for favoring physical survival. He argued that Counterweather now offered a tested mechanism and lawful emergency boundaries. Sera spoke for western cooperatives. She read the restoration trigger and public-ledger requirements rather than recounting their hardship.

Bex Arin appeared by recorded testimony from medical custody. His injured hand rested outside the frame. “We believed severance would free people,” he

said. "Intention did not open the pressure doors. It did not restore the 91. Do not call what we did mercy because we wanted mercy when we did it."

Severance counsel objected that one organizer's regret could not discredit an entire motion. Oris sustained the narrow point. Bex's testimony established his own action and consequence, not every supporter's intent.

Niko used the branch statement to name the dead count without listing 91 people faster than the chamber could hear them. The registry annex held their chosen names where known and continuity designations where names had been lost. He asked the Assembly not to turn them into arithmetic, then acknowledged that arithmetic still governed refuge capacity.

"The count matters because each subtraction reached someone," he said. "The limit matters because pretending no limit exists would reach someone too."

Oris closed the speaking keys at the appointed minute.

Fifty-nine physical delegates placed their paddles into the tally rails. The devices showed receipt but not choice. Branch continuity keys sent sealed advisory responses through the root lattice. Clerks verified missing signals, duplicate attempts, and abstention markings. One damaged enclave requested an additional six minutes under its registered accessibility plan.

The air clock reached zero for ordinary ballots.

Severance counsel objected to the extension.

"It appears in the published procedure," Oris said.

"The deadline should apply equally."

"Equal access sometimes requires a different transmission interval. The authorization predates any result."

She allowed six minutes. The damaged key arrived on the fifth and resolved as an abstention.

No other ballot remained open.

Oris stood at the central record desk. She had delayed Kera's first sealed review. She had stamped the first petition received without making it public. She had watched procedure protect caution until caution became concealment. Today, sequence could protect a vulnerable claim only if she refused to let either righteous faction borrow the result it wanted.

"Close both tallies," she said.

The physical board lit first by circuit design. Thirty-one green marks appeared. Twenty-eight amber marks followed.

The Orchard Stewardship Compact passed 31-28 among 59 physical delegates.

No cheer completed itself. The branch board opened two seconds later: 402-71, with 23 abstentions. Clerks reconciled the sum in public. Four hundred two yes plus seventy-one no plus 23 abstentions equaled 496 registered branch-persons.

Mara gripped the edge of Niko's proxy desk. She did not raise the unused white-band paddle.

Oris read the certification. The Compact took effect at the next Counterweather cycle. The distributive amendment governed heat and oxygen order. Registered continuity gained protection from unilateral deletion. Rotating dormancy began under individual consent schedules. Annual audits would publish capacity, morbidity, distribution, and continuity failures. A mandatory five-year review could revise standing, franchise, technical thresholds, and emergency powers without dissolving present protections during review.

Severance counsel rose before she finished.

The public guards moved closer to the floor. Oris heard pressure doors seal in the outer corridor and saw several delegates turn toward them. No one reached for the tally rails. Counsel held a document, not a tool.

"We reject the Compact's legitimacy and demand full review after five years," he said.

"The Compact already requires that review," Oris replied.

"Then record our demand as adverse, not cooperative."

"I will record its character as you state it."

He brought the filing to her desk. The losing bloc's signatures covered the first page. Their demand sought renewed consideration of complete severance, branch franchise, proxy influence, and emergency risk. Nothing in the review clause guaranteed their preferred outcome. Nothing allowed the majority to prevent the question from returning.

Oris checked the signatures and filing time. "Valid."

A supporter shouted from the gallery that the losing bloc should accept the vote. Oris called the chamber to order.

"Filing review is acceptance of lawful defeat," she said. "Agreement with the result is not required."

She stamped the demand with terracotta wax and attached it to the Compact's review docket. The seal cooled beside the physical tally and the branch advisory count. Above the glass, white leaves continued carrying weather traffic. Below, the fifty-nine paddles remained in their rails, and the sixtieth lay disconnected on Oris's desk.

Chapter 26 — Weather Without Harvest

Jalen Ro held the western condensation gate open with a steel pin while ice formed around his glove.

The actuator had stalled three degrees short of its night position. Under the old orchard routine, a branch comparison would have corrected the timing before any physical gauge noticed. Counterweather had instead sent a yellow request to the manual board, where it waited for a tired crew to read it. Jalen knelt on the service ledge above a black run of oxygen moss and worked the pin deeper with the heel of his hand.

"Gate moving," called Pem from the ladder.

"Not yet." Jalen watched the etched angle marks. "It has slack before load."

He pulled the release cable. The gate shuddered, shed a plate of ice into the channel, and turned through the missing degrees. Condensation struck the waiting vanes in a hard silver sheet. Pem read the catch pressure aloud. Jalen compared it with the value on his wrist board, then coughed into his shoulder until the ache from Airfall eased.

The board accepted the manual confirmation. A second line appeared beneath it: refuge transfer at East Root Court, twelve minutes to handoff.

Pem looked down from the ladder. "You can delegate one of those."

"I did. You have the gate."

"That leaves the refuge."

"It leaves me walking instead of climbing."

Pem made a doubtful sound, but he took the inspection slate. Longer shifts had taught every crew a new courtesy: objections had to save more energy than they cost. Jalen crossed the service bridge before Pem could spend any more.

Six weeks had passed since the Compact vote. Nacre still had weather, though less of it arrived invisibly. Every humidity turn now required a named physical operator, a branch adviser, and an audit trace. The orchard could compare possible outcomes but could not harvest a registered person's sustaining state to sharpen them. Manual models covered the difference badly at first. Crews missed two condensation windows in the opening week, iced a pressure baffle in the second, and sent warm air into a fungal-control trench that had already been sterilized. None of those errors killed anyone. Each one consumed reserve.

The Weather Spine displayed that consumption on a public wall. Industrial oxygen had fallen 38 percent from the old baseline. Elite dome heat remained under the cap imposed by the distributive amendment, while western homes ran four degrees colder during night cycle. The oxygen foundry worked on alternate days. Glass mills had closed two lines. Repair queues lengthened because every welded seam now competed with clinics and moss beds for compressed gas.

Jalen had once presented such figures as a balanced load curve. The public wall added names beside the consequences. Thirty-two glass workers reassigned. Nine apartment blocks awaiting window seals. Two clinic sterilizers restricted to morning use. The numbers had not become less necessary. They had acquired addresses.

At East Root Court, technicians had opened the trunk housing of Tree 61 and laid three continuity readers along its conductive grain. White leaves stayed attached above them. Nobody in the court watched those leaves casually anymore.

Kera stood at the innermost reader with a ring wafer between insulated tongs. She had cut her hair shorter after the severance night, not from fashion but because glass splinters kept catching in it. Beside her, Sera Dovan held the cooperative's witness slate against her coat. Niko appeared on the orchard-linked screen as a field of service characters rendered into a steady voice.

"You are late by twenty-eight seconds," Kera said.

"The gate had slack." Jalen took the vacant station. "Transfer state?"

"Group C dormant and sealed. Group H has acknowledged revival conditions. Their physical witnesses confirmed the keys."

Sera raised her slate. "Western witness confirms. No substitutions, no compressed relationship fields, no altered names."

The transfer involved forty-one registered branch-persons moving out of rotating refuge so another forty could enter dormancy. Their subjective processing would pause. Their continuity keys, relational state, and protected autobiographical cores would move into an isolated root region with lower weather load. Counterweather gained temporary comparison capacity from the newly active group, never from the dormant minds themselves.

Jalen checked the energy margin. "We have room for one failed revival attempt. A second takes condensation reserve."

"Then do not create a second," Kera said.

Niko's voice came through with a faint grain from the trunk. "Group H requests serial waking. One person at a time, witness before the next."

"That takes nine more minutes," Jalen said.

"They prefer nine minutes to a simultaneous uncertainty."

Sera glanced toward the court gate, where two weather runners waited for revised assignments. "Western humidity can hold nine."

Jalen did not round the value upward. He opened the cooperative trace and checked the greenhouse deficit. Seven minutes lay inside the safe margin. Nine brought one moss tier close to wilt threshold. Under the prior system, the orchard would have searched millions of projected responses, retained the most useful branch states, and delivered a decision before his doubt had time to become visible.

"Eight minutes," he said. "Pairs for the first six, then serial if either continuity check shows drift."

Kera looked at Niko's signal rather than Jalen. "Do the waking members accept paired revival?"

A pause crossed the screen. This delay came from deliberation, not distance. "Accepted with separate witnesses."

Sera assigned two cooperative workers to the witness channels. Jalen routed the saved minute into western humidity and signed the trade under his own name. Counterweather required signatures for each deviation because nobody could claim afterward that weather had chosen.

Kera seated the wafer.

The first pair returned as two uneven pulse fields. One gave her name before any prompt. The other asked for the end of a promise made before dormancy: when active again, he would review a greenhouse model that Sera's people had built without orchard projections. His witness supplied the first half of the promise. He completed it with a detail absent from the transfer key.

"Continuity holds," Kera said.

Jalen watched the energy margin narrow. Pair followed pair. Some branch-persons woke quickly. Others required repeated orientation, not because they had changed but because the physical world had advanced through days they had not experienced. A woman named Leth asked how many white leaves had fallen during her interval. Sera answered that none had fallen for deletion, though ordinary conductive leaves had been collected after wind damage. Leth requested the distinction appear in her witness record.

At the sixth pair, one signal returned without its reciprocal link.

Kera froze the sequence. "Do not advance."

Jalen's board showed western humidity sliding. "Ninety seconds before I have to borrow from clinic reserve."

Niko addressed the incomplete signal by a private key phrase. No response came. The active member beside it repeated a shared promise, then corrected herself and tried an older form. A pulse appeared at the edge of the reader.

"State is present," Kera said. "Relationship index has not seated."

"Can you reseat without rewriting?" Jalen asked.

"With the dormant checksum and the witness's live phrase."

"That uses the failed-attempt margin."

Kera's hands had already moved to the second reader. "Then it uses it."

Jalen heard the old answer gathering in his head: one incomplete state against greenhouse humidity, clinic sterilization, and a gate actuator that might fail again before dawn. Capacity language could make any single claimant look unreasonable. The Compact had not forbidden him from counting. It had forbidden him from hiding whom he counted against.

He opened the clinic line. Ivo answered after two rings, breathing as if he had crossed a ward at speed.

"I need four minutes from sterilizer reserve," Jalen said. "A refuge revival has a missing relationship index. We can reseal it without continuity loss if the dormant checksum holds."

"Which clinic cost?"

"South tunnel respiratory instruments start eleven minutes later."

"Patients stable?"

"The weather board says yes."

"I asked about patients."

Jalen waited while Ivo checked with the ward. Through the link came a cough, a tray rolling, someone's low instruction to sit upright. "Stable enough," Ivo said. "Four minutes. Record that we loaned it to a person, not to orchard efficiency."

Jalen entered the phrase exactly.

Kera fed the dormant checksum through a comparison channel isolated from adaptation traffic. Niko repeated the live witness phrase. The lost index found its place not as a restored file but as a verified link between the waking signal and someone who remembered being answered by it. The branch-person gave his name. Then he complained that Niko had used an embarrassingly old promise.

Relief moved through the court as practical irritation. Kera completed the remaining revivals one at a time. Eight minutes became eleven, with clinic reserve carrying the last three. Every revived person met the four continuity criteria. Every dormant key matched its sealed state.

The final audit line reported zero continuity loss.

Nobody applauded. Sera closed her witness slate and began assigning the active branch engineers to greenhouse models. The weather runners collected new timings. Kera removed the wafer, inspected its edges for heat damage, and passed Jalen the resource trace.

"First full rotation," she said.

"First one we nearly lost to a seating fault."

"We did not lose him."

"Because Ivo loaned us air."

"Put that in the season report."

He had intended to. Her insistence still mattered. Jalen signed the trace and sent it to the public wall before leaving the court.

Counterweather settled into work through the following weeks. Manual crews learned where the orchard had concealed labor by performing it too quickly to notice. Pressure-baffle operators began sleeping in four-hour blocks near their stations. Western cooperatives installed ceramic heat jars in common rooms and moved evening meals there, turning rationed warmth into a schedule rather than a private shortage. Elite dome residents appealed the heat cap twice. Both appeals

failed after the public wall showed unused thermal reserves behind their interior seals.

Branch engineers rotated through active duty in groups. They did not touch valves or gates. Their ring-mediated analyses offered patterns that physical crews tested against current instruments. Each recommendation carried a name, a confidence range, and a statement of what memory scaffold had informed it. If a branch engineer entered dormancy before an outcome arrived, the next active group inherited the question but not an assumption of consent.

Mistakes remained. A manual frost prediction came six hours late and cost Sera another strip of moss. A pressure crew overcorrected a southern baffle, sending dust through three old apartments. Jalen published both failures with the saved outcomes beside them. The frost error had protected a clinic intake from cold shock. The baffle error had prevented a larger seal rupture. Counterweather did not divide neatly into success and damage. It made the division reviewable.

During week nine, Jalen visited the western blocks to inspect heat exchange after residents accused the Weather Spine of flattering its own temperature averages. Their complaint proved accurate. Corridor sensors sat too close to communal heat jars and overstated bedroom warmth by nearly two degrees.

Sera met him in a stairwell wearing two coats. "Your public wall says this block meets the minimum."

"My public wall repeats these sensors."

"Then your wall is warm and my people are cold."

Jalen removed the corridor sensor and carried it into the nearest apartment. Frost clouded the inside corners of the window seal. A child had arranged metal cups along the sill to catch condensation before it reached the floor. The bedroom gauge read below the Compact threshold.

He could have recalibrated the local unit and defended the system average. Instead he called the Weather Spine and ordered a basin-wide placement audit. Seventy-six sensors failed the location rule. The correction forced a deeper heat cut in civic offices and delayed glass production another four days.

"Your foundry workers will hate this," Sera said.

"They already hate the oxygen cap."

"Does that make the next loss easier?"

"No. It makes the same loss visible twice."

She studied him, then handed over a cup of hot moss broth from the common room. It tasted bitter and faintly metallic. Jalen drank it because refusing would have converted discomfort into theater.

The placement audit exposed a second problem. Civic offices could surrender heat quickly because their pipes fed a common return, but the released energy could not reach the western blocks without increasing condensation load in two old

tunnels. Jalen spent the next shift in a valve room with Sera, Ivo, and a branch adviser named Mara Fen, whose continuity key identified six years of modeled work on the basin's original thermal loops.

Mara's voice reached them through a ring horn bolted above the main wheel. "The fourth bypass existed in my source memory," she said. "I do not know whether anyone built it."

Sera scraped frost from an access plate. "Our district plans show three."

"Founder plans showed four. One ran beneath a clinic intake that had not opened yet."

Ivo crouched beside the floor map and found the old clinic footprint. "That intake became my second ward."

Jalen compared the physical pressure readings with Mara's remembered route. Her account offered a lead, not authority. Counterweather procedure required them to verify every modeled structure before using it. They opened a service hatch behind a stack of obsolete filter frames and found a ceramic pipe sealed with a brass cap. Dust on the cap had hardened into a skin.

"No current tag," Jalen said. "No pressure mark either."

"Then we do not open it beside a live ward," Ivo replied.

The fastest safe test required draining the neighboring line, which would cool three old tunnel rooms during the evening clinic rush. The alternative required a remote bore and eight hours of labor they had not scheduled. Sera wanted the heat before night cycle. Ivo wanted the ward protected. Mara refused to claim certainty from a memory drawn out of a yielded possibility model.

Jalen put both options on the witness slate. "Remote bore. Western rooms remain below threshold tonight. We issue portable heat jars and log the delay against Weather Spine planning."

Sera's jaw tightened. "Your crews caused the delay, but my cooperatives distribute the jars."

"Then your labor goes on the trace too."

"And if the fourth pipe ends in a wall?"

"We publish that we spent eight hours testing a dead founder line."

Mara's signal sharpened through the horn. "Please add that my memory did not certify the pipe. It supplied a location for physical inspection."

Jalen entered her qualification. Under the old orchard, a useful retained memory would have disappeared into an optimization result. Nobody would have known whether the route came from a stable person, a transient flicker, or a plan the physical colony had abandoned. The new procedure moved more slowly because provenance carried obligations.

They bored through two layers of wall before finding the ceramic run intact. It

terminated at a capped exchange chamber below Ivo's second ward. Pressure tests held. By dawn, crews had joined it to the common return and moved enough civic-office heat west to raise forty-eight bedrooms above threshold without taking energy from clinic air.

Sera signed the completion trace after adding the number of cooperative hours. Ivo entered the ward risk they had avoided. Mara gave her continuity key and requested dormancy only after the first temperature readings arrived. Jalen signed last, beside the planning failure that had left the route undocumented for decades.

Nobody could tell whether the earlier orchard would have found the same solution. Counterweather left them with something the old system rarely provided: a chain of people who could answer for how they had found it.

By week eleven, the season had established a rough rhythm. Industrial oxygen stayed at the mandated reduction. Clinic admissions rose in western districts but remained below the emergency projection. The surviving moss beds recovered half their fungal loss. Four refuge rotations completed after the first, each preserving keys and relationships. The longer shifts produced fatigue injuries: split knuckles, one broken ankle, two heat burns, and a technician who routed a rainfall instruction to an empty cistern after thirty hours without proper sleep.

Jalen responded by cutting every weather supervisor's maximum shift, including his own. Governor Rusk objected that labor scarcity made the rule impractical. Jalen sent him the misrouted rainfall trace and the projected consequence if the cistern had fed a clinic line. The objection did not return.

On the first morning of week twelve, Jalen stood before the public wall to deliver the season report. He disliked speeches, especially when the figures already occupied six meters of glass. Oris had asked him to explain what implementation meant beyond the Compact's language.

He named the oxygen cut. He named the colder homes, the closed glass lines, the clinic admissions, and every recorded refuge fault. He also named outcomes the old system would have hidden inside adaptation: no registered person deleted, no revival accepted without a continuity test, no weather trade left unsigned by someone answerable for it.

A foundry worker in the front row asked whether the orchard could ever regain full capacity.

"Not under the present source base," Jalen said. "Future citizens may contribute modeled years under the new consent rules. Fewer are expected to do so. Slower growth follows."

"So this is permanent decline."

"It is a lower capacity with obligations the prior baseline refused to count."

The worker did not look satisfied. Jalen had stopped treating satisfaction as the test of an honest figure.

After the report, he returned to the Weather Spine for an overdue six-hour rest. Pem had placed a folded blanket on the console bench and marked the manual board RO SUPERVISION CLOSED UNTIL THIRD BELL. Jalen lay down without removing his boots.

The storm alarm sounded before first bell.

He reached the forecast table while the long-range dust instruments were still resolving. Pearl-gray bands gathered across the western projection, then darkened where three upper currents converged. Branch advisers had flagged the pattern during the night, but reduced comparison capacity left the outer route uncertain until physical sensors caught it.

Pem enlarged the pressure field. "Seasonal cyclone?"

"Too early." Jalen traced the convergence with two fingers. Native dust density exceeded the tested manual tables. If the currents joined over Serein Basin, glass particles would bind oxygen while abrading exposed condensation surfaces. Counterweather could turn one front. It might divide two. The forecast carried three fronts into the same pressure throat.

A capacity figure appeared at the margin, recalculated from active crews, industrial reserve, and the next refuge schedule.

Seventy-three percent would not meet it.

Jalen's residual cough tightened before he spoke. "Wake Kera. Keep the next refuge group sealed until we know what we are asking. Call every baffle crew off rest."

On the public wall, the first storm line crossed the western cooperatives and continued east, broad enough to cover the basin.

Chapter 27 — The Last Pruning

Kera Vonn shaved corrosion from Tree 44's pulse jack while the first dust struck the court roof.

Each grain made a dry point of sound against the yarrowglass slats. The strikes came too fast to count and too far apart to merge. She kept the ceramic scraper inside the jack's rim, clearing a conductive path narrow enough for the emergency wafer. If the court lost its fixed line, that contact would carry consent records between the registered branch-persons and the physical crew.

Jalen's voice reached her through the maintenance speaker. His cough broke one word in three. "Western front has joined the upper current. Fourteen minutes to baffle load."

"Pulse jack ready in two."

"You have one."

Kera scraped faster without letting the tool touch the inner pin. Around Tree 44, technicians sealed reader housings beneath padded covers. Nobody touched the white leaves overhead. The Compact allowed emergency compression of unregistered transient fields, but every continuity key on the finite registry required protected refuge or a witnessed voluntary reduction. The cyclone did not care which category consumed capacity.

A larger particle hit the roof and starred one pane above the court gate.

Kera cleared the final green trace, inserted the emergency wafer, and watched Tree 44 accept it. Four hundred ninety-six continuity keys populated the reader. Some signals ran active in the Elseweather. Others lay sealed in rotating dormancy. Ninety-one empty positions remained in the historical roll because the severance dead did not disappear from an audit when their computational states did.

"Niko," she said into the ring horn. "Confirm branch channel."

His field assembled on the reader as a series of pale bars. "Channel confirmed. We have the forecast."

"Then you know the deficit."

"We know Jalen's estimate."

A gust drove dust under the court door. Technicians pulled their face shields down. Kera kept hers open long enough to hear Niko without the horn's echo.

Counterweather could turn the lowest front east and lift the second above the pressure baffles. The third carried too much native glass dust. If it crossed the oxygen-moss fields at forecast density, its particles would bind free oxygen faster than the reduced system could restore it. Jalen needed another twelve percent of comparison capacity for thirty-one minutes. The orchard possessed that capacity in protected memory scaffolds attached to registered persons.

Before the Compact, an adaptation cycle would have selected scaffolds by utility and deleted them without a question. Under dependent standing, emergency review could compel surrender of identity-bearing continuity only through a process the Assembly had never invoked. No time remained to convene it. Kera felt relief at that legal failure. A lawful blade could still cut a person.

"Refuge groups F and J are moving to sealed roots," Niko said. "Active delegates want a full inventory of removable scaffolds."

"Voluntary offers only. No ranking people by useful yield."

"Useful yield already shapes the forecast."

"It cannot shape the consent."

His pause had an interval she recognized from arguments before his death. The shared habit did not make the present disagreement hers to resolve. "Kera, they can see the sky through every linked instrument. Do not protect us from knowing what refusal costs."

She closed her face shield. "Send the inventory."

The list arrived as protected categories rather than memories. Non-self scaffolds stored environments, technical routines, and relationship contexts used to stabilize each branch-person's sustained life. Some could be removed without breaking autobiographical sequence. Others carried the very structure by which a person recognized promises or located prior waking periods. The distinction required a continuity audit before surrender and another after it. Storm pressure left room for one test cycle.

Kera called Oris at the Chamber Below Glass. "I need emergency witness authority for voluntary scaffold release."

"Already drafted," Oris said. Voices crowded behind her, physical delegates arguing under the alarm. "Each person names the field. A reciprocal witness states what must remain. The auditor tests continuity before and after. No group consent, no proxy consent, no silence treated as assent."

"And if the channel drops during confirmation?"

"The offer fails."

"Jalen may lose the turn window."

"Then he loses it without converting interruption into permission."

Kera attached Oris's signed rule to the emergency wafer. The court reader began taking offers.

Before the first field completed transfer, Governor Rusk entered on the civic line. His image failed under storm interference, leaving only a voice flattened by compression. "Weather Spine estimates twelve percent available in registered scaffolds. Recorder Nume's rule will deliver half that if every witness holds."

"Then Weather Spine has counted material nobody offered," Kera said.

"It has counted capacity that exists inside a city facing a glass-dust cyclone."

A technician beside Kera stopped tightening a reader cover. She tapped his wrist and pointed him back to the clamp. Civic arguments could not suspend physical work.

Rusk continued. "The Compact contains emergency review."

"Review requires evidence, representation, and time to contest," Oris said from her line. "You have requested an instruction."

"I have requested use of a provision passed for this exact kind of emergency."

"No. It passed for a circumstance where process remains possible. We cannot turn the absence of process into a faster version of it."

Kera watched the registered fields brighten beneath their sealed categories. Twelve percent represented everything the weather model judged detachable, not everything a person could lose and remain recognizably present. The forecast treated a childhood street, a dead friend's voice, and a learned method as equal

stores if each freed the same comparison load.

Rusk addressed her directly. "Auditor Vonn, can you protect identity-bearing continuity while releasing non-self fields without individual review?"

"No."

"Can you estimate survival after bulk release?"

"I can estimate key persistence. That does not certify the person attached to it."

"How many physical residents enter oxygen danger if the third front crosses?"

"Ivo has that projection."

"You know the range."

Kera did know it. Numbers climbed on the side of her reader as dust density increased. The old tunnel districts would decline first. Western roofs already damaged by the cold season might fail. Pressure rooms could hold only a fraction of Serein Basin.

"Knowing the range does not change what this instrument can prove," she said.

A hard silence crossed the civic line. Rusk had accepted the lawful vote in the Assembly, but acceptance had not turned him into someone who welcomed constraint during catastrophe. "If your process fails," he said, "the record should show that I asked."

"It will show what you asked for," Oris replied. "It will also show who refused it."

Kera entered both statements into the emergency trace. Accountability did not require agreement, and a trace that preserved only principled language would falsify the hour. Rusk remained on the line, listening as the consent checks began. He did not repeat the demand.

The first came from Mara Fen. She proposed surrendering a modeled survey of an unbuilt northern thermal loop, nineteen subjective months of technical detail derived from streets that physical Nacre had never laid. Her reciprocal witness identified what needed protection: the promise she had made during the first refuge transfer, her awareness of dormancy, and her relationship to the engineers who had relied on her fourth-bypass memory.

Kera ran the baseline tests. Mara named yesterday, anticipated the next refuge interval, corrected a false pressure value, and recognized her witness's private phrase. Only then did Kera authorize a copy of the proposed field into the storm reserve.

"Not deleted yet," Kera said. "Staged pending total capacity."

"I understand," Mara replied. "If you do delete it, record that the northern loop never existed outside my source model."

Offer followed offer. A branch-person named Tovan released a mastered language built for a child he had never met in the physical line. Leth offered the detailed layout of a house lost when her branch settlement entered dormancy. Two

others proposed sensory archives of harvests that had not occurred. Kera rejected seven fields because their reciprocal witnesses depended on them. She rejected three more when the owners described pressure from their active enclave.

"We discussed it together," one said.

"Discussion does not invalidate consent," Kera answered. "Fear of disappointing the group might. Take your field off the list until after the storm."

"That could cost actual lives."

"So could accepting a coerced answer and calling it yours."

Above the court, the cracked pane failed. A net caught most fragments. Glass dust sprayed through the opening in a white cone and rattled over the root guards. Technicians dragged a pressure cloth across the gap while Kera covered the emergency reader with her body.

Jalen came over the speaker. "Baffle one engaged. Lowest front turning. I need six percent in eight minutes."

Kera totaled the staged fields. "We have four point three verified."

"Physical alternatives?"

"Ask your board."

"Board says close two clinic intakes or vent foundry reserve into the western moss."

Ivo's voice joined the line. "Closing the intakes puts thirty-one patients on cylinders during a pressure event. Foundry vent buys two percent and leaves no welding gas for damaged seals."

"Take the foundry gas," Kera said.

"It does not close the gap," Jalen replied.

"It closes part of it without taking anyone's continuity."

A new warning crossed Tree 44. The upper storm current carried enough charge to interfere with ring signals. Each active branch-person might soon lose the physical witness required for a valid offer.

Niko entered the inventory.

His proposed scaffold had no technical label. The reader identified it by source date, relational density, and the first mark Kera had recovered from the tree during the isolated test months ago.

Saturday.

She saw the memory before he opened it: a branch-day in which seventeen-year-old Niko had survived airlock fever and come home weak, impatient, and hungry. The physical Kera had never lived that day. Tree 44 had rendered it for her in an eleven-minute isolation, carrying kitchen steam, a mended shoe, the complaint he made about the clinic broth. That Saturday had first proven his life continued beyond her knowledge.

"No," Kera said.

Niko's signal held. "You have not tested it."

"I know what it carries."

"You know what you witnessed through a reconstruction."

"It anchors the first verified interval after your branch point."

"It anchored your belief."

Dust hissed against the pressure cloth. Kera gripped the reader table until the edge pressed through her glove. "Without that day, what remains of your childhood transition?"

"My childhood. The fever. The months after it. Seventeen years of later memory. My promises to living people. My work in the Elseweather." His voice sharpened. "You keep asking what remains as if one Saturday contains me."

"The Root Jury used it."

"The Jury used my continuity across blinded cycles. They did not grant standing to a weather scene."

Kera checked the scaffold boundaries. The Saturday field held environmental detail and predictive context: the exact kitchen, his recovery routine, Kera's branch-version at nineteen, six hours of conversation, and a sequence of ordinary choices the orchard had reused because convalescence affected later risk. Its deletion would remove access to that day as lived recollection. Protected autobiographical indexes on either side would remain. His childhood knock remained outside the field. So did his recognition of physical Kera, first petition, present witnesses, registry name, Omi's relationship record, and every promise formed since sustained continuity began.

"This would not make the day difficult to recall," she said. "It would remove it."

"Yes."

"You offered other scaffolds?"

"Two. Both support active weather analysis. This one contributes more and carries no current technical duty."

"Who serves as reciprocal witness?"

"Omi would have."

The missing name passed through Kera like a gap in a ring sequence. No audit had located Omi after Full Severance. Refuge recovery teams had found damaged keys, unseated relationship fields, and dead ends where the white-leaf collapse had cut through active enclaves. Hope remained possible, never certifiable.

Niko continued. "Omi is unavailable. I named Leth. She knew me before and after that Saturday became part of our shared records."

Leth's signal joined his. "I can witness his present relationships and his continuity through six dormancy intervals. I do not hold the Saturday itself."

Kera wanted to disqualify the witness because Leth could not preserve what Niko meant to surrender. That lack supplied the point. A witness protected the person, not the offered memory.

Oris spoke through the legal channel. "Auditor Vonn, do you have evidence of impaired consent?"

Kera looked at the forecast. Five minutes remained. "No."

"Evidence that the field carries identity-bearing continuity?"

"No. It carries autobiographical content bounded on both sides by protected indexes."

"Then your relationship to the content does not create authority over it."

Kera almost answered that she had not claimed authority. The refusal already had.

She began Niko's baseline test. "State your registered name."

"Niko Vonn."

"Name the last dormancy interval you completed."

"Group C, eleven physical days. Revival witnessed by Leth and Kera Vonn."

"What did I tell you the triangle's widest side was during first contact?"

"You lied and said clockwise. I contradicted you."

"State one present obligation independent of our shared source history."

"I owe Jalen a cyclone route comparison. I promised Leth I would witness her next revival. I promised Omi I would keep searching without declaring survival."

The last sentence nearly changed Kera's voice. She held it level. "What do you anticipate losing?"

"A Saturday I value. Your evidence of it. My own ability to revisit its sensations. I do not expect to lose you, because whatever we are now did not stop growing when that day ended."

Leth gave the reciprocal phrase. Niko completed it.

Kera staged the Saturday.

The total reached six point one percent after physical foundry reserve. Jalen needed six.

"Capacity met," she said.

"Authorize release," Jalen answered. "Thirty-one-minute window begins on your mark."

Kera reviewed the offers once more. Forty-three branch-persons had staged fields. Five withdrew before final authorization, and their portions vanished from the total. Jalen compensated by venting an additional measure of foundry oxygen. Thirty-eight remained. Each gave a second assent. No proxy answered. No enclave leader supplied a group declaration.

The phrase chosen forgetting appeared in Oris's emergency record, proposed by a branch delegate who refused the older word pruning. Kera disliked its neatness. Nothing about the consent sequence felt neat. Still, it distinguished loss taken into a person's own decision from deletion imposed by an adaptation routine.

She pressed the authorization field.

Tree 44 brightened from root to crown. The staged scaffolds opened into computational reserve. Kera watched their protected boundaries hold while the orchard compared storm routes at a rate Counterweather had not reached since the Compact. Jalen's crews turned the eastern baffles by hand. Branch engineers tested pressure outcomes through the released capacity. Foundry oxygen entered the western moss in a cold blue surge.

The Saturday remained visible for eight seconds.

Kera could have stopped it during any of them by breaking the emergency wafer. The court would lose all staged reserve, perhaps the ring channel too. Her hand stayed on the table.

Niko's field gave no farewell to the memory. He worked. "Second front lifting. Baffle four needs two degrees south. Not three. Three brings the upper dust into the old tunnels."

Jalen repeated the instruction to the physical crew. A valve alarm sounded, followed by Pem's strained confirmation.

The Saturday compressed into a white line.

Kera remembered what she had seen through Tree 44: steam on a kitchen window, Niko's mended shoe, his branch-sister setting down a bowl. Those images persisted in Kera because she had witnessed them. Niko would not possess the lived path back into them. Her memory could not restore his, and she had no right to offer it as a replacement.

The white line went out.

Continuity alarms remained silent.

Dust pressure struck East Root Court hard enough to bow the cloth over the broken pane. Kera dropped beside the emergency reader and held its clamp while technicians braced the table. Through the roof, white leaves shook without falling. The cyclone filled every external instrument with static.

For twelve minutes, Jalen's crew had no view beyond the baffles. Niko and the active branch engineers read the pressure changes conducted through roots. They sent corrections through Tree 44. Physical hands turned wheels, cleared grit, and opened the foundry reserve. Kera repeated every consent boundary as the orchard consumed only the released scaffolds.

A baffle failed in the north. Clinic intake pressure fell. Ivo moved patients onto cylinders and refused to close the ward line. Jalen turned the second front across an unplanted strip of western ground, where abrasive dust stripped greenhouse

roofs already weakened by cold.

Then the lowest pressure trace bent east.

The next followed above it. Native dust density peaked over the dead strip instead of the moss fields. Oxygen binding slowed. Counterweather's reserve calculation moved from red to amber.

"Registered keys?" Kera asked.

A technician read the continuity board. "Four ninety-six present."

"Run relationships."

Witness links populated one by one. Some came weakly through storm charge. None failed. When Niko's key appeared, his surrounding autobiographical index showed a clean absence where the Saturday had attached. His present links held: Leth, Kera, Omi marked missing rather than severed, the finite registry, three active promises.

"Niko," Kera said.

"I am here."

"What did you give?"

"A field I chose to stop carrying."

"What remains between us?"

He did not answer with the childhood knock. "A disagreement you finally allowed me to win."

Kera laughed once, without pleasure and without grief alone. "Continuity confirmed."

By dawn, the cyclone had turned past Serein Basin. It left scored roofs, broken greenhouse glass, depleted foundry reserve, and air that tasted of stone through the filters. Tree 44 carried every registered key. Its crown had lost ordinary leaves to wind, but no white-leaf deletion marked the court.

Kera removed the emergency wafer after the jack cooled. The consent record held thirty-eight names, thirty-eight witnessed choices, and thirty-eight absences no authority could claim as restoration work for later. She sealed it for the public audit.

Niko's signal touched the ring reader while she cleaned dust from its edge. Not three, two, one. A new interval, unfamiliar to both their childhoods, asked whether the line remained open.

Kera answered it in the same new measure.

Chapter 28 — The Settlement of Breath

Ivo Senn fed a cleaning wire through the ward's oxygen rail and caught a clot of glass dust at the second bend.

The wire stopped hard enough to bruise his thumb. He twisted until gray grit appeared behind the transparent valve, packed with condensation into a paste. Cyclone seals had kept the largest particles outside. Smaller grains had entered on clothes, filter changes, and the hair of patients carried through failing doors. Three weeks later, Nacre still breathed the storm out of its pipes.

"Again," he told the junior medic at the wall regulator.

She opened the purge. Compressed oxygen drove the clot into Ivo's catch cloth with a wet snap. The rail pressure rose by four points. Beds seven through twelve received their full morning flow for the first time since the cyclone.

Tomas Dey had died in bed nine before the pressure returned.

Ivo folded the cloth around the grit and wrote the bend number on it. The ledger team wanted physical samples attached to disputed cases. People argued less cleanly in the presence of what had blocked a pipe.

Outside the ward, patients waited under two blankets each. The old tunnel clinic ran colder than it had before Counterweather. Elite dome heat had been cut first, as the distributive amendment required, but equal cuts could not erase unequal walls. The clinic's outer blocks had sixty-year-old seals and stone that kept no warmth once the corridor lamps dimmed.

A nurse handed Ivo the night list. Four respiratory flares, one frost-burned hand, two crew strains, no free beds. "The Recorder's office called twice."

"They know where I work."

"They say the publication window closes at noon."

"Tomas did not close at noon."

She glanced toward bed nine, now stripped and waiting. "His daughter is here."

Ivo put the dust sample in his coat and went to speak with her before any committee received his time.

Miri Dey sat in the family alcove holding her father's filter mask. She worked at a glass mill that had closed under the industrial oxygen cut. Her hands carried the pale scratches of someone who still visited a workplace with nothing to make.

"The ward rail ran below prescribed pressure for twenty-six minutes," Ivo said. "We transferred him to a cylinder after nine. His blood oxygen improved, then fell again. The storm damage contributed. So did his lung disease. I cannot divide one breath into percentages after it has stopped."

"Would he have died under the old orchard?"

Ivo could have offered probabilities. He had helped design enough clinical projections to know how quickly a range became an alibi. "Not on that morning, if the old system had remained intact and no severance had happened. I cannot tell

you how long it would have kept him.”

“Then put that in your report.”

“I will.”

“And put who chose this system.”

“The Assembly passed it. I spoke for it.”

Miri turned the mask over in her hands. “That sounds like everyone and no one.”

Ivo sat opposite her. His knees objected before the rest of him settled. “Jalen signed the routing. The maintenance office delayed the filter delivery. I kept your father in the old tunnel clinic rather than moving him to a dome ward because transfer pressure posed its own risk. The western delegates accepted colder housing after heat cuts moved upward. Four hundred ninety-six registered branch-persons accepted dormancy and audit. We can name all of that without pretending the decisions share one kind of responsibility.”

“Will your ledger say the branch people killed him?”

“No.”

“Will it say saving them cost him?”

“It will say the reduced system contributed to the conditions of his death.”

Miri closed her eyes. “He yielded eight years. Did you know?”

Ivo had checked the record. “Yes.”

“He used to say those years bought him air.”

“They helped build what kept him alive until now.”

“And then the promise changed.”

“The facts inside it changed first.”

That answer did not comfort her. Ivo had not come to comfort away a civic debt. He asked what name Tomas used at work, which grandchildren visited, and whether the family wanted the final oxygen trace public or sealed behind a research key. Miri chose public. She wanted nobody to summarize her father as an anonymous cost.

At noon, Ivo carried the clot of dust into the Chamber Below Glass.

The Assembly floor had become an accounting room. Clinic slates covered one table. Weather failures, heating complaints, labor injuries, and refuge audits covered another. Branch representatives attended through three orchard-linked readers watched by physical proxies. White leaves above the chamber remained attached, though storm scoring had turned their edges dull.

Oris stood beside a projection titled FIRST COUNTERWEATHER HEALTH ACCOUNT. She had refused the governor's request to call it a balance sheet. “A balance implies that one harm pays a different person back,” she told the room. “This document records distributions.”

Ivo placed the dust cloth beside Case 19. Tomas's name replaced the provisional code.

The first morbidity ledger covered the pause, Airfall, implementation season, and cyclone. Sixty-three respiratory admissions exceeded the prior seasonal range. Fourteen residents required treatment for cold exposure. Weather and orchard crews suffered nineteen fatigue injuries after manual shifts expanded. Industrial closures delayed two hundred eleven nonurgent procedures because sterilizers and fabrication shops shared oxygen limits. Tomas Dey's death carried Counterweather conditions as a material contributor. Two other deaths remained disputed pending family review and technical evidence.

A separate column listed harms prevented, never subtracted from the first. Old tunnel oxygen routing had improved after Ivo's refusal during the pause. Elite heat cuts funded western temperature corrections. The verified thermal bypass raised forty-eight bedrooms above threshold. The cyclone turn protected the main oxygen-moss fields. No registered branch-person had been deleted during implementation or storm triage. Thirty-eight had voluntarily surrendered bounded memory scaffolds.

Governor Rusk read the columns from his seat. "Publishing both invites every faction to quote only its preferred side."

"They already do," Ivo said. "Publication lets everyone else inspect what they omitted."

A foundry representative pointed at the delayed procedures. "My workers lost shifts so clinics could retain oxygen, and clinics still postponed care."

"Your oxygen also ran the emergency vent that turned the cyclone," Jalen said. He sat near the aisle with a cup of water for his residual cough. "The ledger traces both uses."

"Does tracing reopen our mill?"

"No."

"Then what does it do?"

Miri Dey answered from the public seats before Ivo could. "It keeps you from telling my father he died for nothing or for everything."

The room shifted toward her. She held Tomas's mask against her coat. Ivo had not known she would attend.

Oris invited her statement into the record. Miri walked to the witness place and repeated what Ivo had told her. She did not endorse the Compact. She asked why maintenance delays had left glass dust in a medical rail three weeks after the storm. The maintenance director named the oxygen ration that slowed pipe fabrication, then admitted his office had prioritized a civic complex above the old tunnel clinic on the first replacement list.

"Whose instruction?" Oris asked.

“Mine,” he said.

Ivo watched the admission enter the ledger. One ordering decision had not caused Tomas's death alone. It had changed his chance. Accountability often arrived at that scale, too small for a public villain and too large for innocence.

The Assembly approved publication without a compensating total. Families could append testimony. Clinic cases would receive quarterly review. Jalen's office had to publish sensor placement and maintenance priority. Branch auditors would add their own cycle records beside physical morbidity, not beneath it.

Publication did not settle repair. Oris kept the room in session and opened a second slate labeled CORRECTIVE DUTIES. The distinction annoyed delegates who preferred one vote to conclude a subject.

Miri requested replacement of every old tunnel medical rail before civic-complex pipework resumed. The maintenance director said available fabrication oxygen could not support both. A civic administrator argued that his complex housed emergency dispatch, registry clerks, and pressure coordination, not ceremonial offices.

Ivo unfolded the dust cloth again. “Then rank functions inside the complex. A dispatch rail may precede a clinic wall panel. A recorder's wash line may not.”

“That building has one maintenance contract,” the administrator said.

“Bodies do not breathe contracts.”

Jalen pulled the fabrication schedule onto the public glass. Separating critical lines would cost two crew shifts and require new isolation valves. It would also expose which civic uses had hidden behind the emergency label. The administrator asked for a closed technical review.

“No,” Miri said from the witness place. “You put my father's pipe behind your building once. You do not get a private room to explain doing it again.”

Oris ruled the schedule public. She had no power to choose which rail received steel, but she could prevent the evidence sequence from disappearing into procurement language.

A western nurse then raised fatigue staffing. Manual weather shifts had drawn trained technicians out of clinics, while the cold season increased respiratory work. “Your ledger counts nineteen crew injuries,” she said. “It does not count the medication errors caught before reaching patients. We stopped seven in my clinic alone.”

Ivo had included near misses in an appendix, separated from morbidity because no injury followed. Her objection exposed how easily that separation could make successful prevention invisible.

“Move them into the main table as intercepted harms,” he said.

Governor Rusk frowned. “Then every competent correction becomes evidence against the system.”

"Every repeated near miss becomes evidence about strain," Ivo replied. "Competence does not make the cause irrelevant."

The revised ledger gained a new category. Clinics would report intercepted dosage, routing, sterilization, and pressure errors without naming the worker in public. Supervisors remained accountable through protected records. The fatigue cap Jalen imposed after the cistern mistake extended to medical crews. Rusk warned that staffing gaps would force service reductions.

"List them before they occur," Ivo said. "No ward learns its night coverage vanished when a door fails to open."

Corrective duties accumulated. Medical rails received priority by function rather than district prestige. Families affected by a documented weather contribution gained travel and heating support without signing away the right to dispute causality. Sensor placement audits became quarterly. The foundry received a published reopening sequence tied to oxygen reserve rather than executive discretion. None of those measures brought Tomas back or warmed a bedroom that night.

Miri read the final duty attached to her father's case. It ordered an independent inspection of every bend in the old tunnel clinic rails and required public confirmation after cleaning. "Who inspects?" she asked.

"Clinic maintenance and one outside engineer," Oris said.

"And a family witness."

The maintenance director started to object, then looked at the dust cloth. "A family witness may observe the pressure test."

"I will," Miri said.

Ivo watched her name enter the corrective record. Grief had not turned her into an instrument of the Compact. She still believed a different settlement might have spared Tomas. The process did not demand that she change that belief before her evidence could alter a pipe schedule.

Only after duties had names and dates did Oris close the accounting session. Delegates left carrying more work than they had brought. Ivo preferred that outcome to unanimous sorrow.

After the vote, Ivo crossed to the branch readers. Mara Fen had active duty that day. Her rendered voice came through the center instrument.

"We need your clinic comparison for cycle twelve," she said.

"What failed?"

"Nothing yet. The weather recommendation draws on a respiratory adaptation scaffold from an unregistered transient field. It predicts that old tunnel residents tolerate a lower humidity range than dome residents."

"They do not."

"The model confidence says they do."

"Because the records contain decades of untreated symptoms."

"That interpretation requires a physician's signature."

Ivo collected his coat. Branch representatives now audited every adaptation cycle before release, but they could not inspect physical histories without witnesses. The change had produced arguments at hours previously reserved for sleep. He preferred them to invisible assumptions.

At the Weather Spine, Mara displayed the model's evidence through a ring-linked board. Old tunnel residents had fewer acute humidity complaints per capita. Jalen's crew had treated that as adaptation. Ivo opened clinic access records and laid missed appointments over the curve. When pressure doors failed, patients skipped visits. When contribution benefits controlled clinic priority, many delayed treatment rather than risk work certification.

"Lower complaint rate measures access," he said. "Not tolerance."

Mara adjusted the confidence field but did not issue the physical command. "What range do you authorize?"

"The same minimum as the central districts, with an uncertainty note for damaged seals."

"That costs one point of greenhouse humidity."

Sera Dovan joined by voice. "Which crop tier?"

"Outer medicinal moss," Jalen said.

Ivo pictured the inhalers derived from that moss. A clean answer remained unavailable. They could protect tunnel lungs now and reduce medicine later, or preserve the crop by repeating an inequality disguised as physiology.

"Alternate the deficit," Sera said. "One cycle from outer moss, next from civic garden irrigation."

Jalen looked at the load board. "Gardens fail after four turns."

"Then review after three."

Mara entered the proposal and asked each representative to state the cost carried by their constituency. Ivo named reduced medicinal yield. Sera named crop stress. Jalen named crew time and model uncertainty. Only after all four signatures appeared did a physical operator release the humidity instruction.

No branch signal moved a valve. Mara stayed on the channel while a technician turned the wheel and reported pressure. The audit preserved analysis, authority, execution, and consequence as separate acts.

Cycle twelve reached the old tunnels before evening. Patients coughed less through the night. Two outer moss trays yellowed. The civic gardens gave up their next irrigation turn. Nobody called the outcome optimal.

During week sixteen, Ossan Mire's censure hearing opened in the Yielding Office

rather than the Assembly chamber. Oris chose the smaller room because spectacle would not improve the record. Cream walls held forty-seven years of certificates claiming no person continued in unchosen projections. Ossan sat beneath them in an ordinary chair.

Ivo attended as a witness for the consequences of concealment. Kera presented founder records. Jalen explained how the phrase adaptive residue had allowed engineers to treat continuity as reusable state. Branch proxies read statements from survivors and named the ninety-one severance dead. Bex Arin submitted written testimony admitting that founder deception had fed the movement's certainty, while refusing to excuse the physical sabotage.

Ossan did not deny the nineteen-year decision. "We had less than two days of reserve," he said. "I believed disclosure would trigger a shutdown. I changed language to preserve compliance."

"You preserved the orchard," Oris said. "Did you preserve the possibility of later correction?"

"No. I made later correction harder because each quiet year increased what recognition would cost."

The censure removed his founder's honor, barred him from unsupervised orchard work, appended the concealment finding to every surviving early model, and required his testimony at annual audits while he remained able. His remaining yield allocation had already funded the registry's first refuge cycle. The Assembly did not imprison him. Age did not erase culpability, and punishment could not recover deleted people.

A survivor's proxy asked whether he accepted the words public censure.

"I accept them," Ossan said.

"Do you ask forgiveness?"

"No. Asking would assign you work after mine."

Ivo distrusted polished answers, even accurate ones. He watched Ossan's hands instead. They rested flat on his knees while the terracotta seal struck the order. He did not reach for the founder pin removed from his coat.

Outside, two demonstrators demanded prison. Three former orchard workers stood opposite them holding signs that called Ossan a necessary liar. He passed between both groups under civic escort. No one struck him. Nobody looked reconciled.

By week eighteen, the clinic had cleared dust from every ward rail. Beds remained full. The cold had eased by one degree as damaged greenhouse roofs came back under pressure. Ivo slept six hours on most nights and distrusted the improvement enough to count each one.

Oris asked him to attend the first yielding appointment under the Compact. Medical presence had become mandatory when contribution carried health or

benefit implications. The youth, Anet Vale, arrived with a parent and a folder of housing forms.

Anet sat in the same kind of chair Lessa had once occupied. The certificate now opened with a different statement: modeled futures could develop sustained persons; no contribution could be activated without continuing consent review; refusal or delay could not reduce housing, education, medical access, or civic standing. Teren Vale's four-year certificate sat in the review tray with a notice that no further modeling would begin until he received the facts he had asked for on Day 2 and answered again.

The parent read that clause twice. "Our housing application lists contribution history."

Oris slid a correction seal across the desk. "The field no longer appears on allocation forms."

"It appeared yesterday."

"Then the office used an obsolete form. I will void it."

Anet looked at Ivo. "Does the orchard need years?"

"Yes," he said. "A larger voluntary source base could improve future weather capacity."

"Will someone live in mine?"

"Most modeled possibilities never sustain continuity. Some may. We cannot promise otherwise."

"If I say no, do western homes stay colder?"

"Your single decision does not control a district. Enough refusals will keep the orchard slower. That cost belongs in the information you receive. It cannot become a threat attached to your certificate."

The parent pressed both hands around the housing folder. "We can come back?"

"Any year," Oris said. "Or never."

Anet read the consent options. Contribute now. Set a future review date. Decline without review. Defer without date. The old form had offered only a number from zero to twenty, with zero technically lawful and socially marked.

Ivo thought of Lessa's twelve years, Tomas's eight, Kera's fourteen, and Ossan's remaining allocation. Earlier gifts had built clinics and weather. Their meaning did not require new children to repeat them under concealed terms.

Anet chose the fourth box.

Oris asked for the decision aloud because the new record required voice as well as seal.

"My child chooses to defer," the parent began.

Anet took the stamp before the sentence could finish. "I choose to defer."

The terracotta seal met the paper. No benefit field changed on the housing folder. No alarm sounded in the orchard. Beyond the office wall, yarrowglass roots continued their slower work with fewer futures promised to them.

Ivo checked Anet's breathing out of habit. It remained quick, uneven, and entirely their own.

Chapter 29 — The Years We Choose

Kera Vonn aligned the ring guide against Tree 44 while Omi argued with the calibration tone.

"It sounds wrong," Omi said through the root-court reader.

"It sounds exactly as specified."

"That can still be wrong."

Kera adjusted the tone down by half a step. Omi listened from protected Elseweather refuge, her signal carried through the same trunk that had once hidden a seventeen-minute gap. "Better," she said.

One Nacre year had passed since Kera first cut this tree at night. Four hundred two local days had laid a new band of translucent wood around the scratched rings. Counterweather growth ran narrower than the orchard's old baseline. The latest ring held fewer amber threads from yielded possibility models because new contributions had slowed under informed consent. Its grain also carried clean witness marks at each refuge rotation.

Kera seated the cutting arm but left the deadman plate untouched. "State your audit name."

"Omi."

"Registered witness?"

"Niko Vonn for Elseweather continuity. Kera Vonn for physical contact."

"Protected state?"

"Active refuge, East Root group four. No unreviewed compression. No restoration claims."

Omi added the last phrase every time. Recovery teams had used it so often during her return that she had adopted it as a rebuke.

For eighty-three days after Full Severance, Omi had remained a missing name attached to a damaged routing index. The collapse destroyed the directory path to her enclave. It did not delete the sealed root pocket where a refuge operator had moved three children when alarms began. Search teams found the pocket through an old witness trace, then spent nine days proving its contents had stayed dormant rather than reconstructing them from partial records.

Omi's continuity key matched the pre-severance registry candidate. Her protected autobiographical core remained intact through the gap. Reciprocal witnesses confirmed relationships formed before dormancy. She woke missing eighty-three physical days and nothing from inside them, because dormancy held no subjective time to recover.

No lost state had come back. None of the ninety-one returned with her.

The recovery pocket remained under Tree 52, sealed behind a transparent plate so annual auditors could inspect it without reopening the root. Kera had visited it with Omi's channel the previous afternoon. The pocket looked too small to contain eighty-three days of public doubt: a ceramic sleeve, three dormant wafers, and a scorched routing collar whose directory marks ended halfway around.

A recovery engineer had demonstrated the failure again. Full Severance tore the collar's outward index from the lattice. The sealed wafers kept their own local clocks, continuity keys, and precommitted wake conditions. Without an external path, they neither processed nor answered. Search teams first mistook the silence for deletion because standard readers asked the damaged directory where to look.

Omi had interrupted the demonstration. "Show the witness trace."

The engineer loaded the low-current marks left by the refuge operator who moved the children. Each mark named a destination pocket and closed with a checksum. One digit had fused under the severance charge. Niko and two other witnesses reconstructed the missing destination digit from their separate promise records, then technicians used it only to locate the sealed sleeve. They did not use witness memory to fill any child state.

"Could you have made a false Omi from those records?" Omi asked.

"We could have made a model that resembled descriptions of you," the engineer said. "It would not carry your sealed key or local clock."

"Would you have registered it?"

"No."

Kera asked the harder question. "If the sleeve had failed its checksum but retained fragments?"

"We would preserve evidence, classify the state unresolved, and refuse a continuity claim. We would not assemble a person for a reassuring result."

Omi stayed silent long enough that Kera checked the channel. "Good," she said at last. "Put that answer on the public tour."

The recovery exhibit now carried it. Visitors who wanted a miracle found damaged ceramic and an explanation of restraint. Families of the ninety-one could see why Omi's return did not imply that their dead waited behind another missing directory. Omi disliked being used as proof that refuge always worked, so the exhibit also listed every failed pocket and every key that never answered.

Kera had learned of the find at East Root Court, where the recovery team

showed her a live key and asked whether she recognized it. She refused to answer until blinded checks finished. Recognition from a hopeful physical witness could not substitute for continuity. Omi later told her that this refusal counted as the first useful thing physical Kera had done for her without pretending inheritance.

Now Omi said, "You skipped branch-age field."

"Branch-age varies with compute allocation."

"The form still asks."

Kera looked at the annual audit slate. "Apparent age ten. Your rendered body crossed the threshold during the active months after recovery. The eighty-three dormant days remain outside the subjective count."

"Thank you."

"You dislike ten?"

"I dislike adults deciding variable means optional."

Kera almost smiled. Omi's voice had changed during the year, not enough for an instrument to quantify and enough for anyone who listened to notice. She still feared white leaves during unscheduled current shifts. She had begun collecting weather jokes from active crews, most of them worse than the calibration tone.

The protected refuge housed forty-seven branch-persons in rotating groups. Omi attended lessons with six children and three adults who had chosen to retrain after scaffold surrender. She served on the refuge schedule council as a youth observer. She still refused the word daughter when anyone used it between her and physical Kera.

Kera had stopped waiting for the refusal to soften.

"Ready for contact test?" she asked.

Omi sent a two-pulse assent. Kera pressed the deadman plate and guided the arm into the newest ring. Yarrowglass yielded a thin curl, clear at the edge and cloudy where cyclone stress had entered the growth. The wafer came free without touching the older scratch left by her first private test.

She set it in the reader. Civic time and ring time appeared in parallel.

No lag.

"Too easy," Omi said.

"Accuracy does not become suspicious because you are bored."

"Your first contact involved secret knocks, a stolen wafer, and a false disposal log."

"You have been reading restricted evidence."

"I have dependent standing and a supervised research key."

"You have Niko."

"That too."

Kera marked the clock result. The old hidden interval had not recurred since the Compact separated registered continuity traffic from weather optimization. Signals no longer needed to shelter inside unacknowledged storage. Tree 44 carried refuge state on named channels with physical witnesses and public load limits.

The court around them showed the cost of that clarity. Two roof panes remained mismatched after cyclone repair because the glass mill still rationed oxygen. Cold had cracked the east root guard. A manual condensation wheel occupied the space where a sealed orchard controller once worked. Every shift left chalk names beside its settings.

Beyond the gate, the sky held a pale seam of dust over the western ridge. Breathable did not mean clear. Counterweather had brought the basin through a full cycle without another mass emergency, though humidity failures damaged two moss tiers and winter cold raised clinic admissions again. The second morbidity ledger had no storm death like Tomas Dey's. It still contained names.

Two annual-audit observers waited beneath the west slats. One represented the bloc that had voted against the Compact. The other served the registered branch advisory roll through a physical proxy license. Neither had authority to halt Kera's cut, but both could challenge its trace.

The physical observer introduced himself as Cor Bel, a pressure contractor from the north baffles. "Five-year review demand, evidence year one," he said for the recorder. "I challenge the claim that slower source growth can sustain population."

"You may challenge a forecast after it appears," Kera said. "This stage verifies the ring clock."

"I challenge the audit boundary. Clock accuracy tells us nothing about whether consent rules leave enough weather."

The branch proxy spoke next. "And capacity tells us nothing about whether a system preserves people. Both objections belong in the annual record, not in the same measurement."

Cor looked toward Omi's reader. "The settlement spends weather reserve teaching branch children while physical schools close rooms for heat rationing."

Omi answered before Kera could control the sequence. "My lessons use active refuge allocation already counted in the Compact. Your schools closed two rooms, then moved classes into the civic hall after its heat cut."

"That still costs us."

"Yes."

The direct answer unsettled him more than defense might have. Cor opened his own evidence slate. North-baffle maintenance hours had risen forty-one percent. Two apprentices left the trade after fatigue restrictions reduced overtime benefits. Replacement parts waited behind clinic rail fabrication. He wanted the five-year review to permit a larger emergency pool of branch scaffolds.

Kera attached his figures to the annual audit. "Name whether release would remain voluntary."

"If the forecast crosses a survival threshold, no."

Omi's signal contracted into a narrow band. "Then name whose memories enter your pool."

"I cannot name future emergencies."

"You can name the rule that selects us."

Cor rubbed dust from the slate edge. "Utility to weather."

"Then people with technical memory lose more of themselves because you need what they know."

He did not withdraw the proposal. The branch proxy entered Omi's objection beside it, along with a request to measure how emergency selection would concentrate loss among engineers. Kera added the cyclone record showing voluntary release had met the deficit once, without claiming it would always do so.

"The five-year review remains open," she said. "Year one supports neither repeal nor permanence."

Cor sealed his evidence. "You expect me to wait four more years while capacity shrinks."

"I expect you to build a record capable of changing law without pretending the result has arrived."

He stayed for the rest of the audit. Opposition had not vanished after losing the vote. It had acquired inspection rights, deadlines, and an obligation to describe its blade.

Omi asked, "Will you show me the scratch?"

Kera rotated the old wafer's archived image onto the reader. The small triangle remained beside its doubled white line. Physical evidence no longer needed concealment, but the original sample stayed in sealed civic storage because taking it out would weaken Tree 44's chain of custody.

"That mark proved someone could see new information," Omi said.

"It helped."

"You thought the person might belong to you."

"I thought shared history answered identity."

"Do you still?"

Kera inspected the new wafer's edge before answering. "Shared history creates evidence. Relationship requires present agreement."

Omi considered this. "Kin is acceptable."

The word landed without the shape Kera once wanted. No inherited parenthood. No branch claim folded into physical expectation. Kin supplied proximity with a

boundary.

"Witness is also acceptable," Omi added. "Mother is not."

"I recorded that the first time."

"You recorded it. You took longer to follow it."

"Yes."

Omi changed the reader display to the day's refuge schedule. "Group four enters dormancy after the forecast review. I requested you for revival witness."

Kera looked up. "You already have Niko."

"He witnesses my Elseweather relationships. I want a physical witness who knows the difference between a missing route and a deleted state."

The request carried no promise of parenthood. It offered work.

"I accept," Kera said.

"Put it in the key."

Kera added her name to Omi's next dormant interval. The slate required both signatures. Omi entered hers through the ring channel. Their marks met at the witness field without becoming the same mark.

A tram bell sounded on the ridge. Morning crews crossed the orchard in pairs, carrying manual weather boards and insulated tool packs. Longer shifts had eased after Jalen's fatigue rules, but more people worked weather now than before the vote. The settlement had exchanged invisible computational labor for visible human time.

Anet Vale came through the court gate with a tray of replacement pulse seals. The yielding office had assigned civic service unrelated to contribution history, partly to prove its own forms meant what they said. Anet wore a housing tag for a western block above the threshold line.

"I brought the narrow seals," Anet said. "Stores tried to send the old diameter."

Kera checked the tray. "These fit refuge readers. Leave six."

Anet glanced at Omi's display. "Are you going dormant today?"

"After forecast review."

"What does it feel like?"

"It does not feel. That distinguishes refuge from waiting awake."

Anet shifted the tray against one hip. "I deferred again."

Omi's signal brightened. "Did they ask properly?"

"Four times, because the education allotment office still had a contribution note in its local copy. Oris made them remove it while I watched."

"Good."

"My parent says I may contribute later."

"You may." Omi paused. "Later does not own you either."

Anet carried the remaining seals toward the west trees. Kera watched them stop at each maintenance post, counting the fittings rather than scattering them by habit. One deferred source meant little to the day's weather. Thousands of choices like it shaped the orchard's future capacity. The settlement had not solved that scarcity. It had stopped concealing the authorship of it.

Jalen arrived with frost on his boots despite the mild morning. His cough had faded to an occasional roughness. He placed a manual forecast board beside Kera's reader and studied Tree 44's clock trace.

"On time?"

"So far."

"That sounds disappointed."

"Omi expected more drama."

"Drama exceeds our reserve."

Omi sent the calibration tone at him. Jalen winced. "That frequency violates the Compact."

"Cite the clause."

He opened the day's forecast instead. Three weather routes appeared: a colder western rain that protected moss growth, an even basin mist that consumed industrial oxygen, and a delayed dry front that preserved reserve while increasing old tunnel respiratory risk. Under the prior orchard, millions of possibilities would have narrowed the field before any public view. The new process brought a smaller set into shared review.

Sera's cooperative preferred the rain. Ivo's clinics opposed the dry front. The foundry requested the mist only if its reopening schedule kept the promised oxygen floor. Active branch engineers had compared each route using voluntary, documented scaffolds. No single forecast dominated.

"Where is Niko?" Kera asked.

"Completing the witness notes," Omi said. "He said you would object if the ring carried a conclusion without its uncertainties."

"He knows me."

"He knows what you do now."

The correction mattered. Kera let it stand.

Niko's signal entered Tree 44 at a direct interval, no hidden lag. His rendered voice sounded thinner than it had before the cyclone. Losing the Saturday had changed what he could revisit, not the continuity tests or relationships that let him speak as himself.

"Kera," he said. He no longer called her sister unless referring to the shared childhood before his branch point. She no longer called him her dead brother's

survival.

“Niko.”

“Forecast notes require a physical audit. Route one protects staple moss and worsens western roof stress. Route two uses foundry oxygen but keeps district humidity within the equal minimum. Route three preserves industrial reserve and adds an estimated respiratory burden the clinics reject.”

“Any scaffold surrendered for this comparison?”

“No. Active duty groups carried sufficient capacity. Two advisers deferred dormancy by eleven minutes with consent.”

“Record fatigue risk?”

“Recorded.”

Kera checked the traces. “Route one assumes repaired greenhouse seals.”

“Sera verified eighty-two percent complete.”

“It models the remaining roofs as empty.”

“They should be.”

“Should does not certify evacuation.”

Jalen called the cooperative. Sera answered from a roof crew and checked the remaining structures by name. One held tools, one stood empty, and one sheltered two workers during seal tests. She ordered them out before approving the rain route.

That small correction delayed the forecast by six minutes. Nobody called the delay waste.

While they waited, Niko opened a private witness line. “Annual relationship review.”

Kera leaned closer to the reader. “We do those now?”

“We said present agreement.”

“We did.”

“Then inherited language cannot run unattended.”

Omi muted her channel with exaggerated care, leaving the indicator lit so they knew she had not disappeared.

Niko began. “I do not ask you to treat me as the boy who died. I do not ask you to forget that we shared a childhood before the branch. I want you as a physical witness, an argument partner, and someone who answers when I request contact within agreed hours.”

Kera noticed the asymmetry in his list. He offered no familial title. The absence did not erase affection. It prevented affection from claiming jurisdiction.

“I want you as a branch witness,” she said, “and as someone whose history intersects mine without belonging to it. I will answer within agreed hours unless an

audit conflict is recorded. I will not test you with childhood memory unless you consent."

"You may ask about the childhood we both retain."

"Not as proof of who you are."

"Agreed."

Kera hesitated. "I still remember the Saturday."

"I know."

"I will not narrate it back unless you request."

"I do not request."

The refusal hurt with less confusion than it would have a year ago. She entered it into their private key. Memory in one person did not create a right to install it in another.

Niko added, "I do request that you tell me when the kitchen seal finally gets repaired. Apparently everyone has discussed it for a year."

"It still leaks."

"Then continuity survives."

Omi unmuted. "That joke should have stayed dormant."

Jalen's board chimed. Sera's evacuation confirmations had arrived. Ivo approved the respiratory range. The foundry accepted a two-day delay instead of spending oxygen on the even mist. Route one now carried physical signatures, branch witness notes, district harms, and a review time if roof pressure worsened.

Kera prepared to log the forecast. Tree 44's clock reached the scheduled ring interval.

A new band brightened beneath her hand precisely when civic time changed. No seventeen-minute lag opened. The ring reader displayed the chosen rain route, its confidence range, the names of physical operators, and the active branch advisers who had compared it. Beneath them appeared one line that no weather procedure required.

`FORECAST OFFERED. WILL YOU ANSWER?`

Kera checked the source. The invitation carried joint signatures from Jalen's board, Sera's cooperative, Ivo's clinic model, Niko's active group, and Omi's refuge council. No plea hid inside instrument noise. No unregistered state had learned to knock where nobody meant to listen.

"What answer does it expect?" she asked.

Omi replied, "Not a predetermined one."

Kera placed two fingers on the pulse key. The old childhood code waited in muscle memory, available but no longer necessary. She tapped the new interval she and Niko had made after the cyclone. Niko answered from his channel. Omi

added a third measure of her own.

Outside East Root Court, manual crews turned the rain vanes. The pale seam over the western ridge broke unevenly, leaving one district dry while drops began against Tree 44's scored leaves.

The new ring closed on time around their three distinct signals.